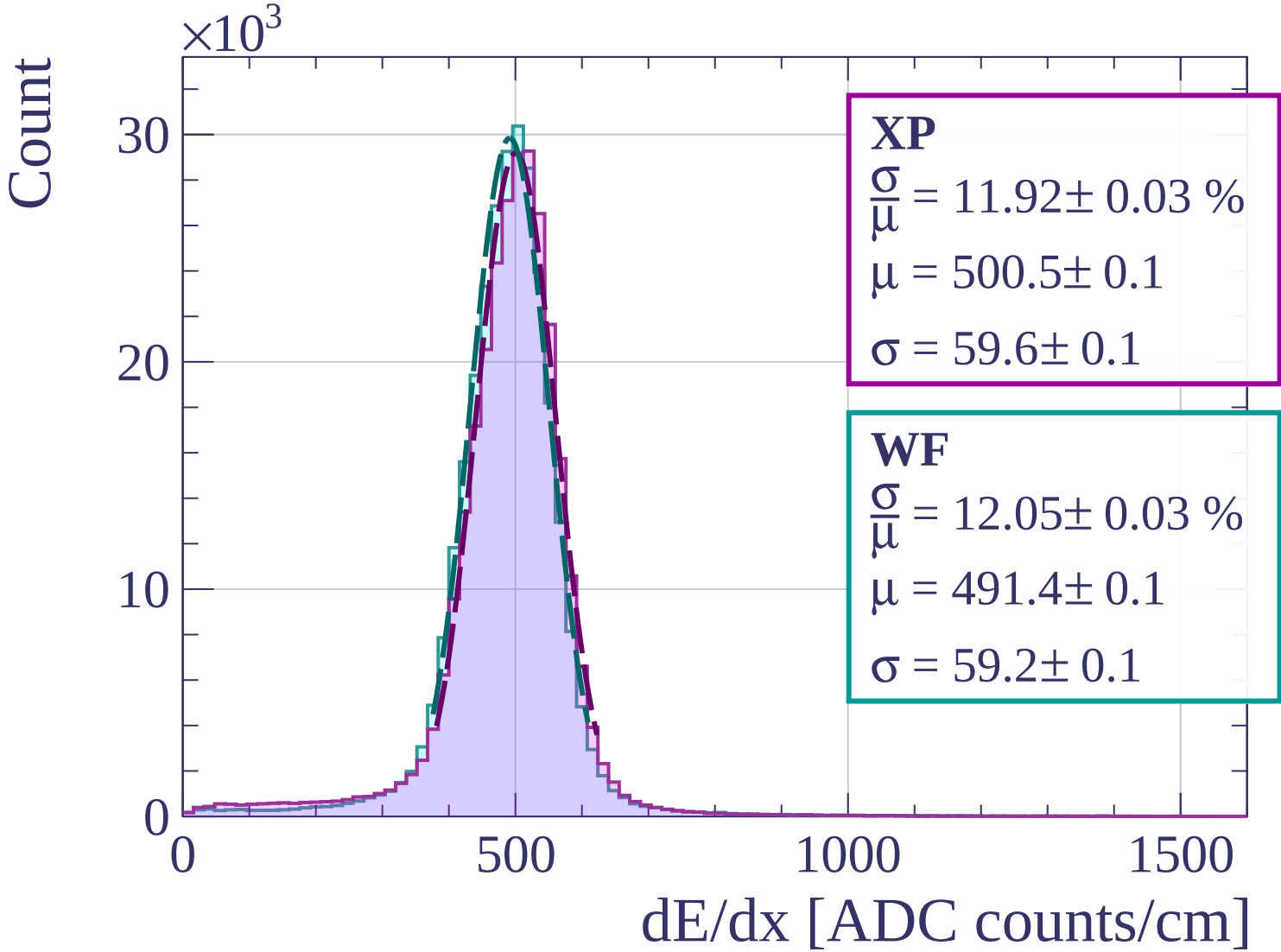
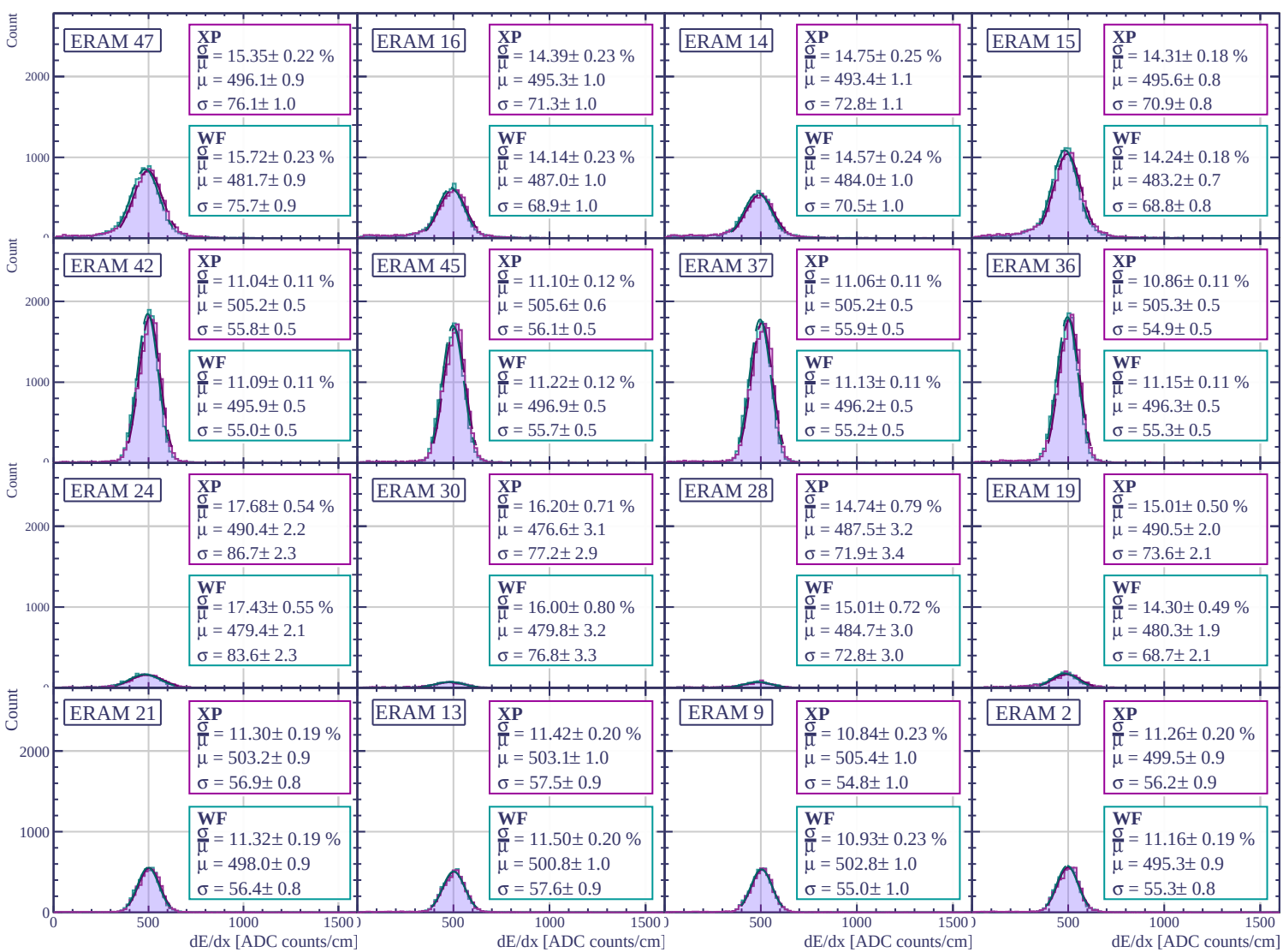
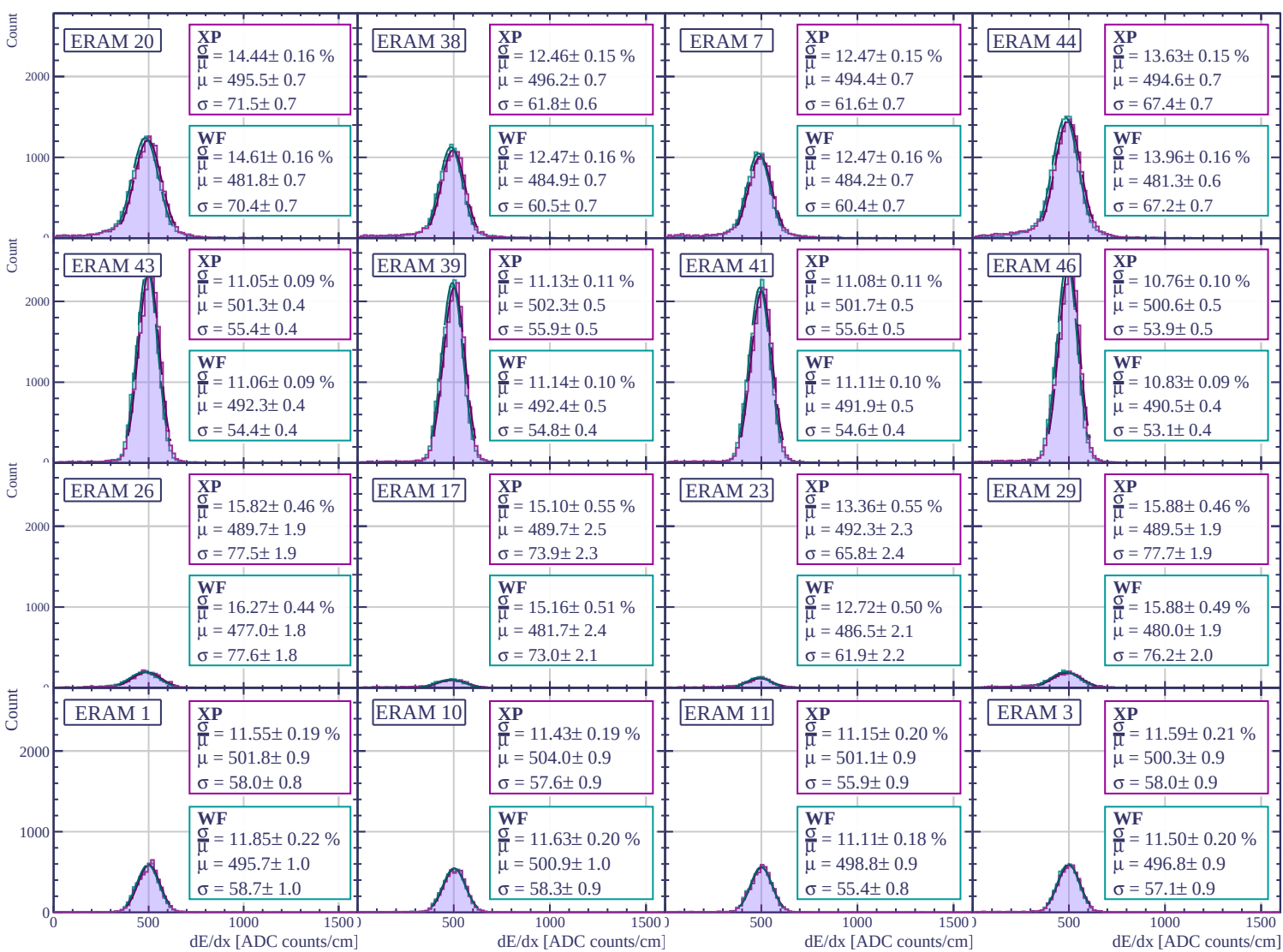
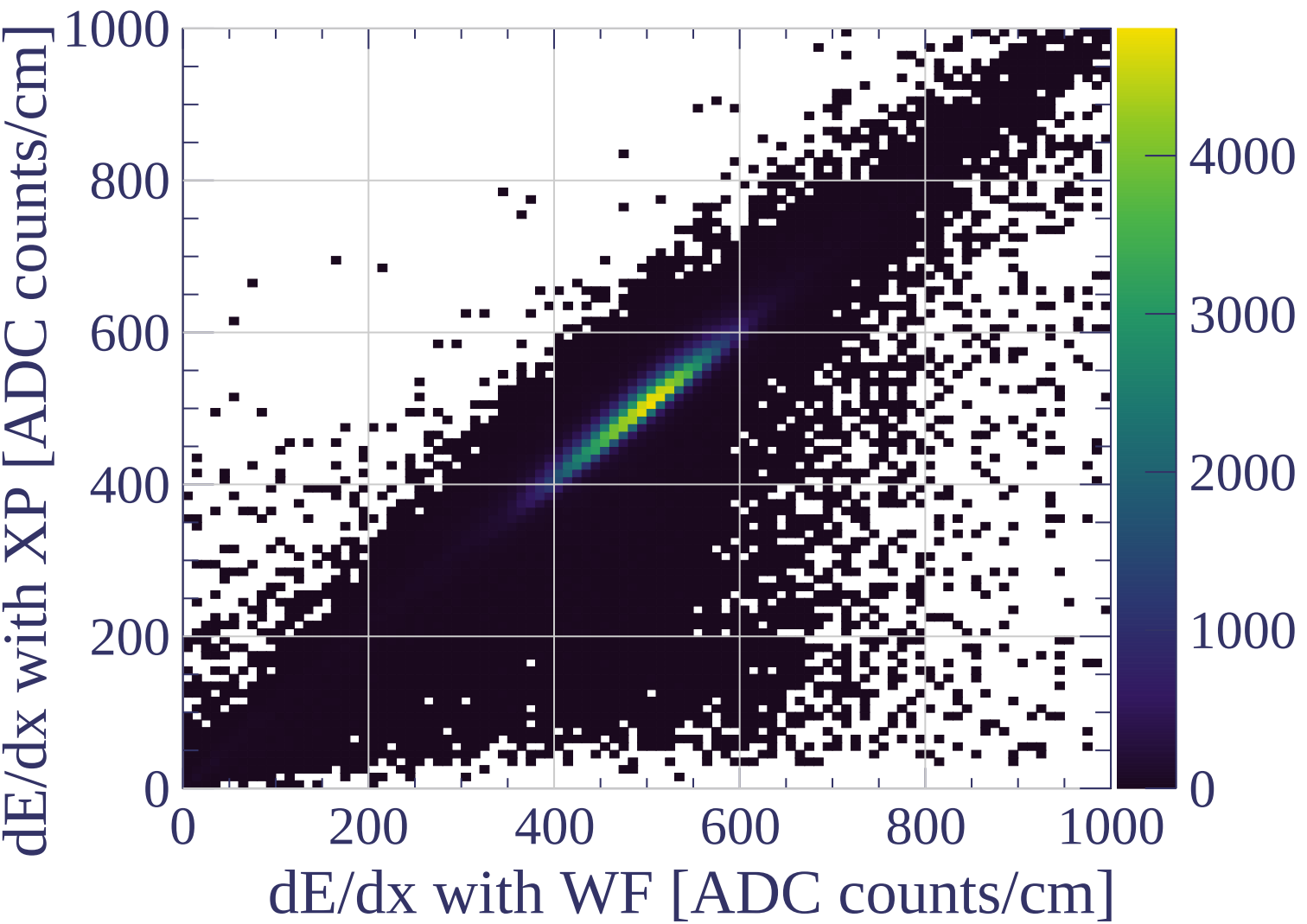
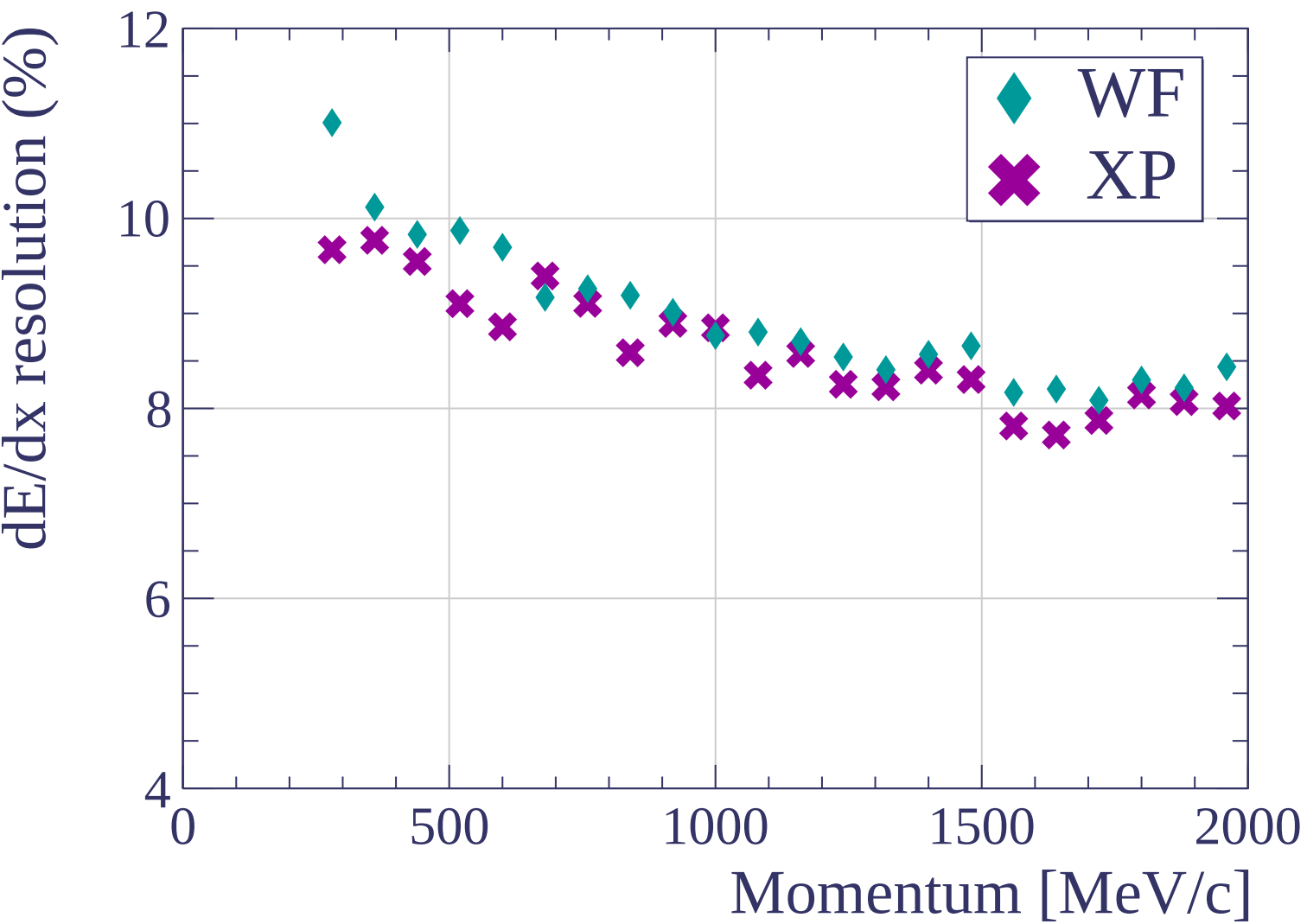


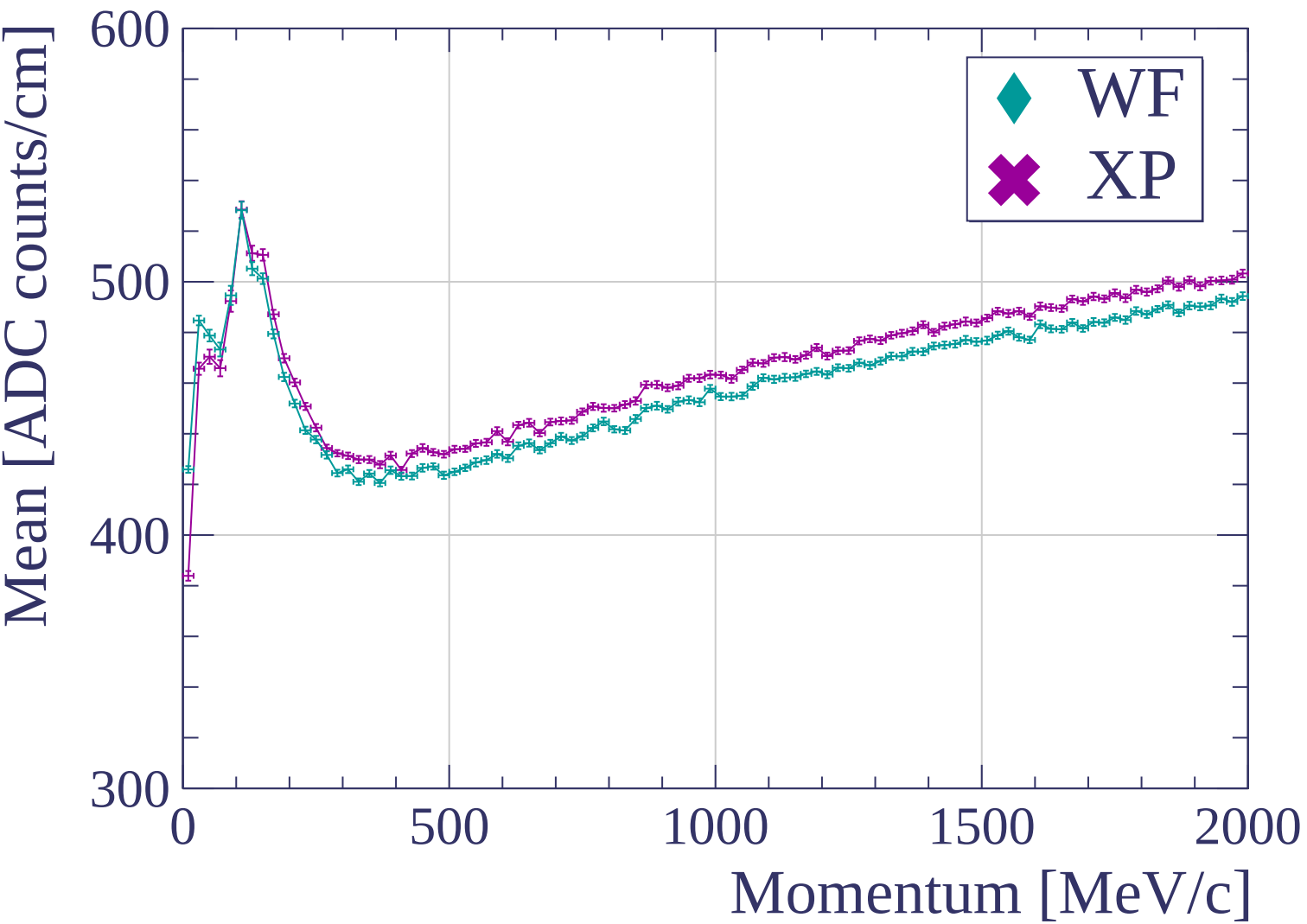


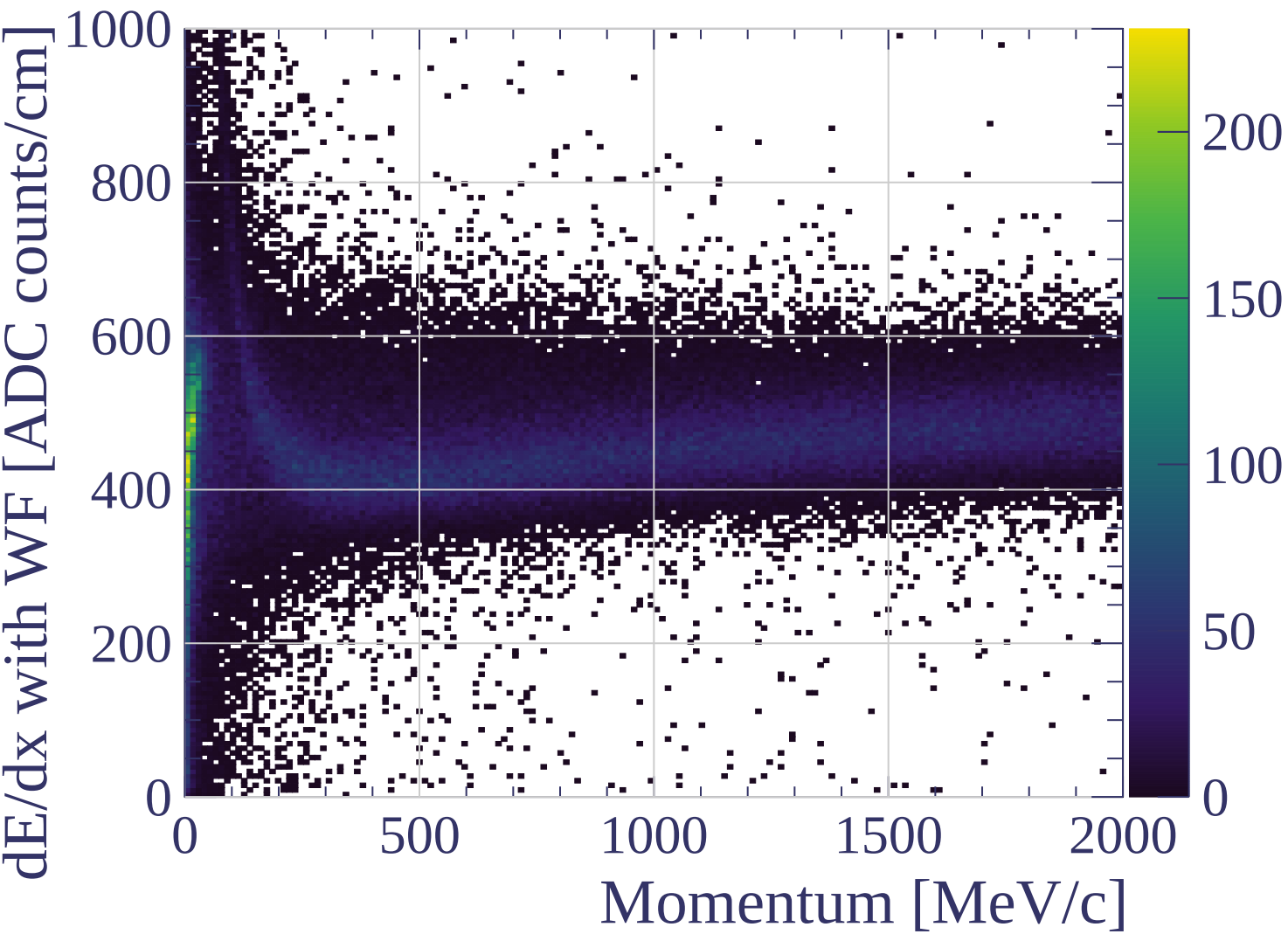


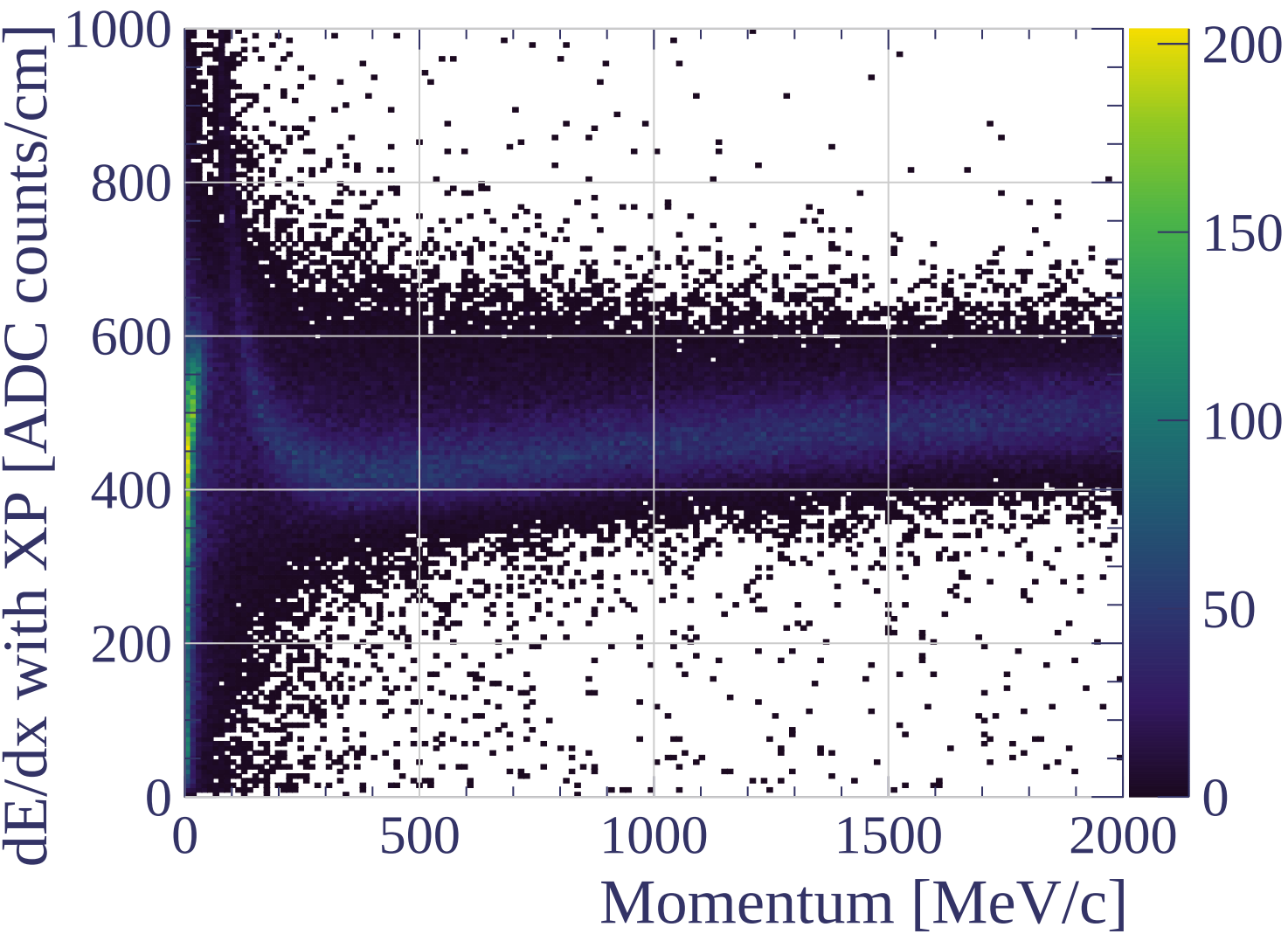




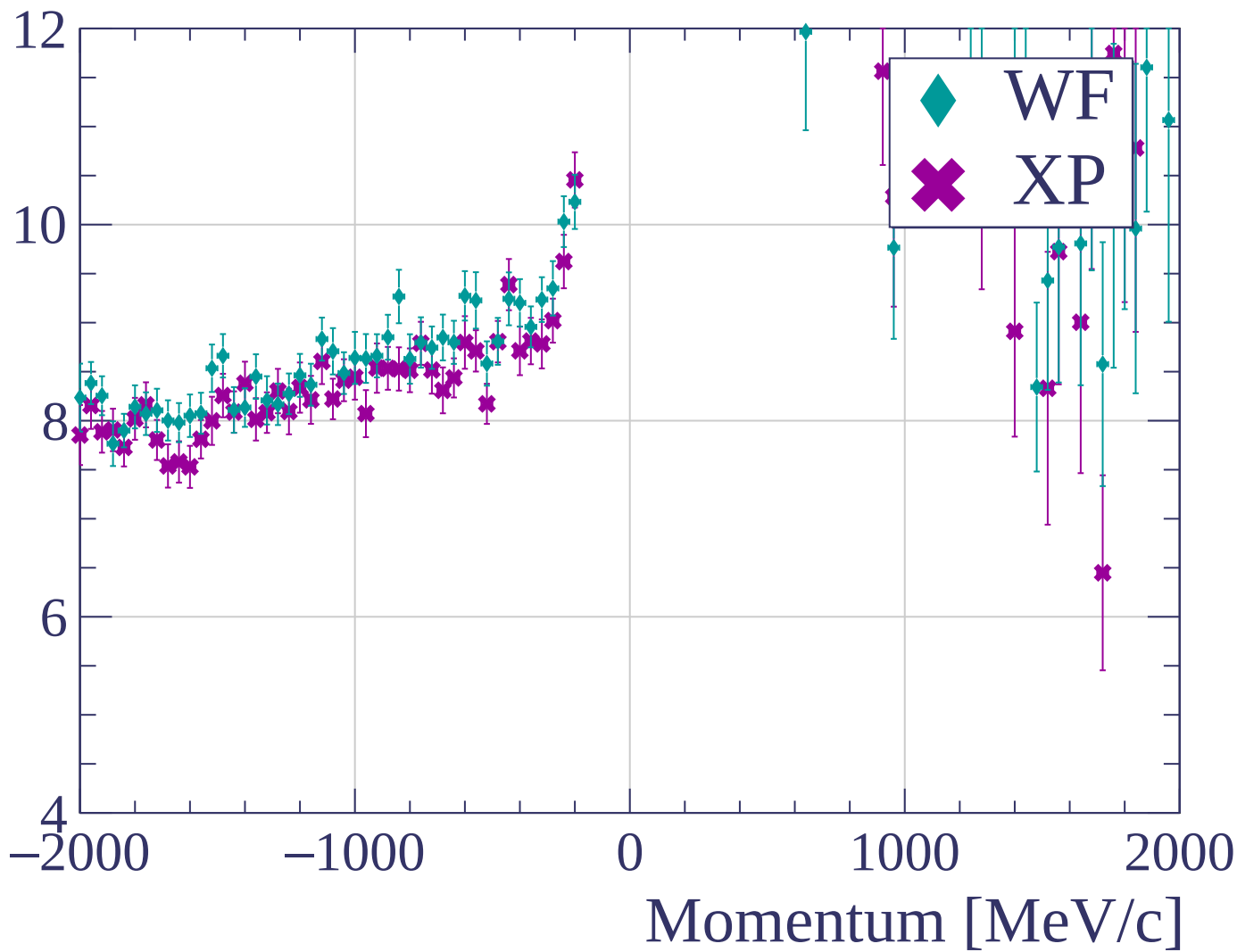


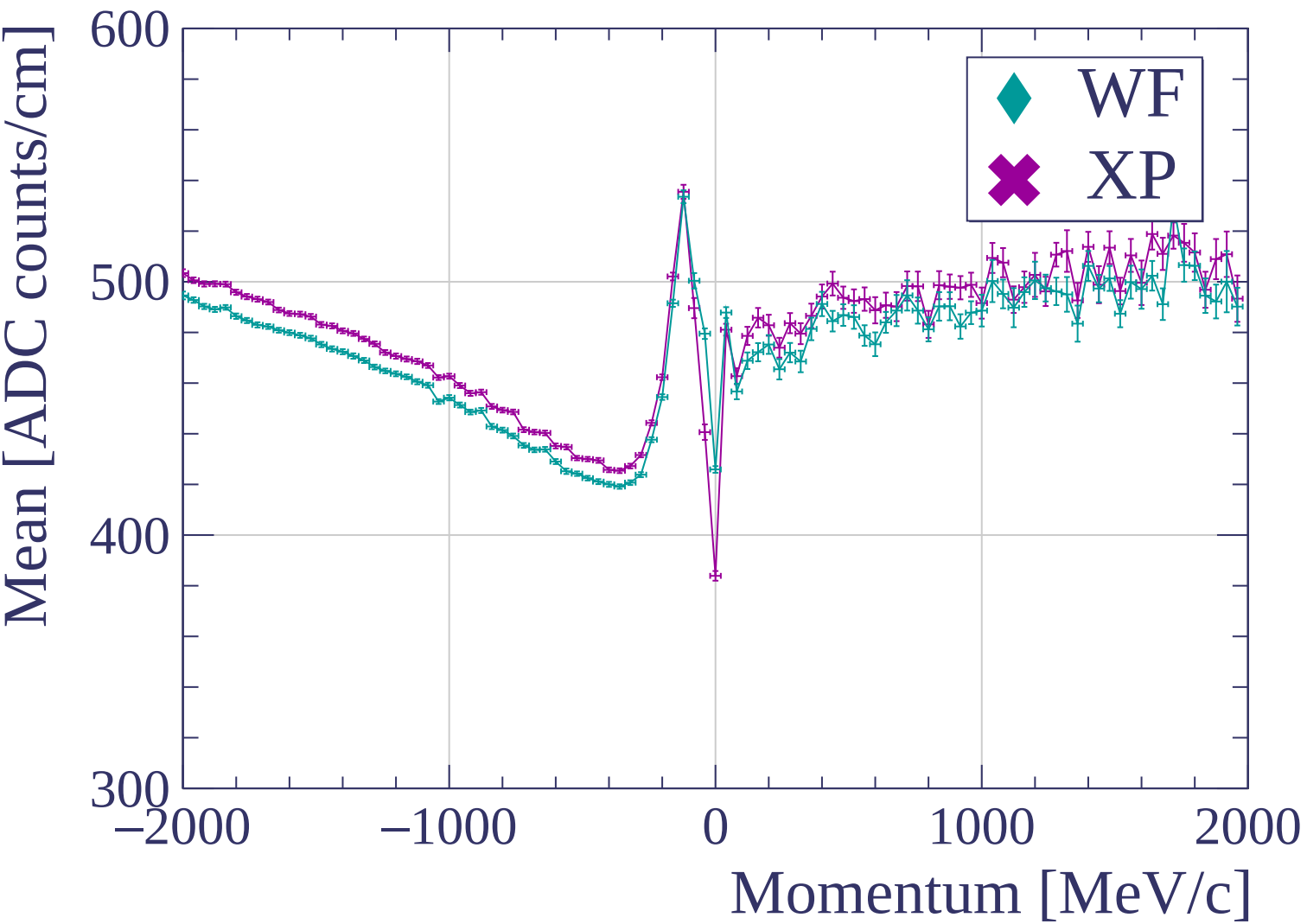


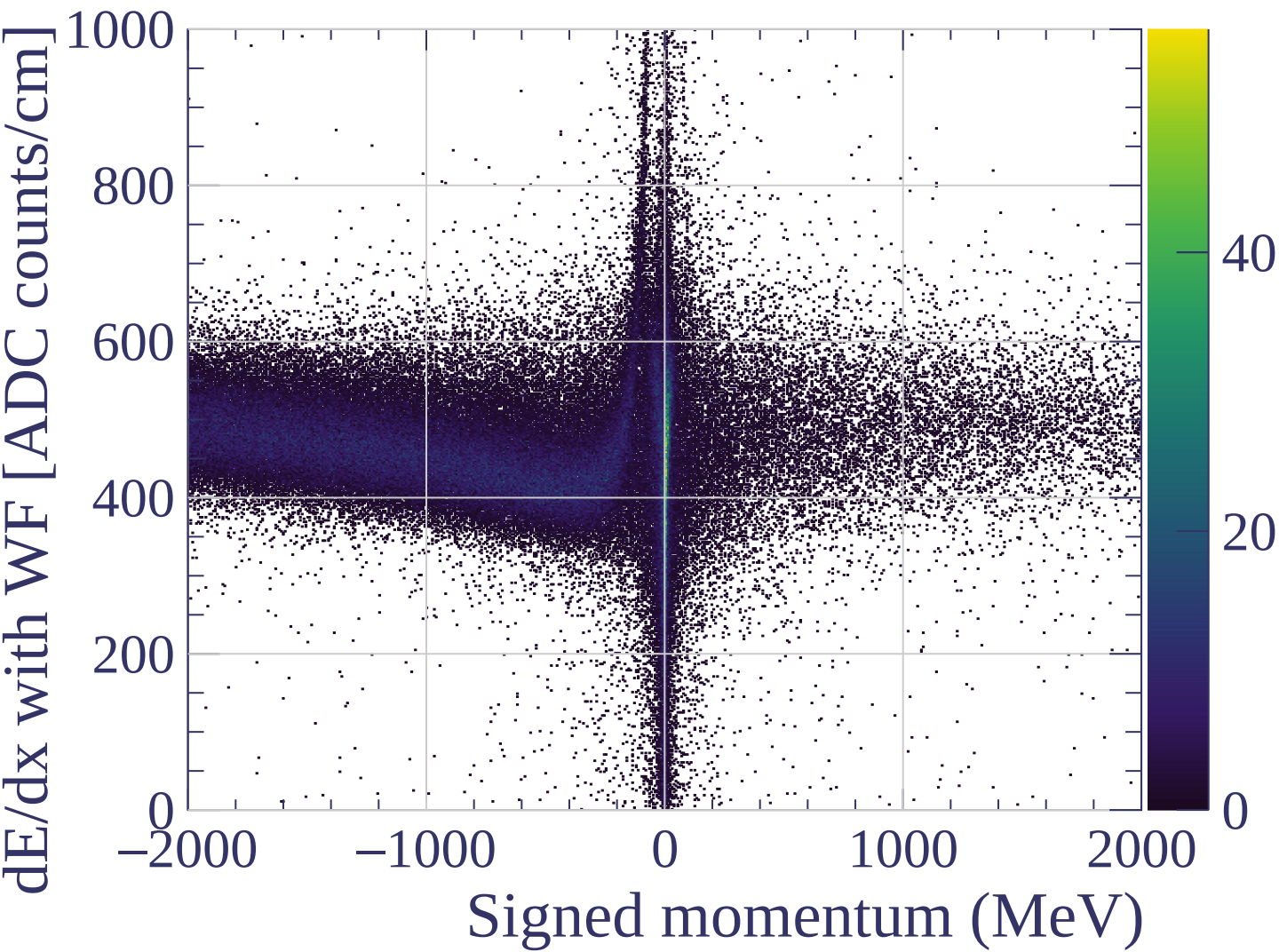


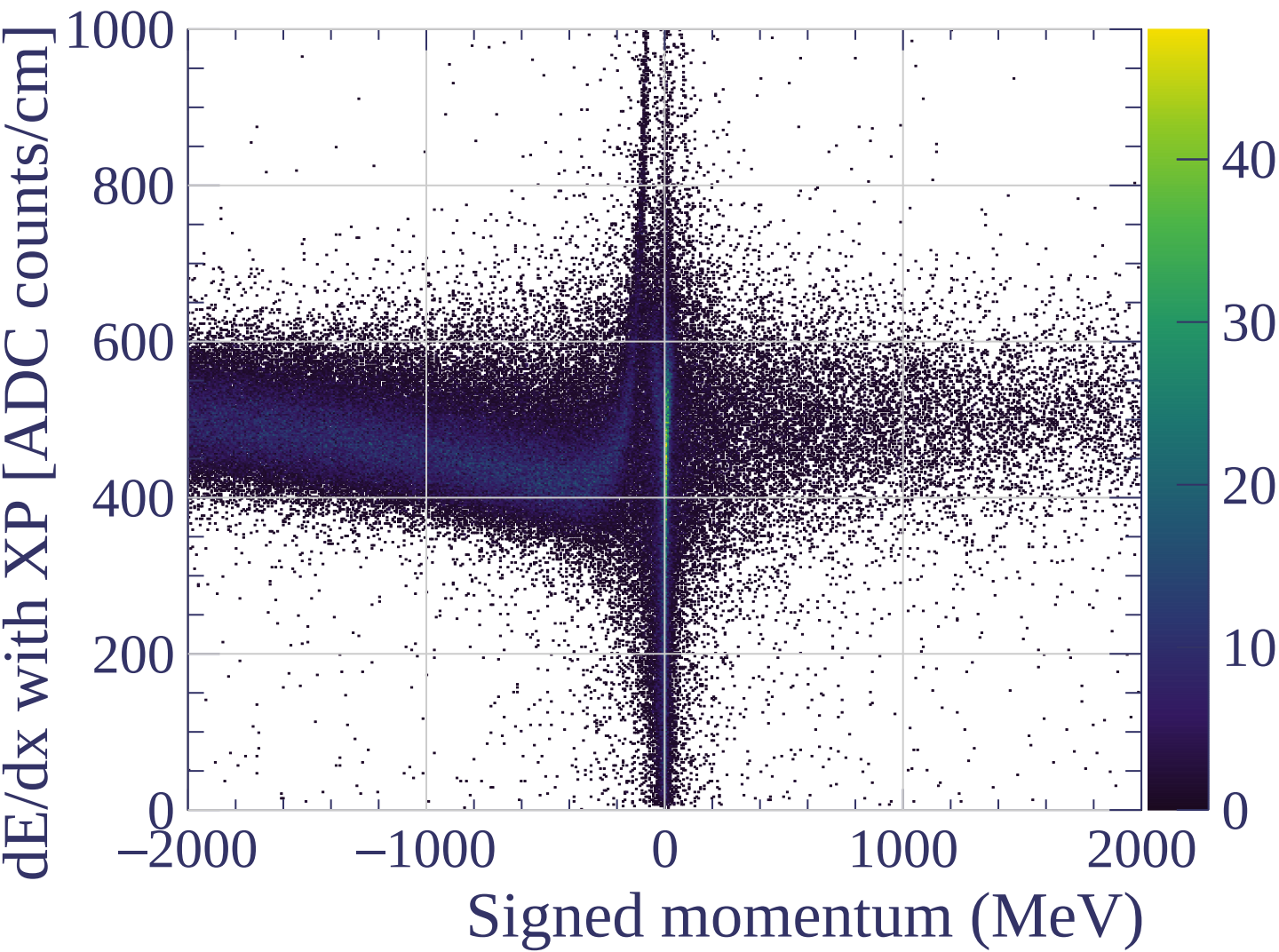


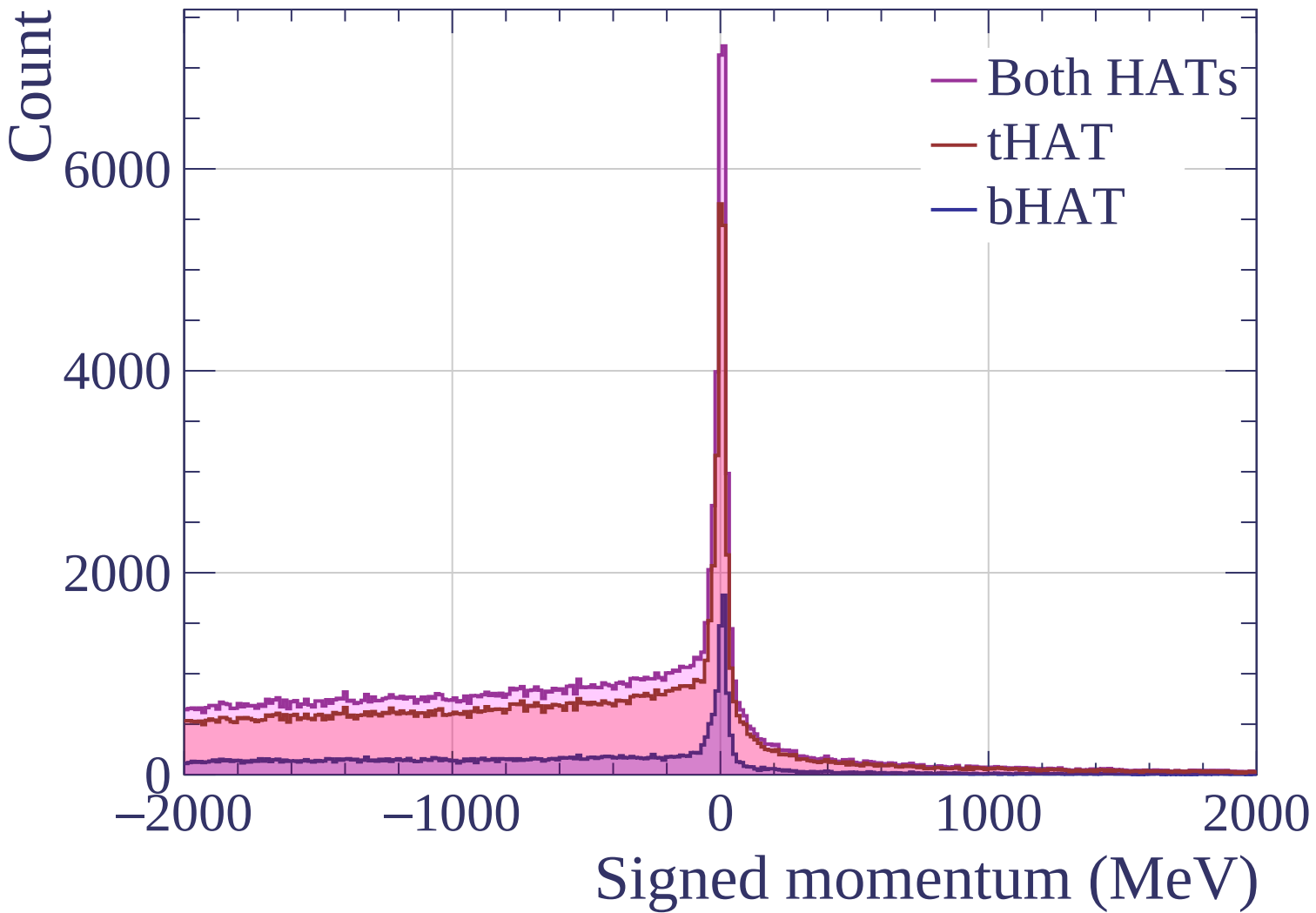
dE/dx resolution (%)

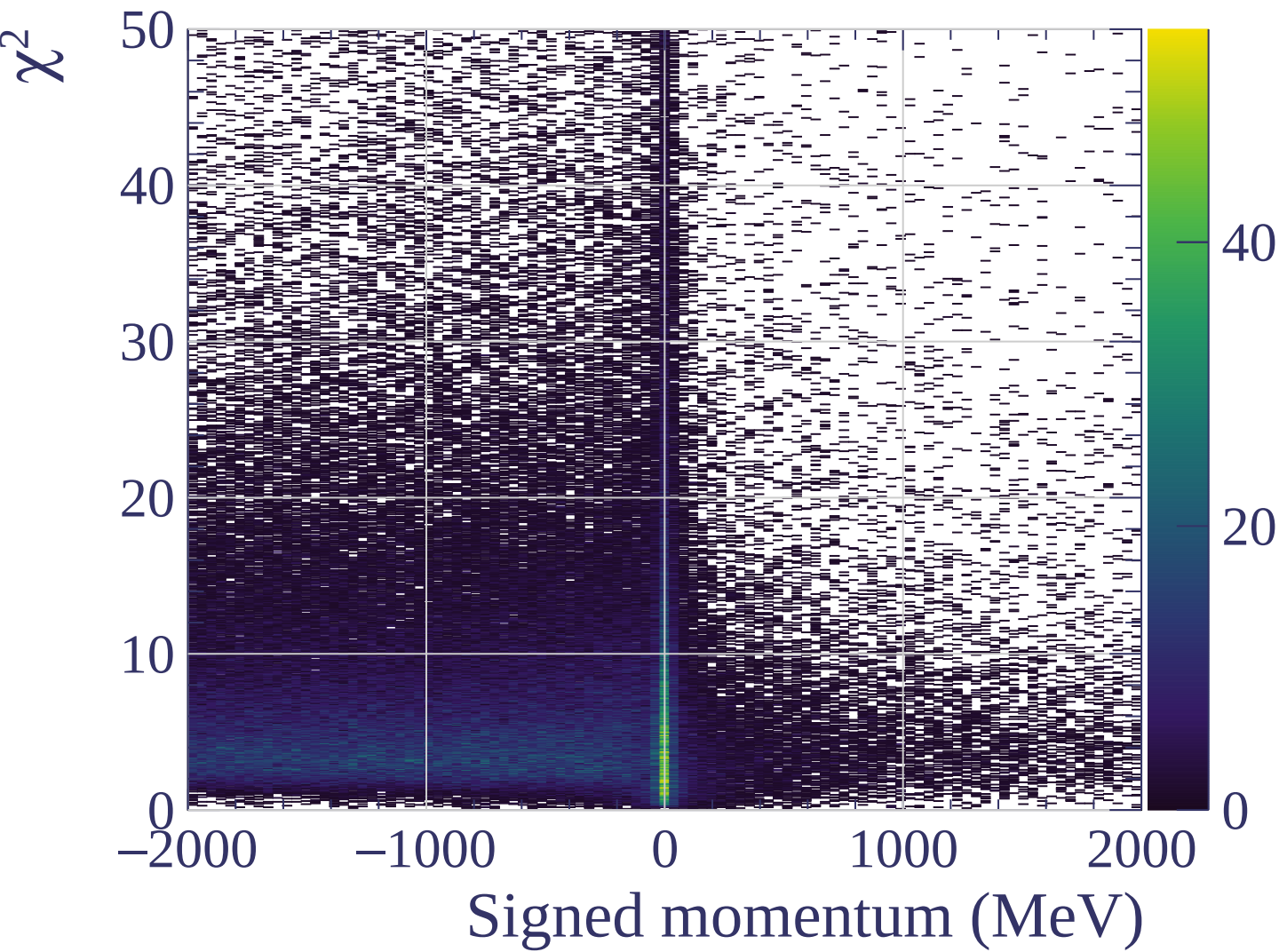


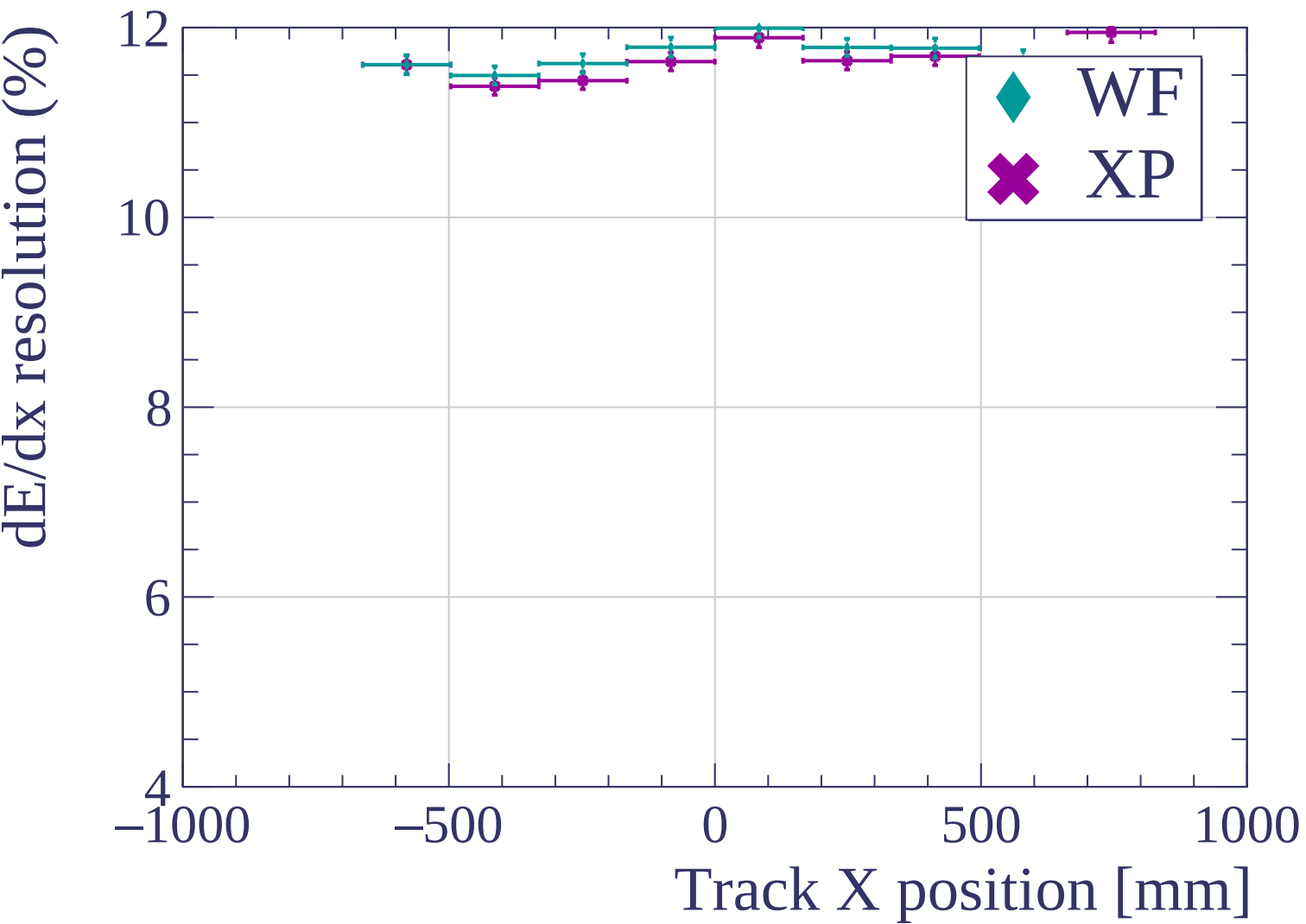


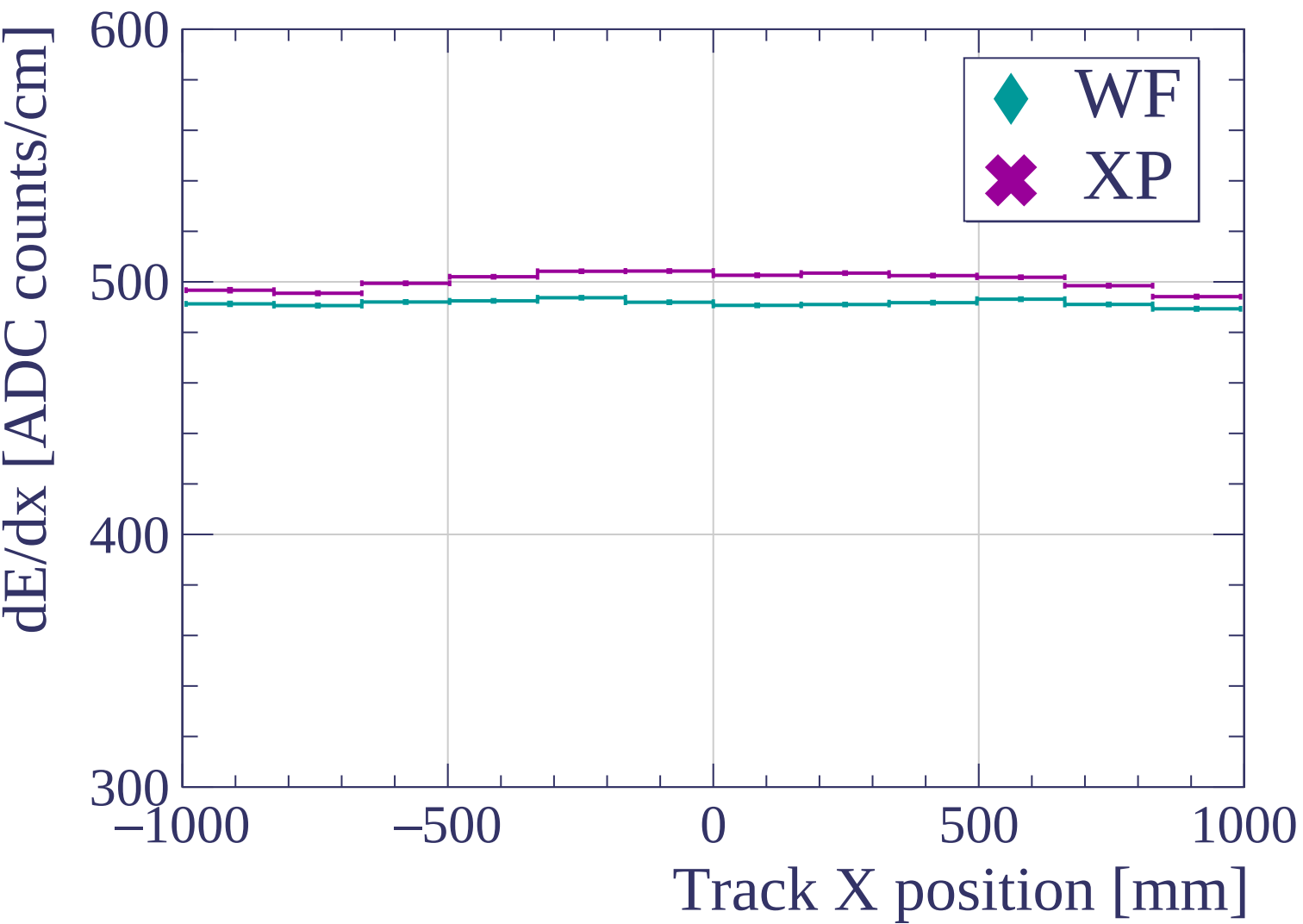


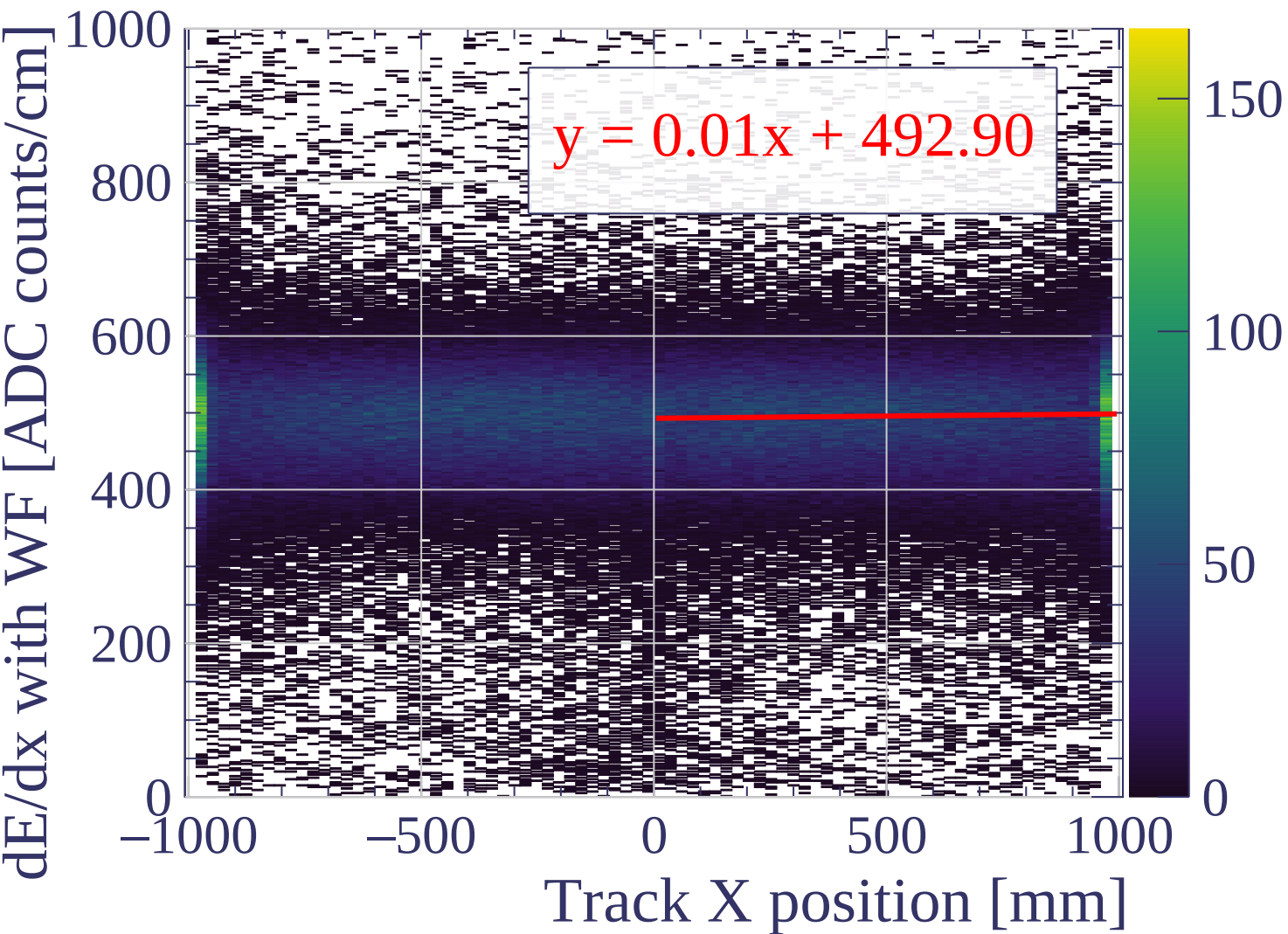


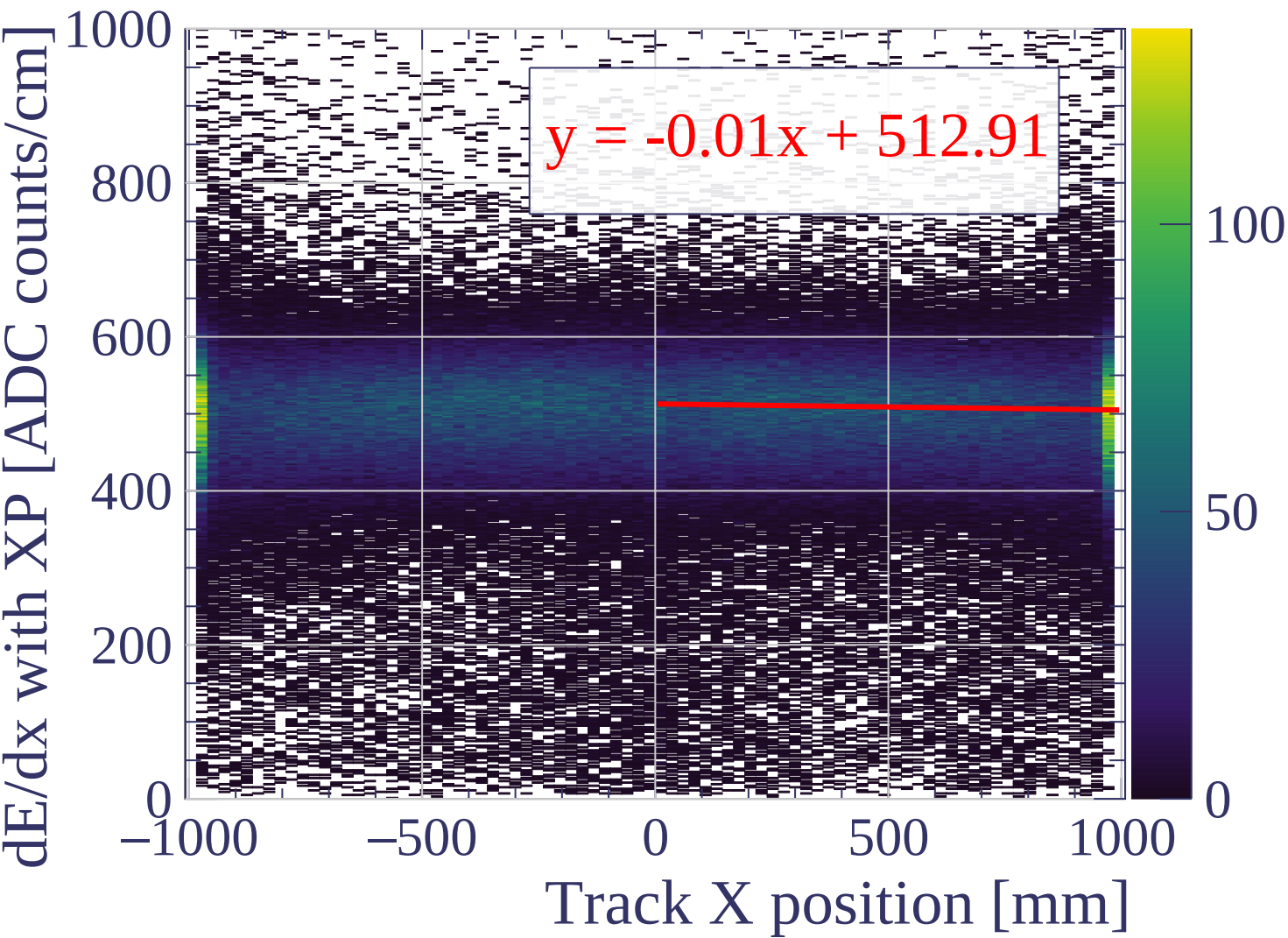


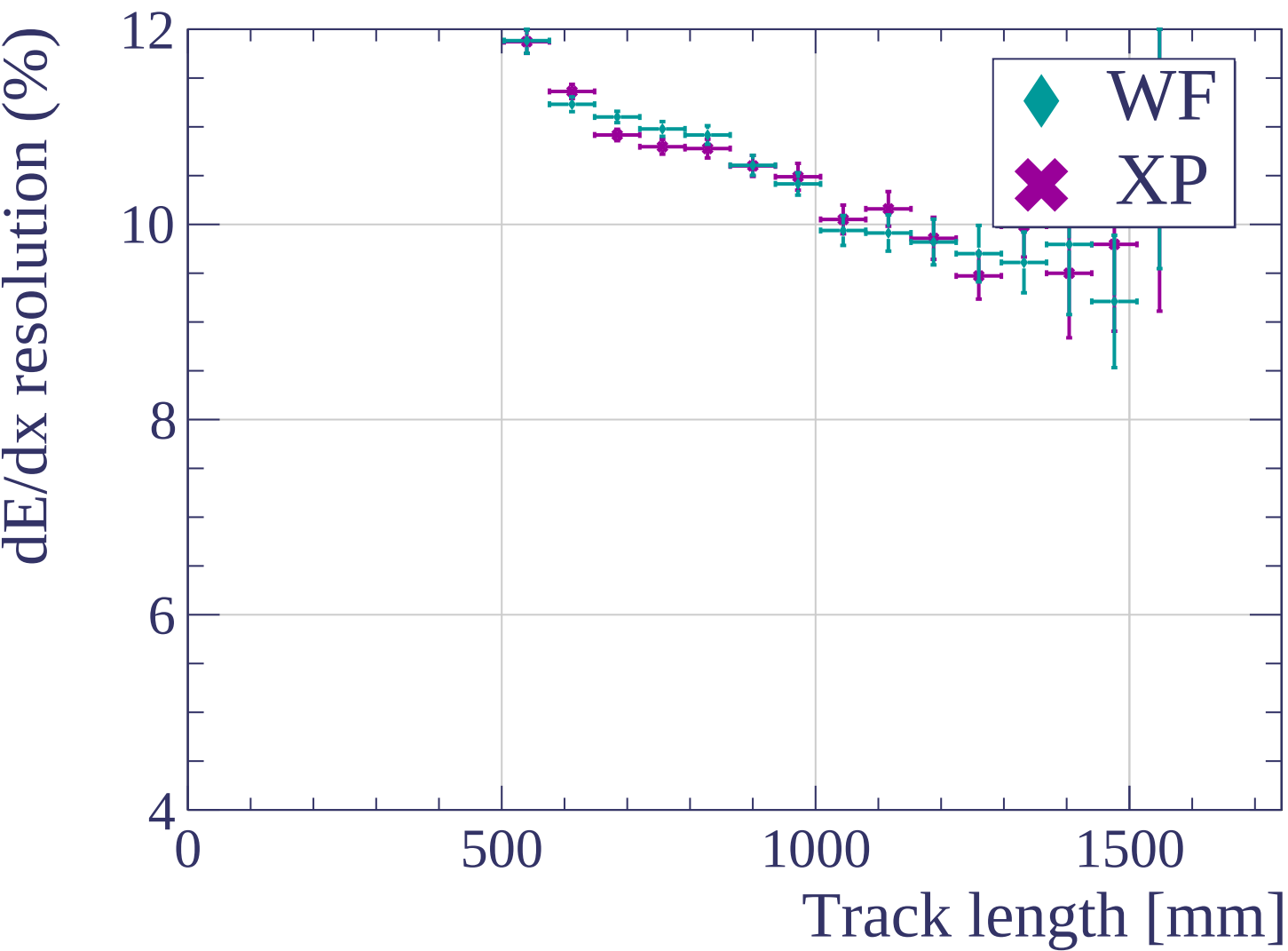


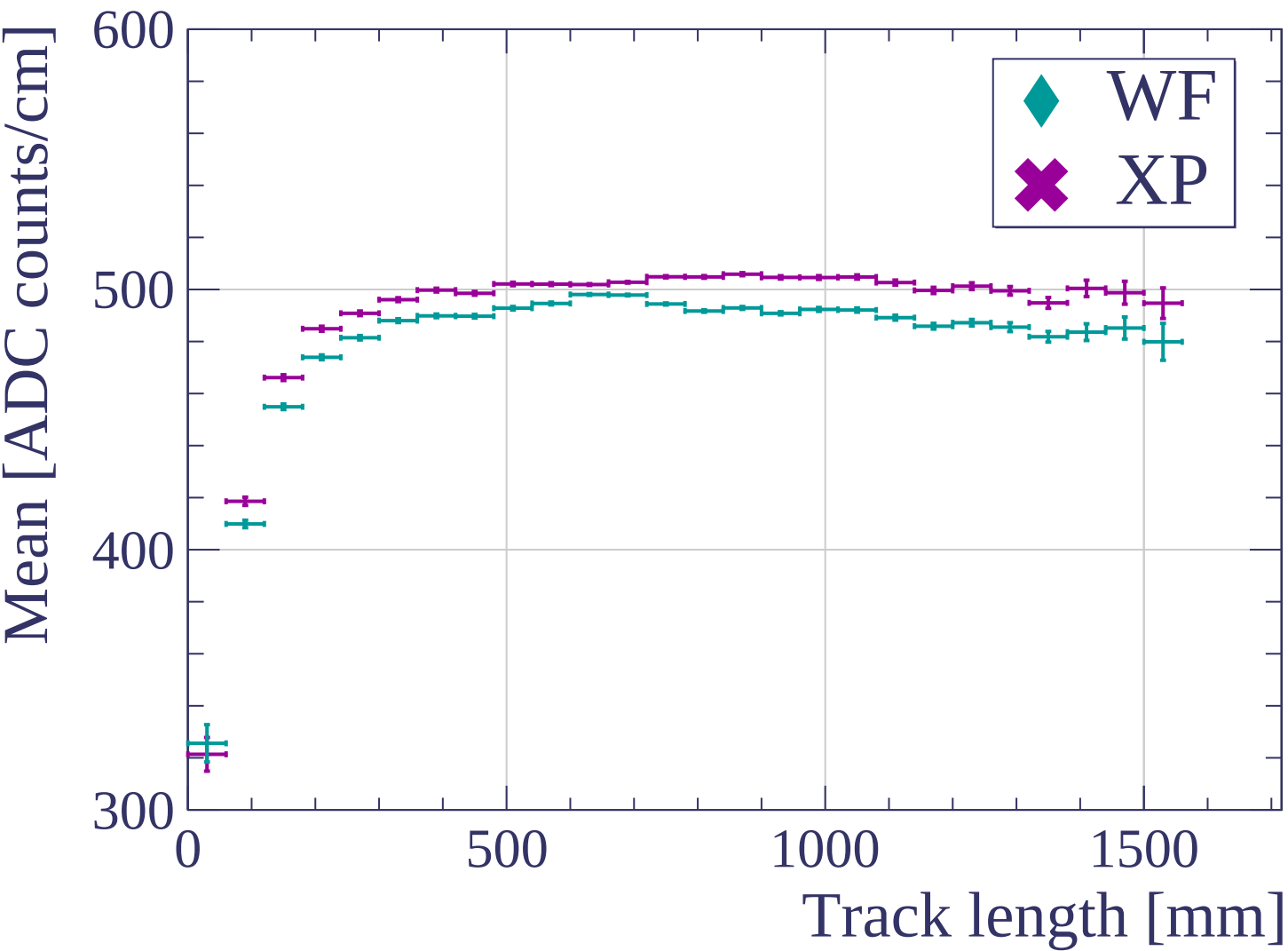


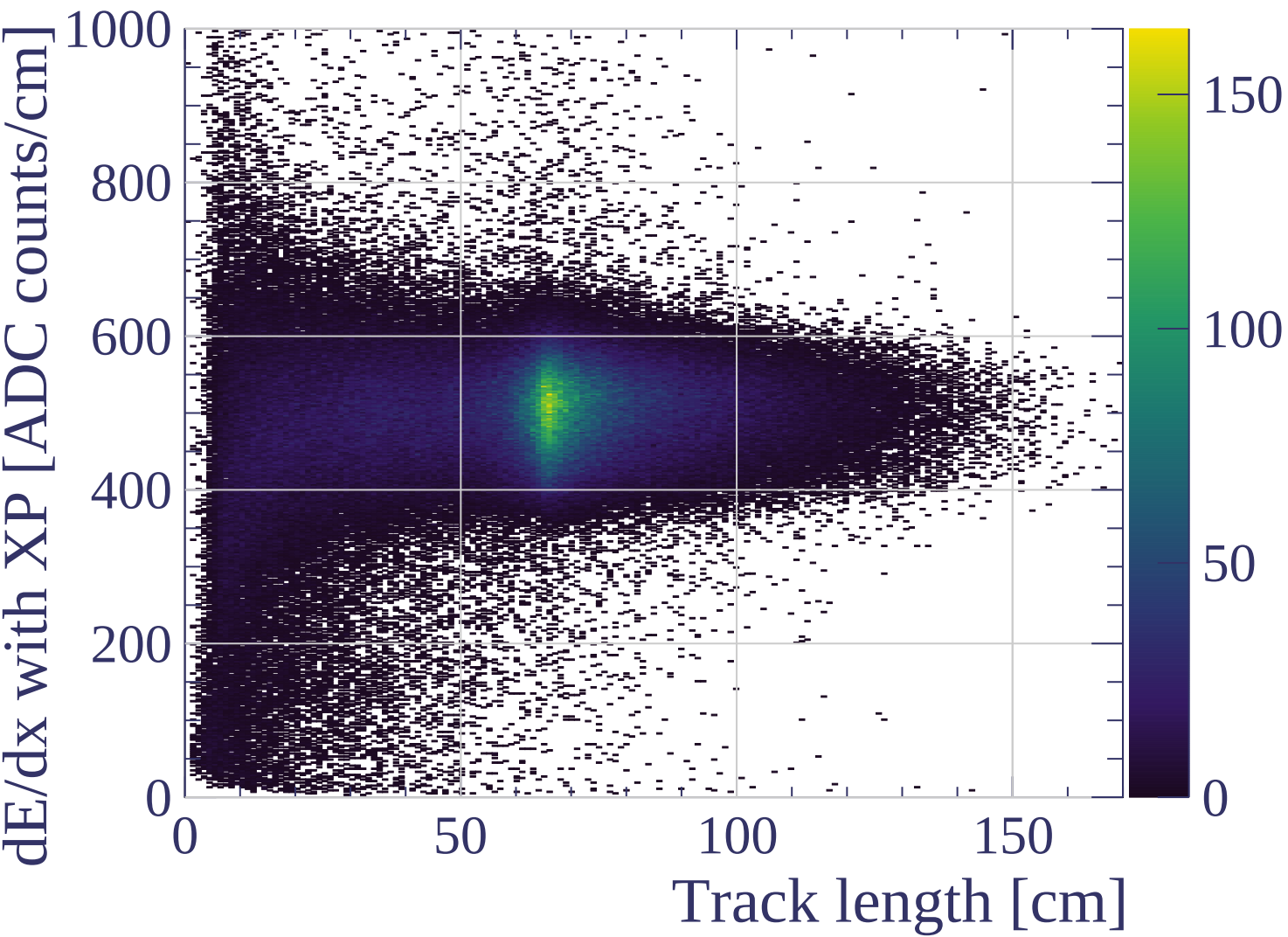


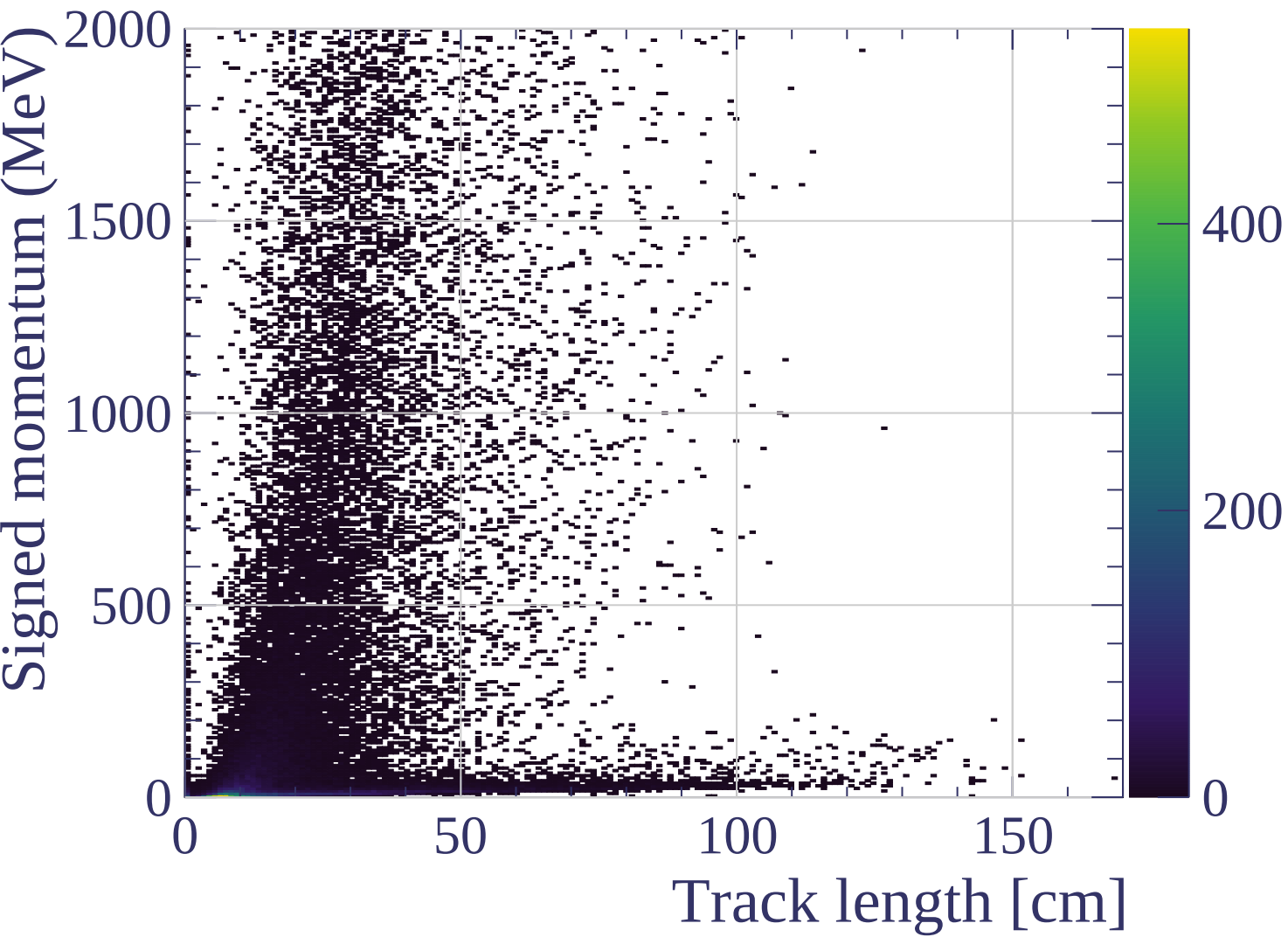


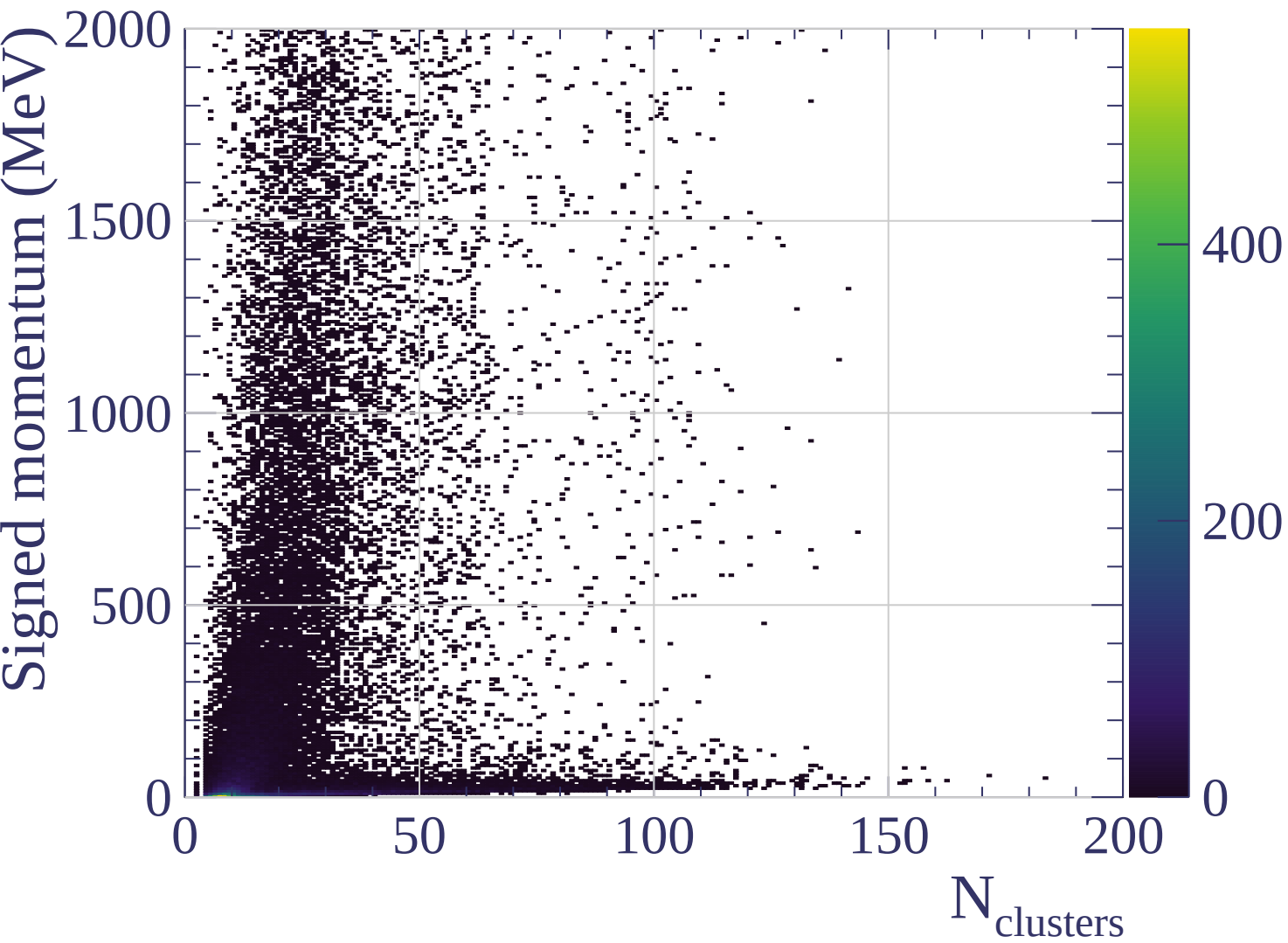


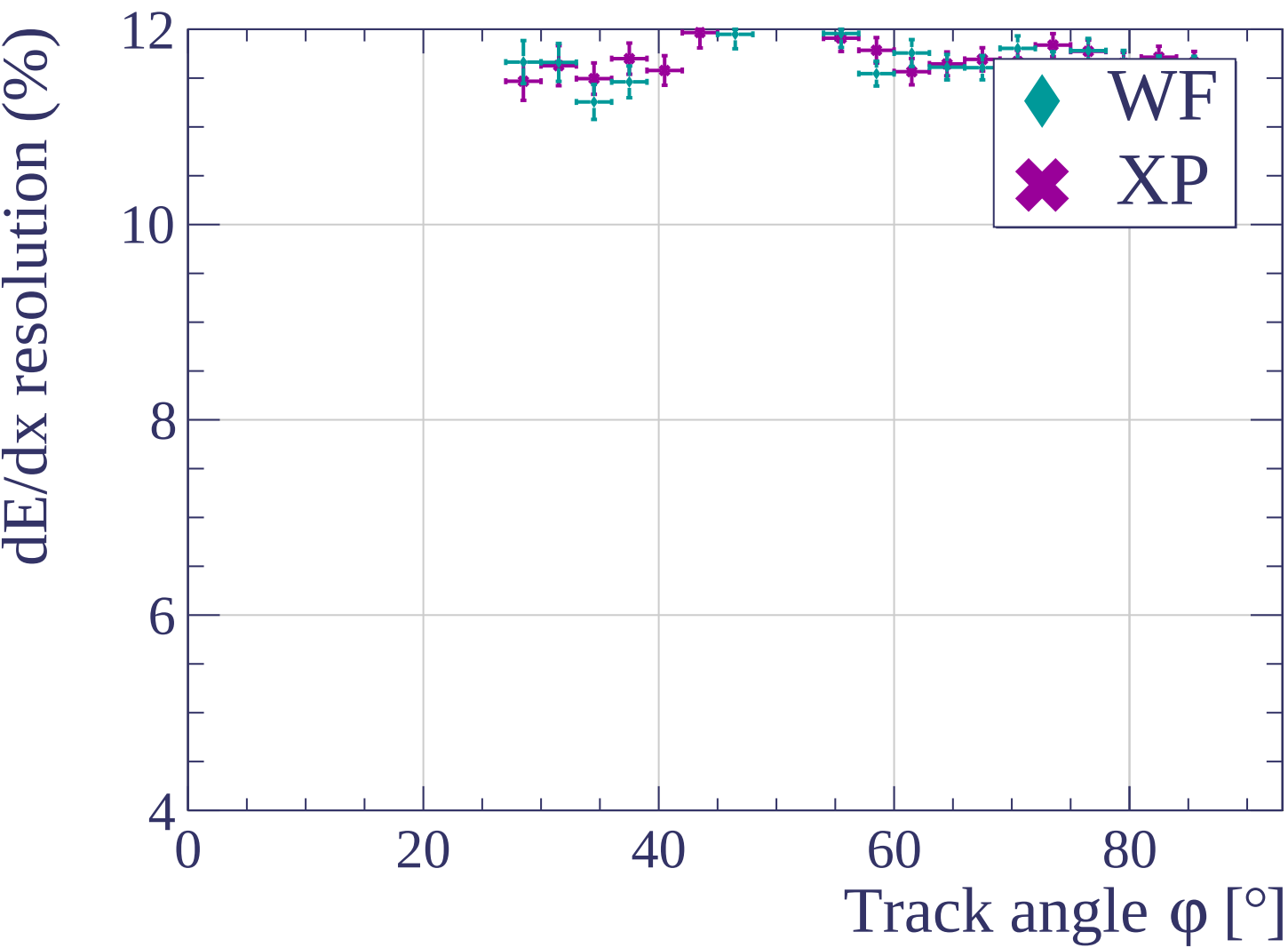


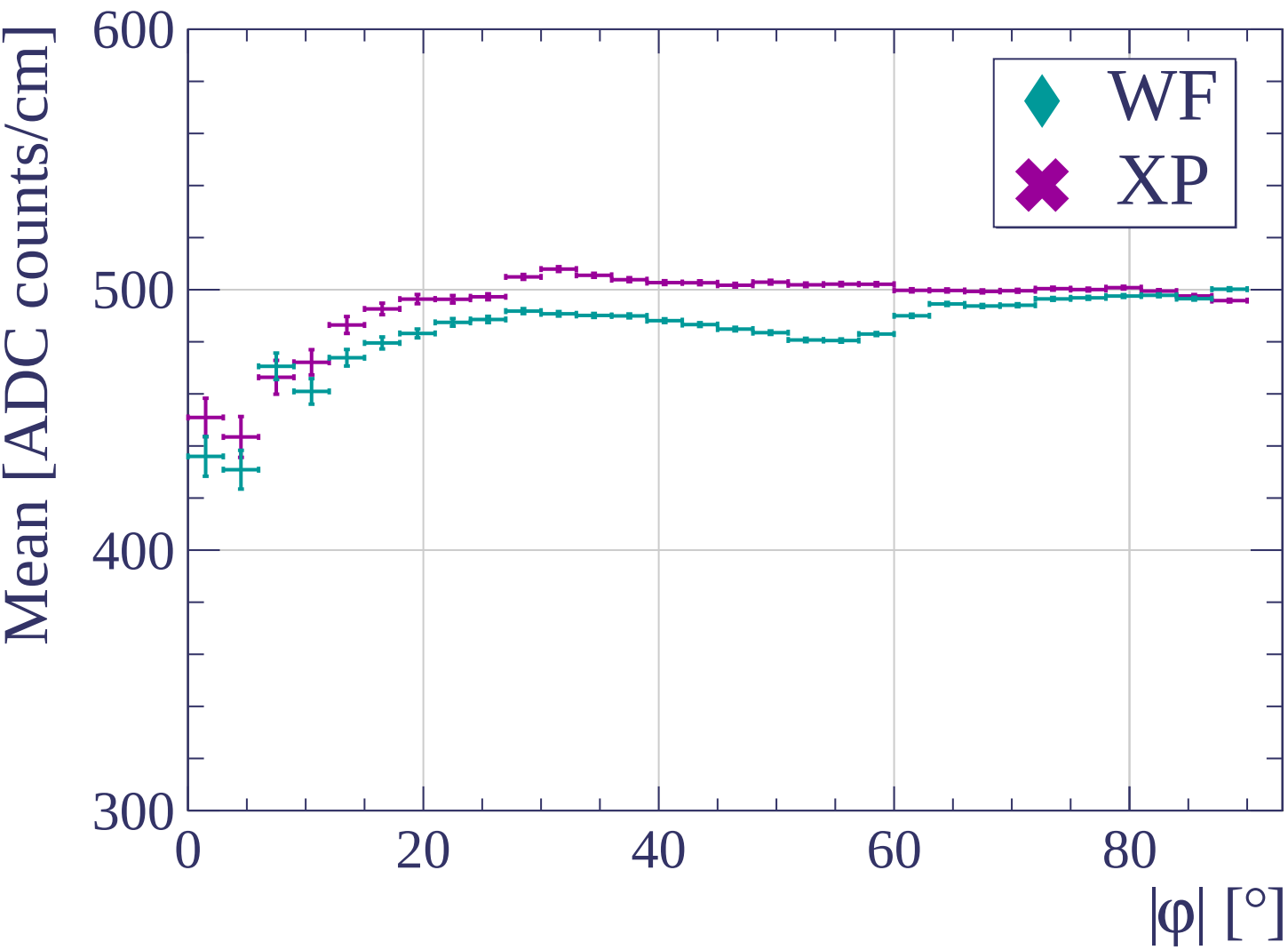




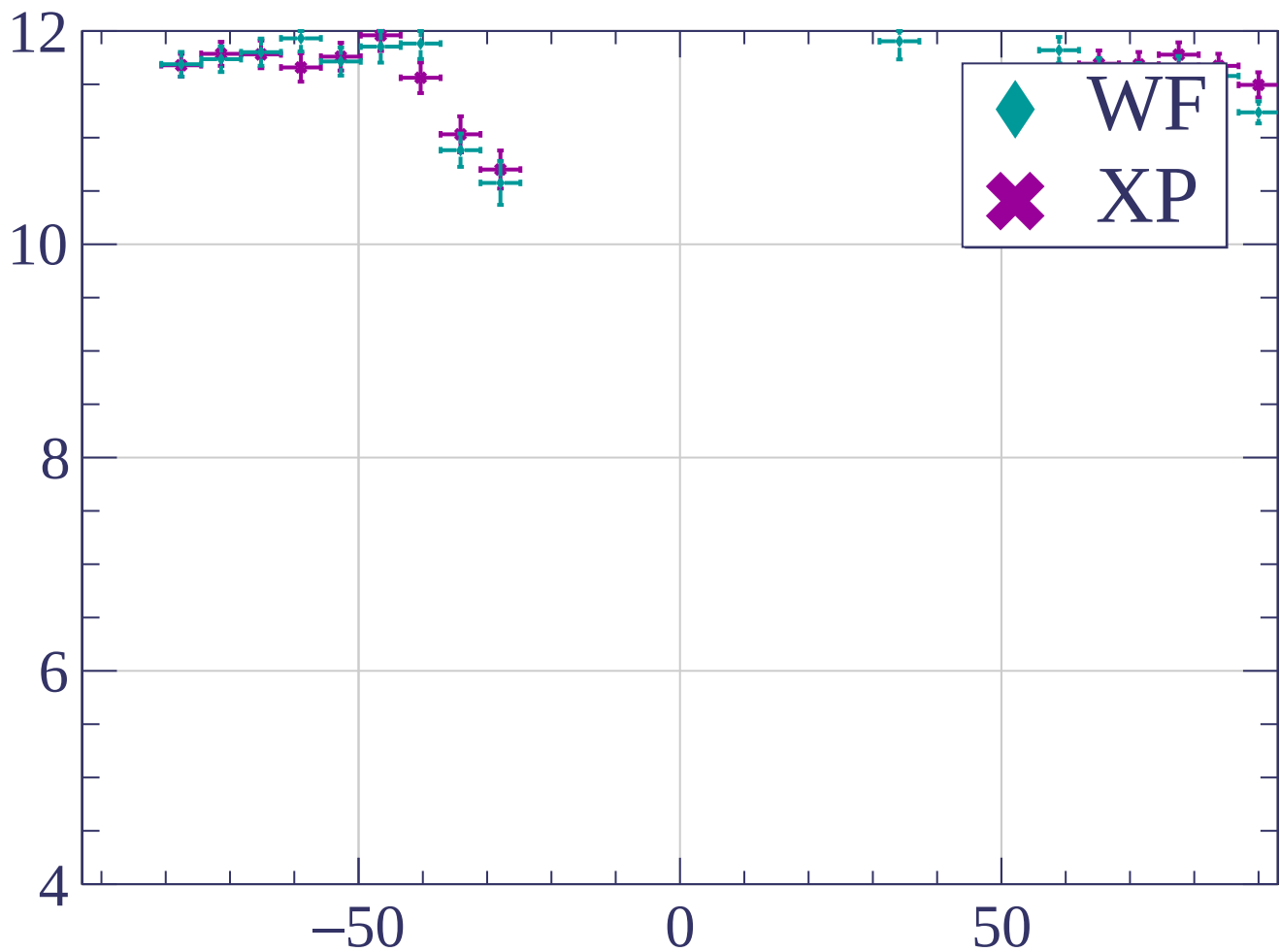




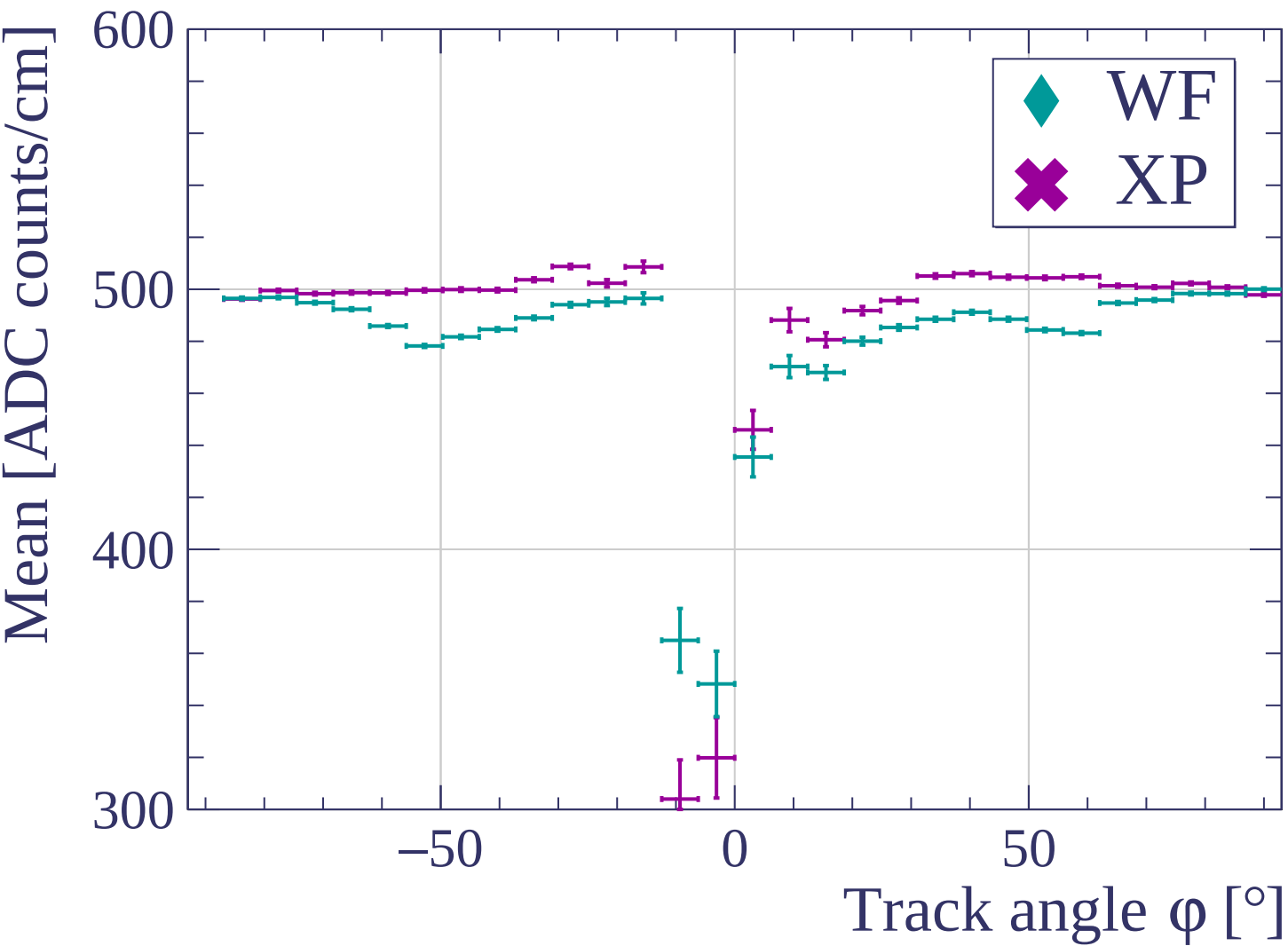


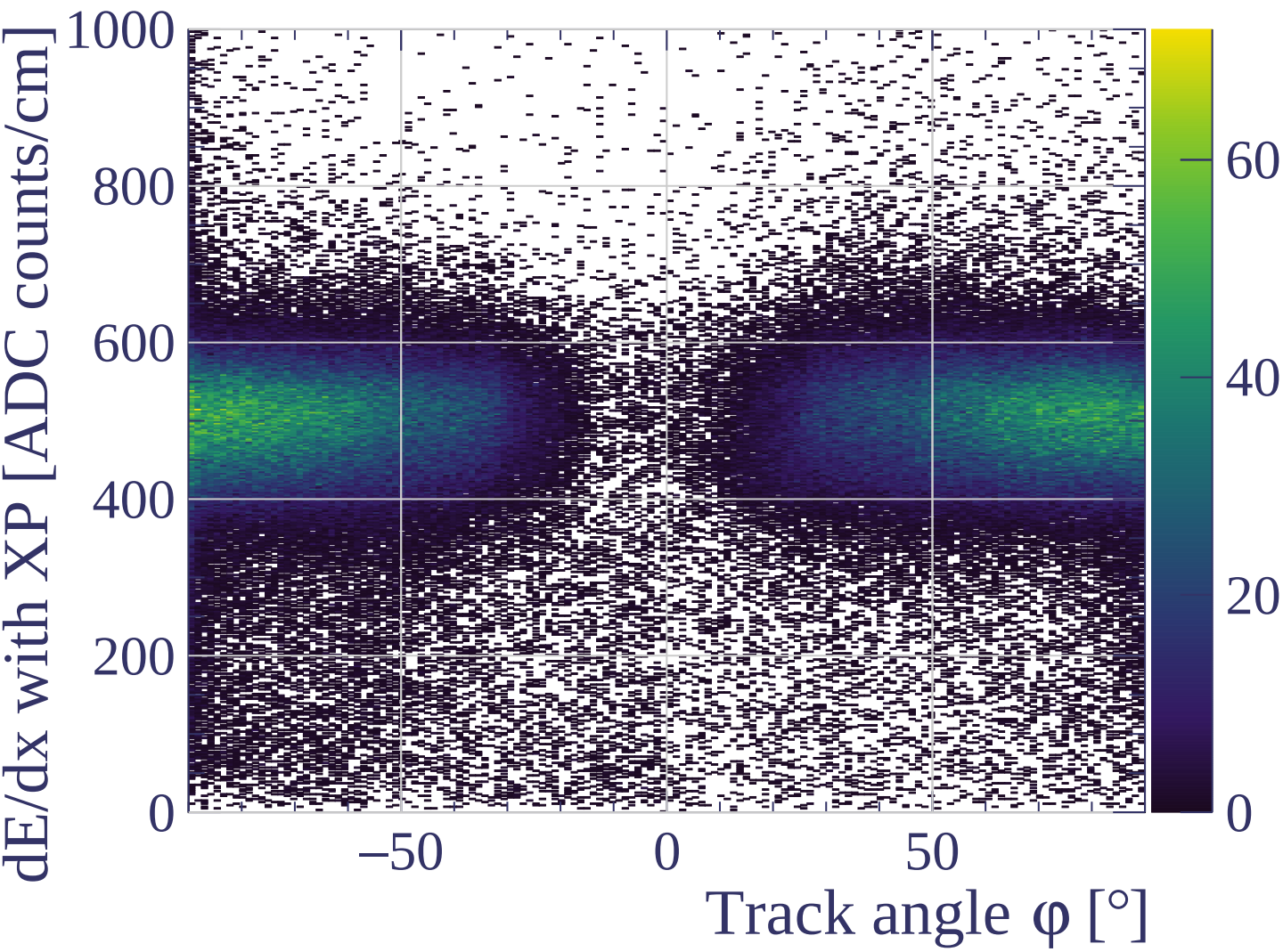


dE/dx resolution (%)

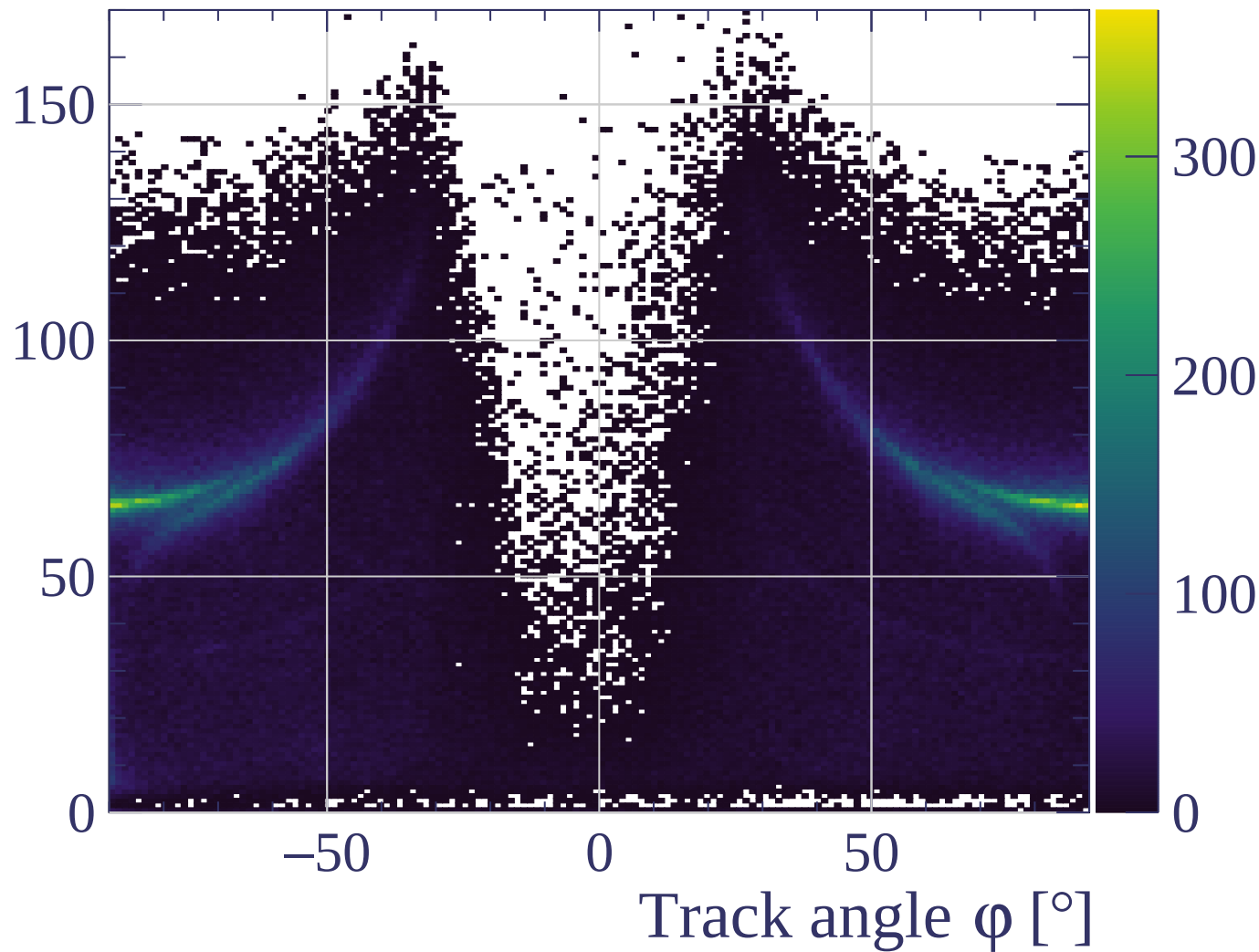


Track angle ϕ [°]

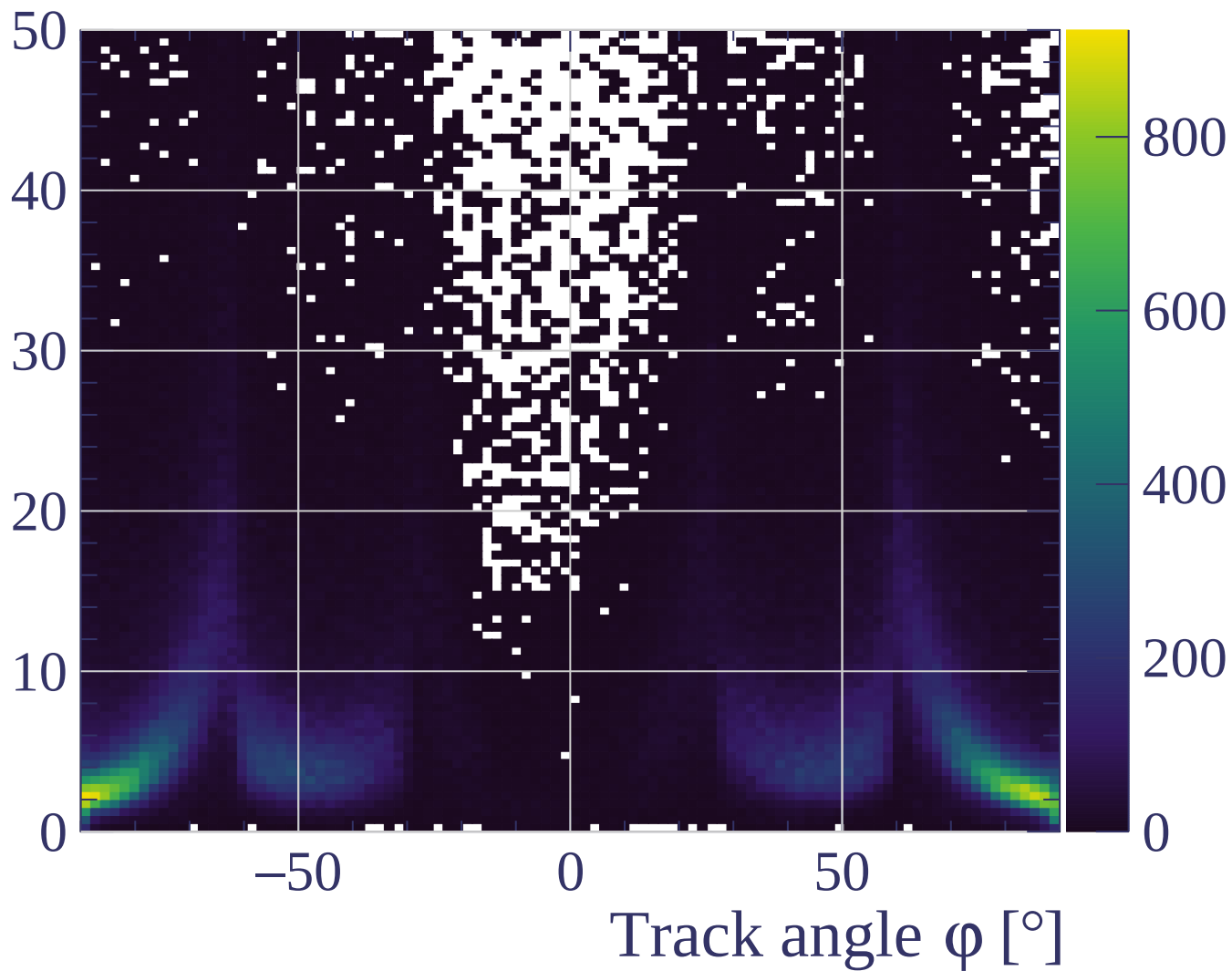


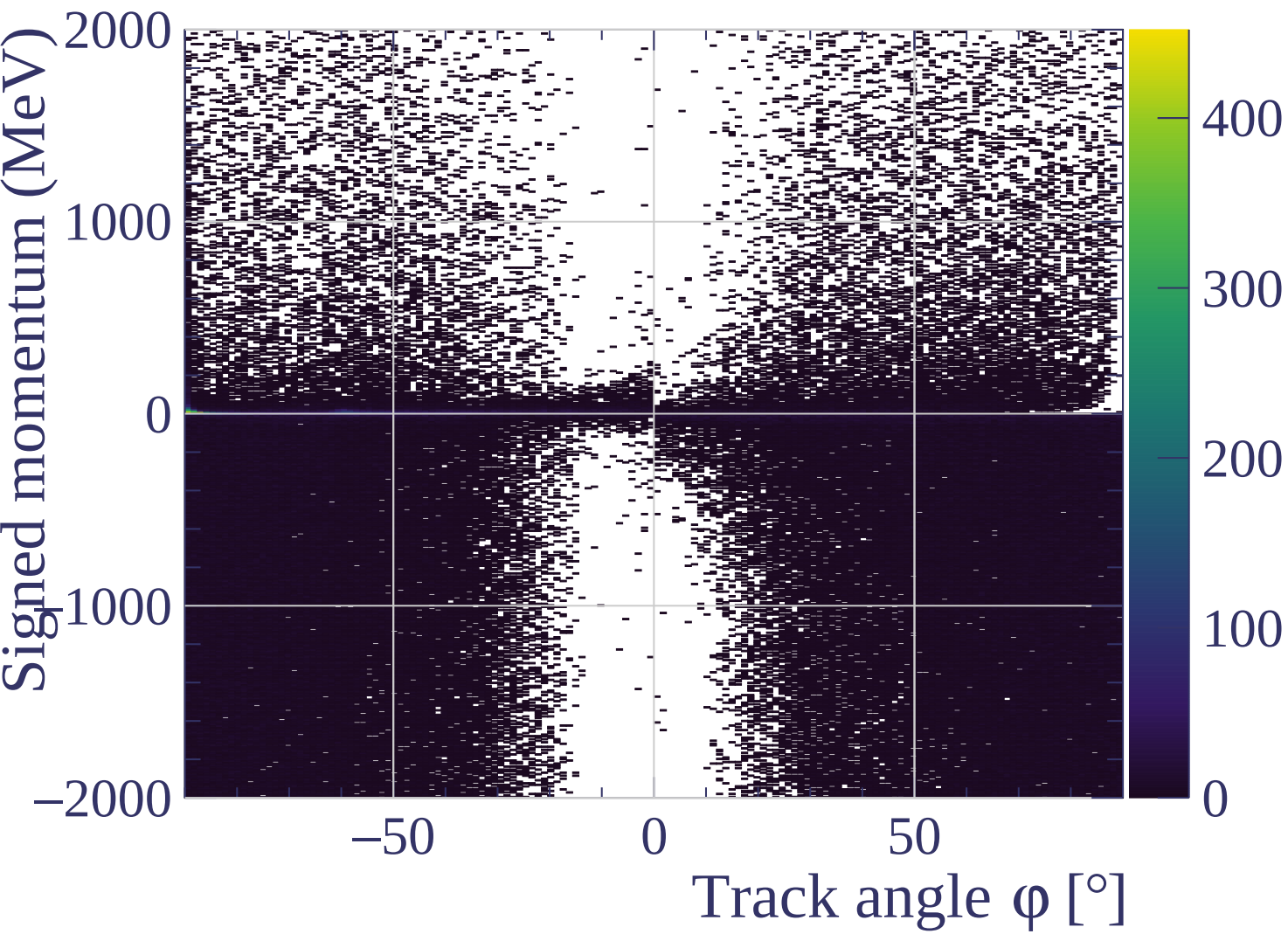


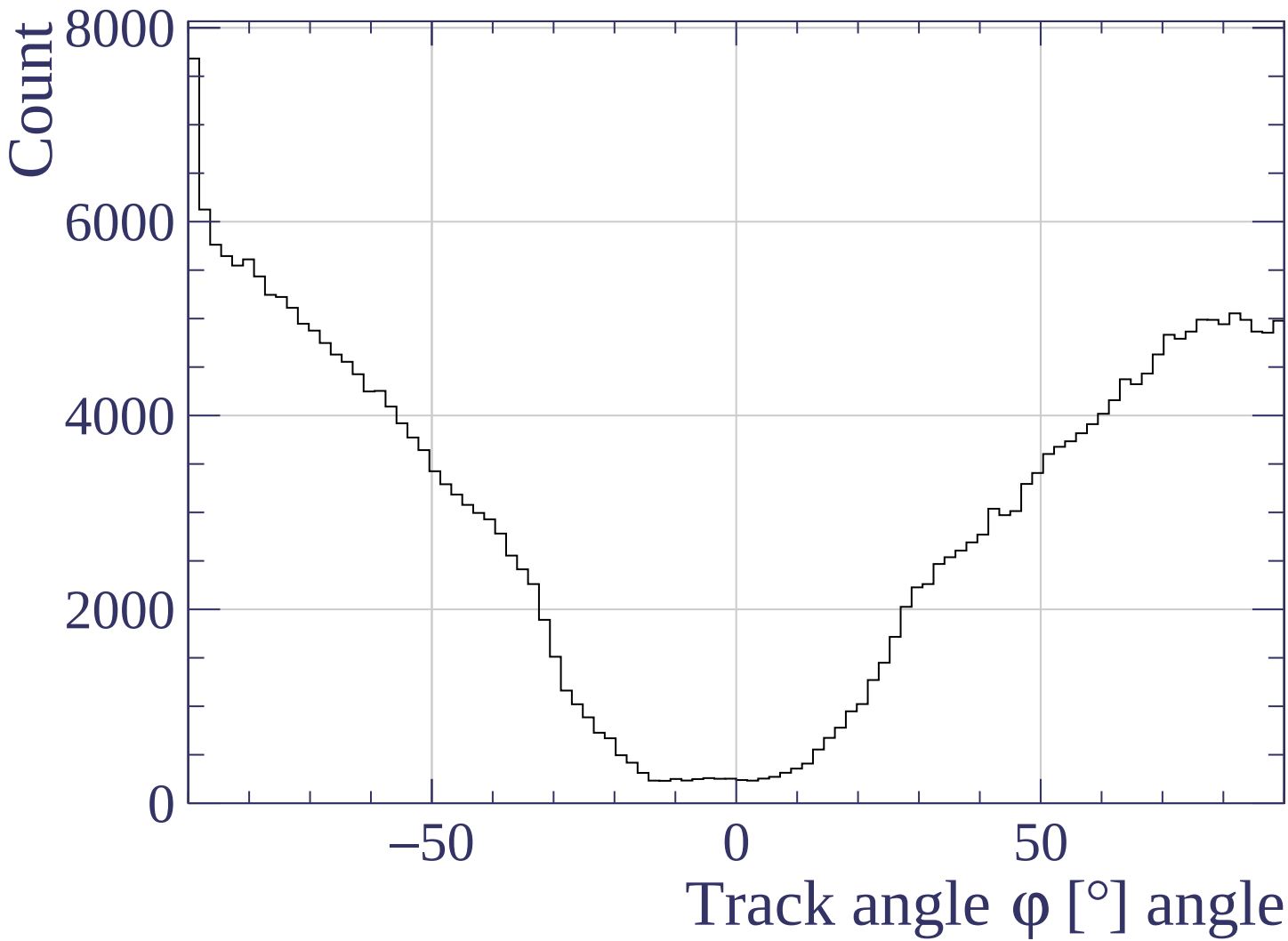
Track length [cm]

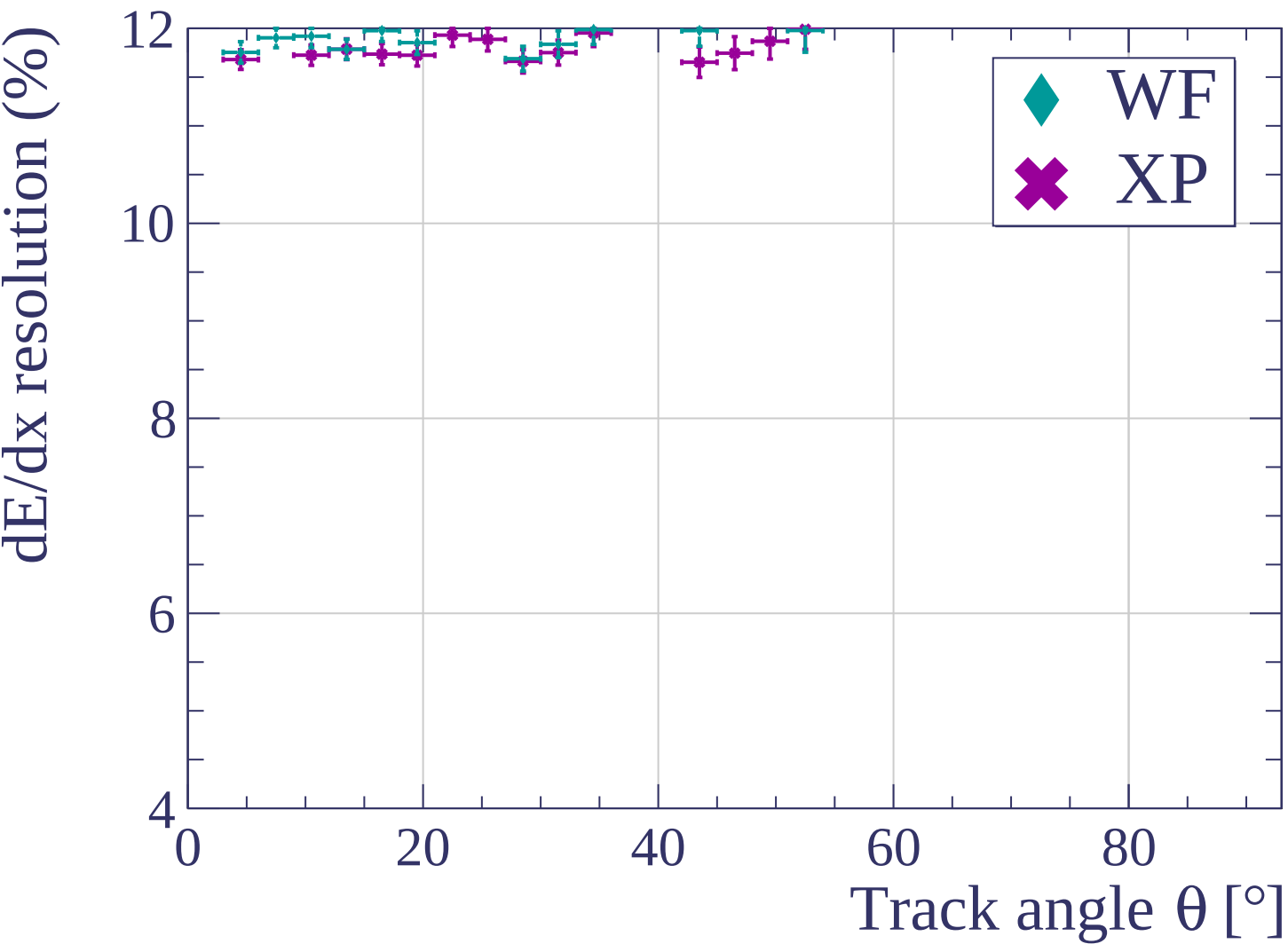


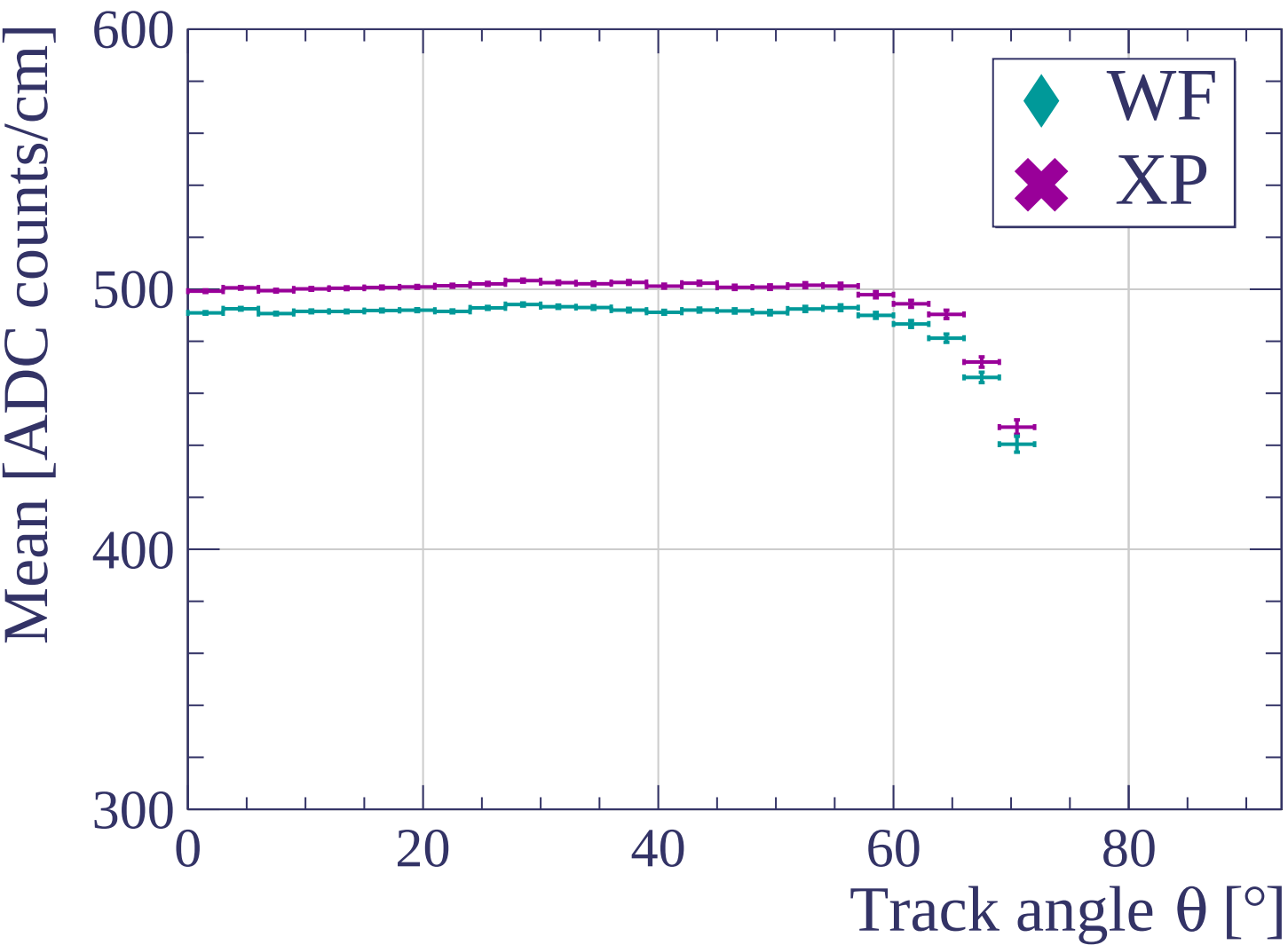
χ^2/NDF

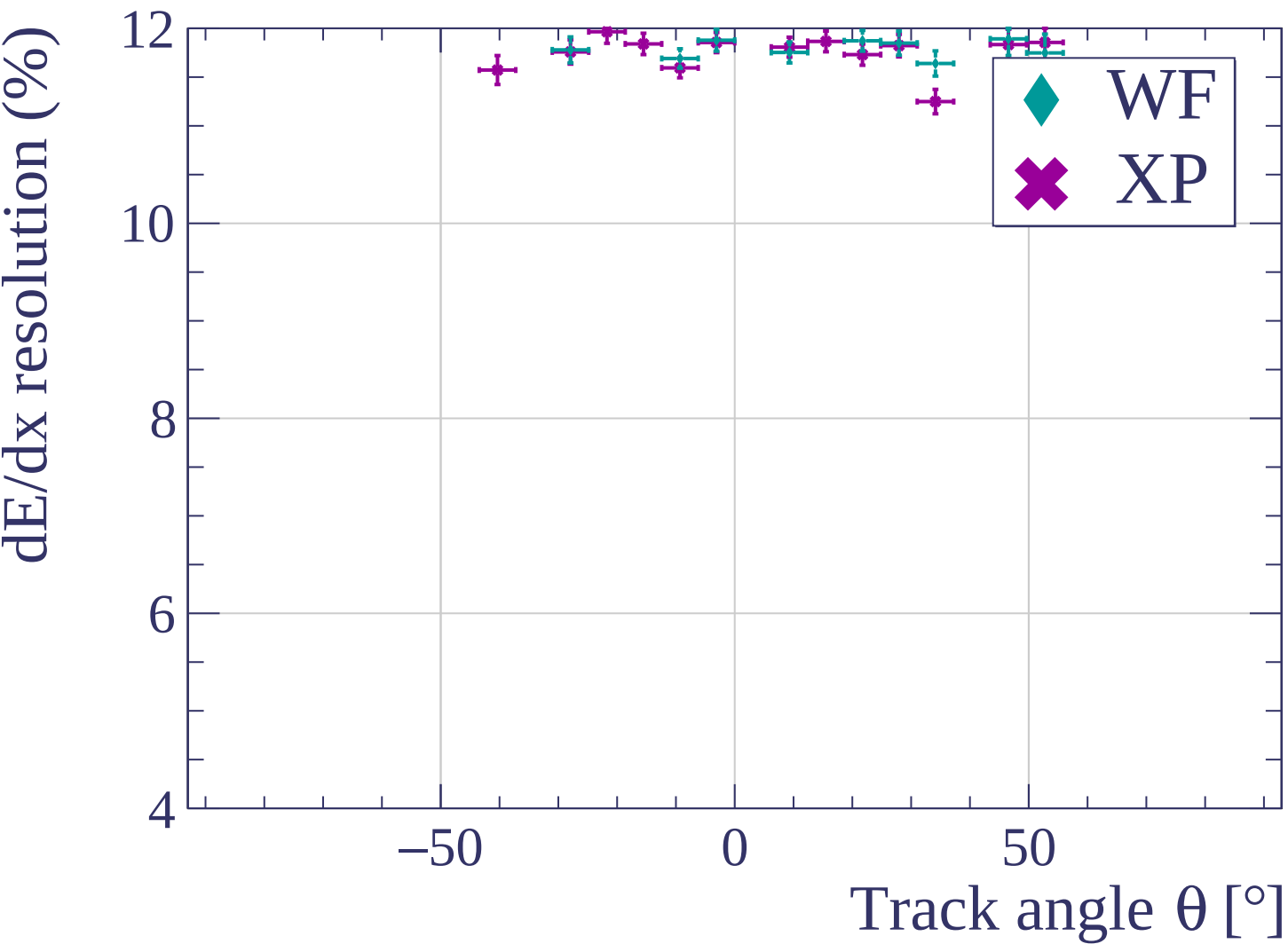


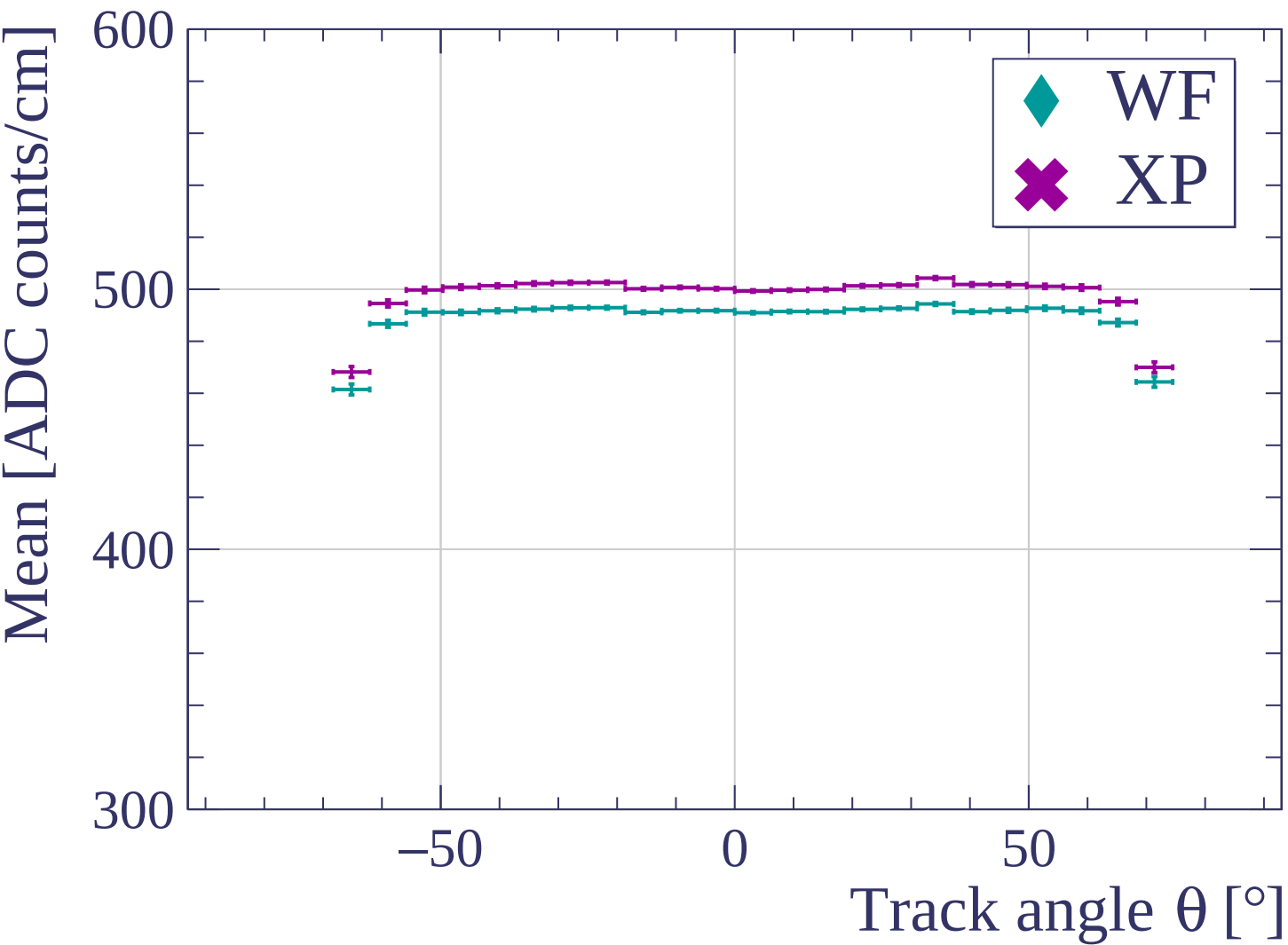


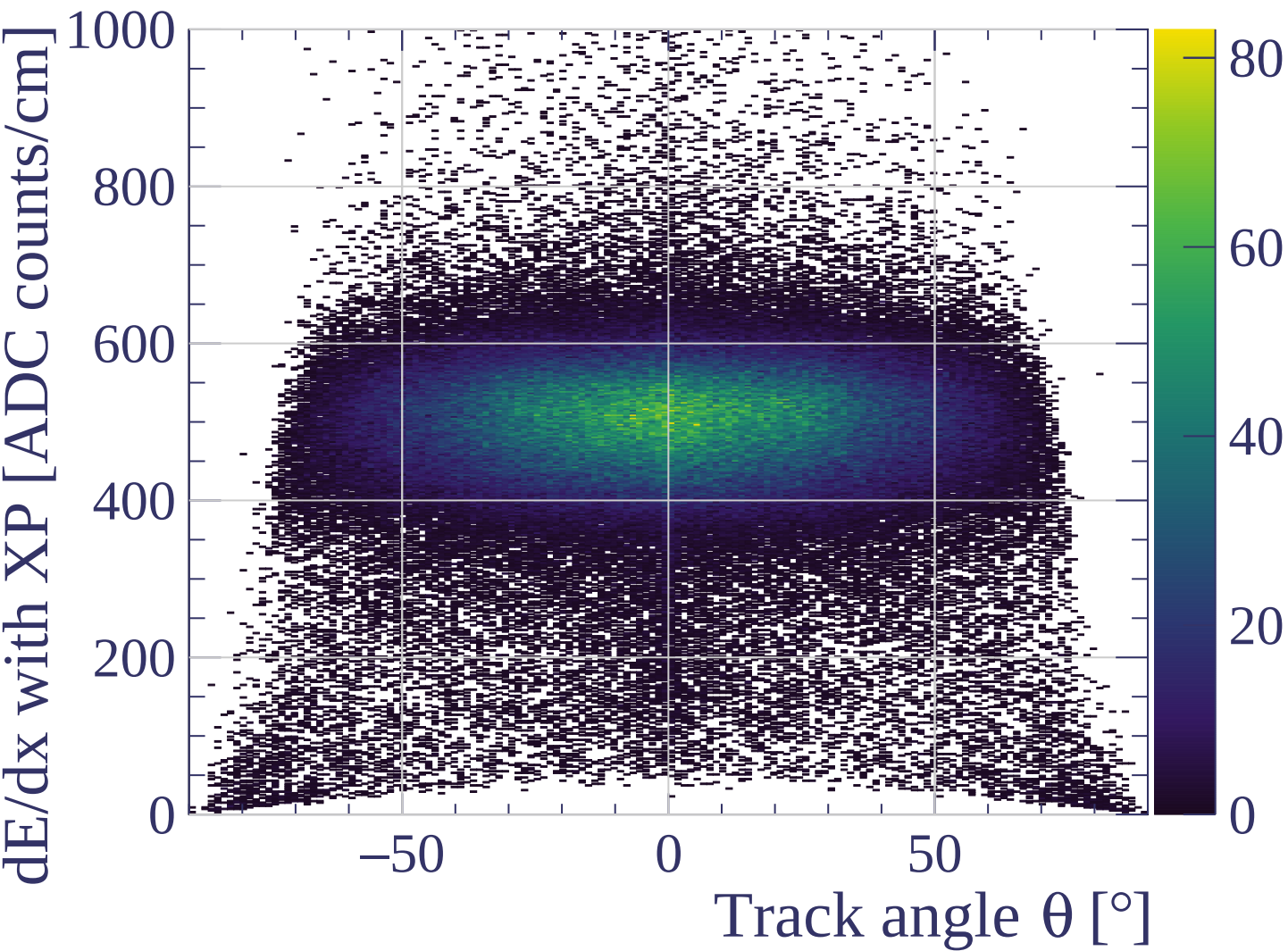


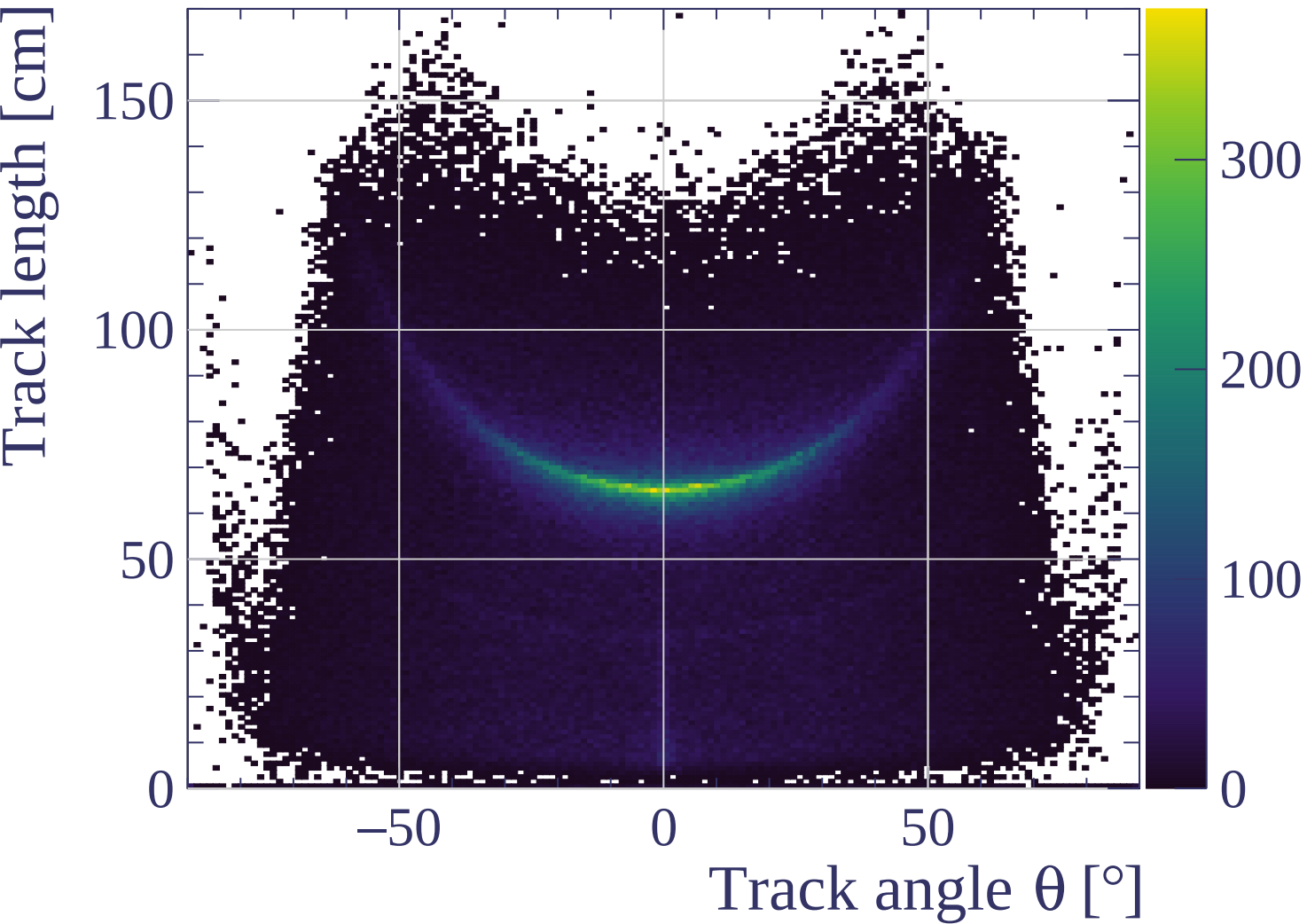


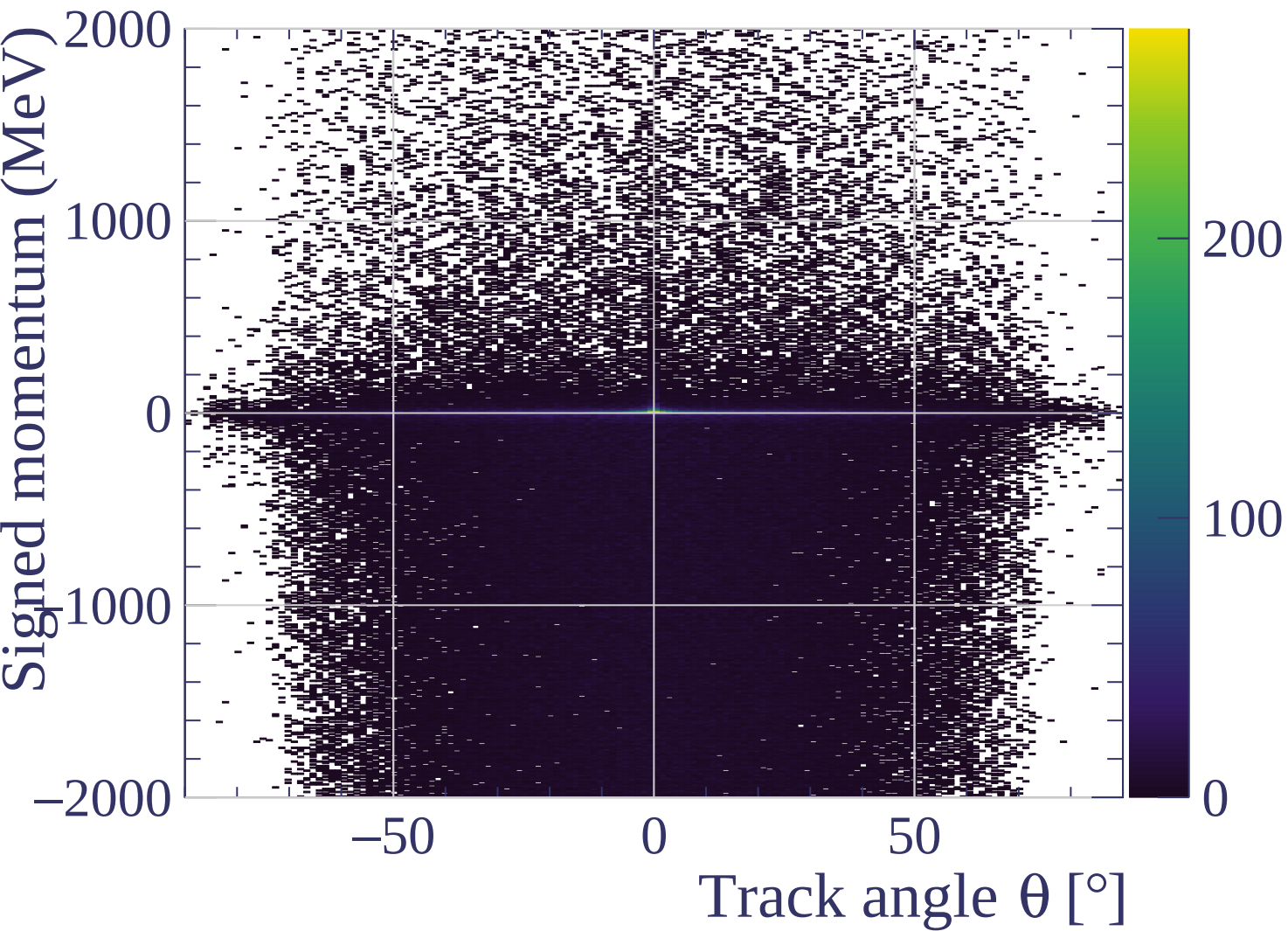




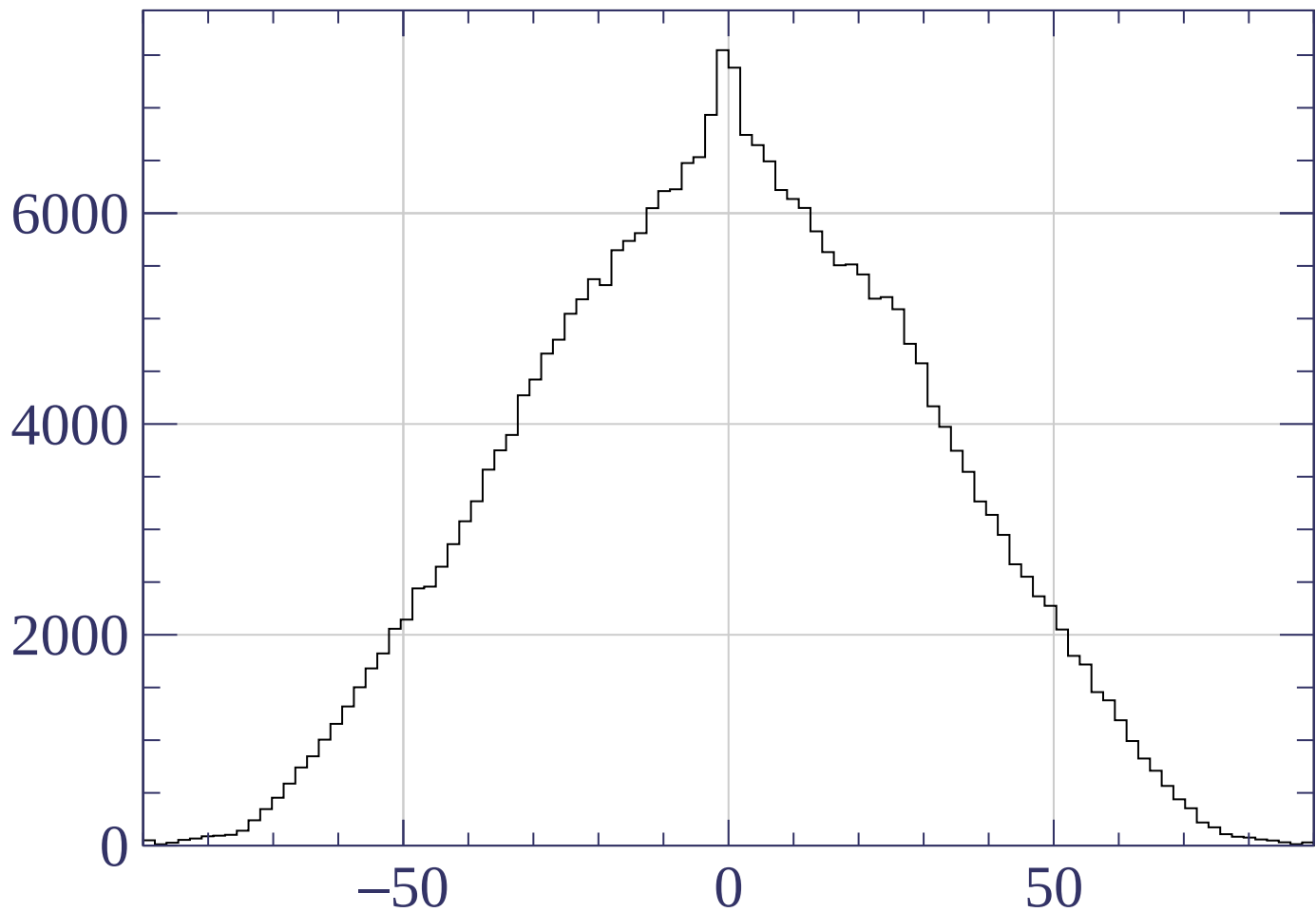






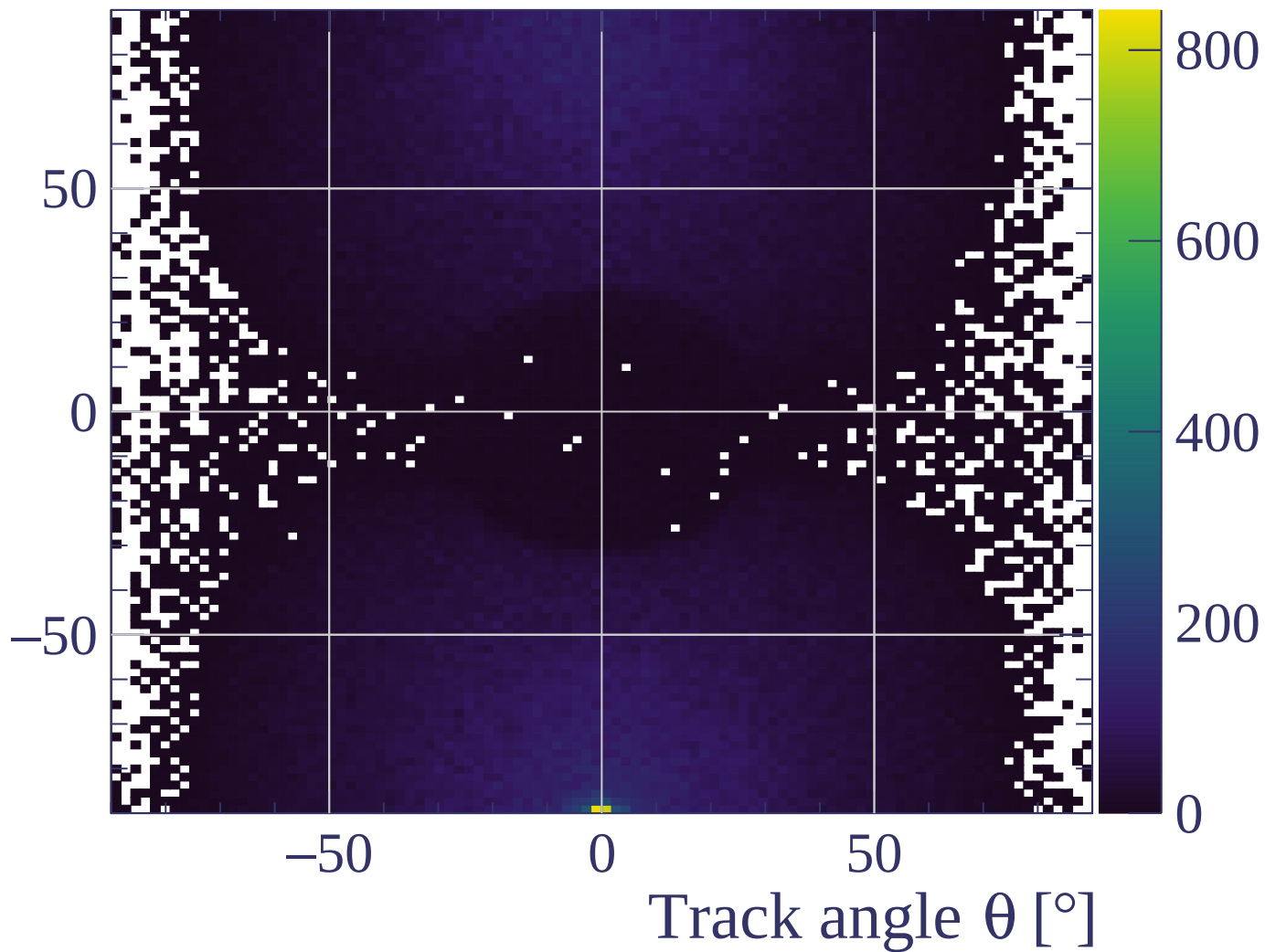


Count

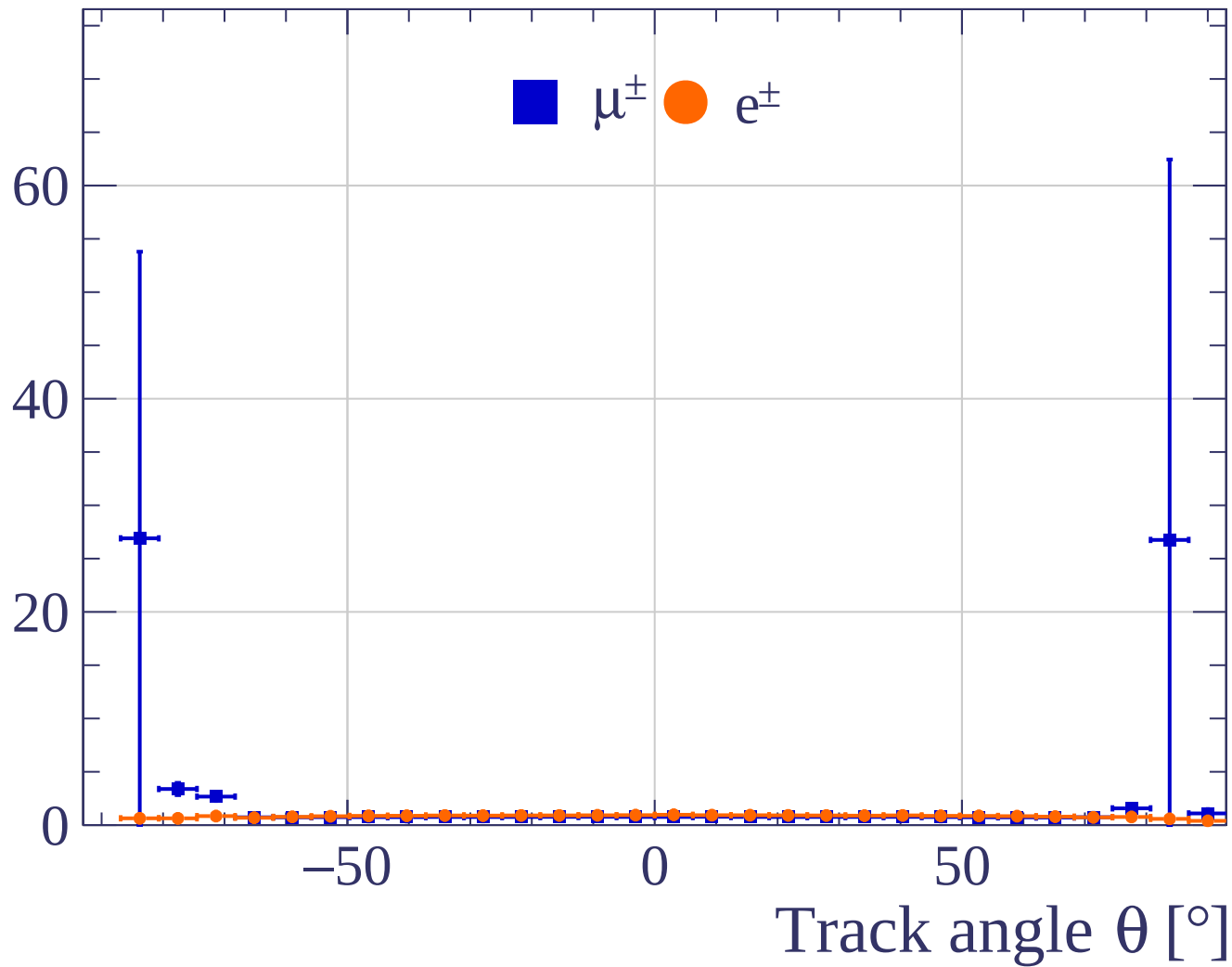


Track angle θ [°]

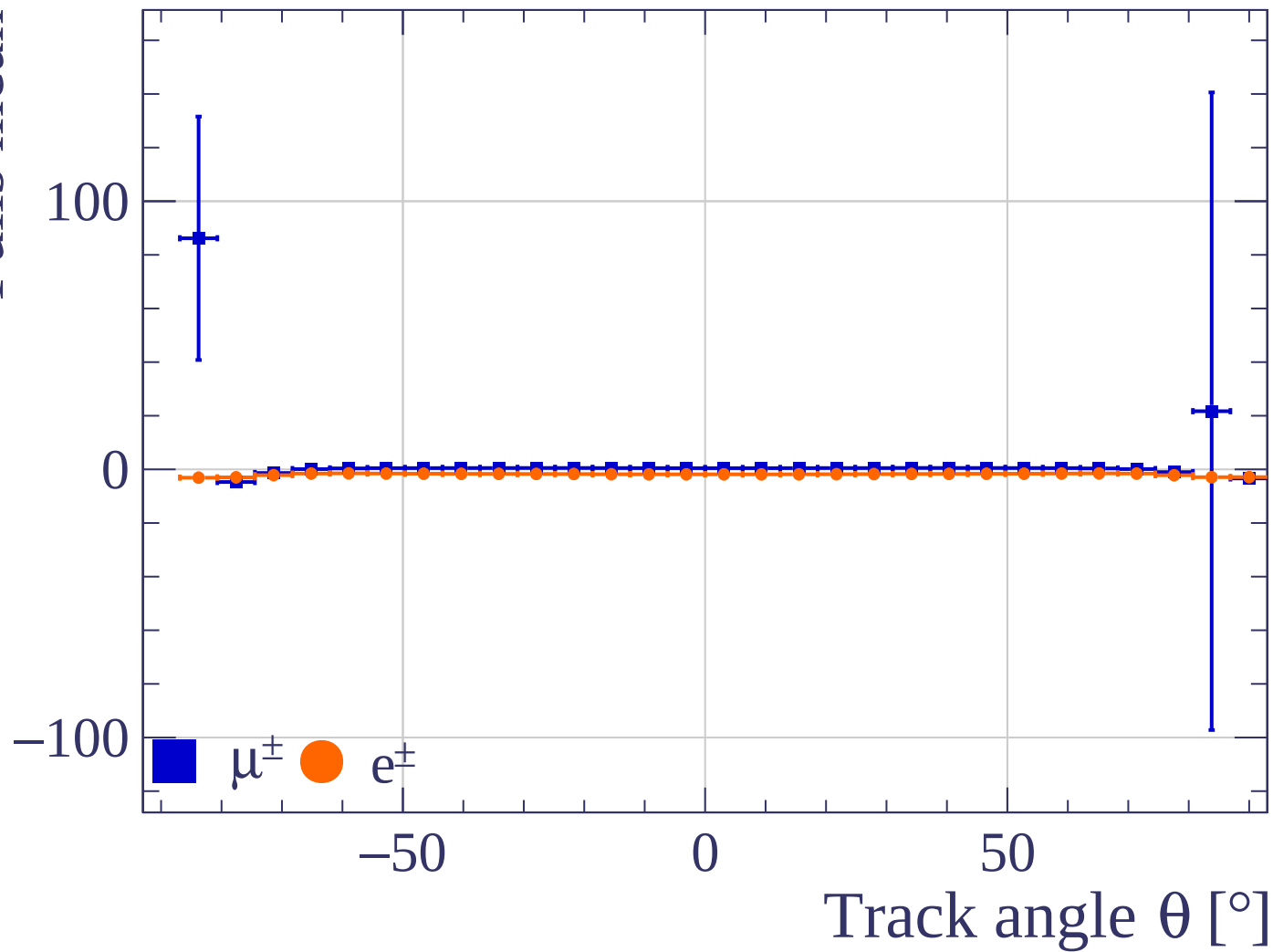
Track angle ϕ [°]



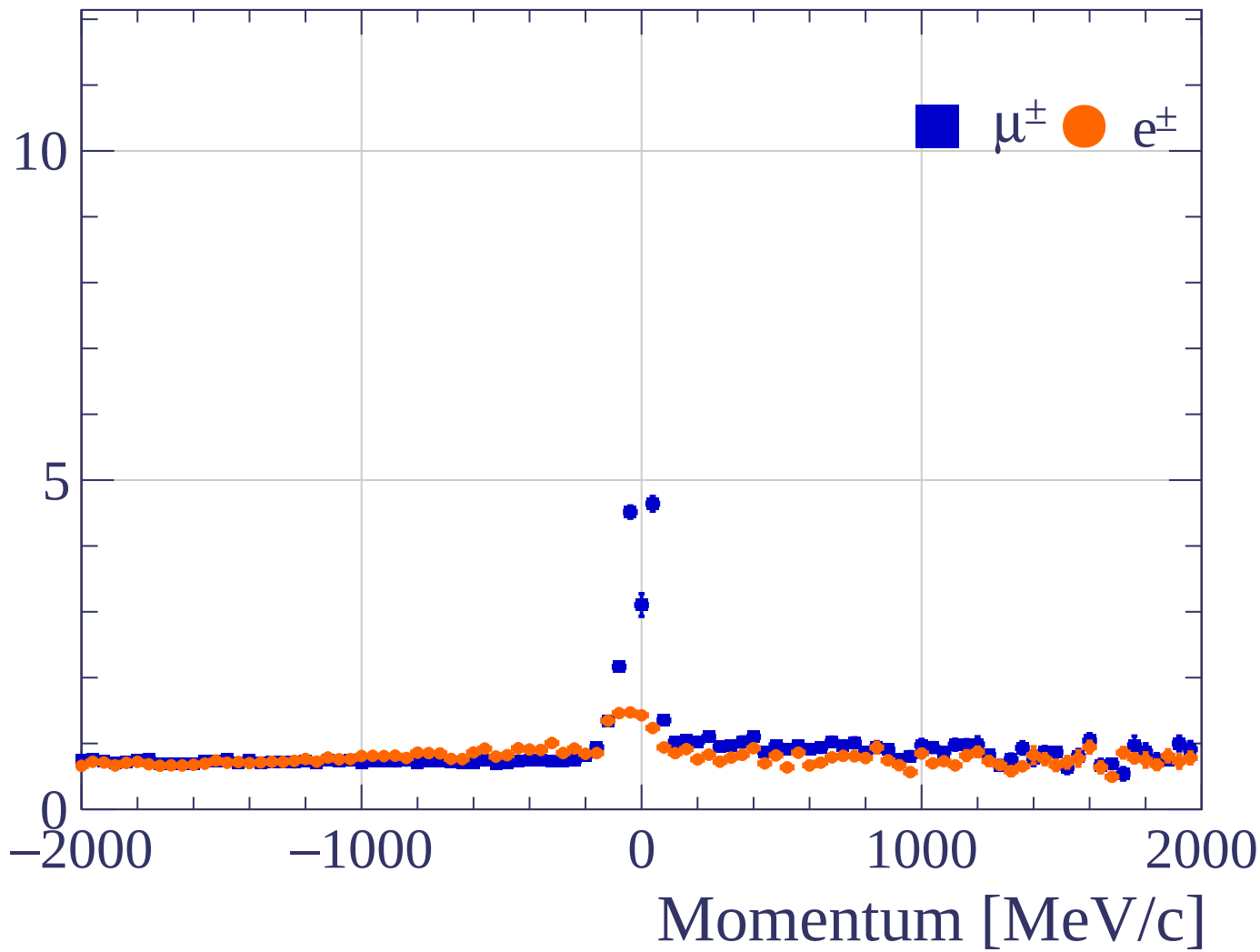
Pulls standard deviation



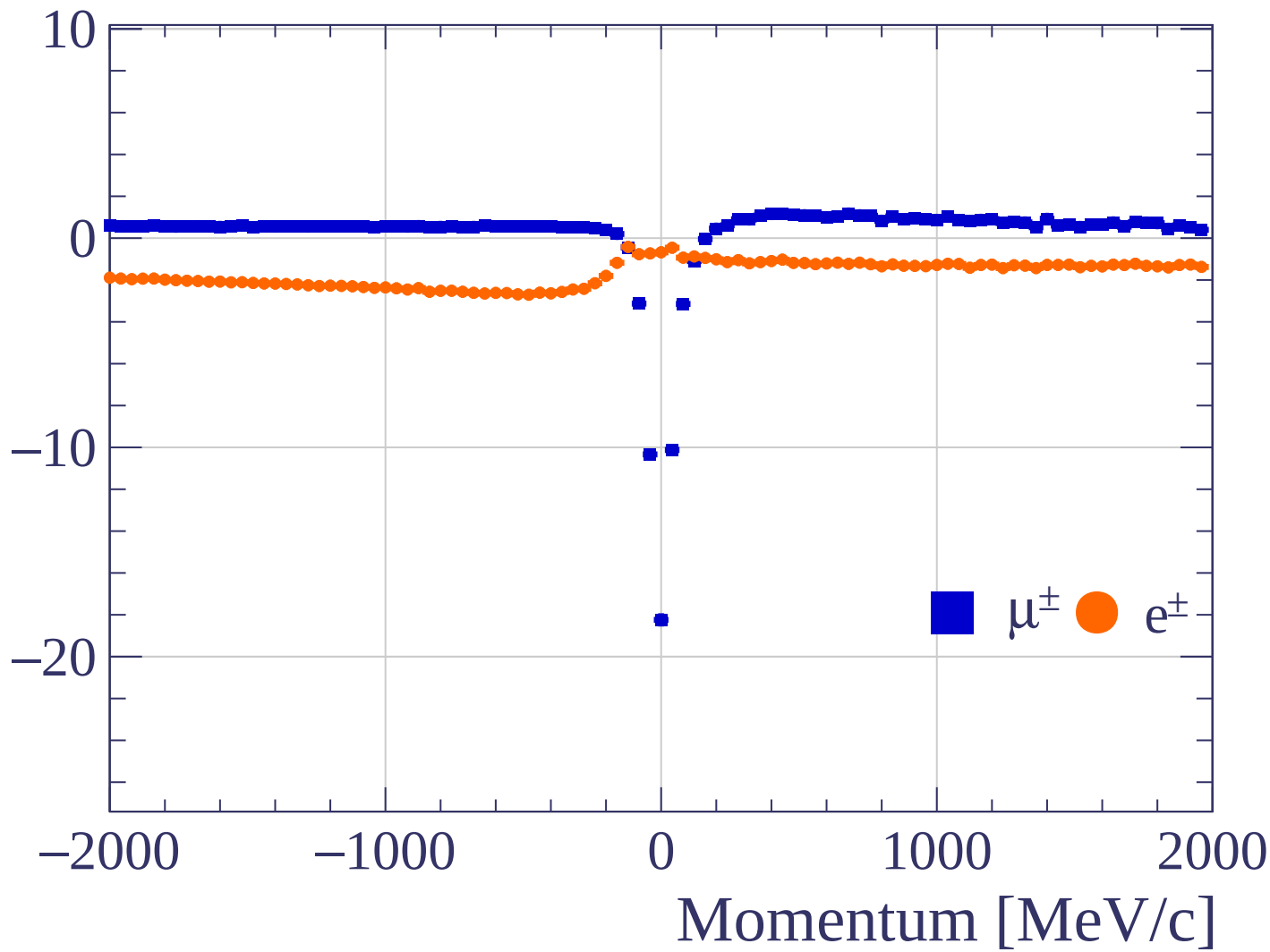
Pulls mean

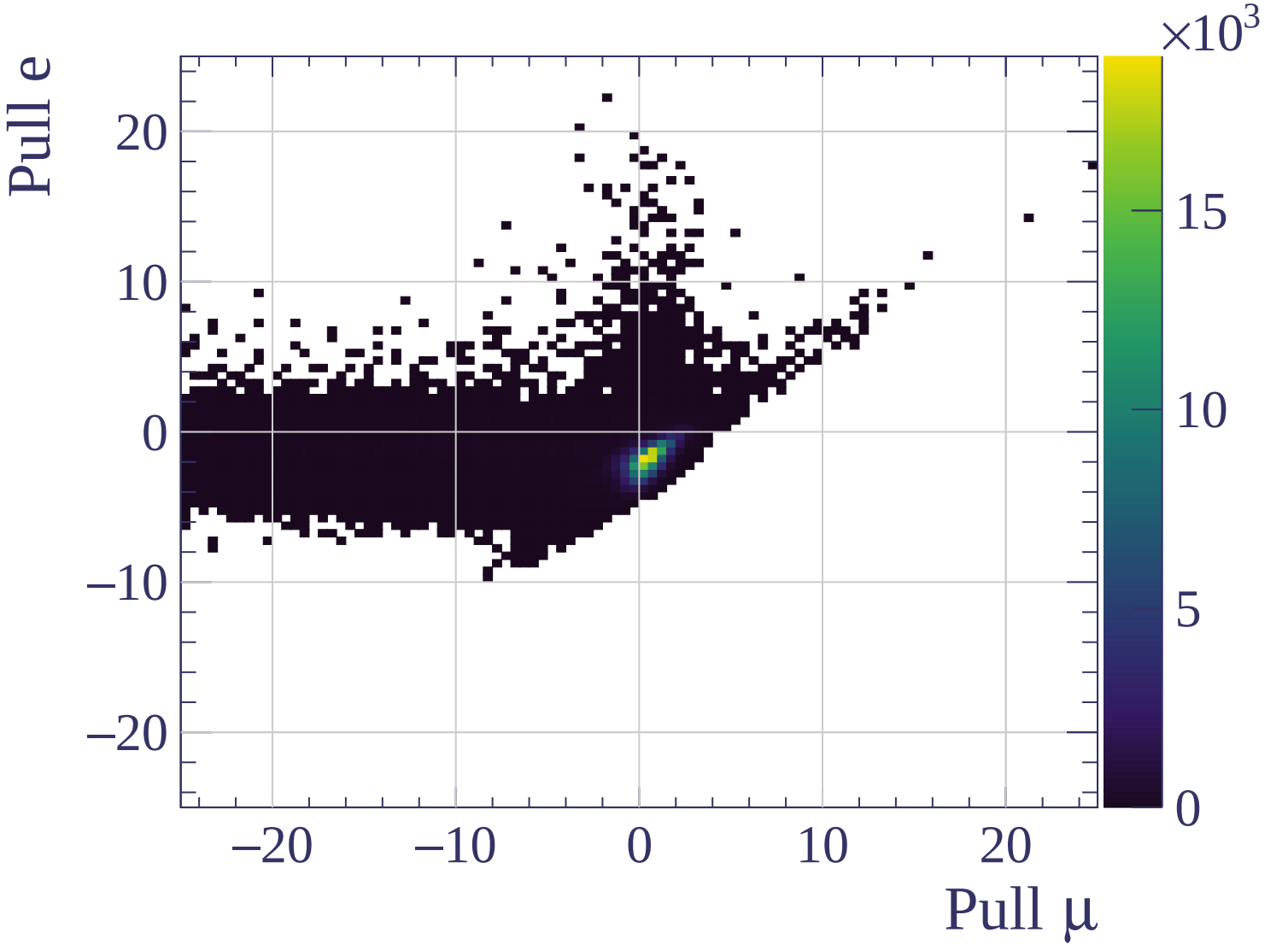


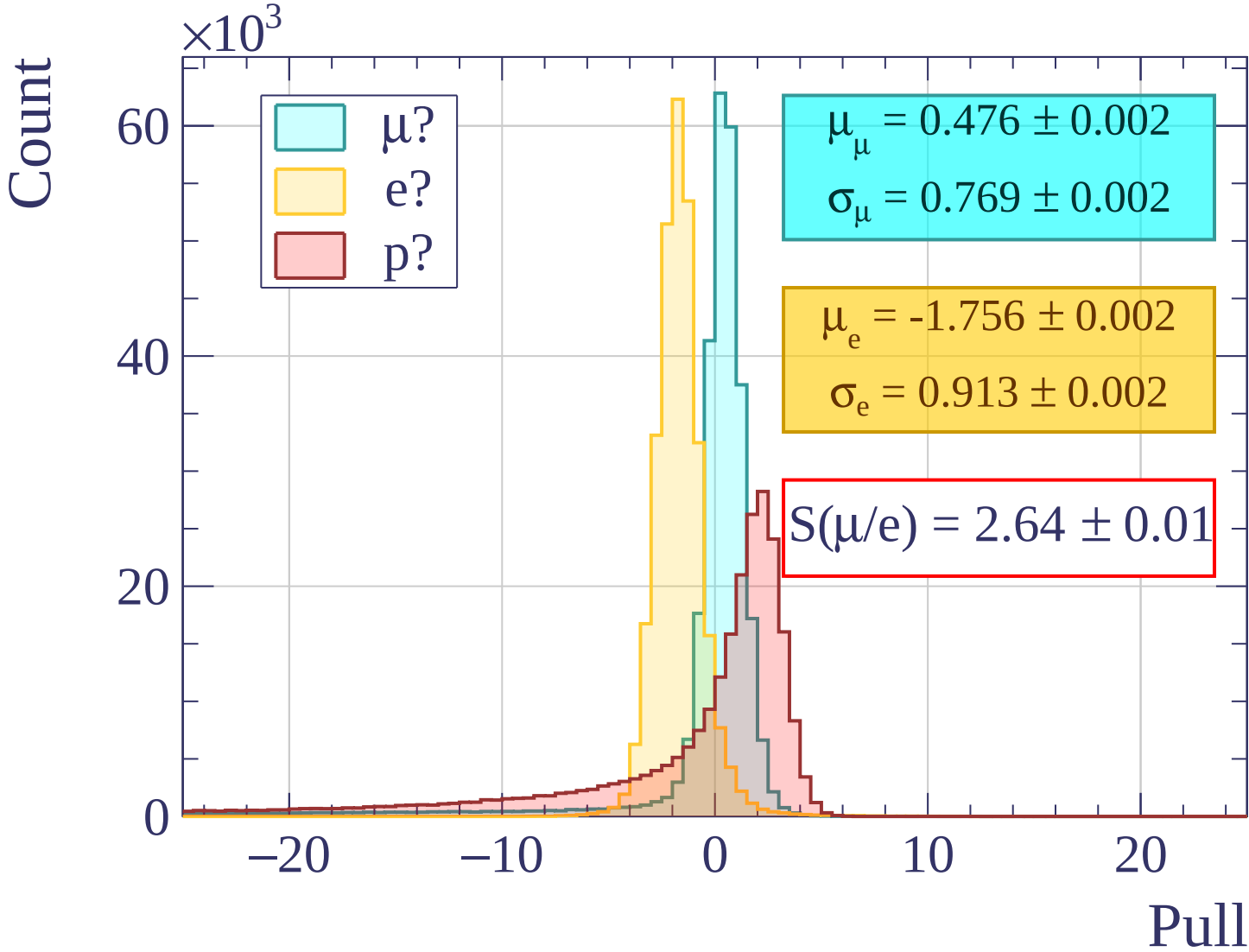
Pulls standard deviation

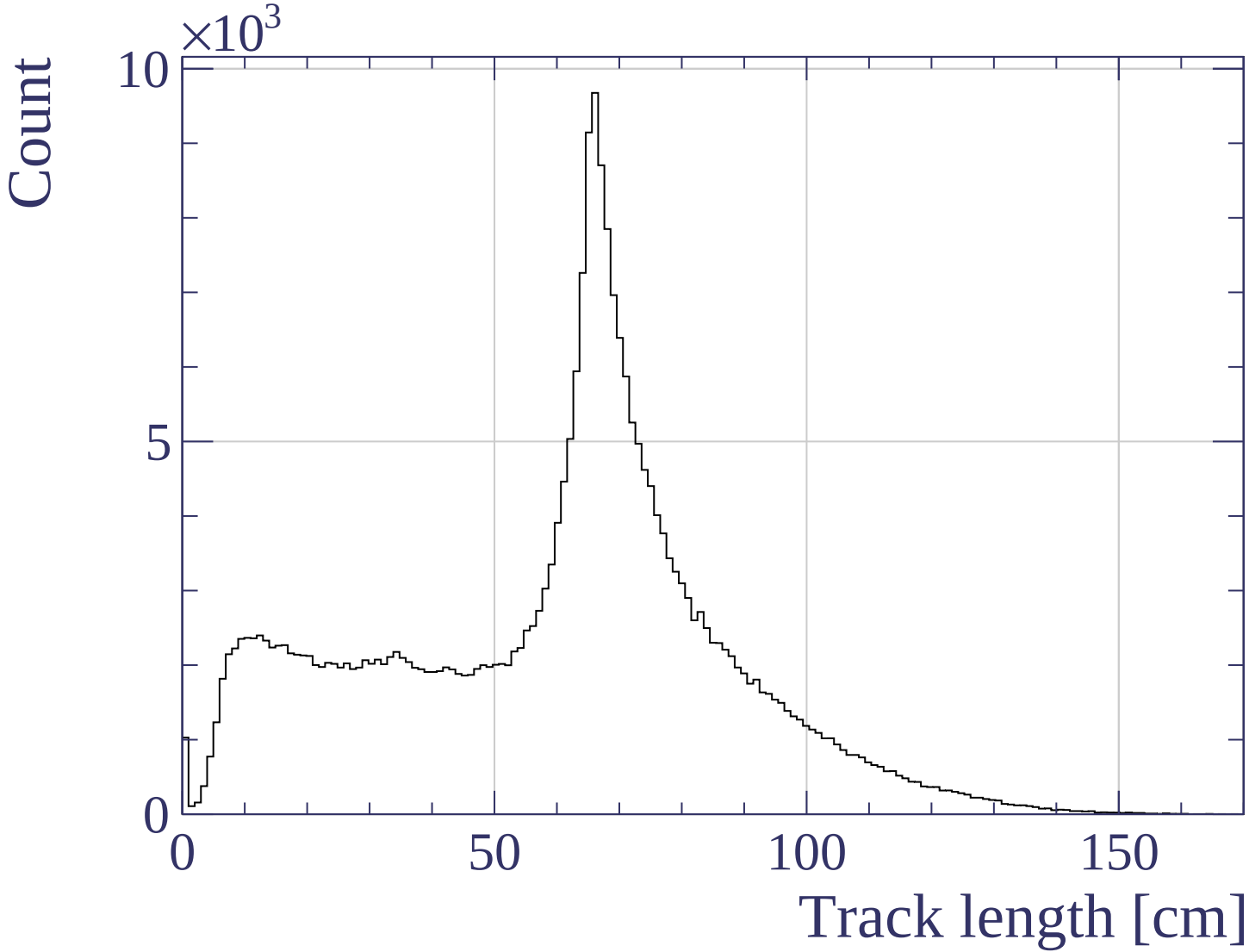


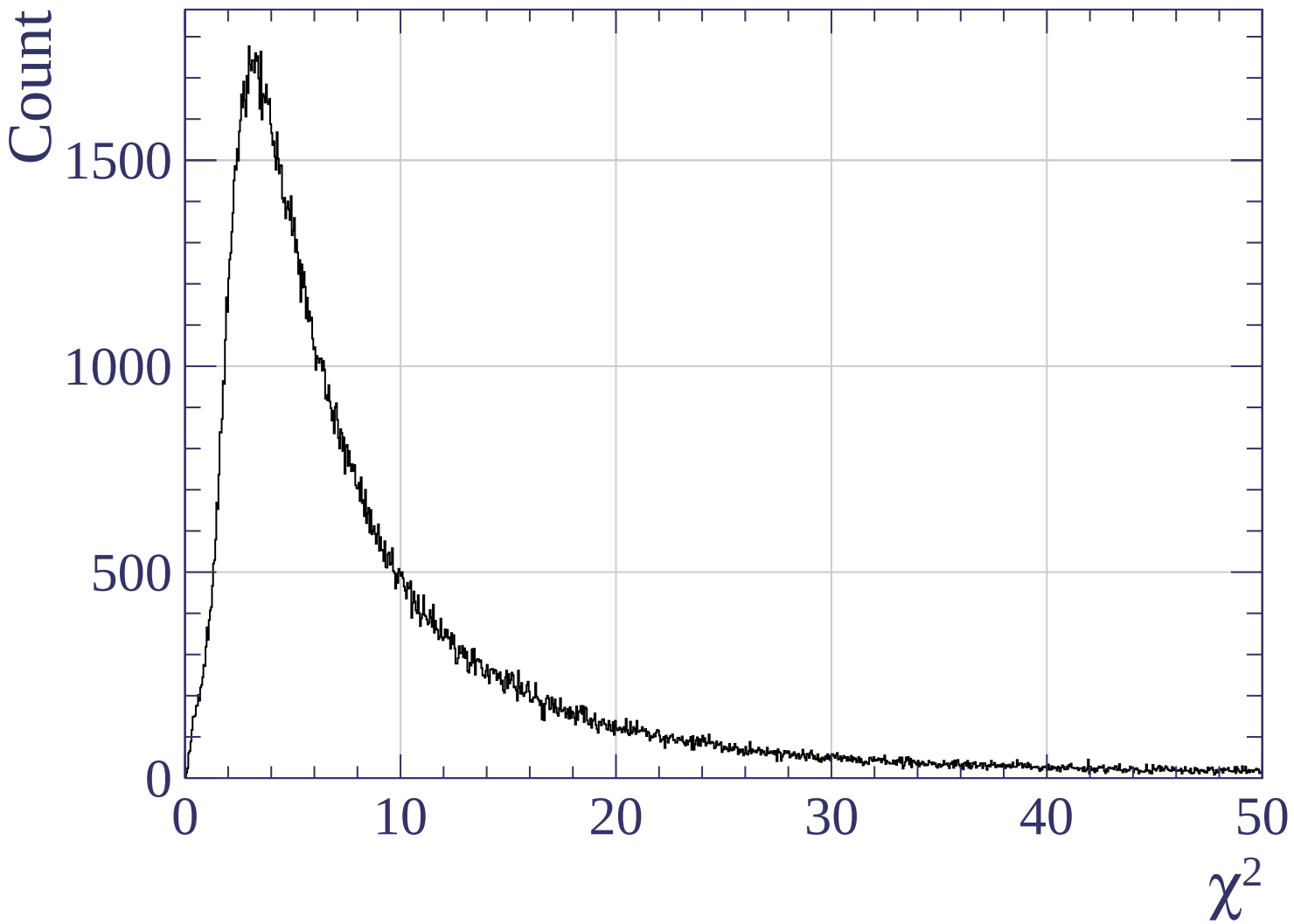
Pulls mean



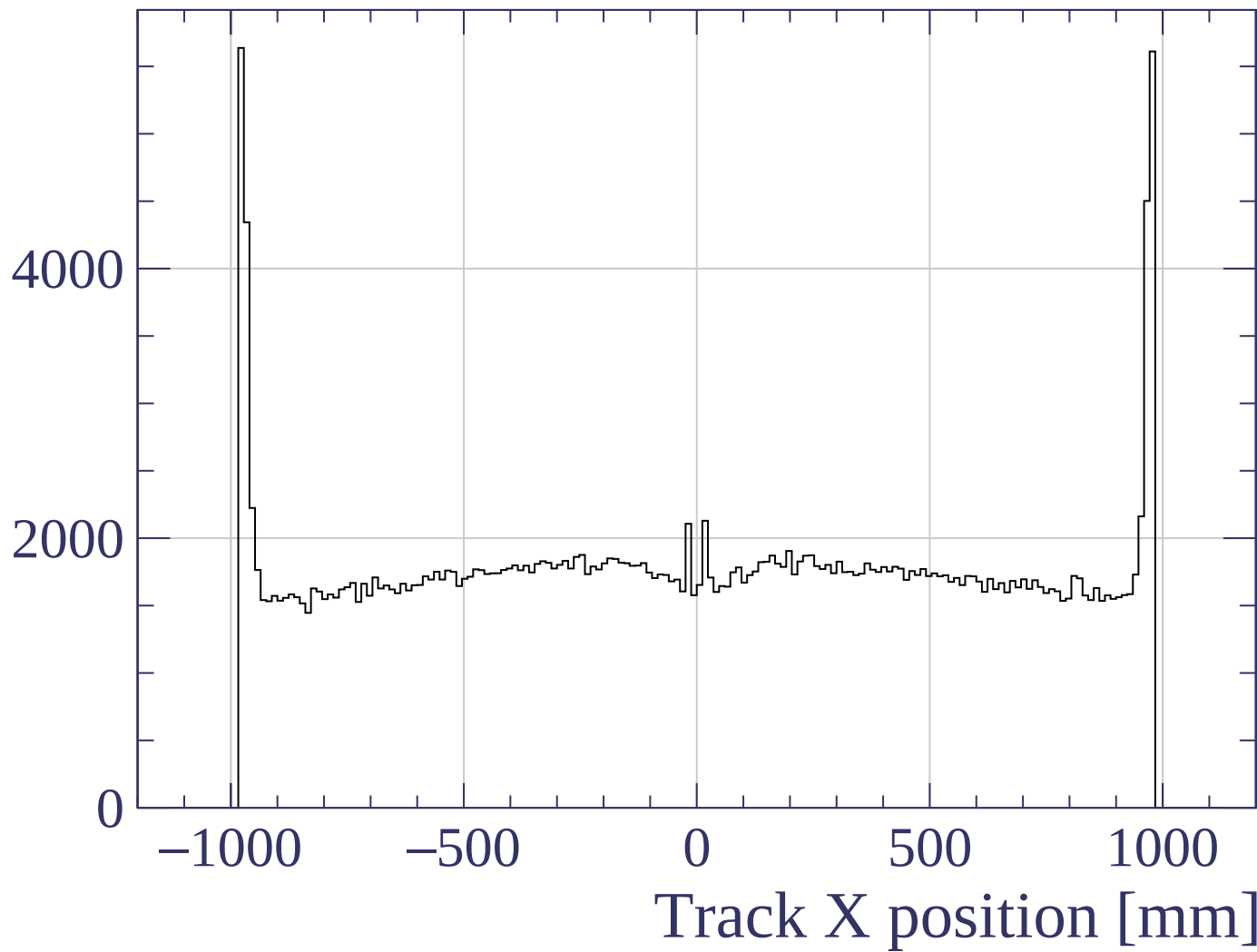


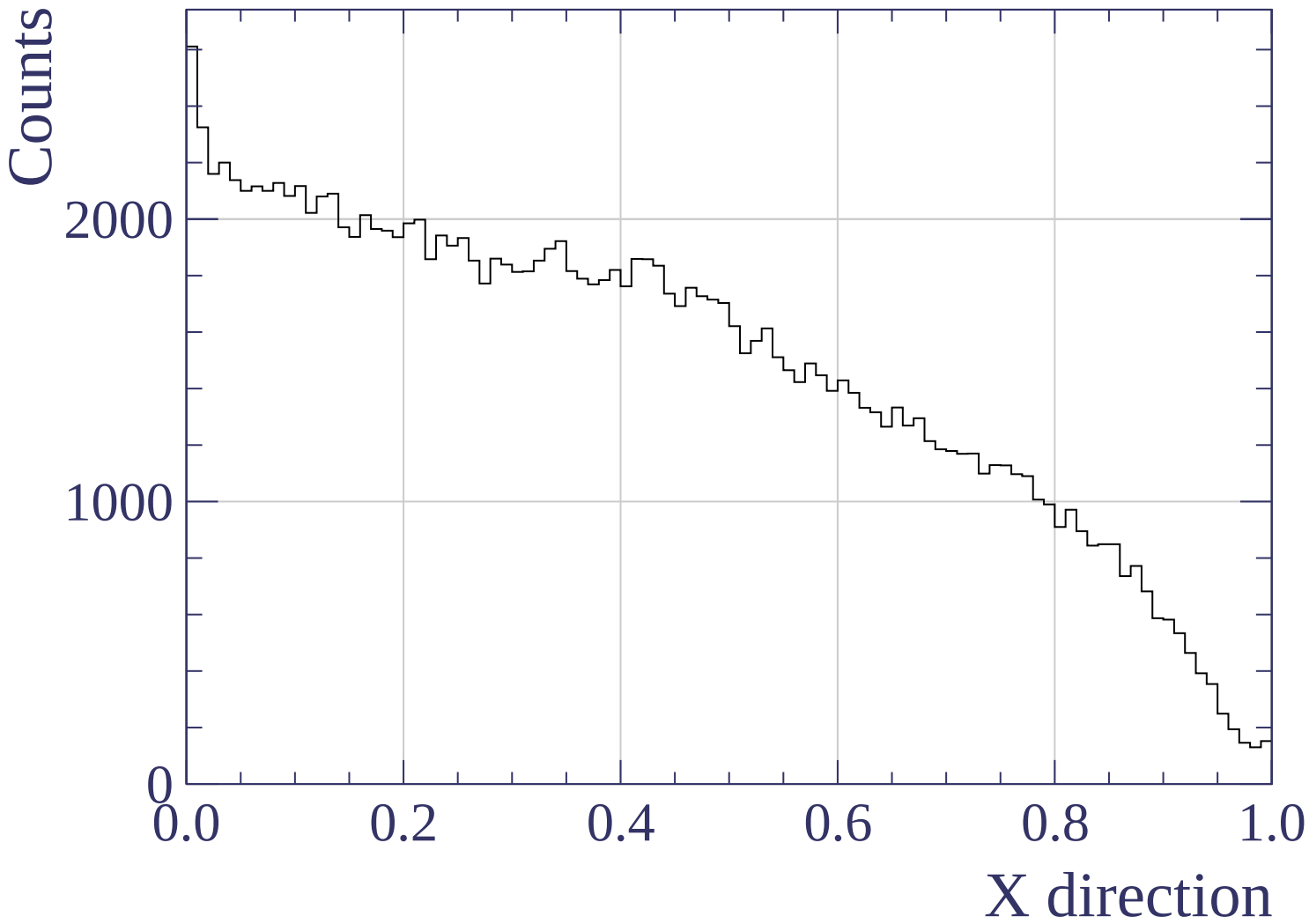


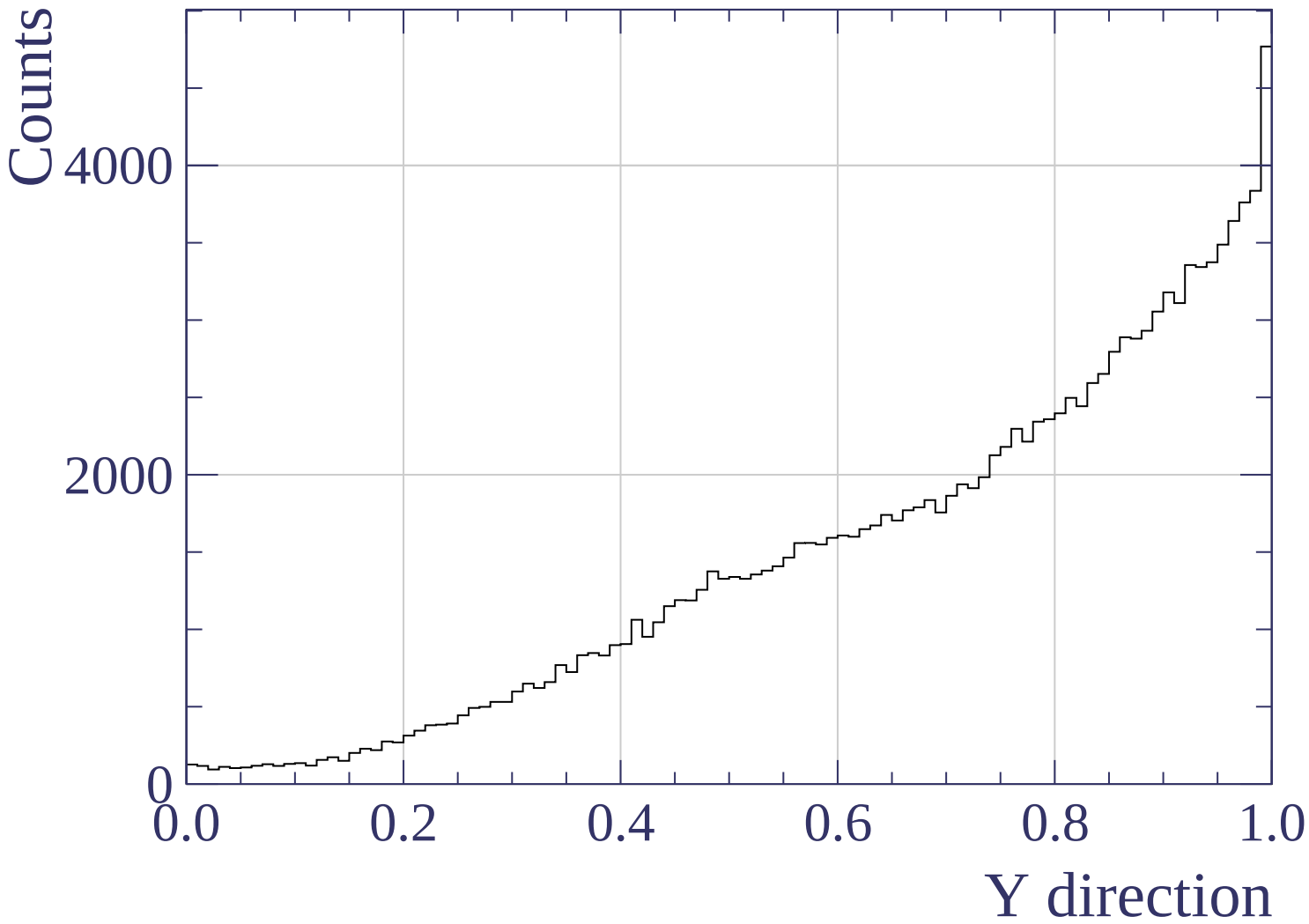


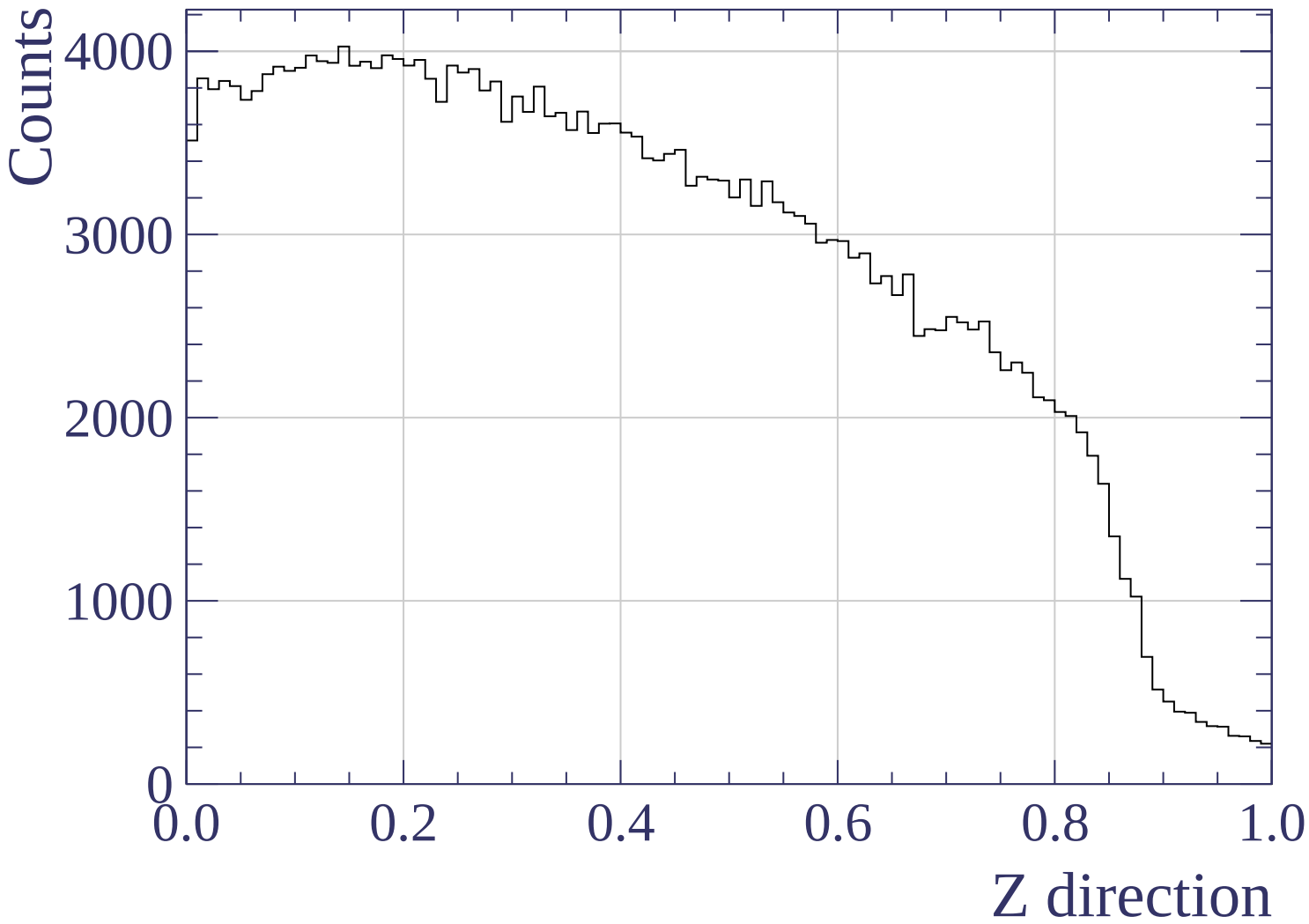


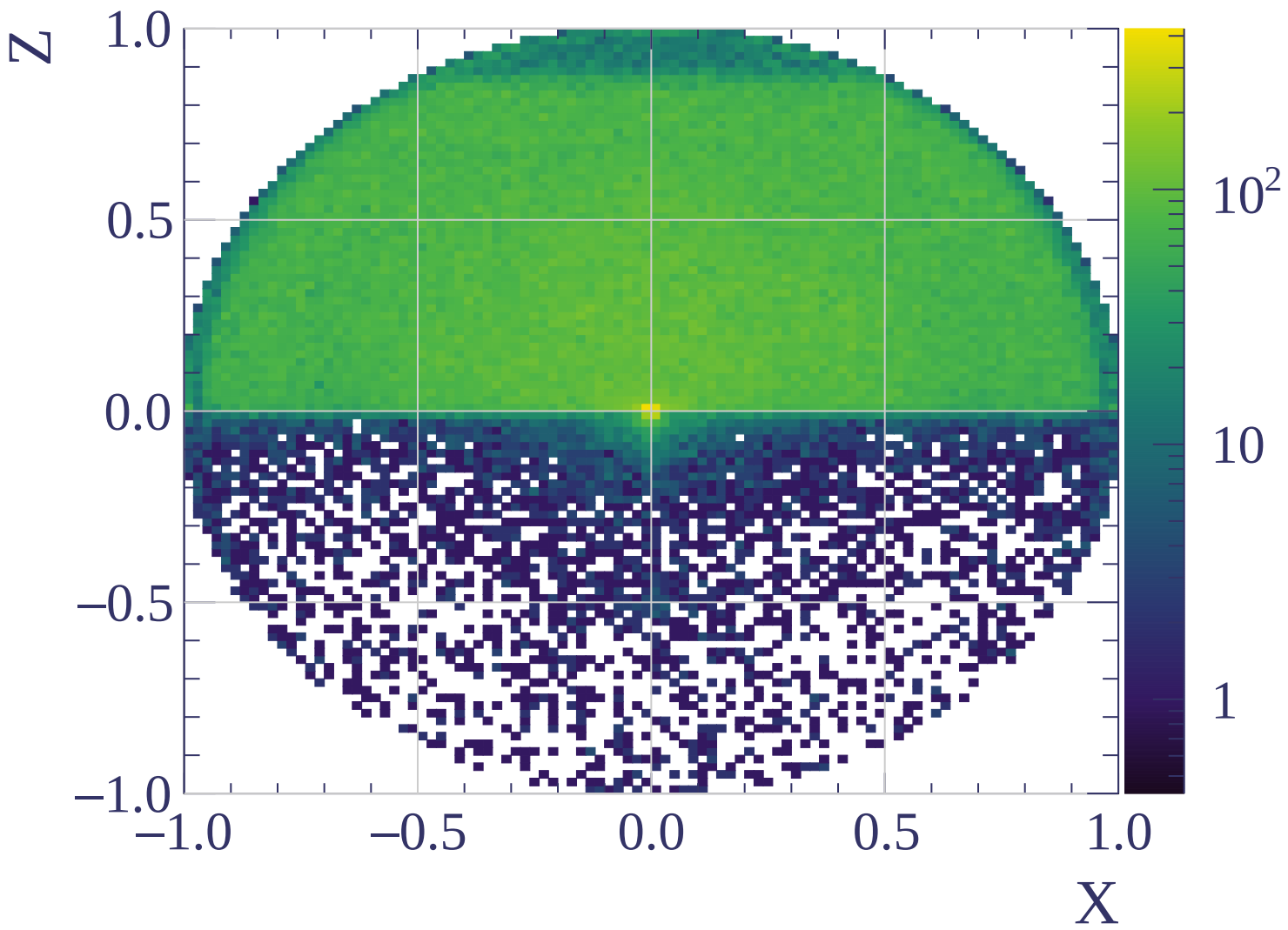
Count

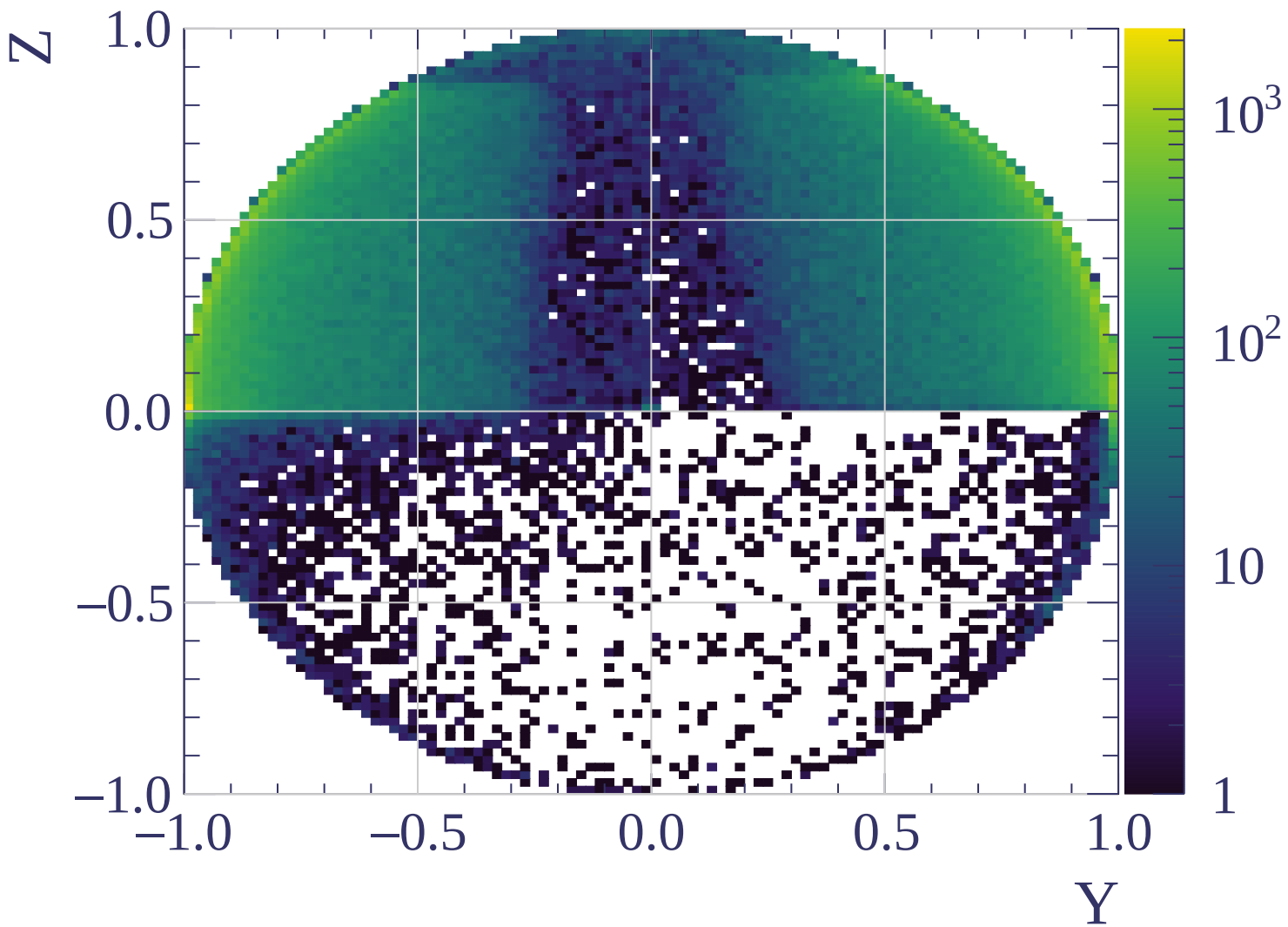


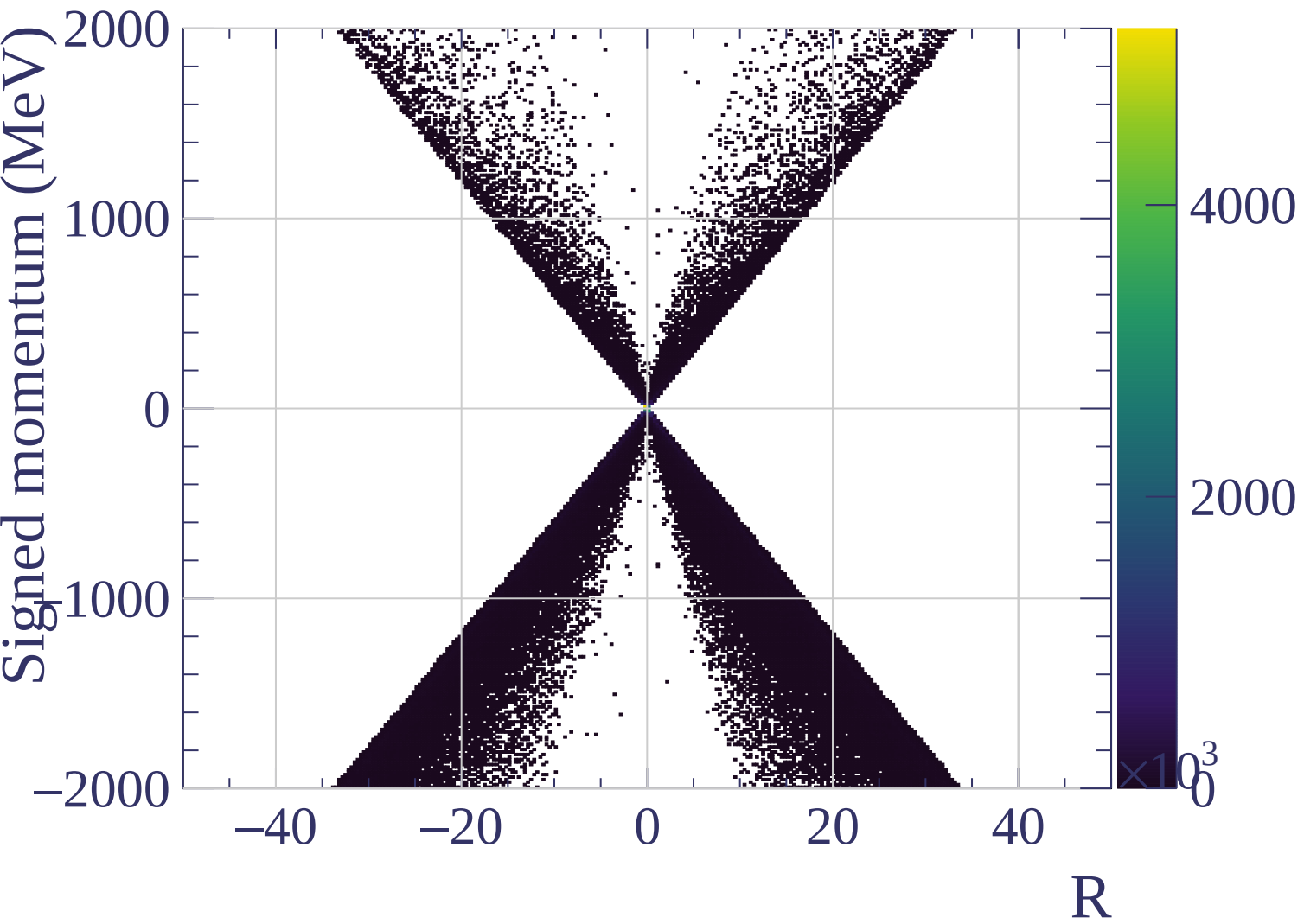


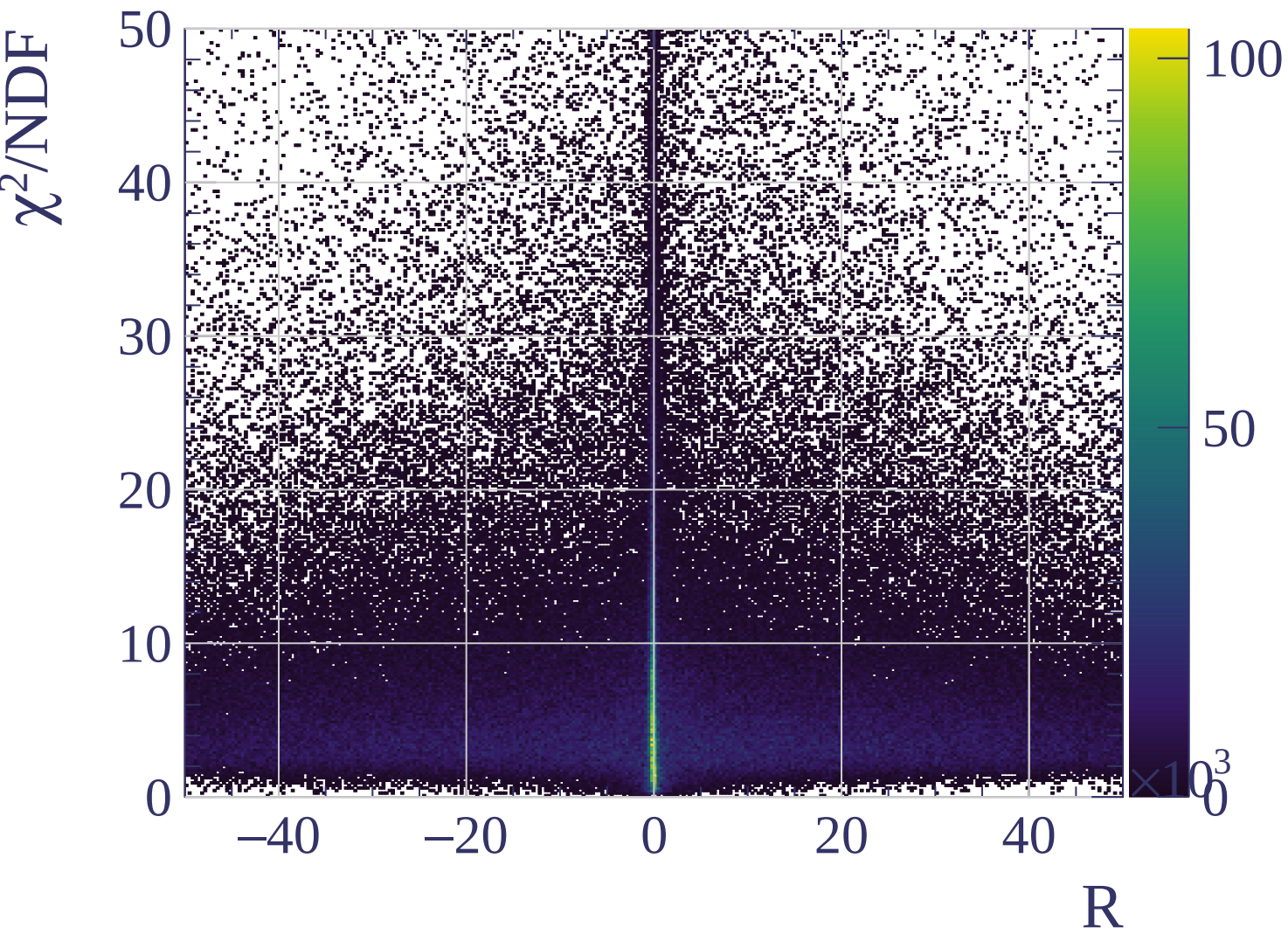






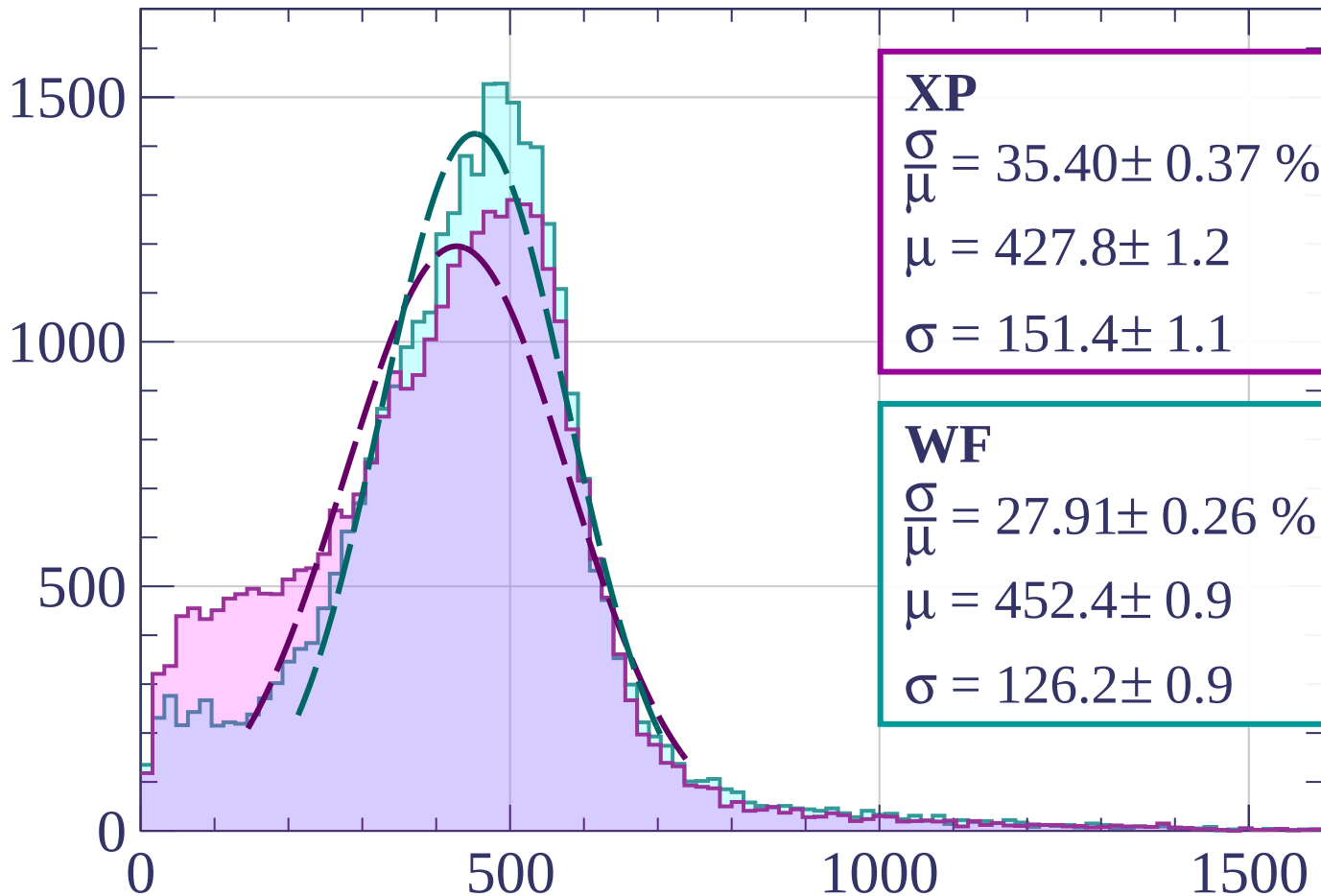






Energy loss | $0 < p < 80$

Count



XP

$$\frac{\sigma}{\mu} = 35.40 \pm 0.37 \%$$

$$\mu = 427.8 \pm 1.2$$

$$\sigma = 151.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 27.91 \pm 0.26 \%$$

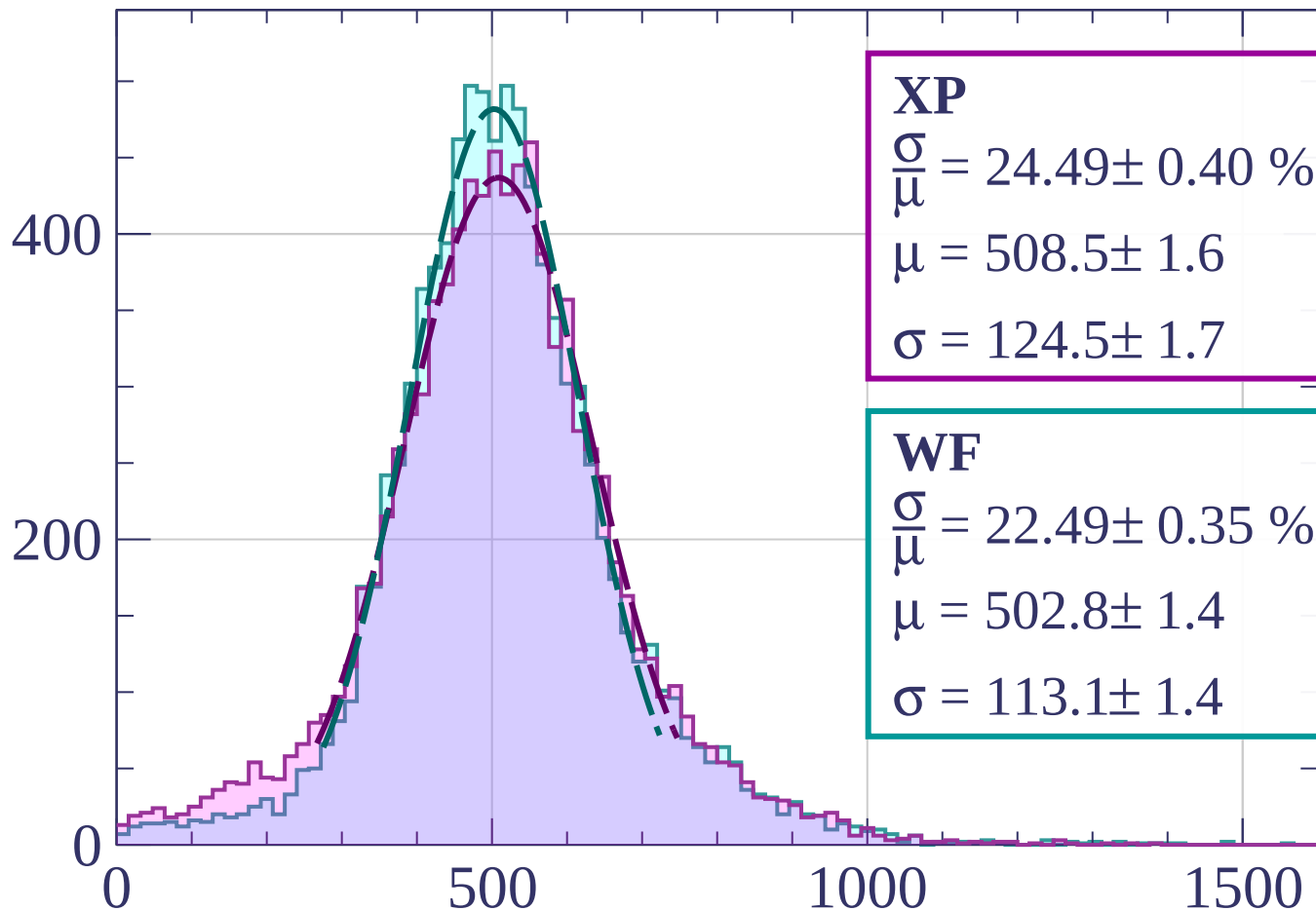
$$\mu = 452.4 \pm 0.9$$

$$\sigma = 126.2 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $80 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 24.49 \pm 0.40 \%$$

$$\mu = 508.5 \pm 1.6$$

$$\sigma = 124.5 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 22.49 \pm 0.35 \%$$

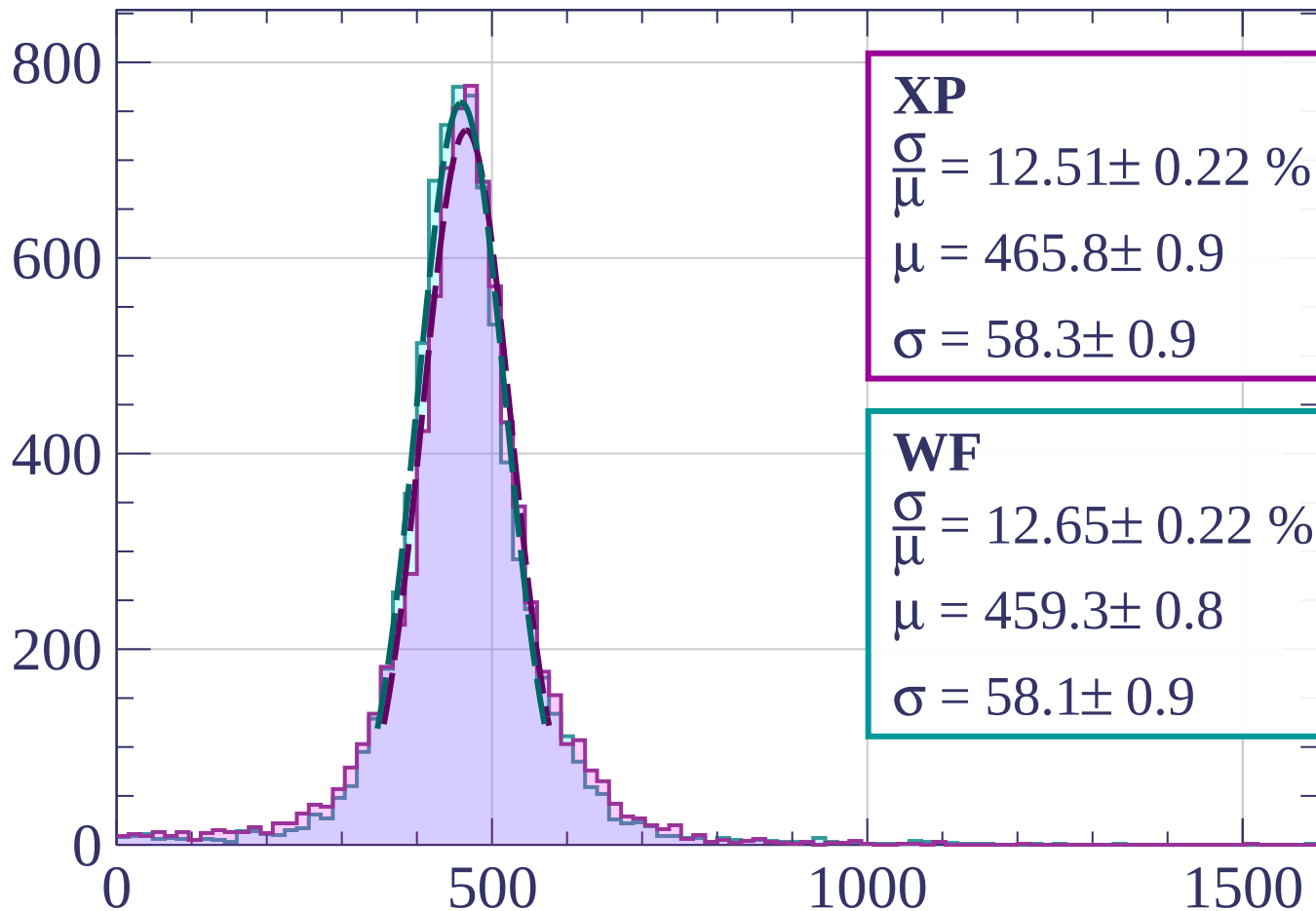
$$\mu = 502.8 \pm 1.4$$

$$\sigma = 113.1 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $160 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 12.51 \pm 0.22 \%$$

$$\mu = 465.8 \pm 0.9$$

$$\sigma = 58.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 12.65 \pm 0.22 \%$$

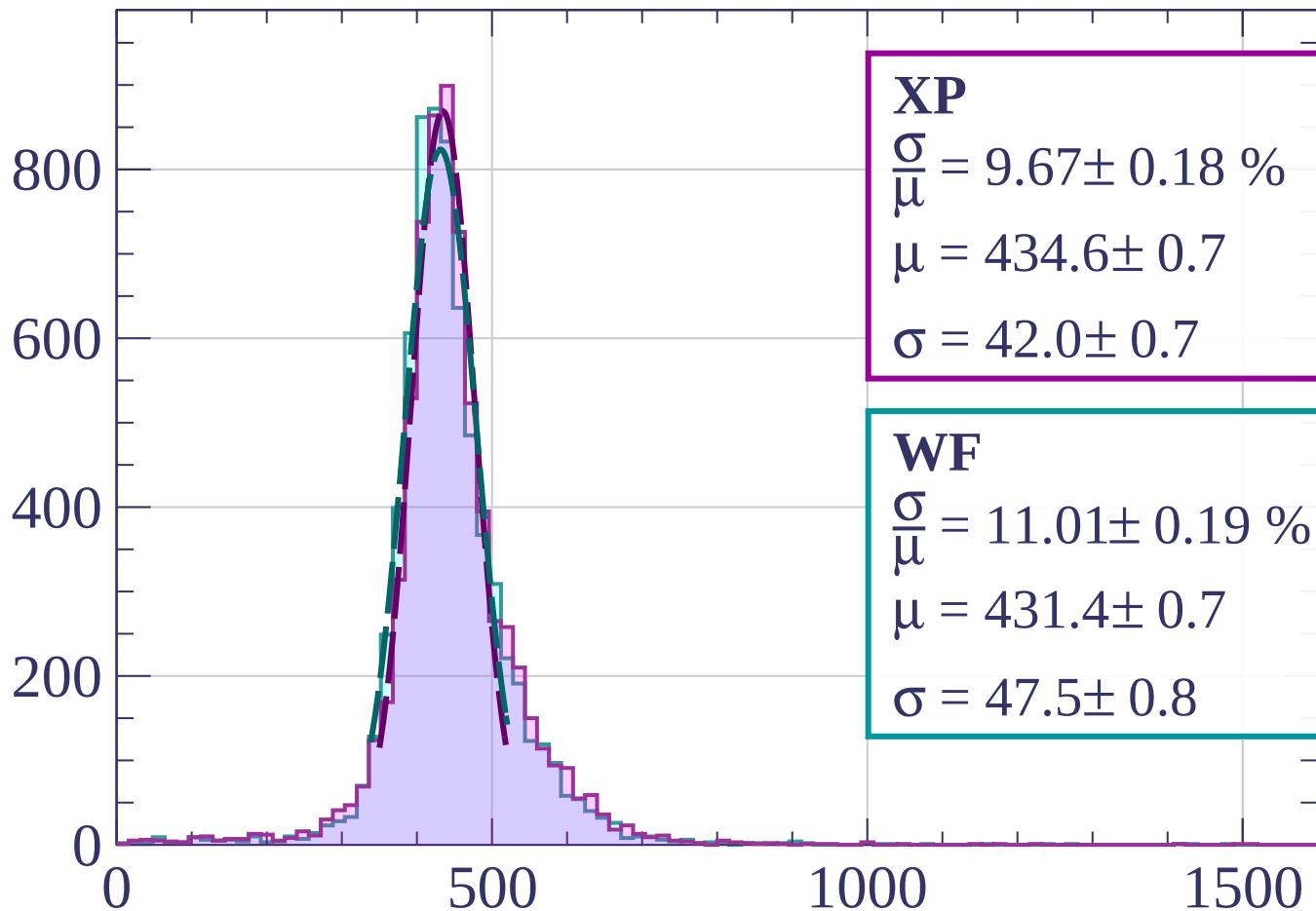
$$\mu = 459.3 \pm 0.8$$

$$\sigma = 58.1 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 9.67 \pm 0.18 \%$$

$$\mu = 434.6 \pm 0.7$$

$$\sigma = 42.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.01 \pm 0.19 \%$$

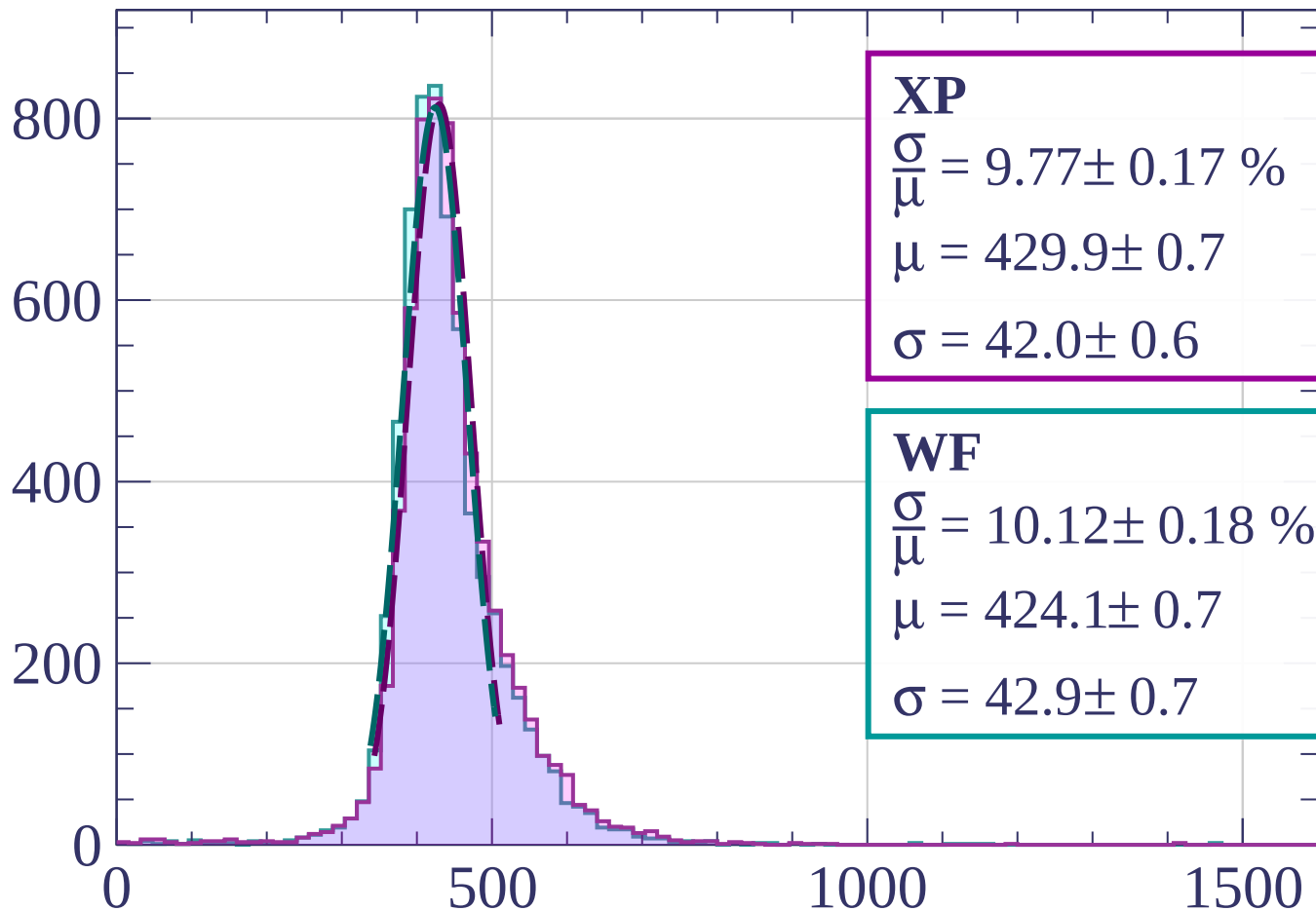
$$\mu = 431.4 \pm 0.7$$

$$\sigma = 47.5 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $320 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 9.77 \pm 0.17 \%$$

$$\mu = 429.9 \pm 0.7$$

$$\sigma = 42.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.12 \pm 0.18 \%$$

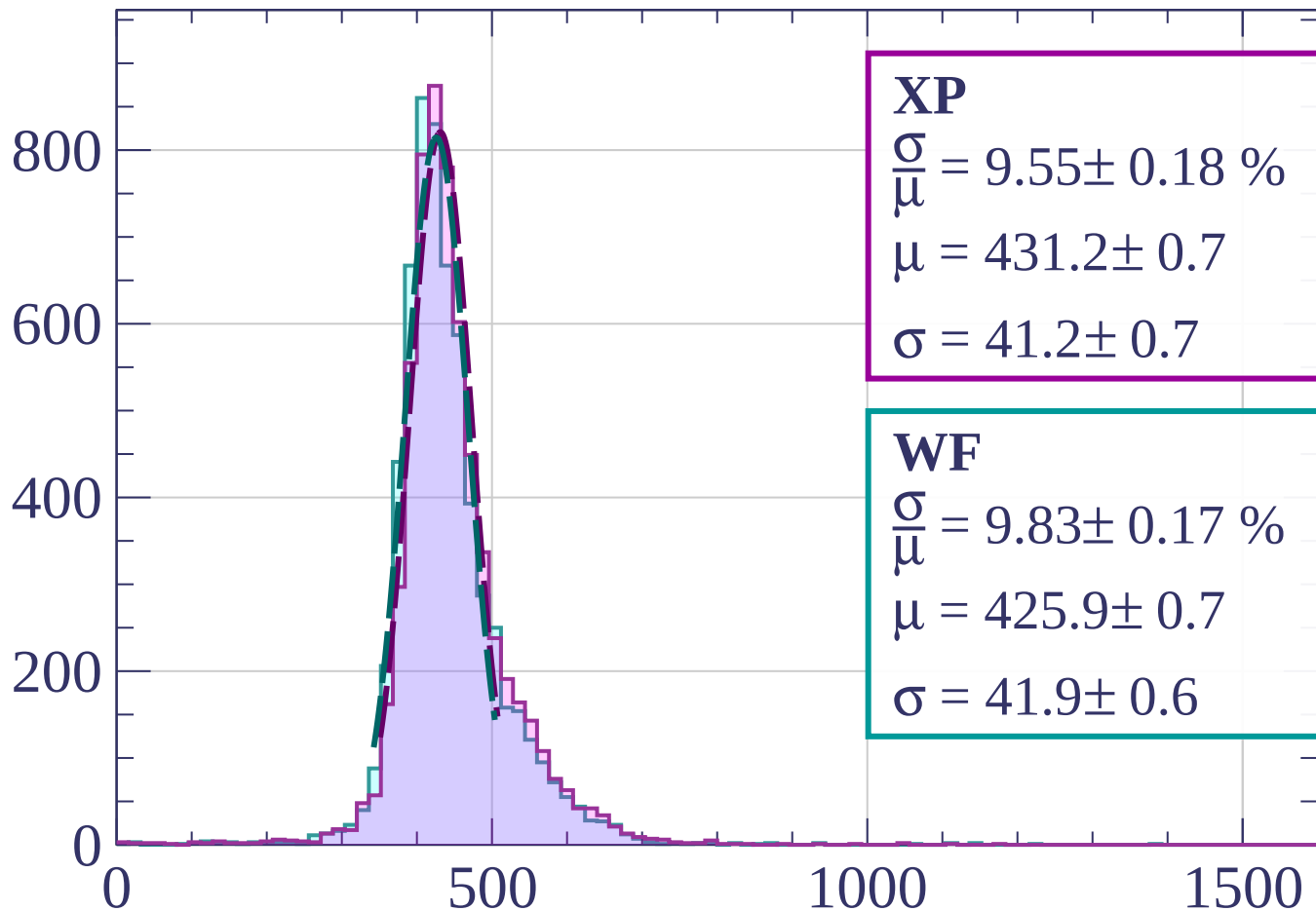
$$\mu = 424.1 \pm 0.7$$

$$\sigma = 42.9 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 9.55 \pm 0.18 \%$$

$$\mu = 431.2 \pm 0.7$$

$$\sigma = 41.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.83 \pm 0.17 \%$$

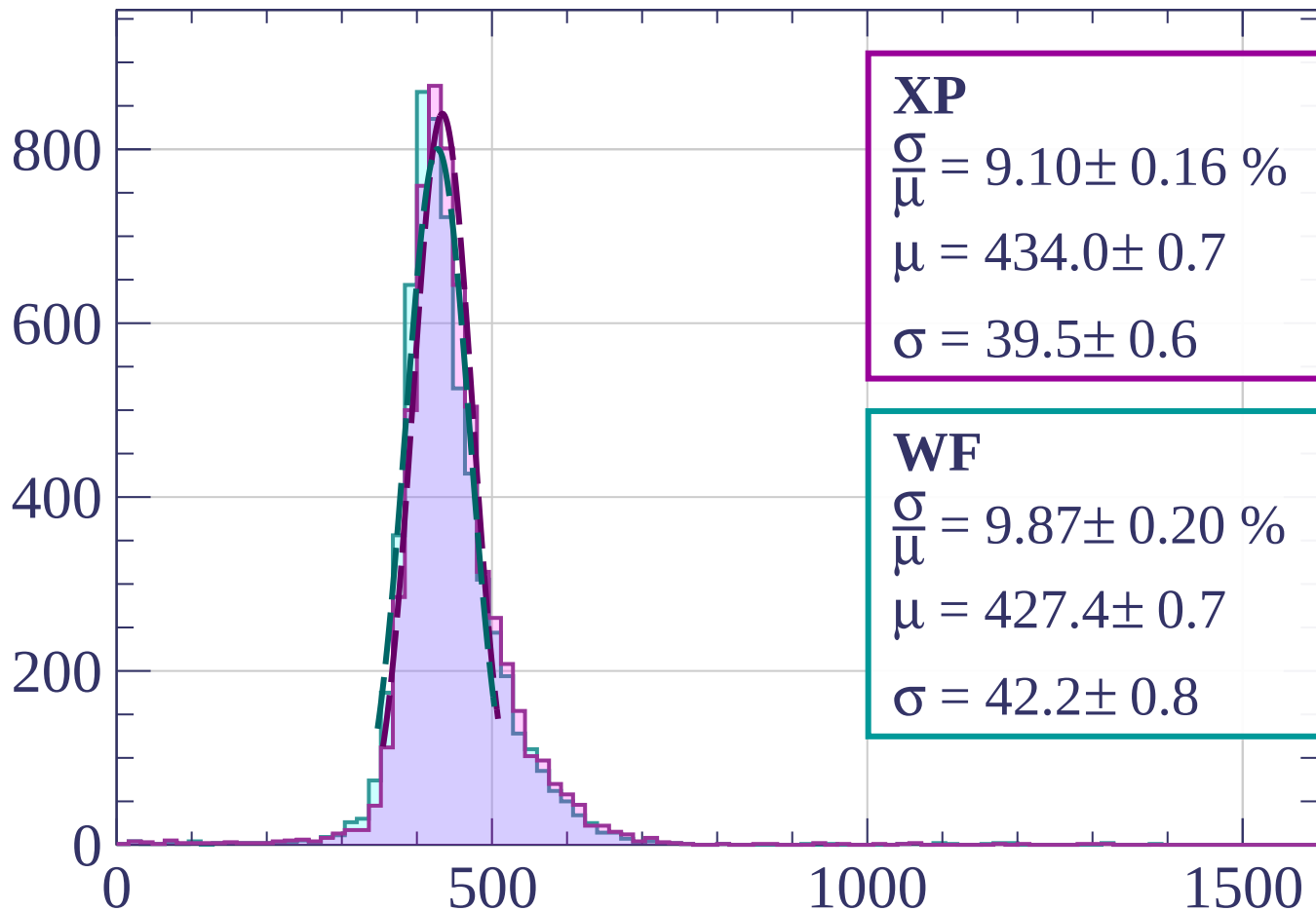
$$\mu = 425.9 \pm 0.7$$

$$\sigma = 41.9 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 9.10 \pm 0.16 \%$$

$$\mu = 434.0 \pm 0.7$$

$$\sigma = 39.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.87 \pm 0.20 \%$$

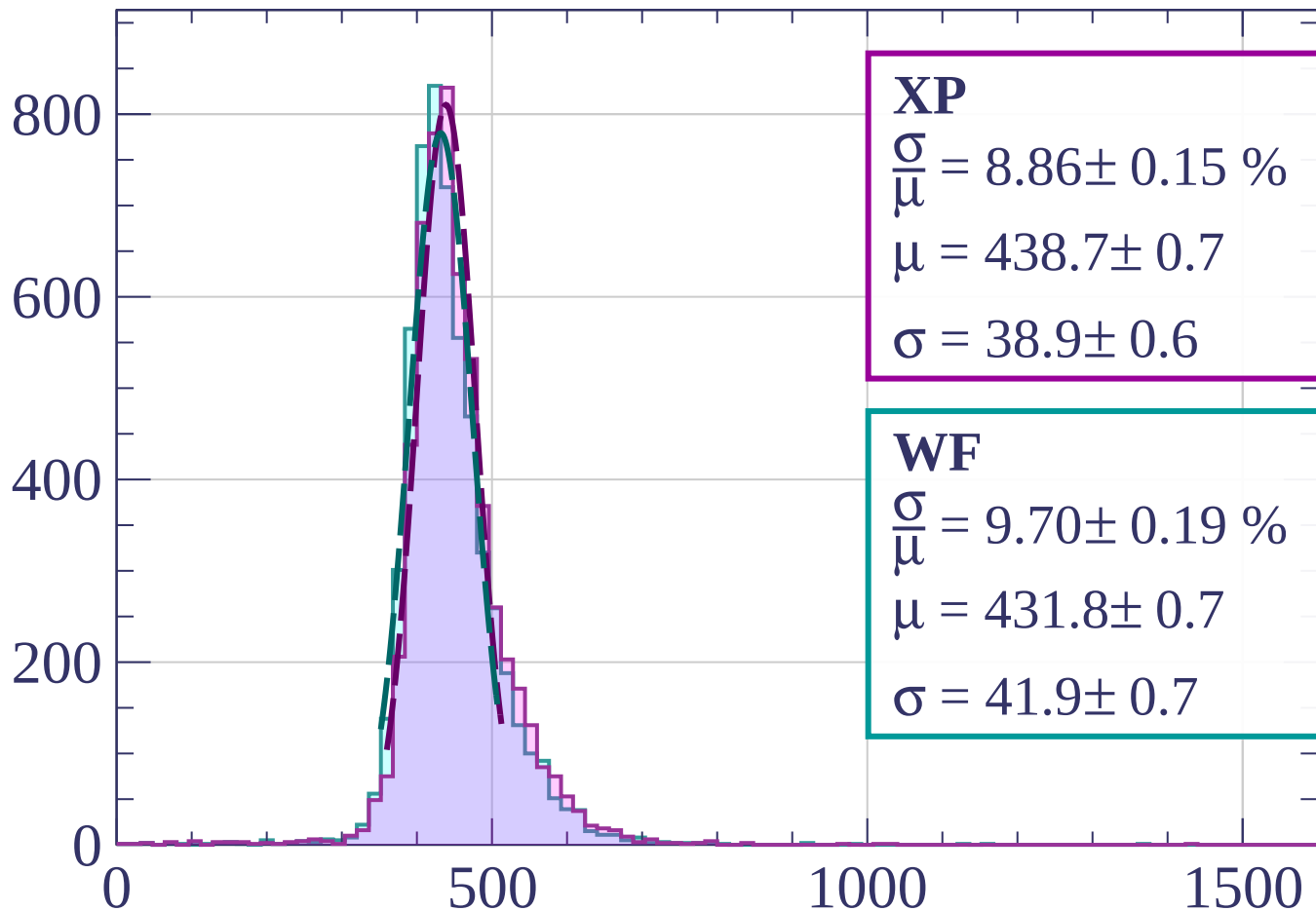
$$\mu = 427.4 \pm 0.7$$

$$\sigma = 42.2 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 8.86 \pm 0.15 \%$$

$$\mu = 438.7 \pm 0.7$$

$$\sigma = 38.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.70 \pm 0.19 \%$$

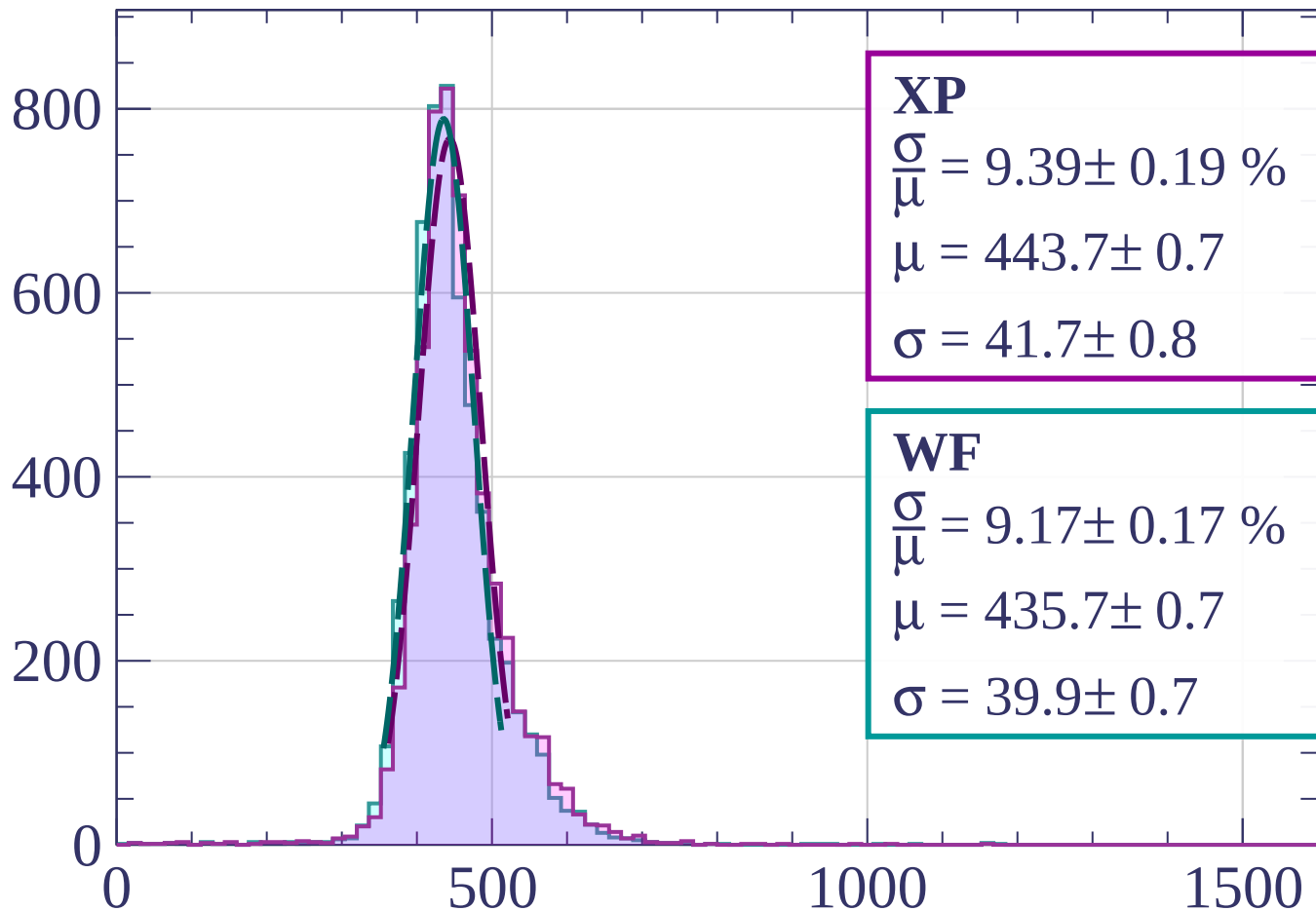
$$\mu = 431.8 \pm 0.7$$

$$\sigma = 41.9 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $640 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 9.39 \pm 0.19 \%$$

$$\mu = 443.7 \pm 0.7$$

$$\sigma = 41.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.17 \pm 0.17 \%$$

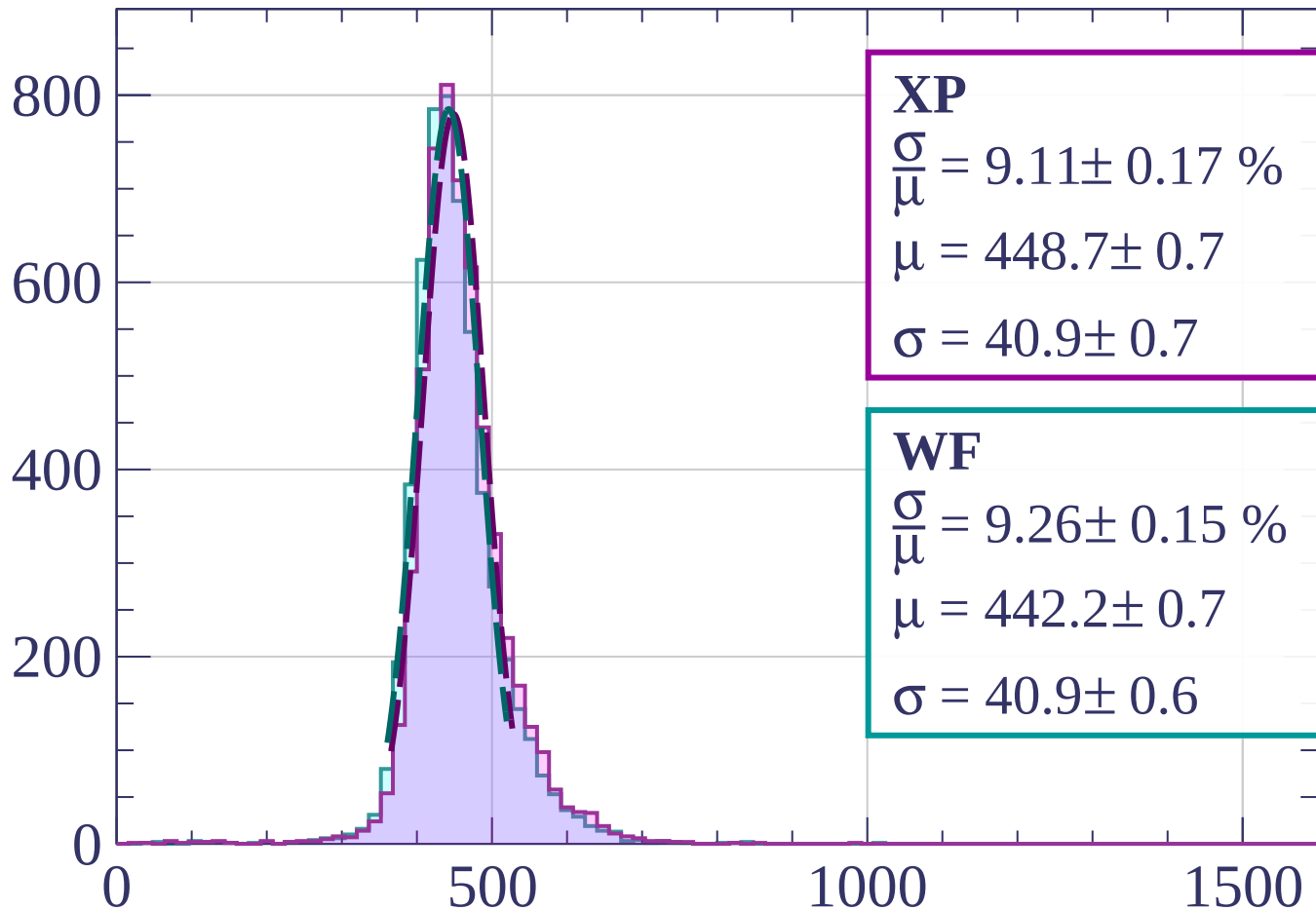
$$\mu = 435.7 \pm 0.7$$

$$\sigma = 39.9 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 9.11 \pm 0.17 \%$$

$$\mu = 448.7 \pm 0.7$$

$$\sigma = 40.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.26 \pm 0.15 \%$$

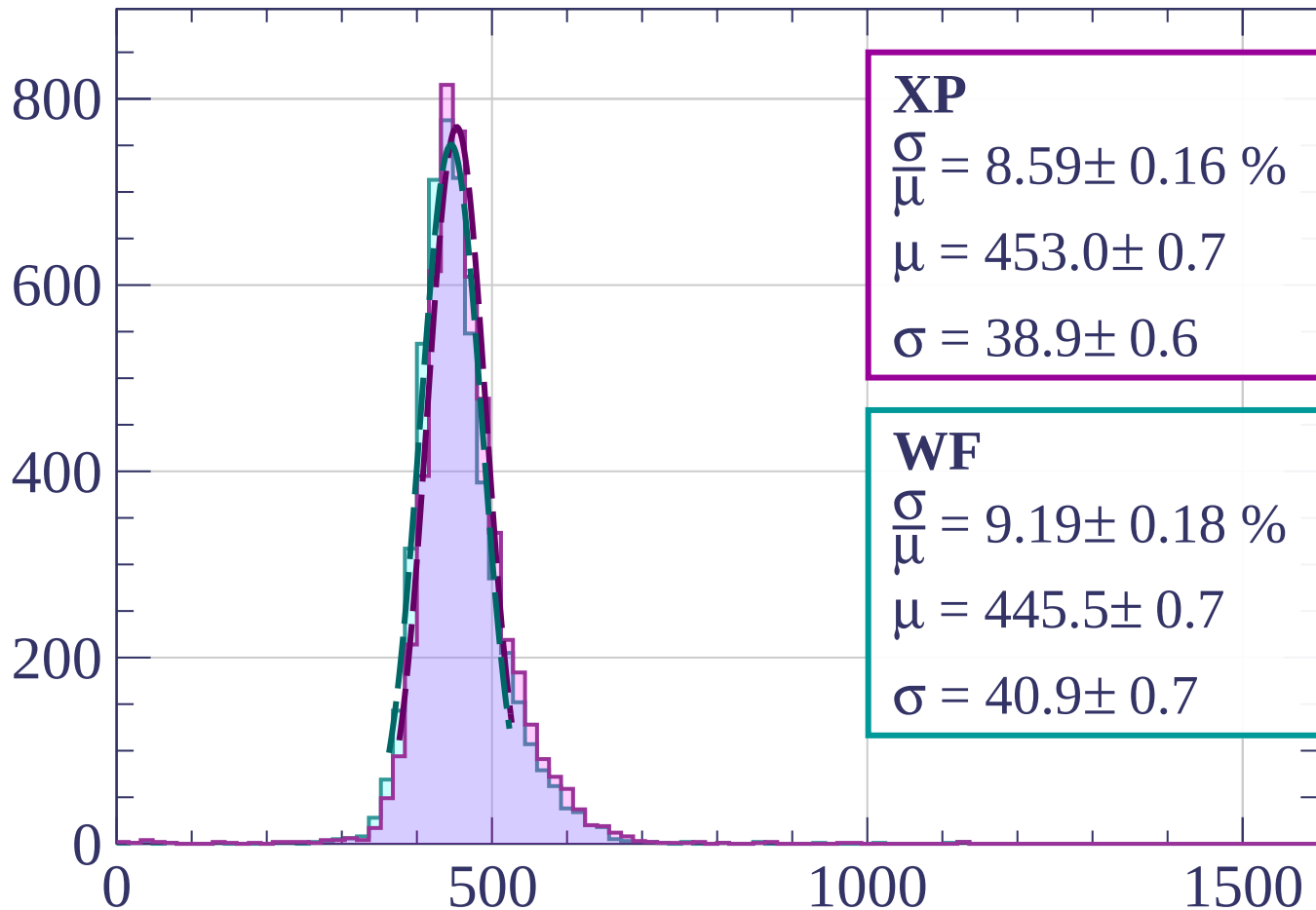
$$\mu = 442.2 \pm 0.7$$

$$\sigma = 40.9 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $800 < p < 880$

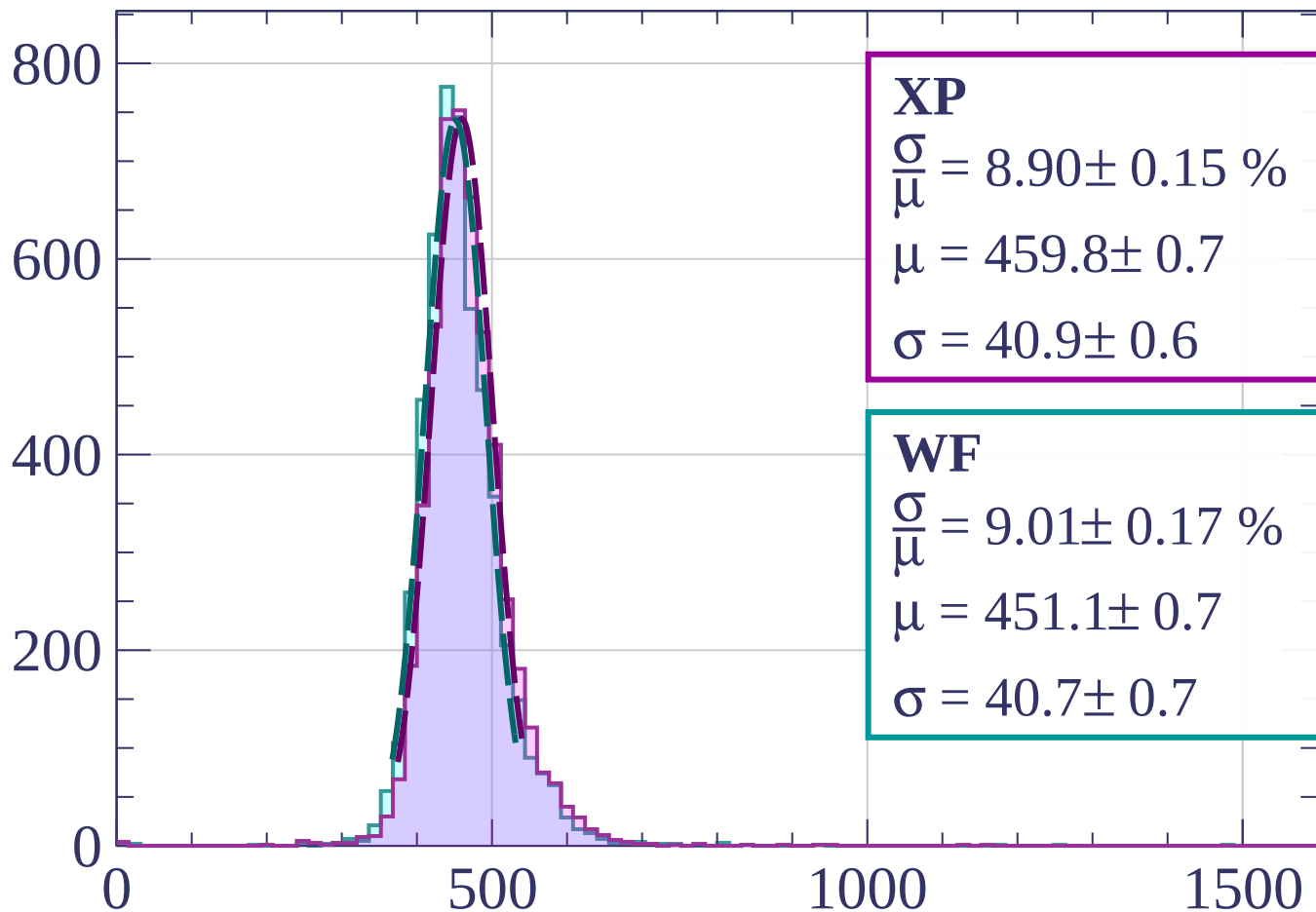
Count



dE/dx [ADC counts/cm]

Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.90 \pm 0.15 \%$$

$$\mu = 459.8 \pm 0.7$$

$$\sigma = 40.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.01 \pm 0.17 \%$$

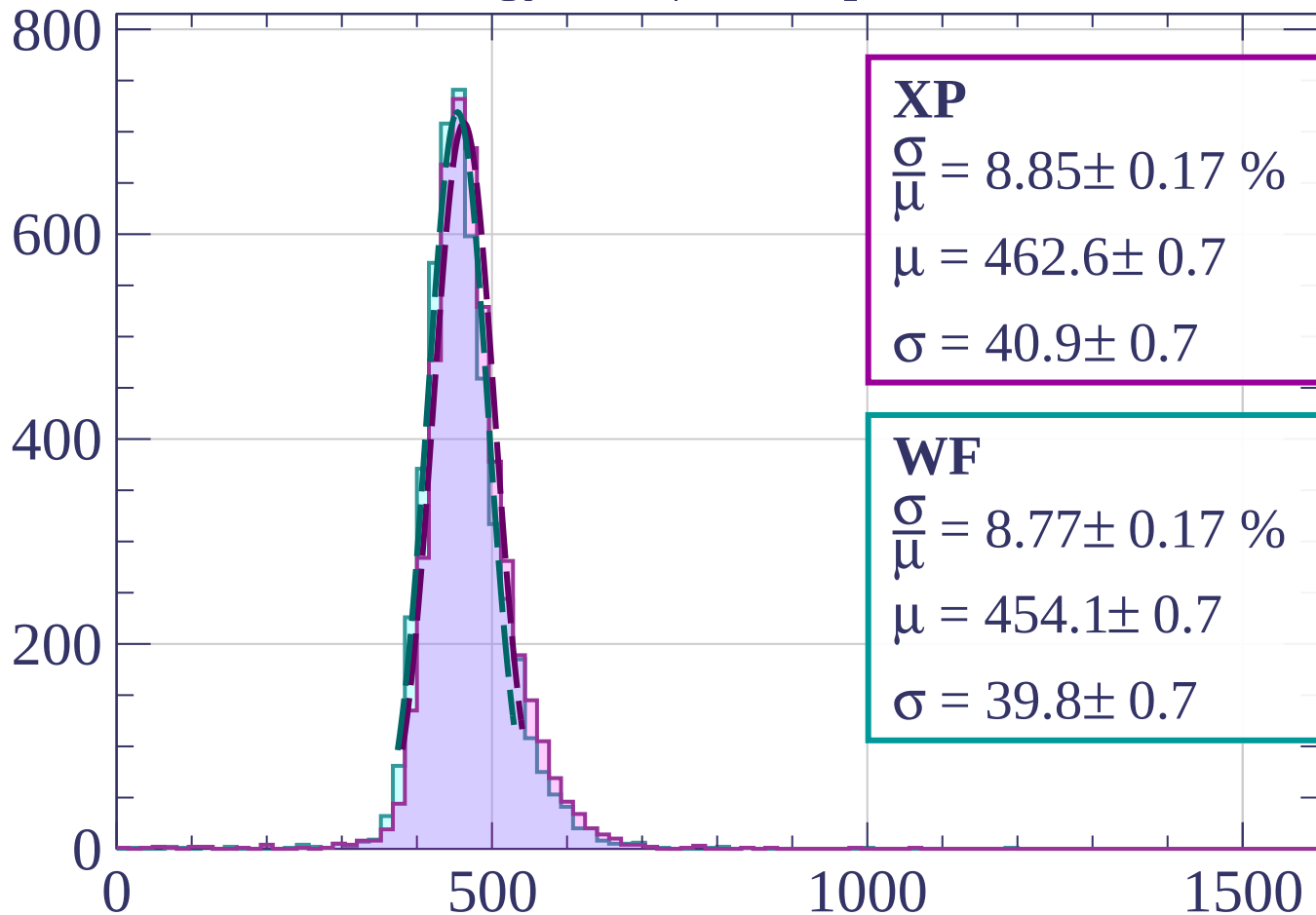
$$\mu = 451.1 \pm 0.7$$

$$\sigma = 40.7 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $960 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 8.85 \pm 0.17 \%$$

$$\mu = 462.6 \pm 0.7$$

$$\sigma = 40.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.77 \pm 0.17 \%$$

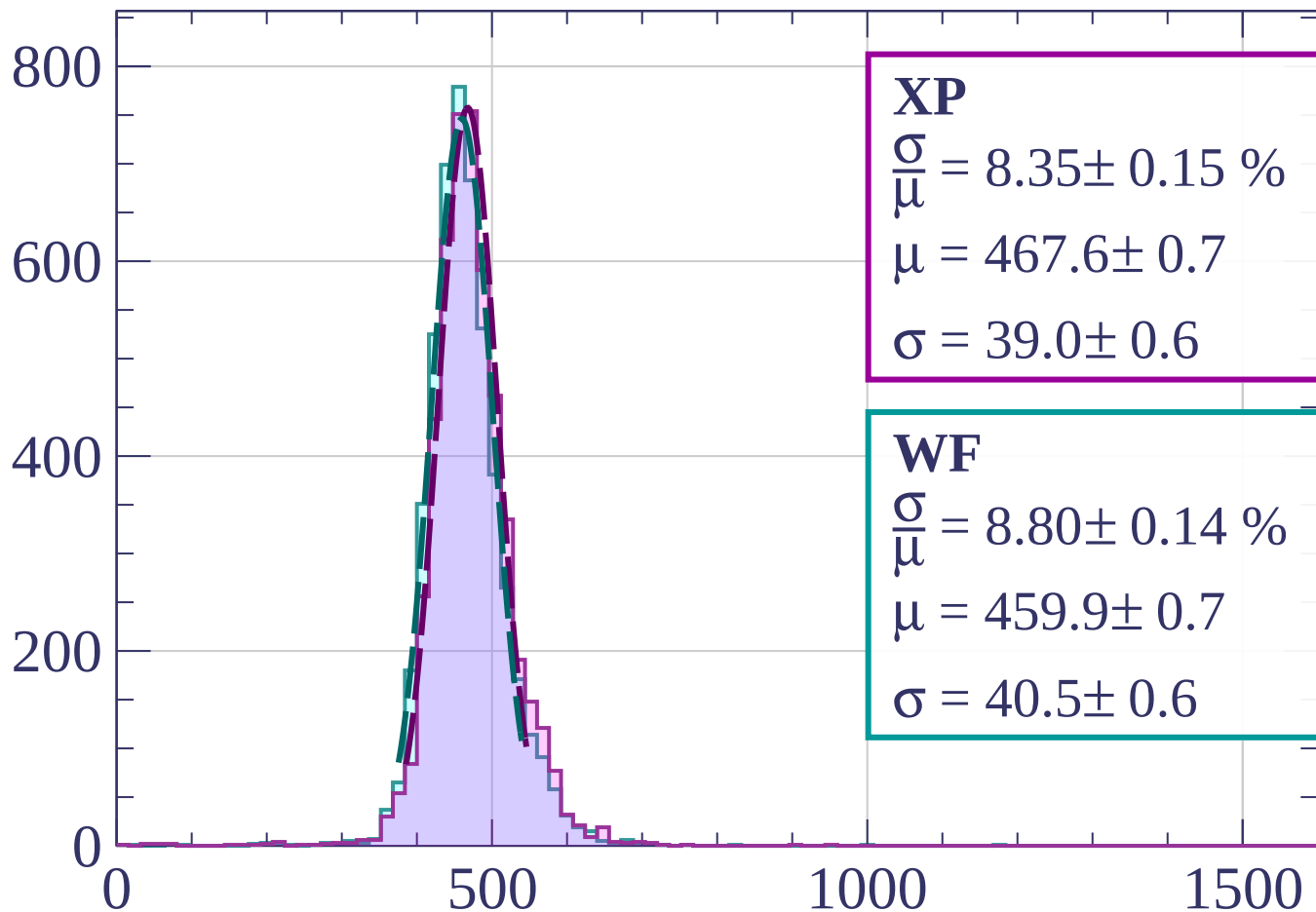
$$\mu = 454.1 \pm 0.7$$

$$\sigma = 39.8 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 8.35 \pm 0.15 \%$$

$$\mu = 467.6 \pm 0.7$$

$$\sigma = 39.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.80 \pm 0.14 \%$$

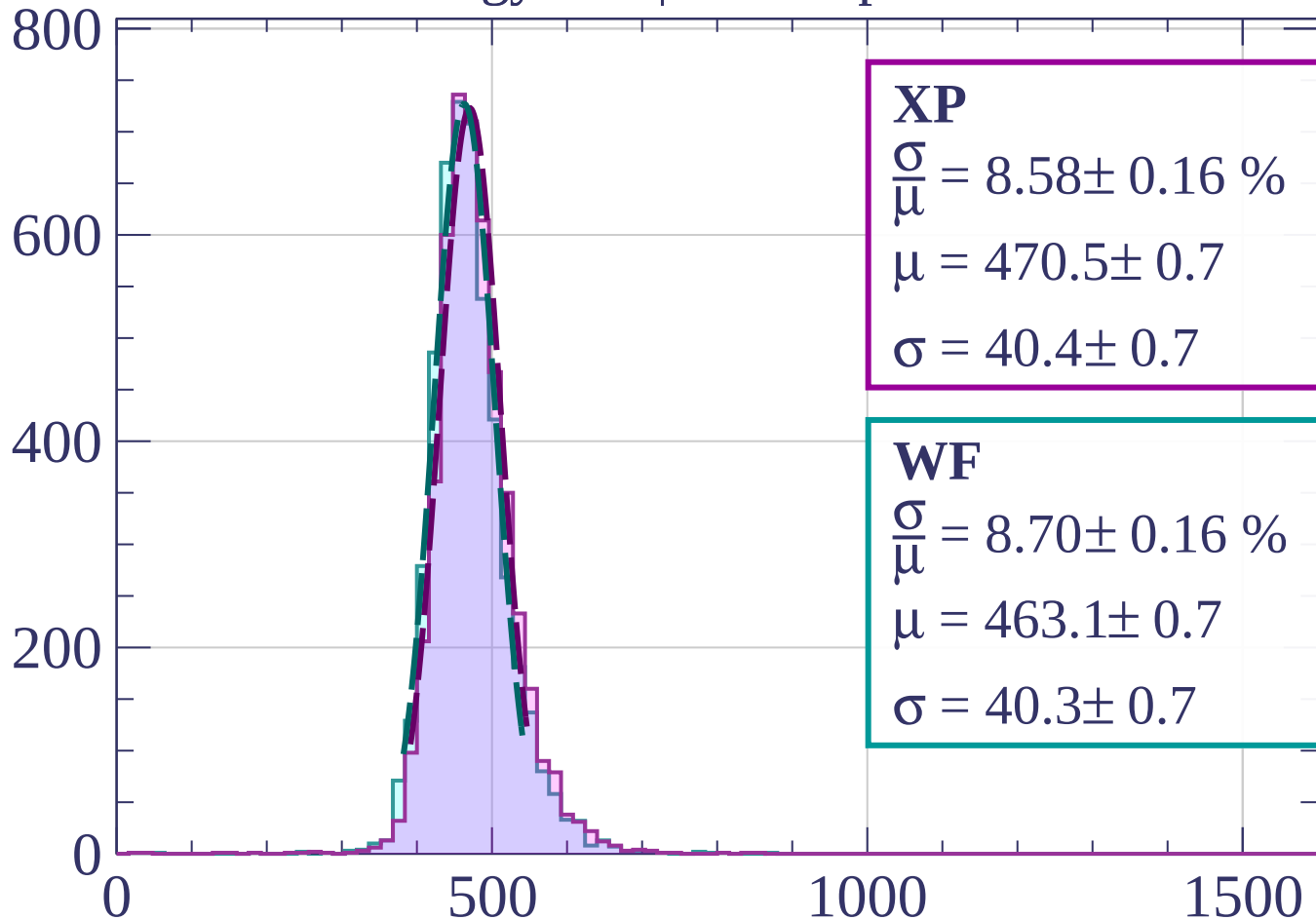
$$\mu = 459.9 \pm 0.7$$

$$\sigma = 40.5 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 8.58 \pm 0.16 \%$$

$$\mu = 470.5 \pm 0.7$$

$$\sigma = 40.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.70 \pm 0.16 \%$$

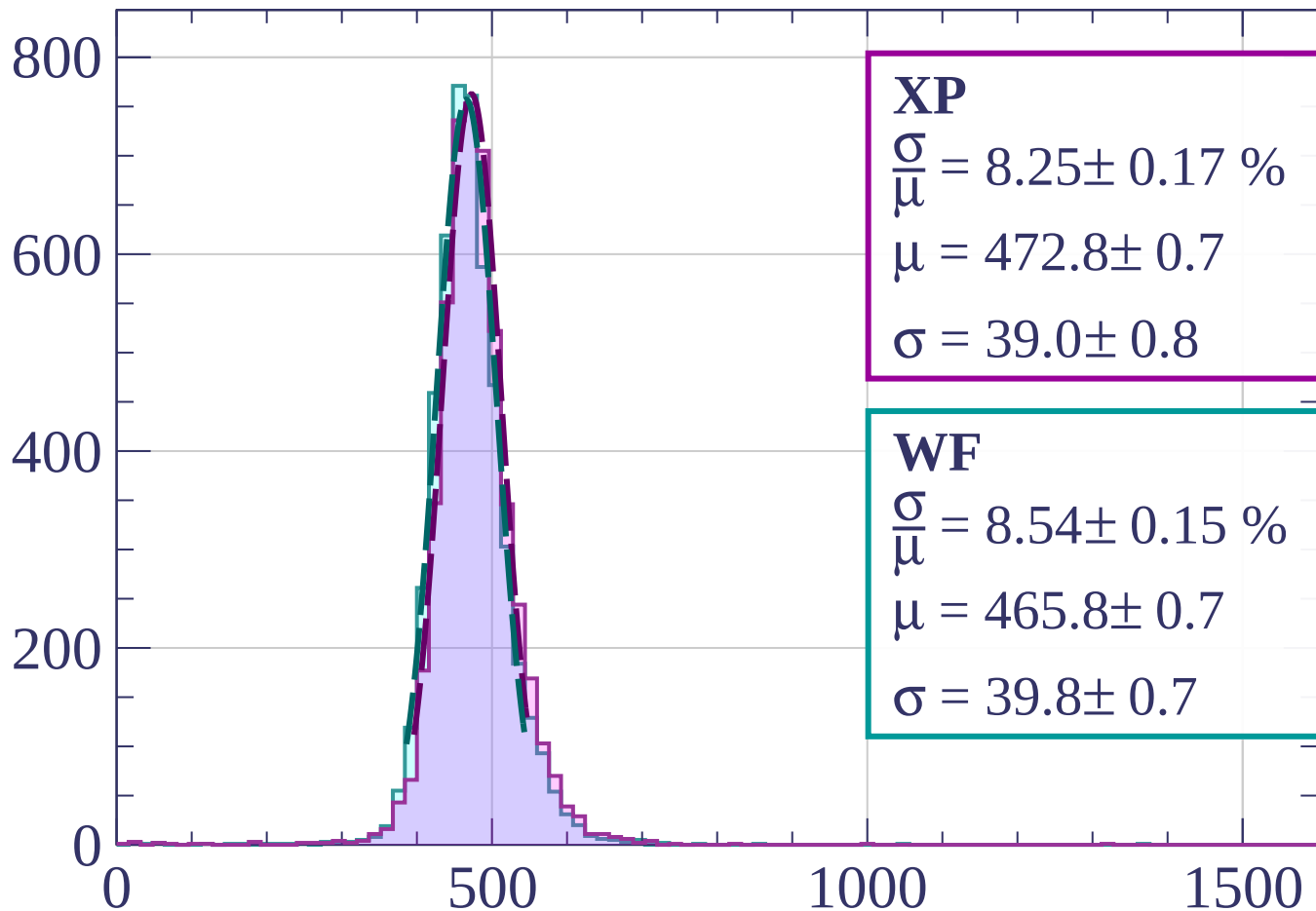
$$\mu = 463.1 \pm 0.7$$

$$\sigma = 40.3 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1200 < p < 1280$

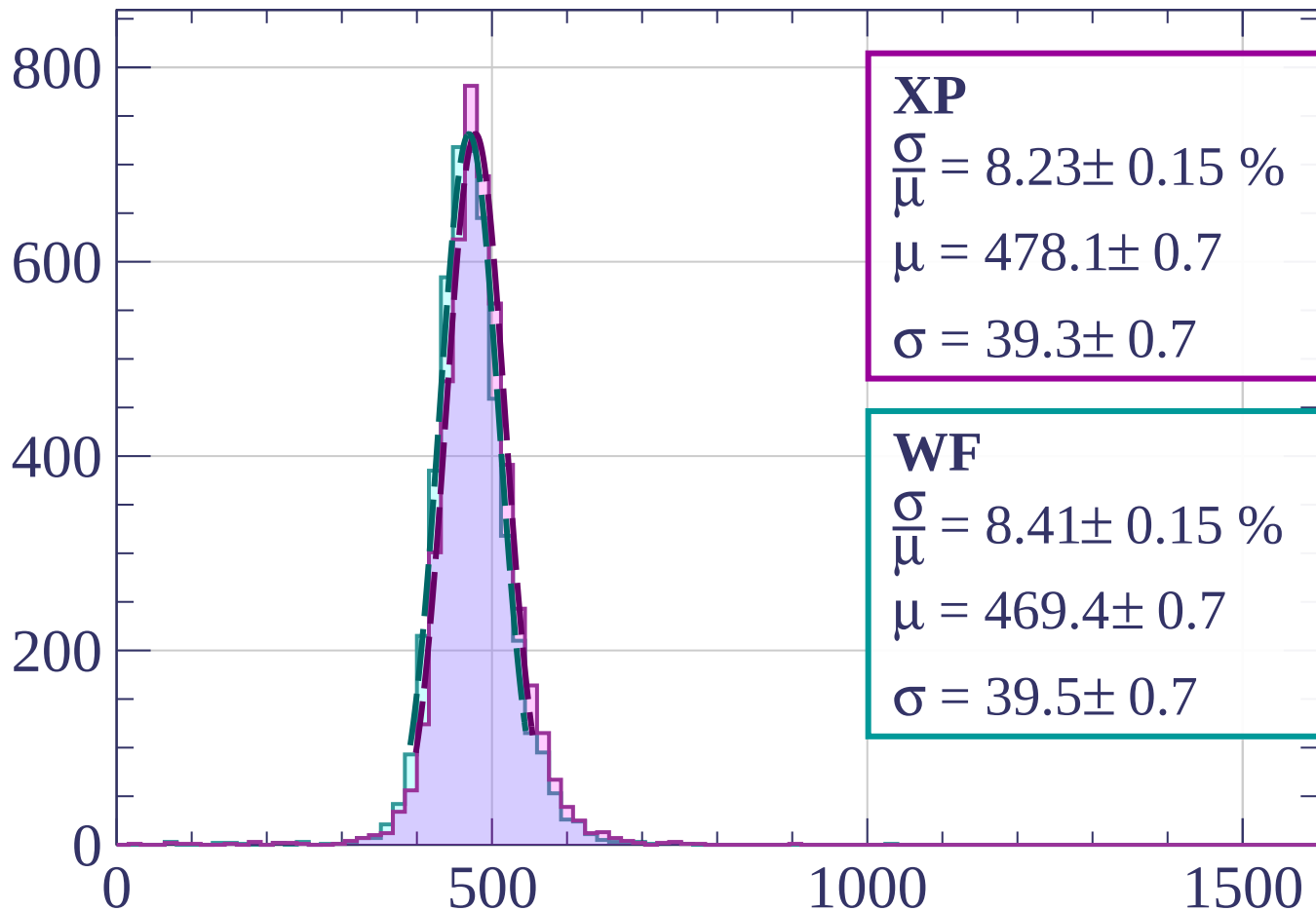
Count



dE/dx [ADC counts/cm]

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 8.23 \pm 0.15 \%$$

$$\mu = 478.1 \pm 0.7$$

$$\sigma = 39.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.41 \pm 0.15 \%$$

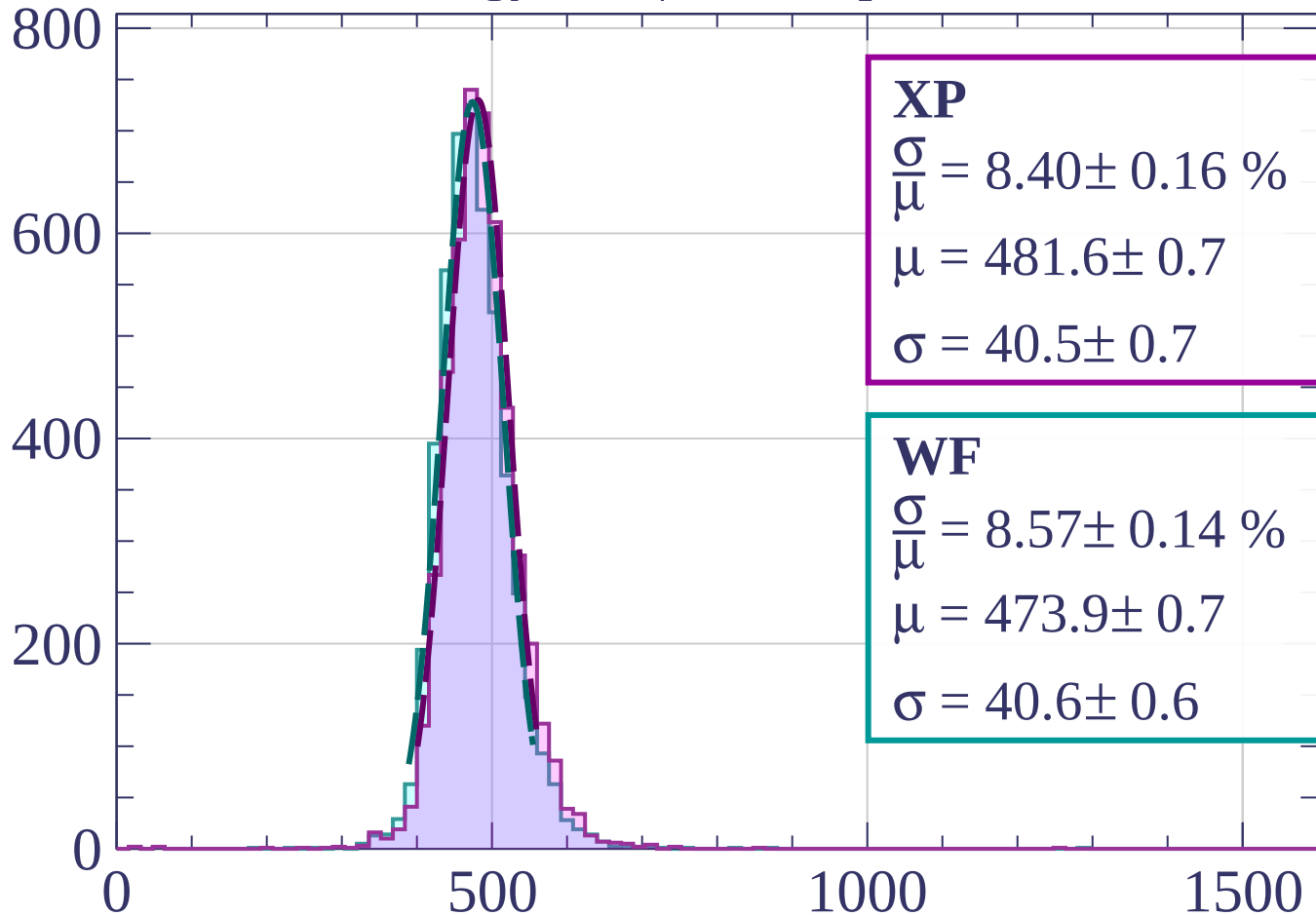
$$\mu = 469.4 \pm 0.7$$

$$\sigma = 39.5 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 8.40 \pm 0.16 \%$$

$$\mu = 481.6 \pm 0.7$$

$$\sigma = 40.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.57 \pm 0.14 \%$$

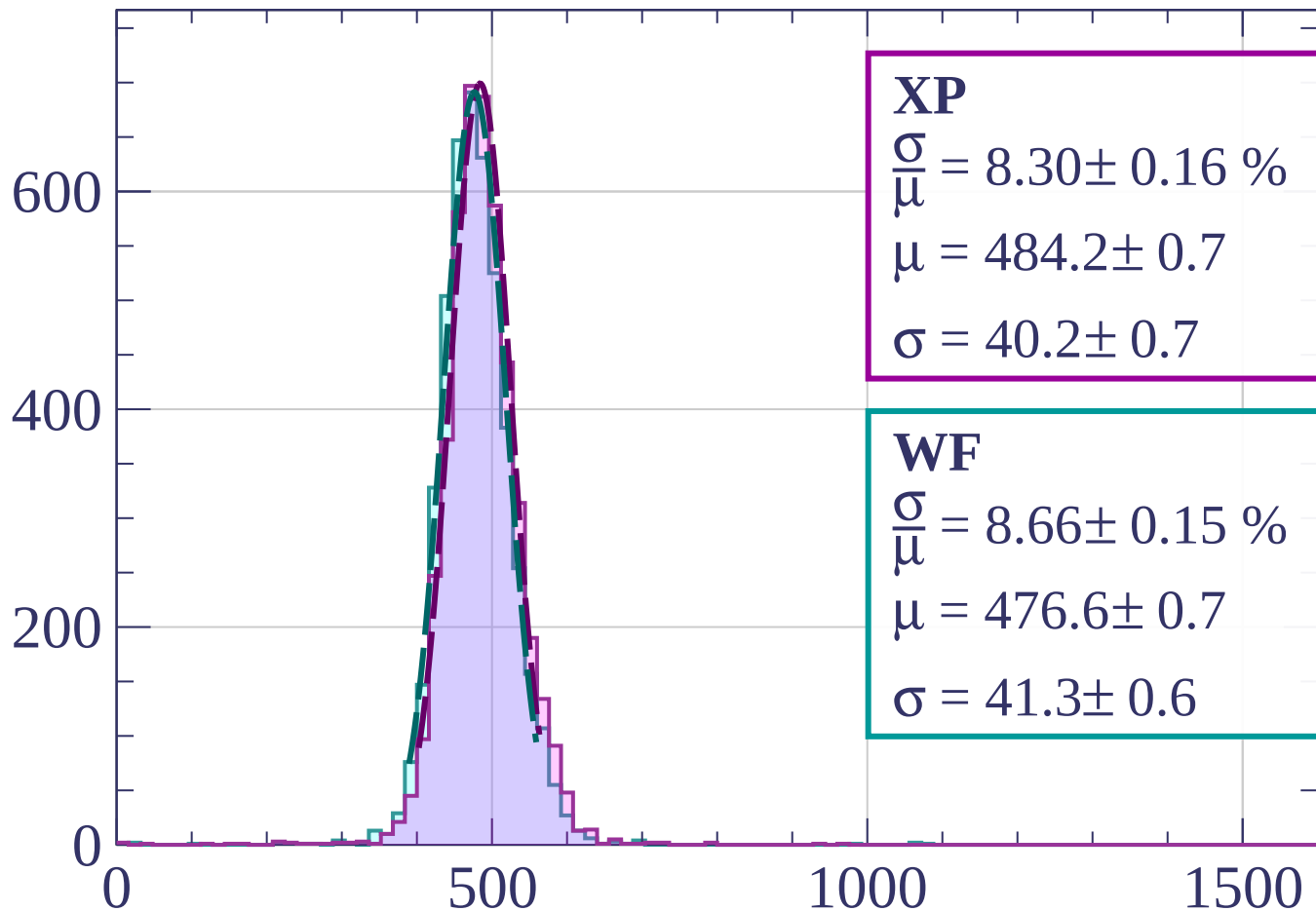
$$\mu = 473.9 \pm 0.7$$

$$\sigma = 40.6 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 8.30 \pm 0.16 \%$$

$$\mu = 484.2 \pm 0.7$$

$$\sigma = 40.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.66 \pm 0.15 \%$$

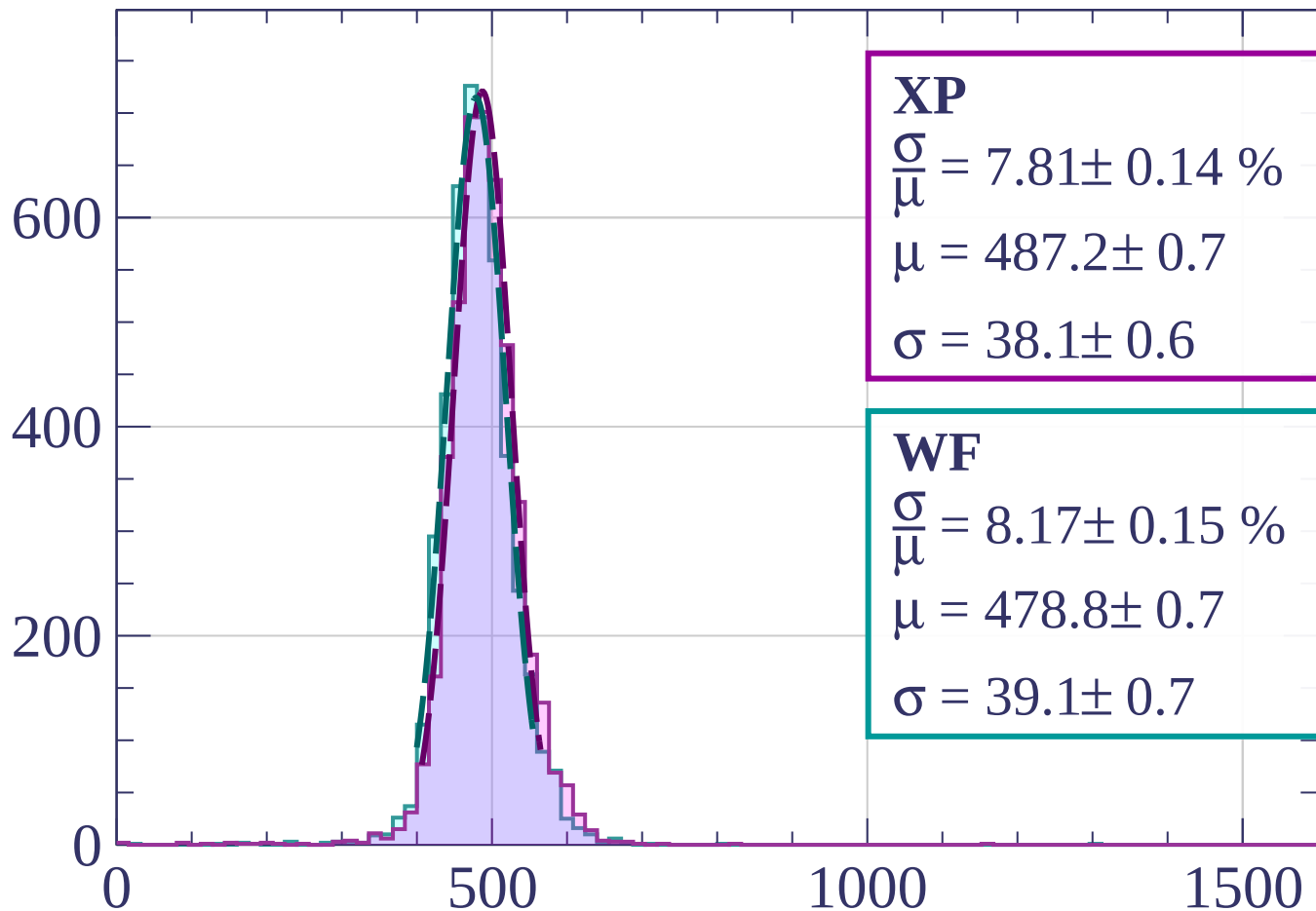
$$\mu = 476.6 \pm 0.7$$

$$\sigma = 41.3 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $1520 < p < 1600$

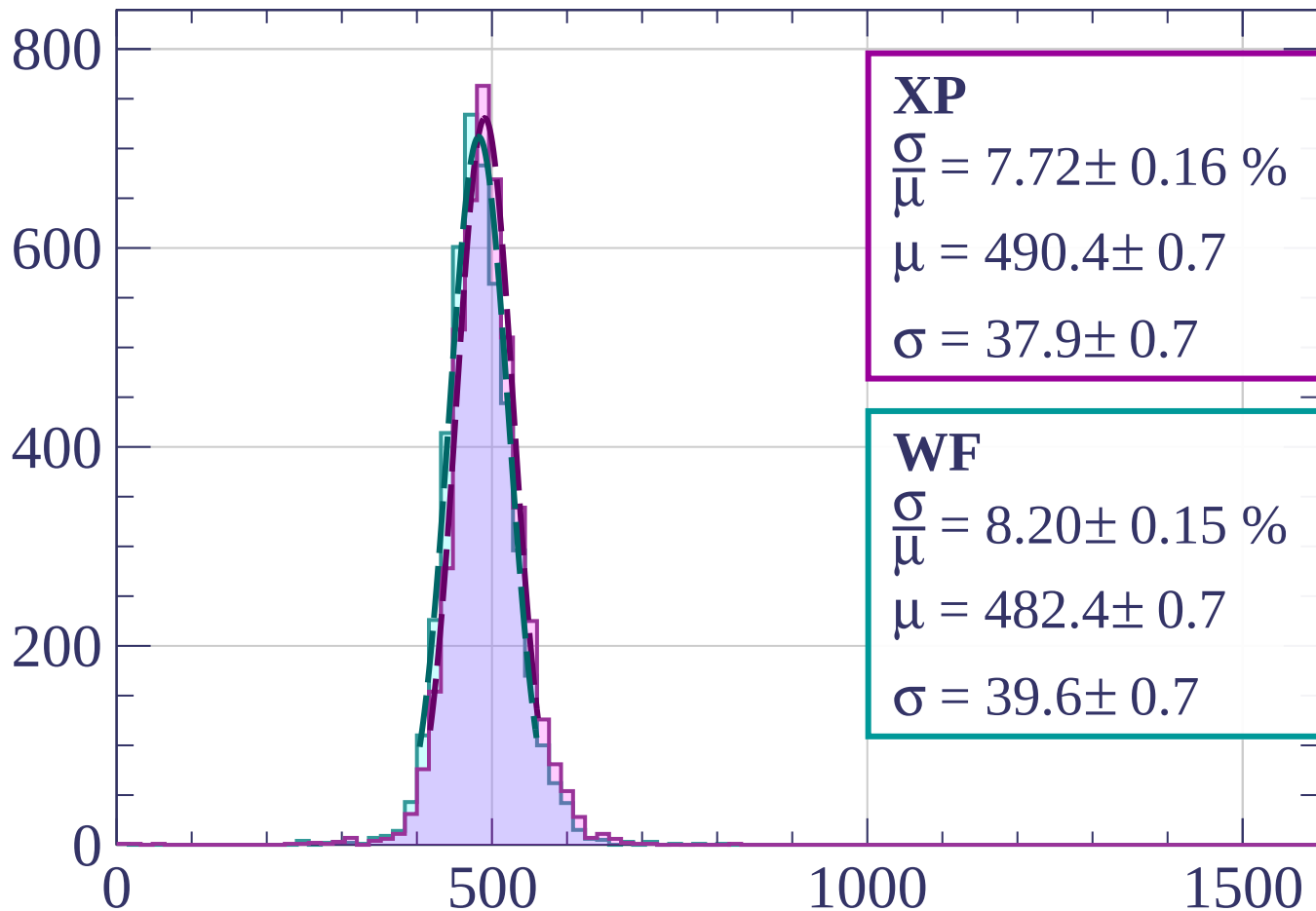
Count



dE/dx [ADC counts/cm]

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 7.72 \pm 0.16 \%$$

$$\mu = 490.4 \pm 0.7$$

$$\sigma = 37.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.20 \pm 0.15 \%$$

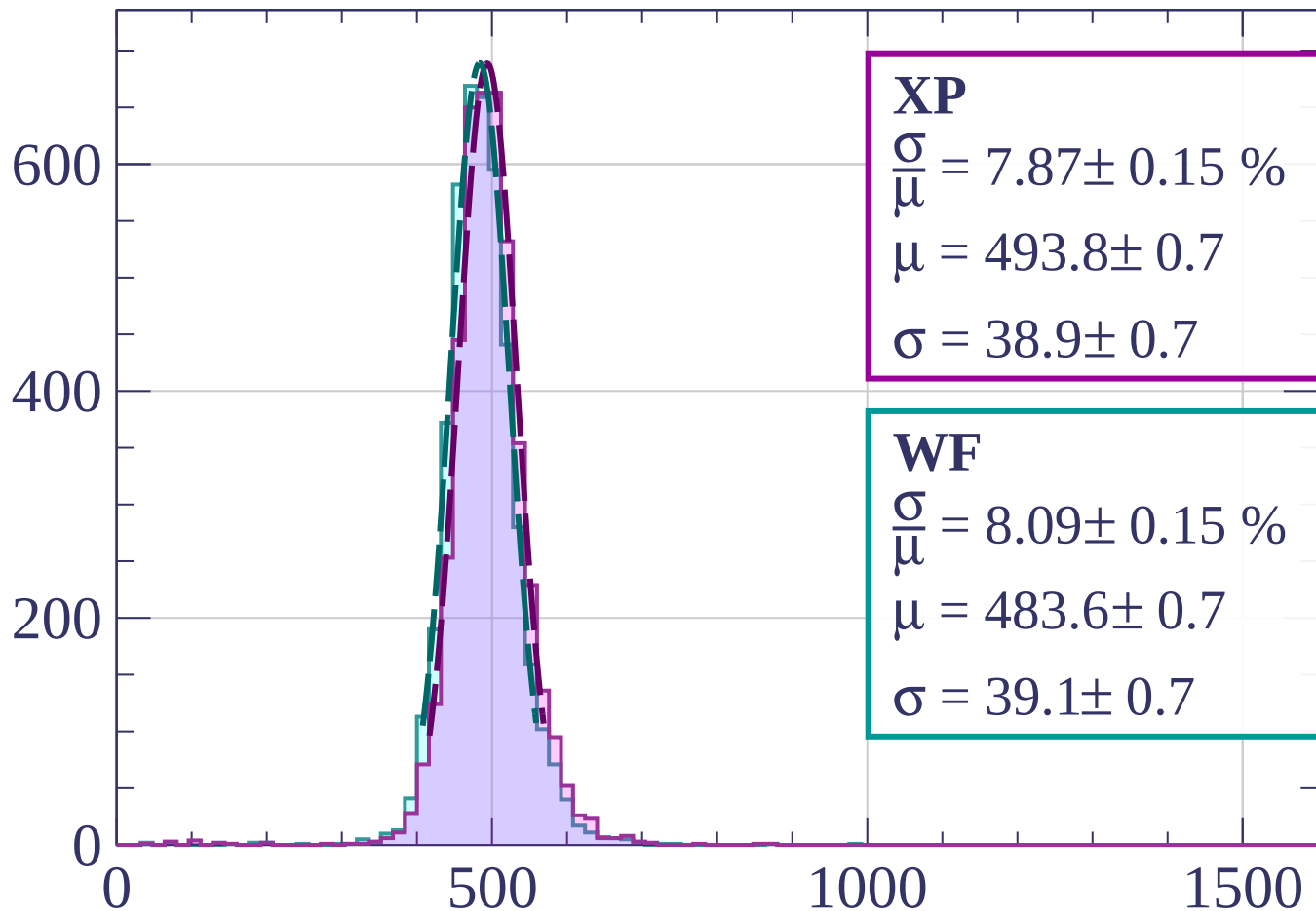
$$\mu = 482.4 \pm 0.7$$

$$\sigma = 39.6 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 7.87 \pm 0.15 \%$$

$$\mu = 493.8 \pm 0.7$$

$$\sigma = 38.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.09 \pm 0.15 \%$$

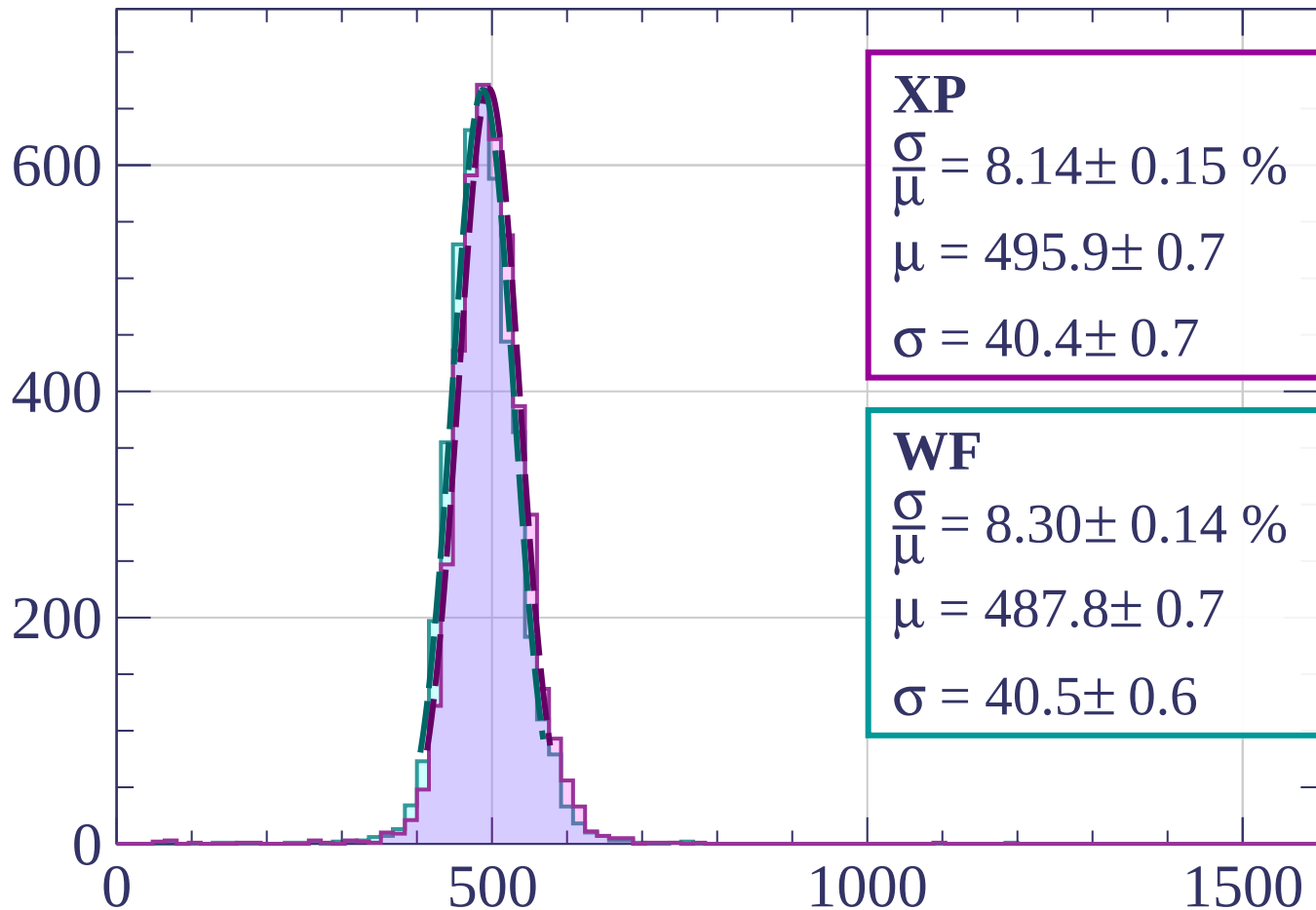
$$\mu = 483.6 \pm 0.7$$

$$\sigma = 39.1 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.15 \%$$

$$\mu = 495.9 \pm 0.7$$

$$\sigma = 40.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.30 \pm 0.14 \%$$

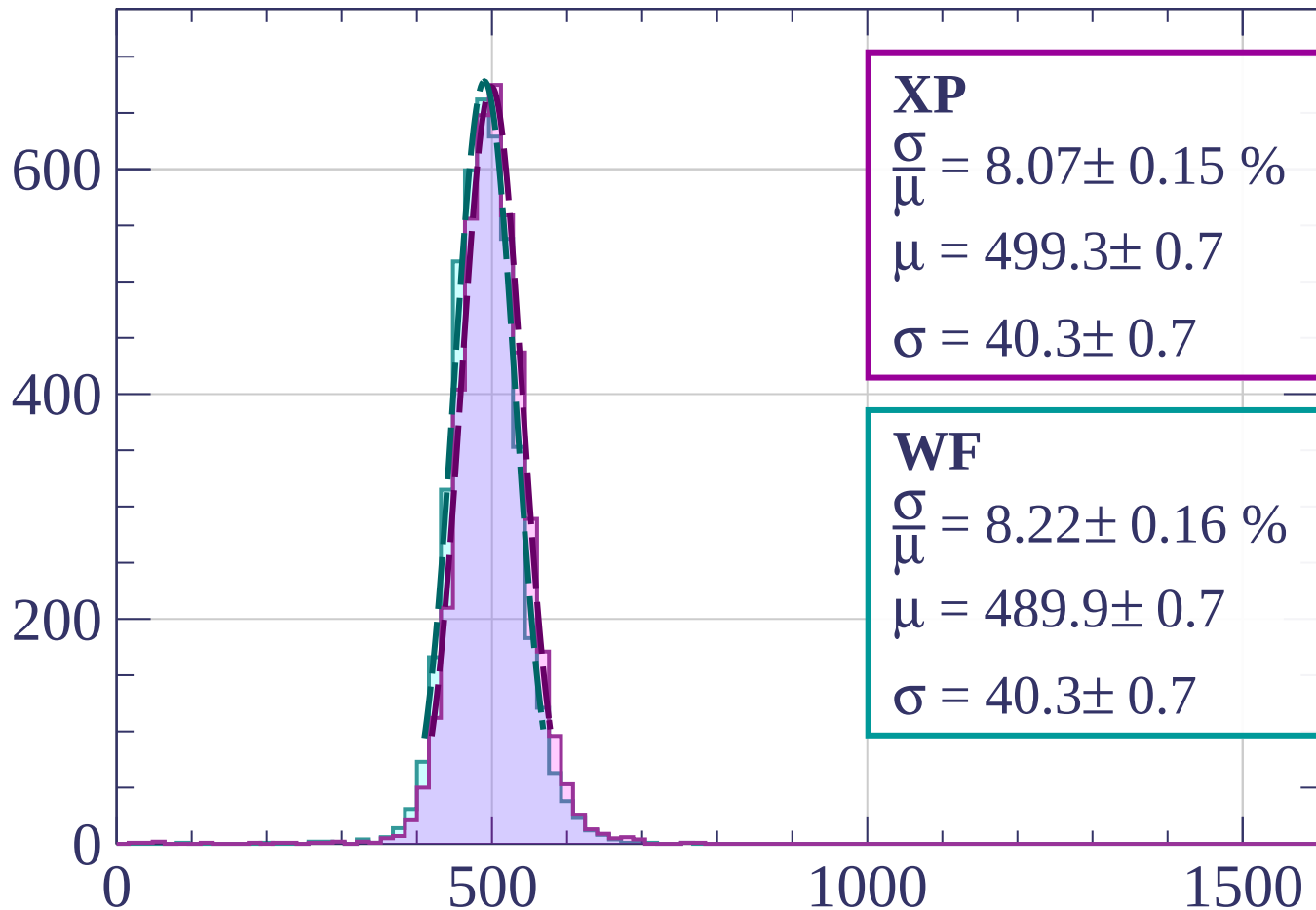
$$\mu = 487.8 \pm 0.7$$

$$\sigma = 40.5 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 8.07 \pm 0.15 \%$$

$$\mu = 499.3 \pm 0.7$$

$$\sigma = 40.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.22 \pm 0.16 \%$$

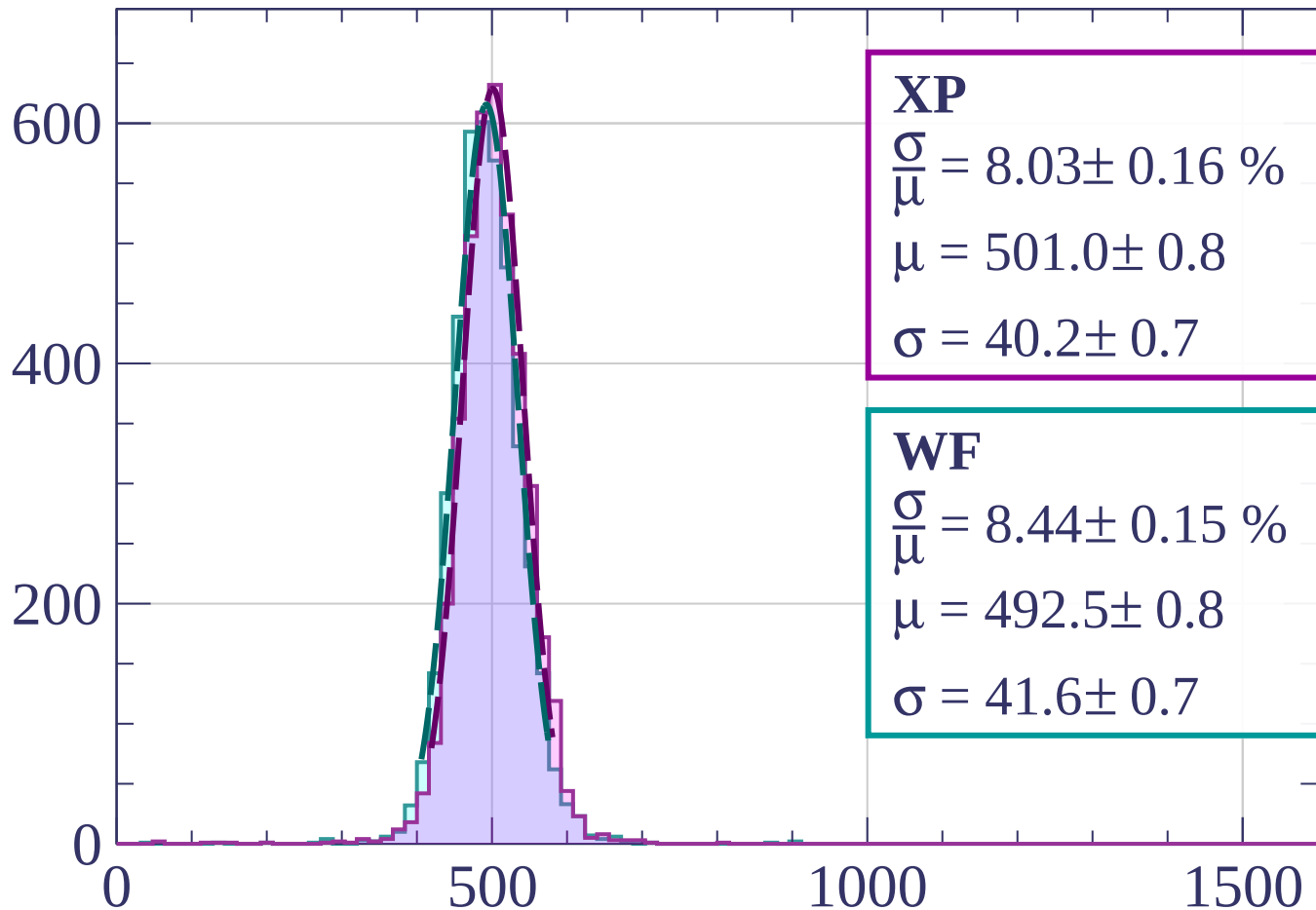
$$\mu = 489.9 \pm 0.7$$

$$\sigma = 40.3 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 8.03 \pm 0.16 \%$$

$$\mu = 501.0 \pm 0.8$$

$$\sigma = 40.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.44 \pm 0.15 \%$$

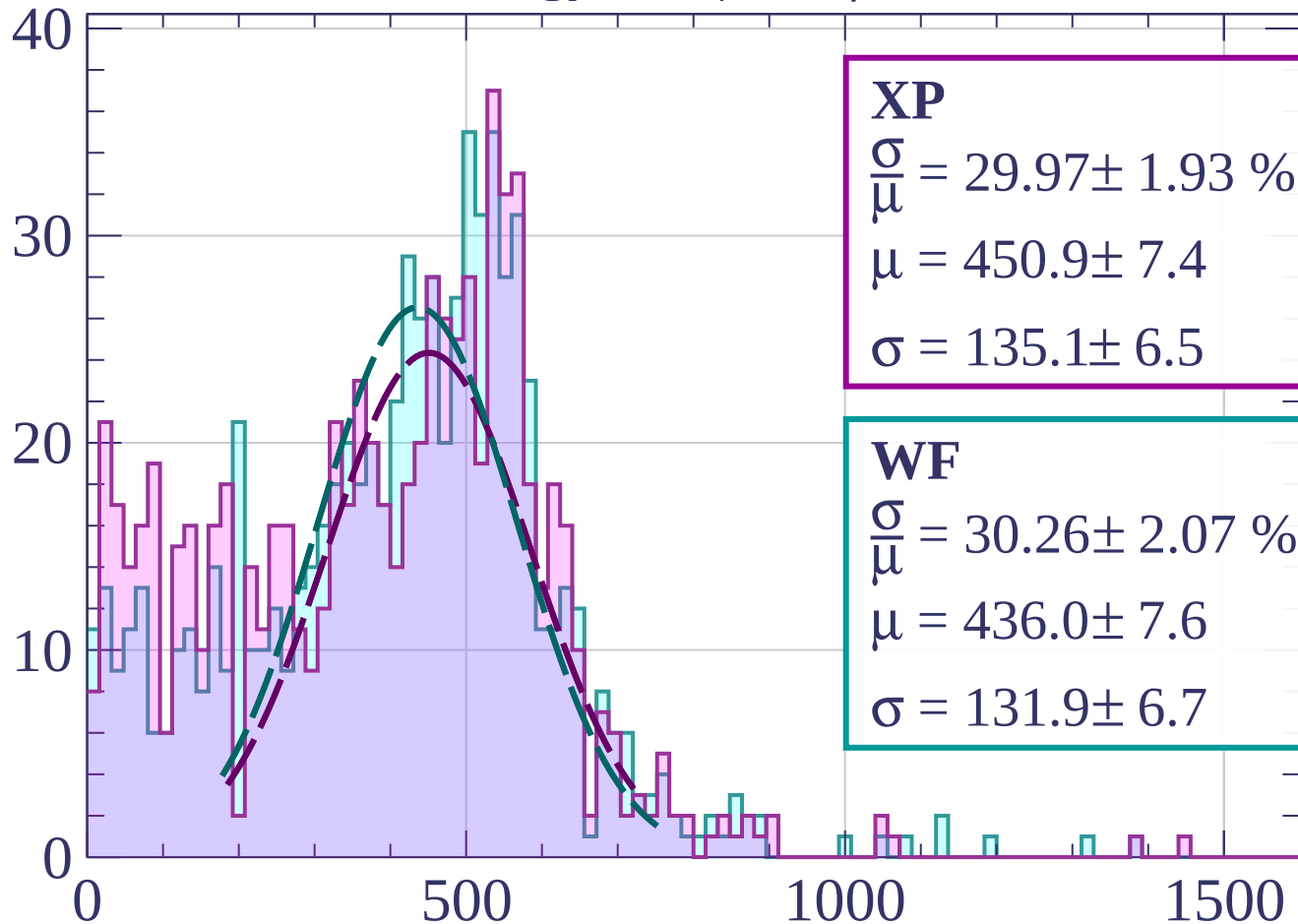
$$\mu = 492.5 \pm 0.8$$

$$\sigma = 41.6 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $0 < \phi < 3$

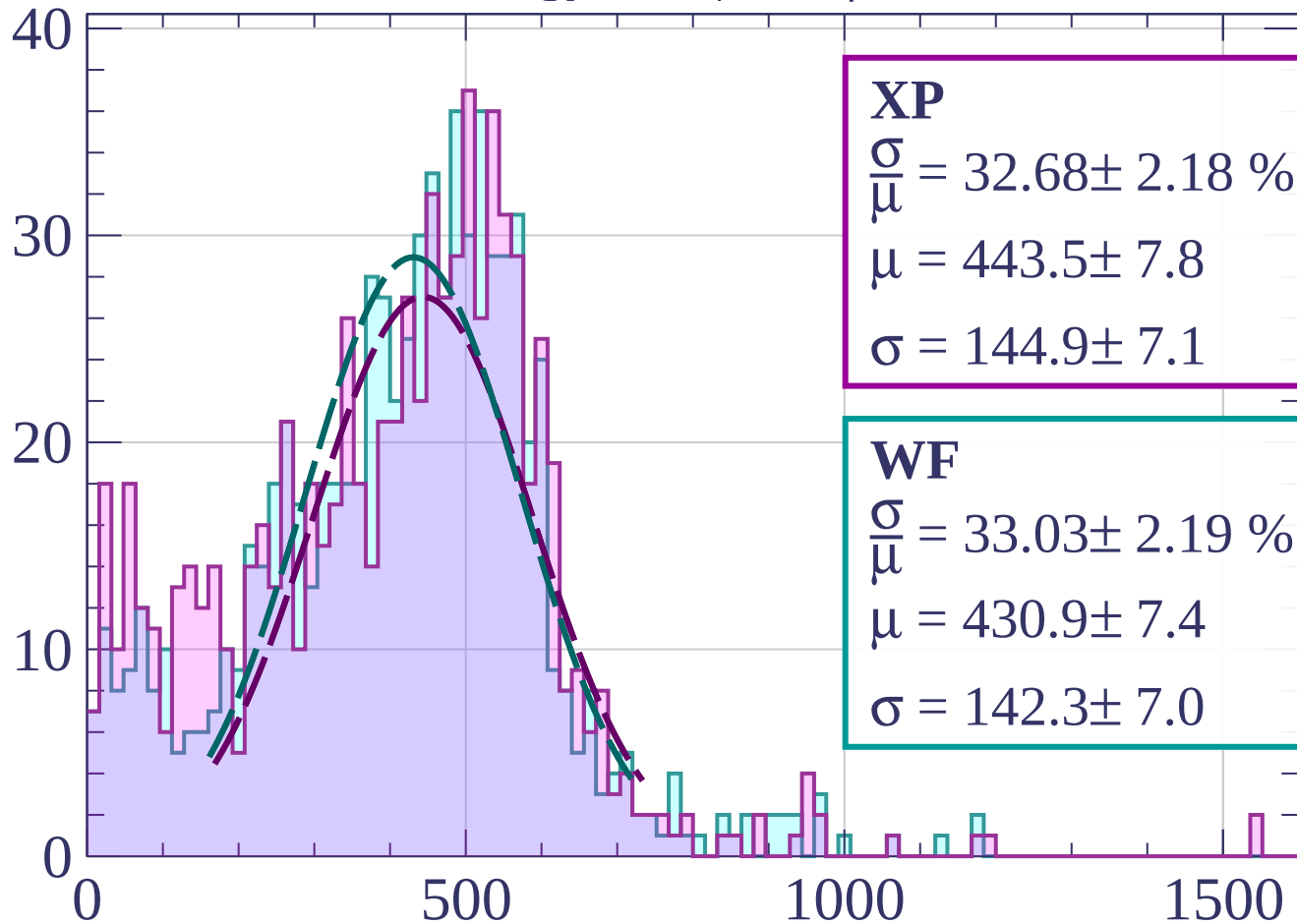
Count



dE/dx [ADC counts/cm]

Energy loss | $3 < \phi < 6$

Count



XP

$$\frac{\sigma}{\mu} = 32.68 \pm 2.18 \%$$

$$\mu = 443.5 \pm 7.8$$

$$\sigma = 144.9 \pm 7.1$$

WF

$$\frac{\sigma}{\mu} = 33.03 \pm 2.19 \%$$

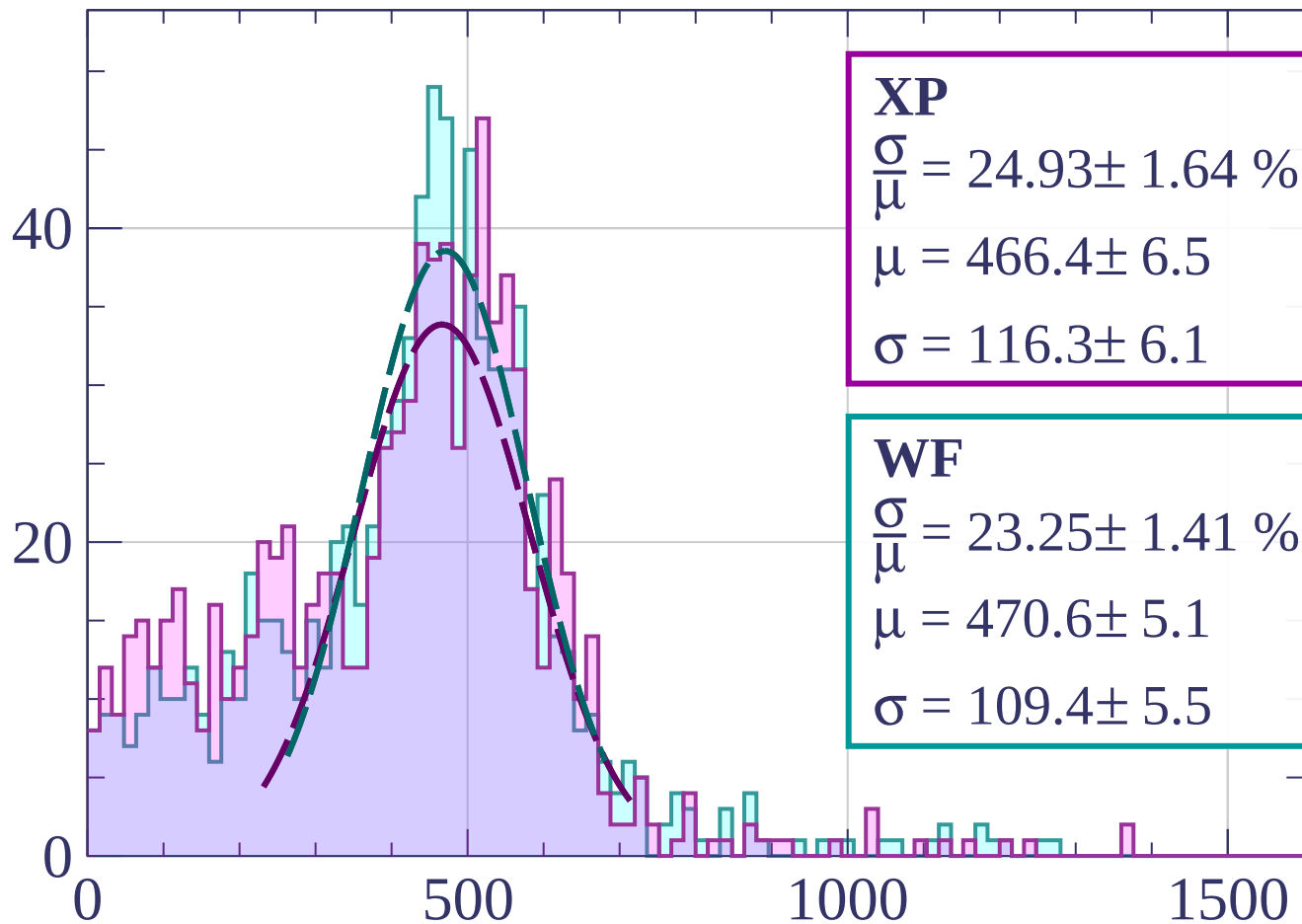
$$\mu = 430.9 \pm 7.4$$

$$\sigma = 142.3 \pm 7.0$$

dE/dx [ADC counts/cm]

Energy loss | $6 < \phi < 9$

Count



XP

$$\frac{\sigma}{\mu} = 24.93 \pm 1.64 \%$$

$$\mu = 466.4 \pm 6.5$$

$$\sigma = 116.3 \pm 6.1$$

WF

$$\frac{\sigma}{\mu} = 23.25 \pm 1.41 \%$$

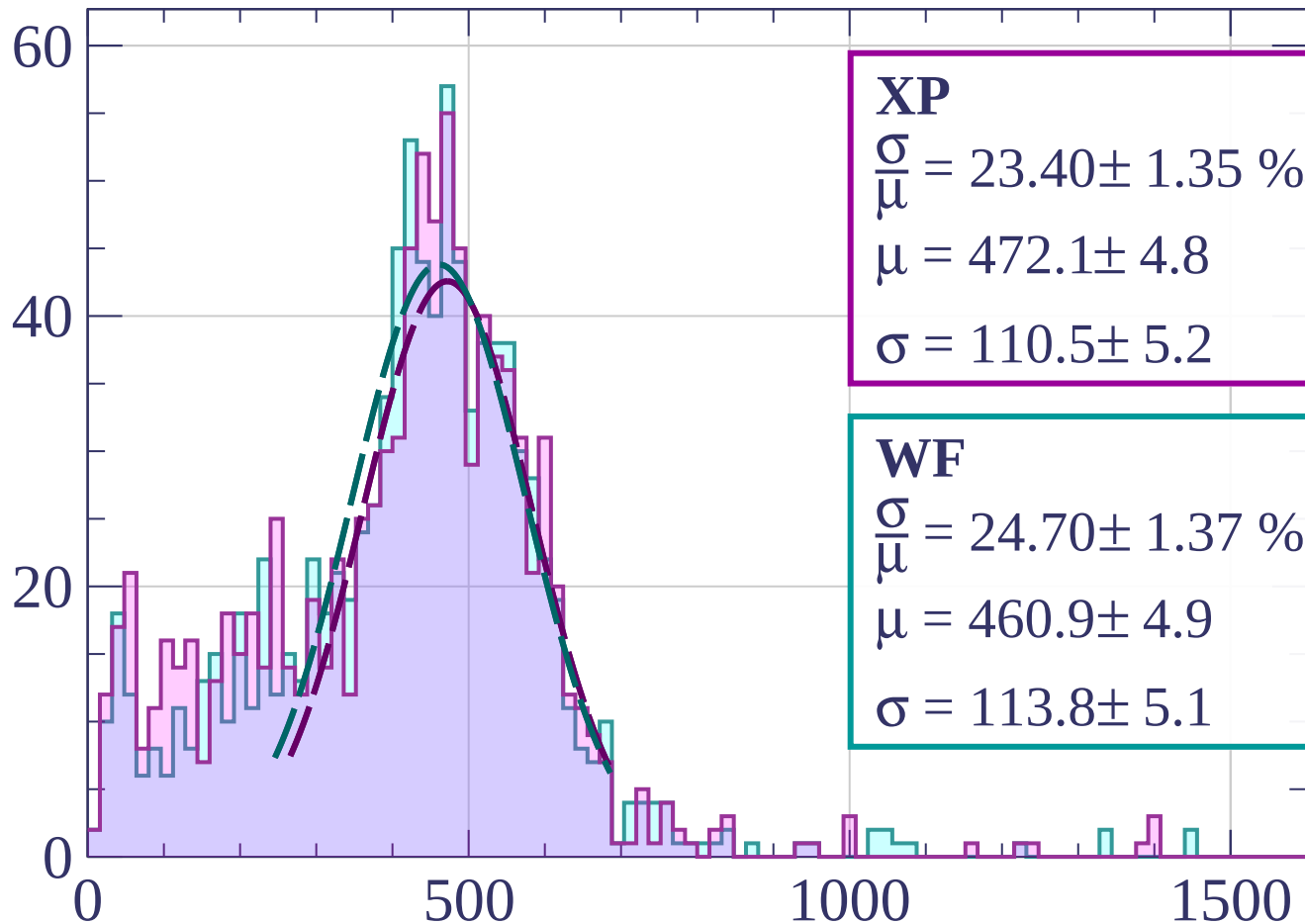
$$\mu = 470.6 \pm 5.1$$

$$\sigma = 109.4 \pm 5.5$$

dE/dx [ADC counts/cm]

Energy loss | $9 < \phi < 12$

Count



XP

$$\frac{\sigma}{\mu} = 23.40 \pm 1.35 \%$$

$$\mu = 472.1 \pm 4.8$$

$$\sigma = 110.5 \pm 5.2$$

WF

$$\frac{\sigma}{\mu} = 24.70 \pm 1.37 \%$$

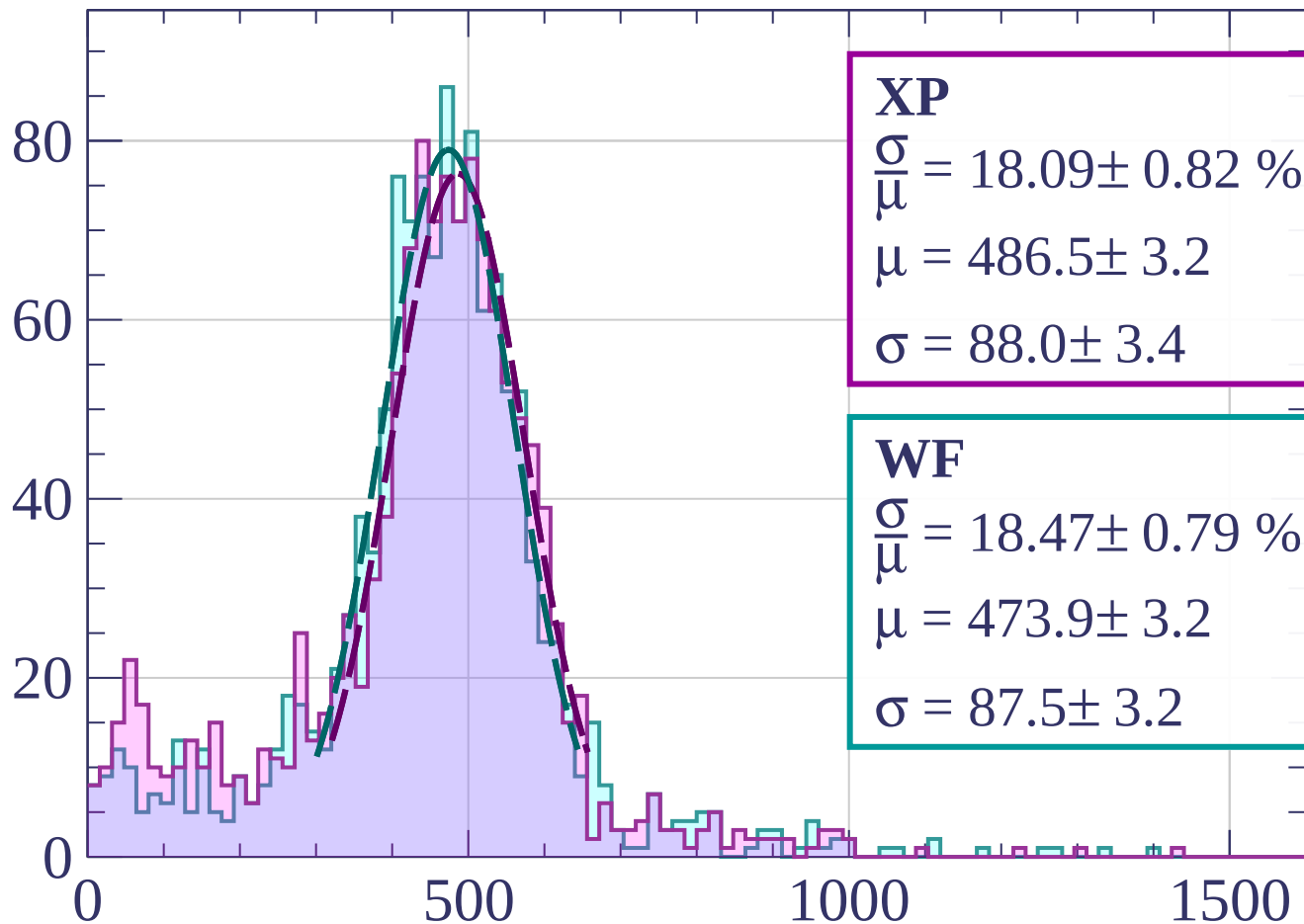
$$\mu = 460.9 \pm 4.9$$

$$\sigma = 113.8 \pm 5.1$$

dE/dx [ADC counts/cm]

Energy loss | $12 < \phi < 15$

Count



XP

$$\frac{\sigma}{\mu} = 18.09 \pm 0.82 \%$$

$$\mu = 486.5 \pm 3.2$$

$$\sigma = 88.0 \pm 3.4$$

WF

$$\frac{\sigma}{\mu} = 18.47 \pm 0.79 \%$$

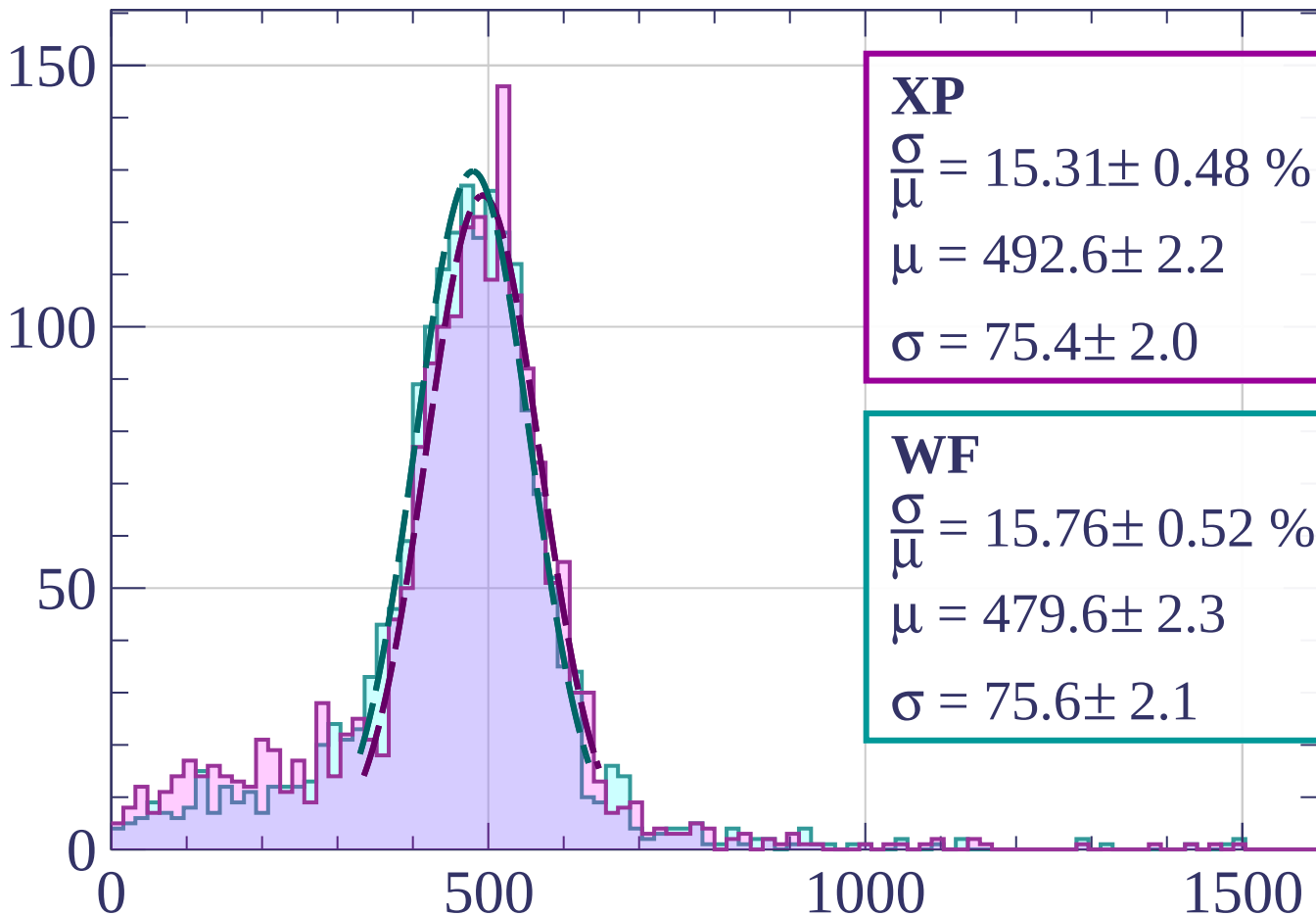
$$\mu = 473.9 \pm 3.2$$

$$\sigma = 87.5 \pm 3.2$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \phi < 18$

Count



XP

$$\frac{\sigma}{\mu} = 15.31 \pm 0.48 \%$$

$$\mu = 492.6 \pm 2.2$$

$$\sigma = 75.4 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 15.76 \pm 0.52 \%$$

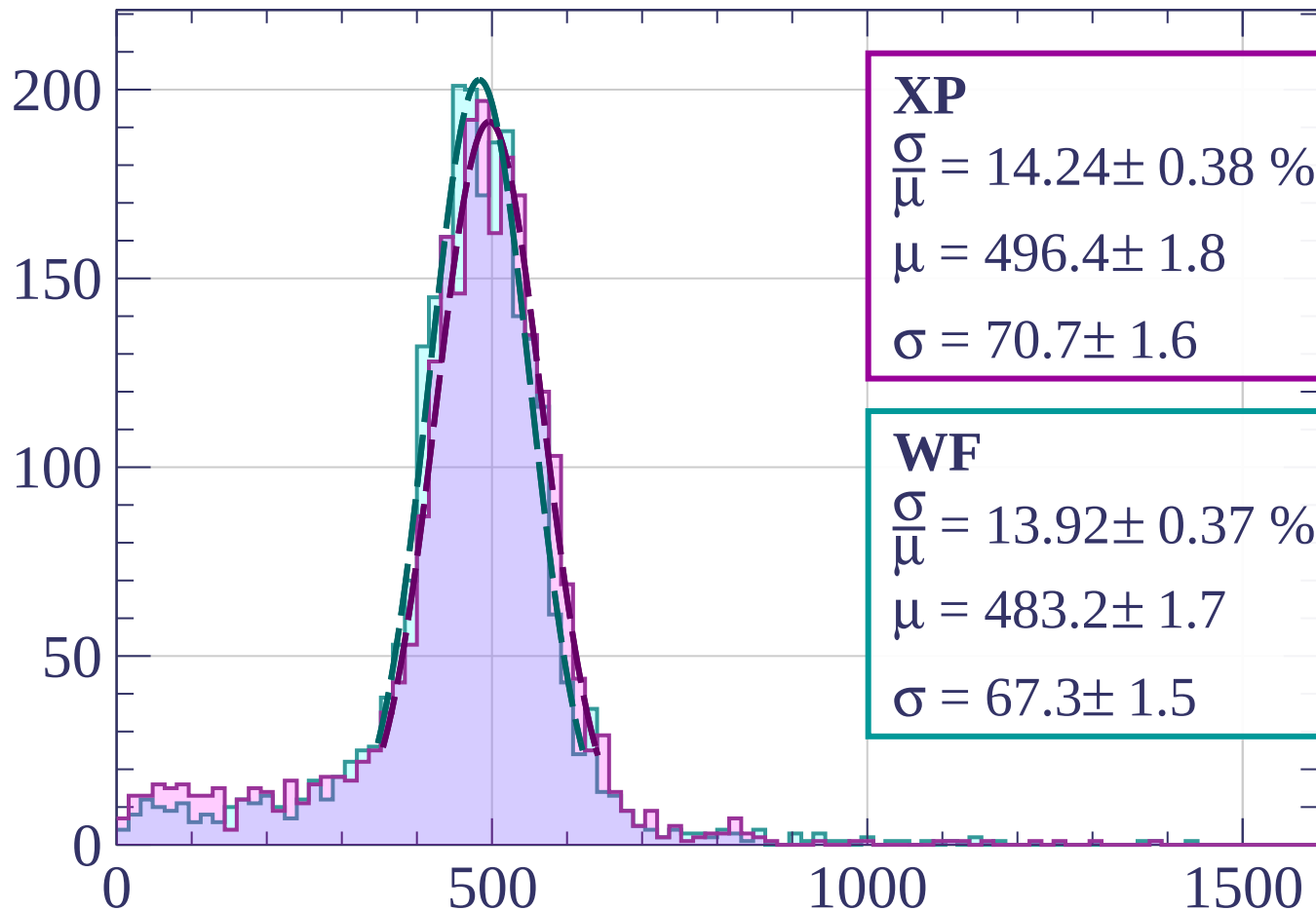
$$\mu = 479.6 \pm 2.3$$

$$\sigma = 75.6 \pm 2.1$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \phi < 21$

Count



XP

$$\frac{\sigma}{\mu} = 14.24 \pm 0.38 \%$$

$$\mu = 496.4 \pm 1.8$$

$$\sigma = 70.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 13.92 \pm 0.37 \%$$

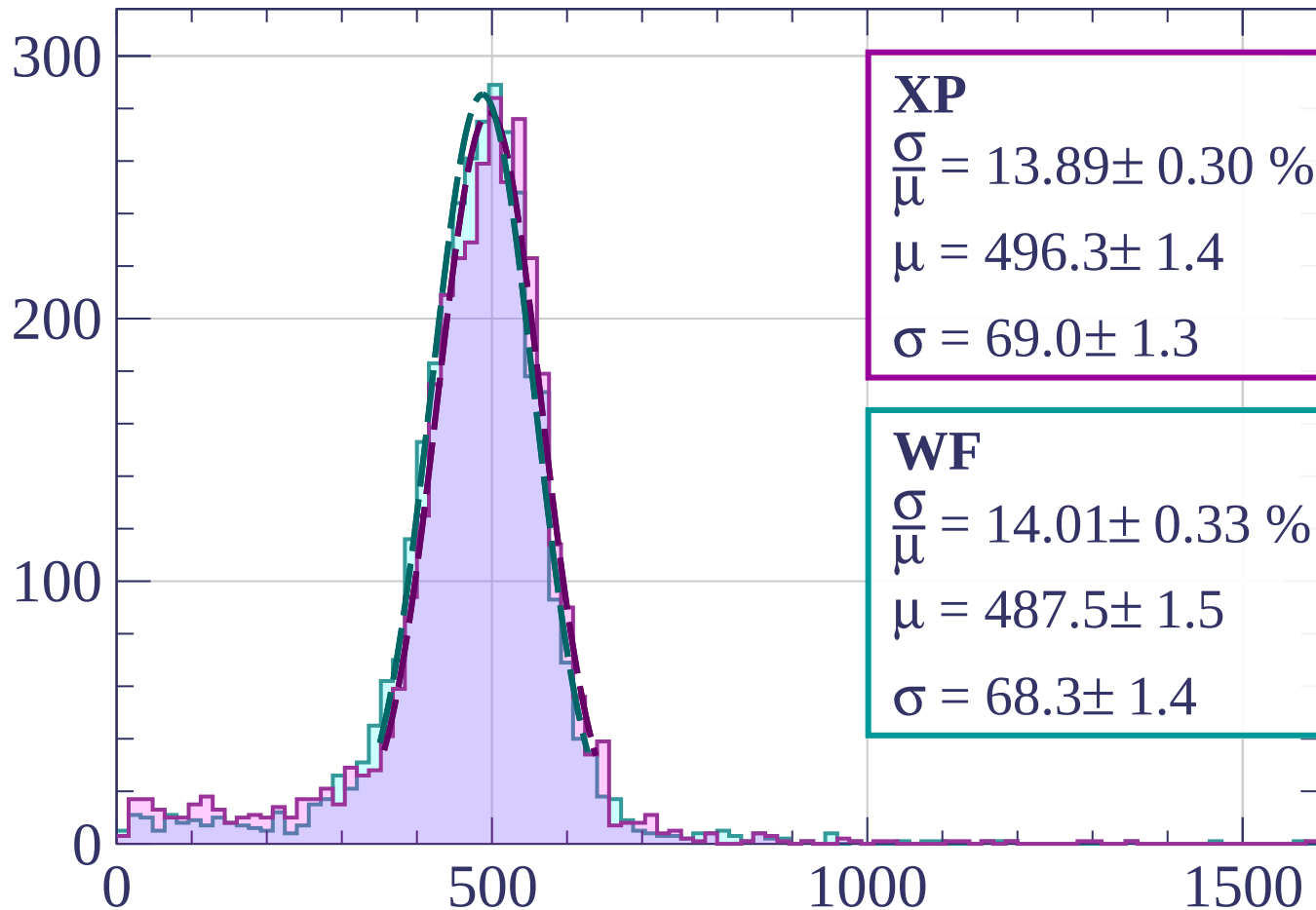
$$\mu = 483.2 \pm 1.7$$

$$\sigma = 67.3 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \phi < 24$

Count



XP

$$\frac{\sigma}{\mu} = 13.89 \pm 0.30 \%$$

$$\mu = 496.3 \pm 1.4$$

$$\sigma = 69.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 14.01 \pm 0.33 \%$$

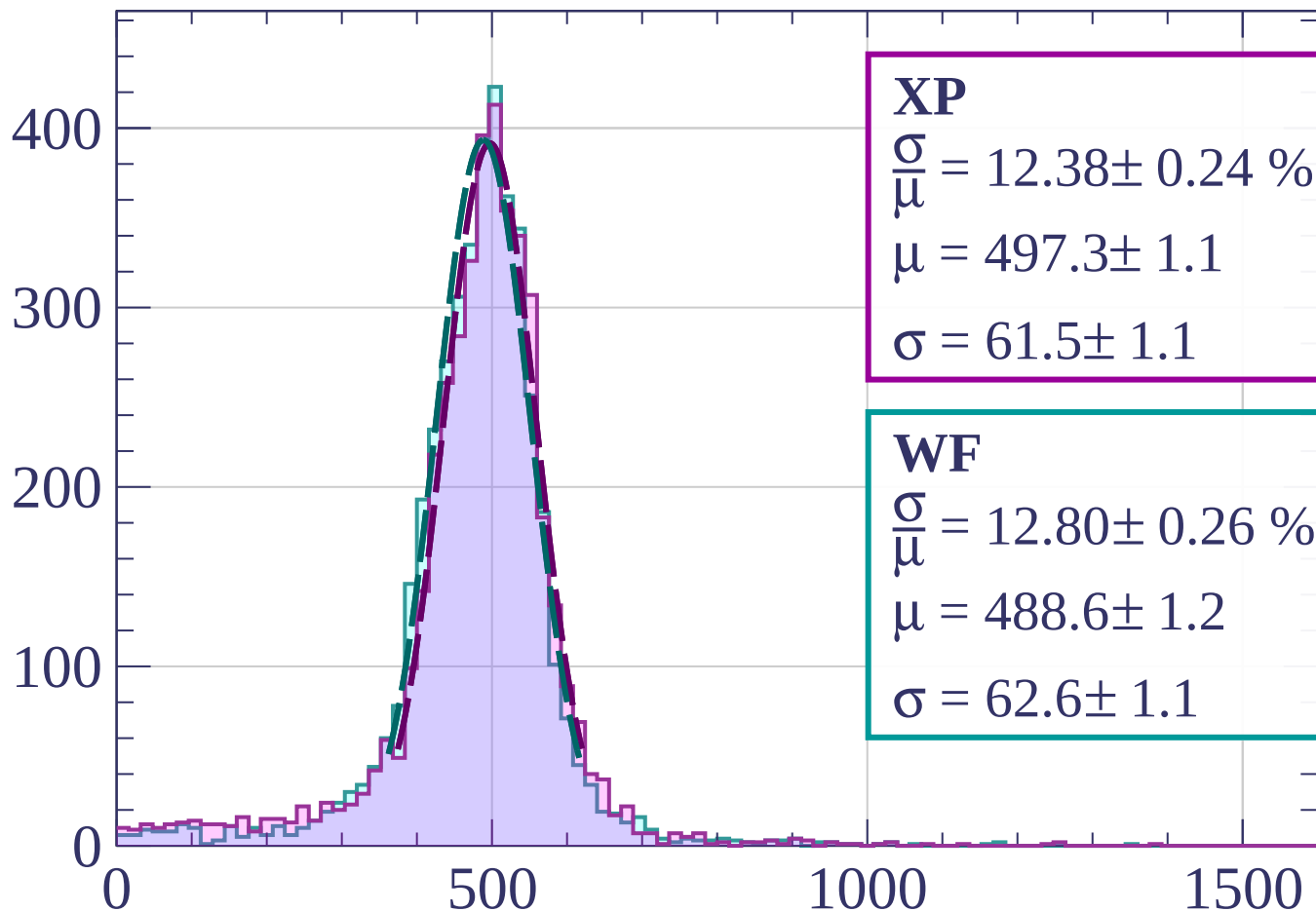
$$\mu = 487.5 \pm 1.5$$

$$\sigma = 68.3 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $24 < \phi < 27$

Count



XP

$$\frac{\sigma}{\mu} = 12.38 \pm 0.24 \%$$

$$\mu = 497.3 \pm 1.1$$

$$\sigma = 61.5 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 12.80 \pm 0.26 \%$$

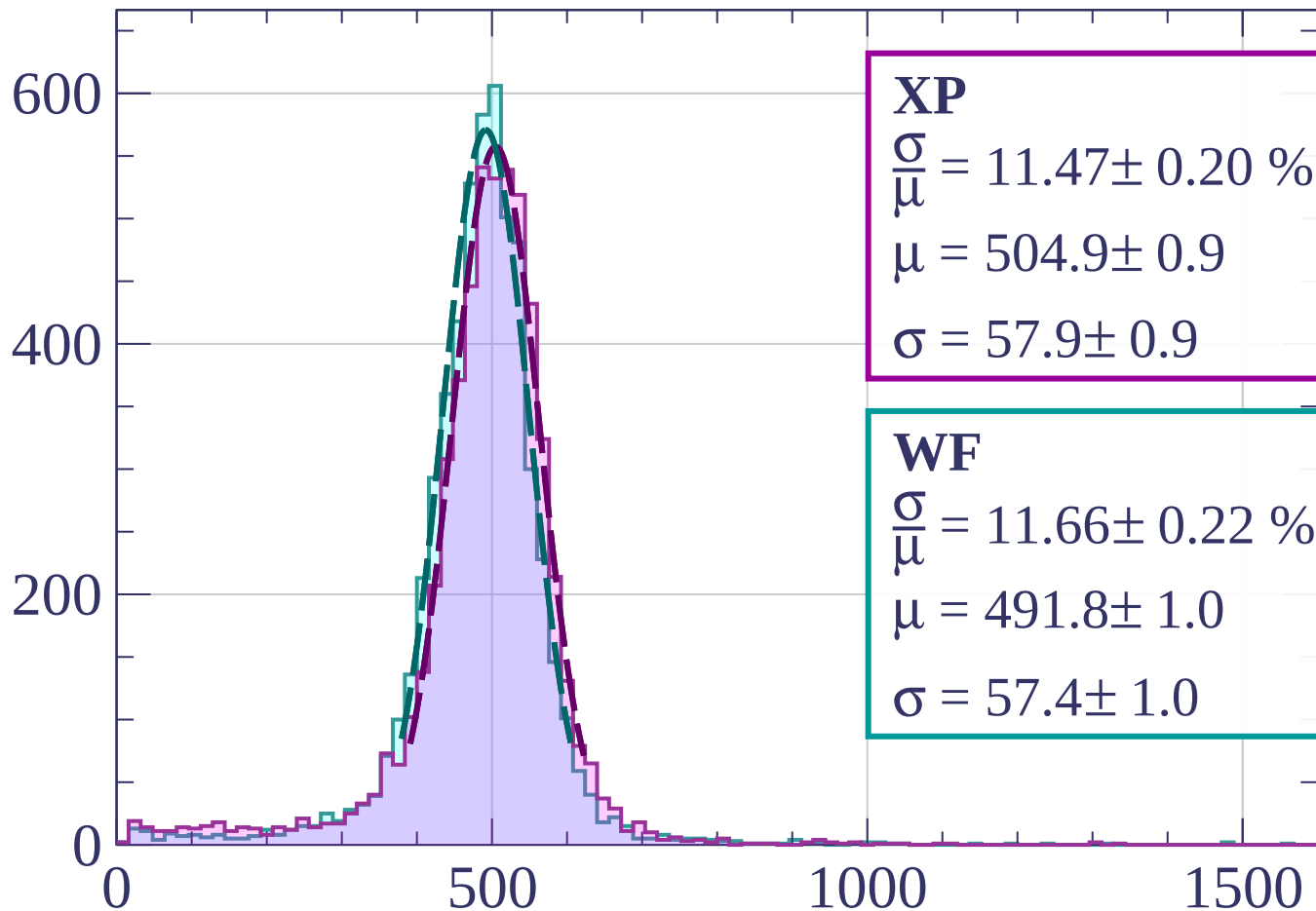
$$\mu = 488.6 \pm 1.2$$

$$\sigma = 62.6 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \phi < 30$

Count



XP

$$\frac{\sigma}{\mu} = 11.47 \pm 0.20 \%$$

$$\mu = 504.9 \pm 0.9$$

$$\sigma = 57.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.66 \pm 0.22 \%$$

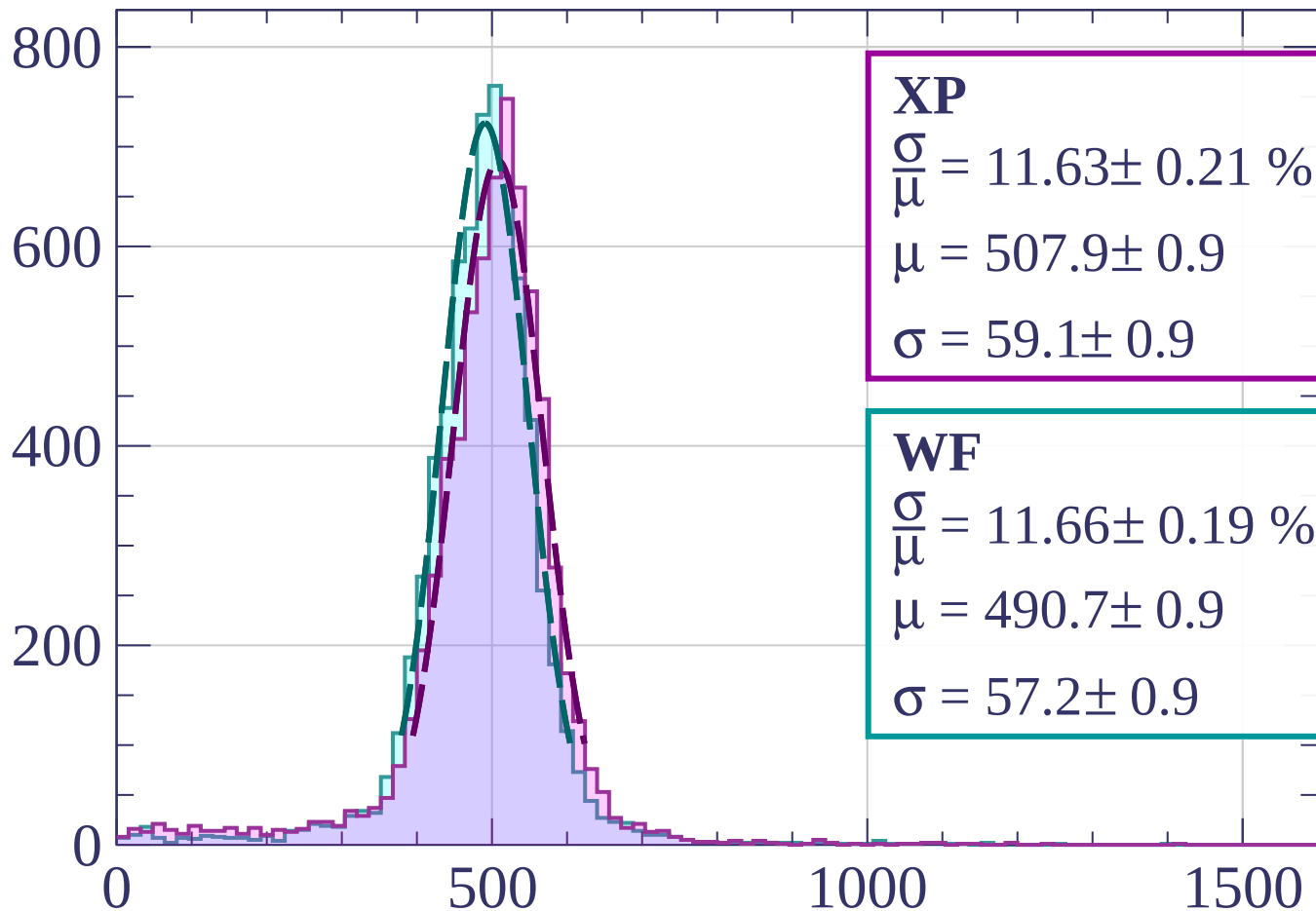
$$\mu = 491.8 \pm 1.0$$

$$\sigma = 57.4 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 11.63 \pm 0.21 \%$$

$$\mu = 507.9 \pm 0.9$$

$$\sigma = 59.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.66 \pm 0.19 \%$$

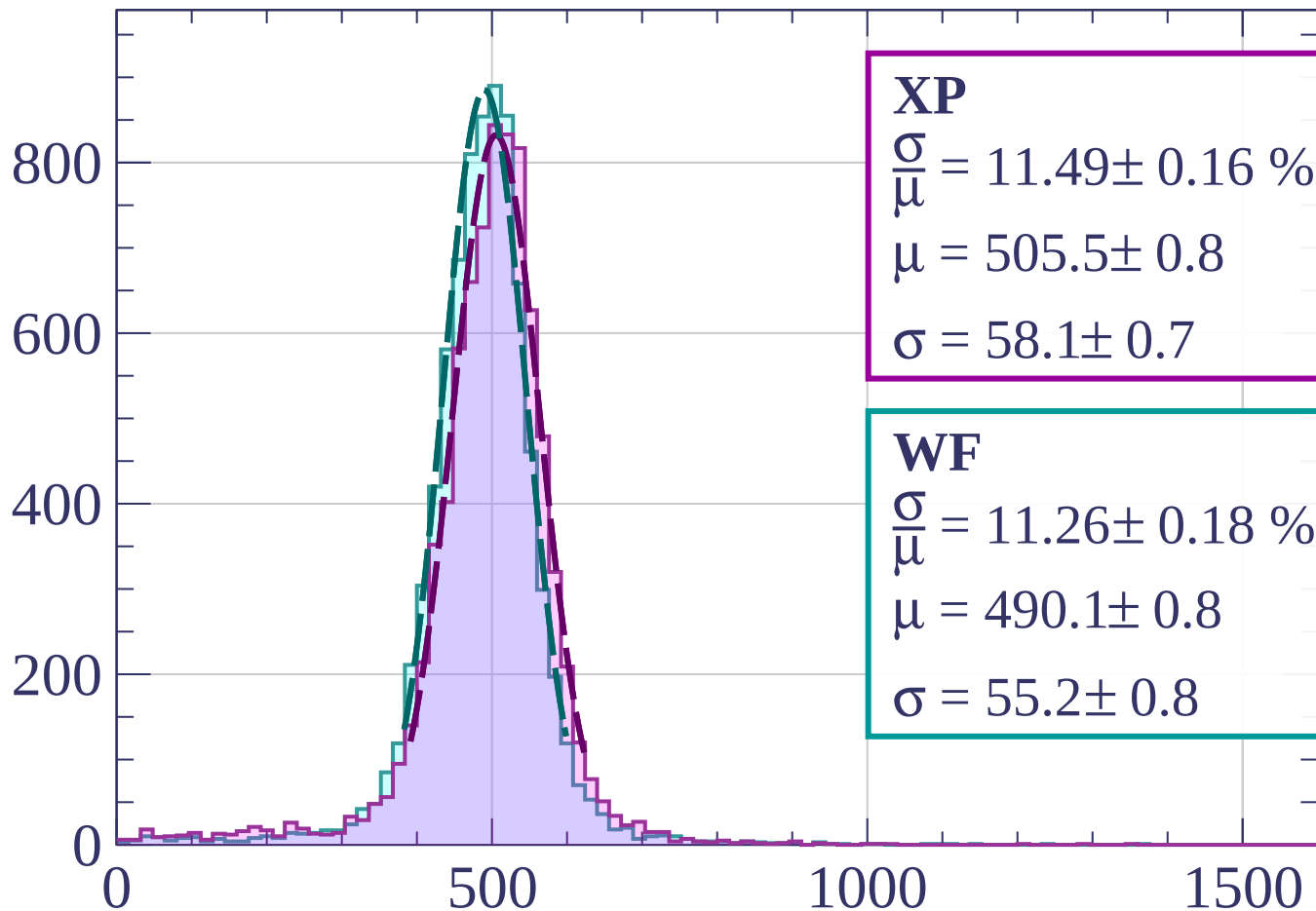
$$\mu = 490.7 \pm 0.9$$

$$\sigma = 57.2 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \phi < 36$

Count



XP

$$\frac{\sigma}{\mu} = 11.49 \pm 0.16 \%$$

$$\mu = 505.5 \pm 0.8$$

$$\sigma = 58.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.26 \pm 0.18 \%$$

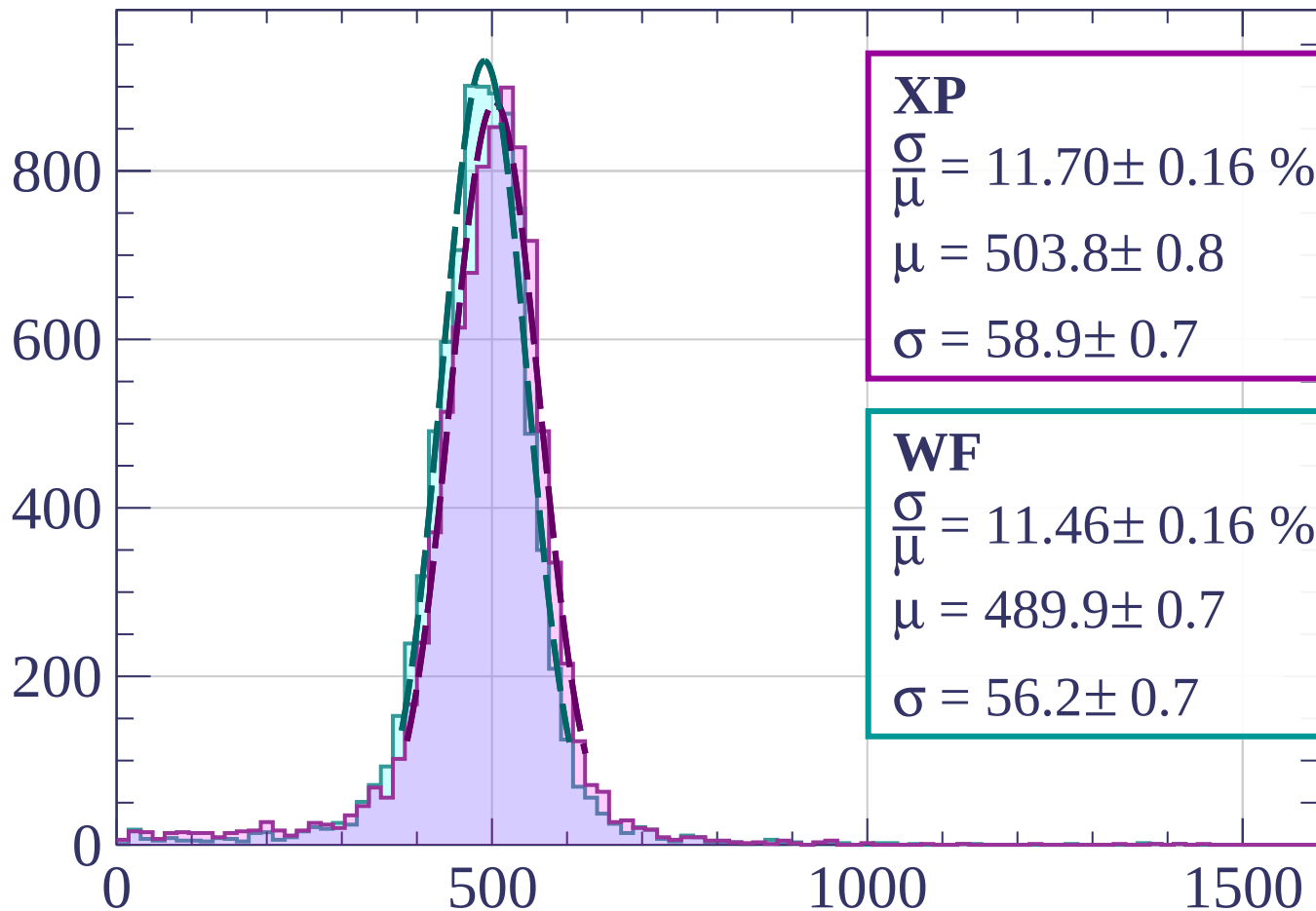
$$\mu = 490.1 \pm 0.8$$

$$\sigma = 55.2 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \phi < 39$

Count



XP

$$\frac{\sigma}{\mu} = 11.70 \pm 0.16 \%$$

$$\mu = 503.8 \pm 0.8$$

$$\sigma = 58.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.46 \pm 0.16 \%$$

$$\mu = 489.9 \pm 0.7$$

$$\sigma = 56.2 \pm 0.7$$

dE/dx [ADC counts/cm]

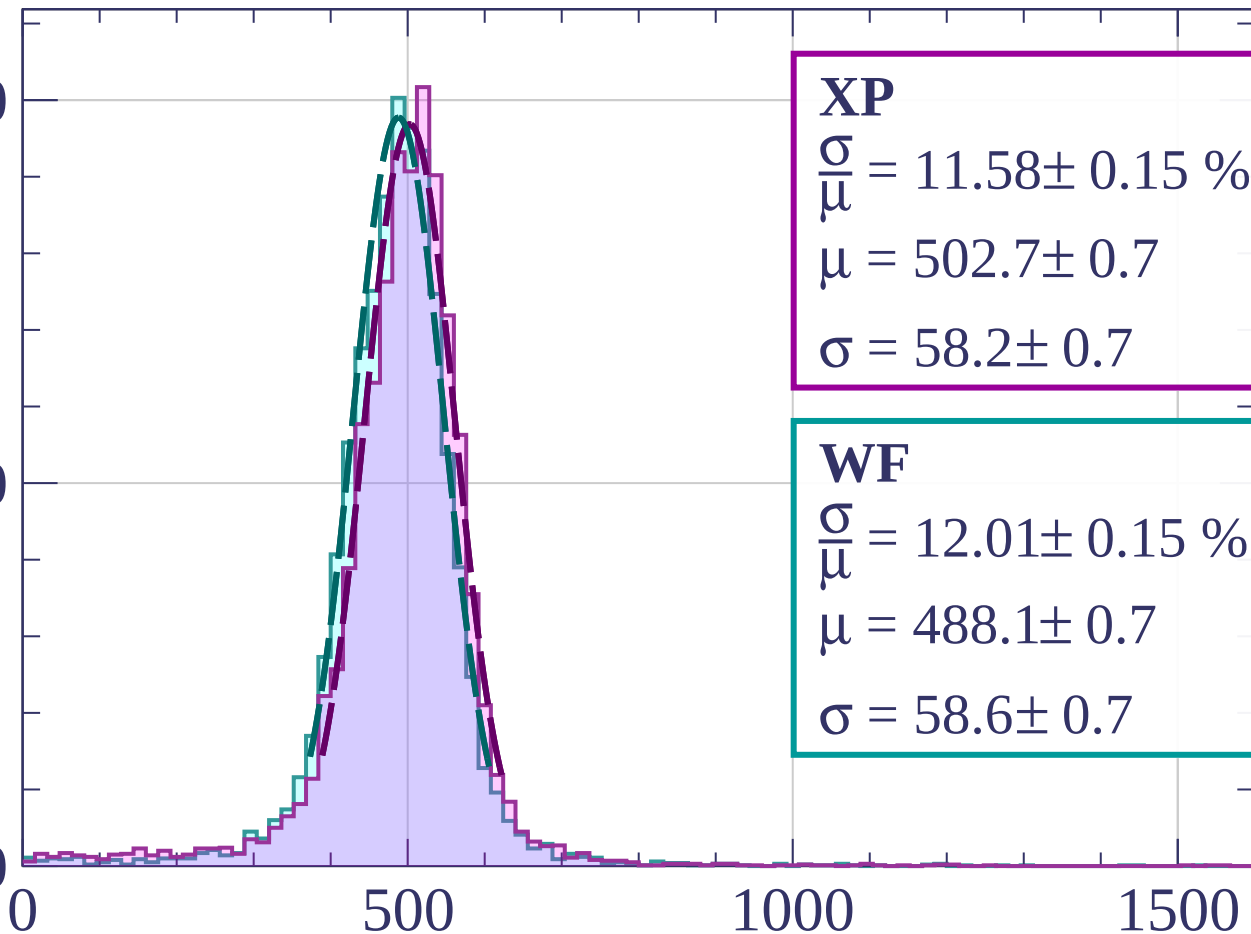
Energy loss | $39 < \phi < 42$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 11.58 \pm 0.15 \%$$

$$\mu = 502.7 \pm 0.7$$

$$\sigma = 58.2 \pm 0.7$$

WF

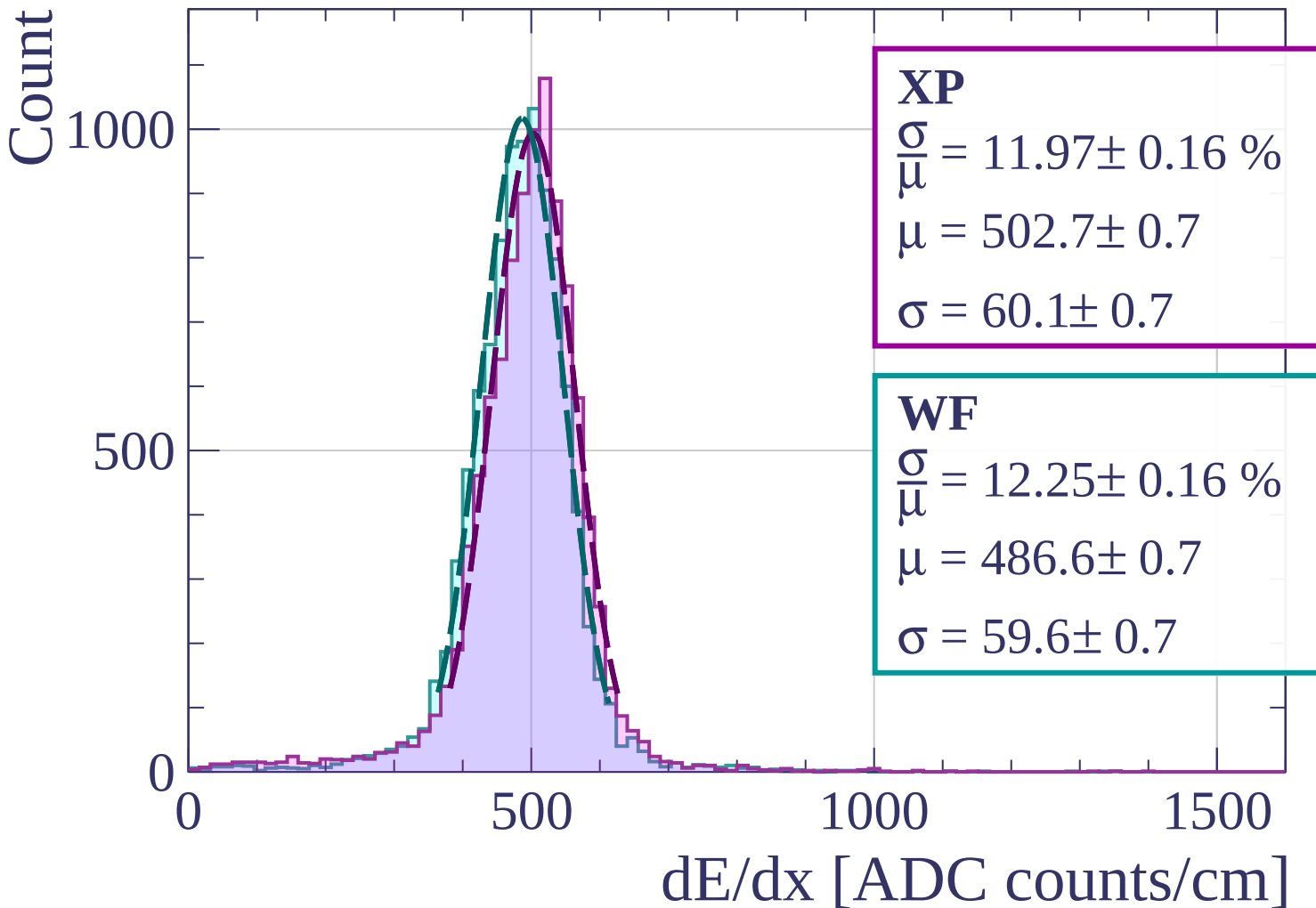
$$\frac{\sigma}{\mu} = 12.01 \pm 0.15 \%$$

$$\mu = 488.1 \pm 0.7$$

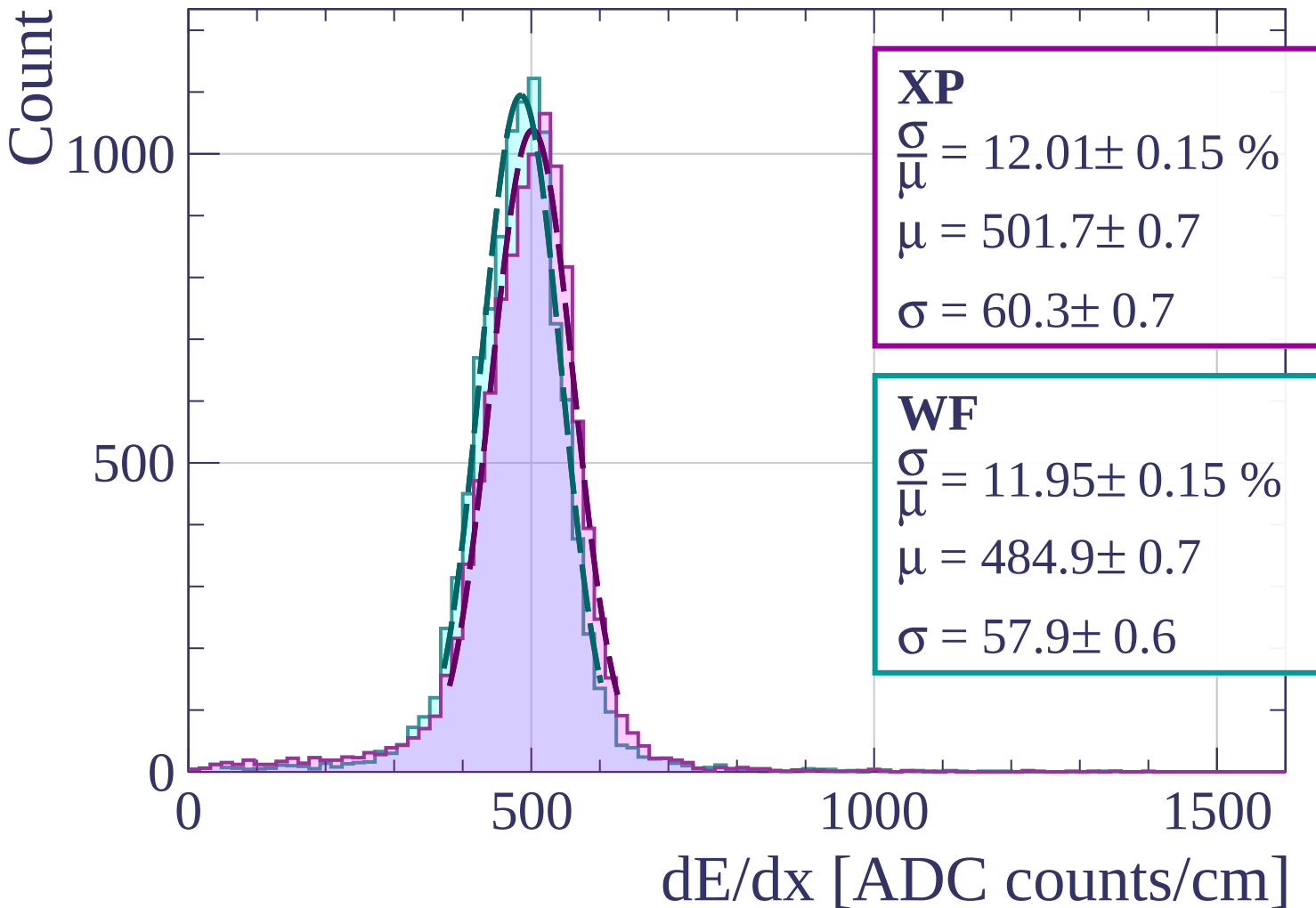
$$\sigma = 58.6 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \phi < 45$

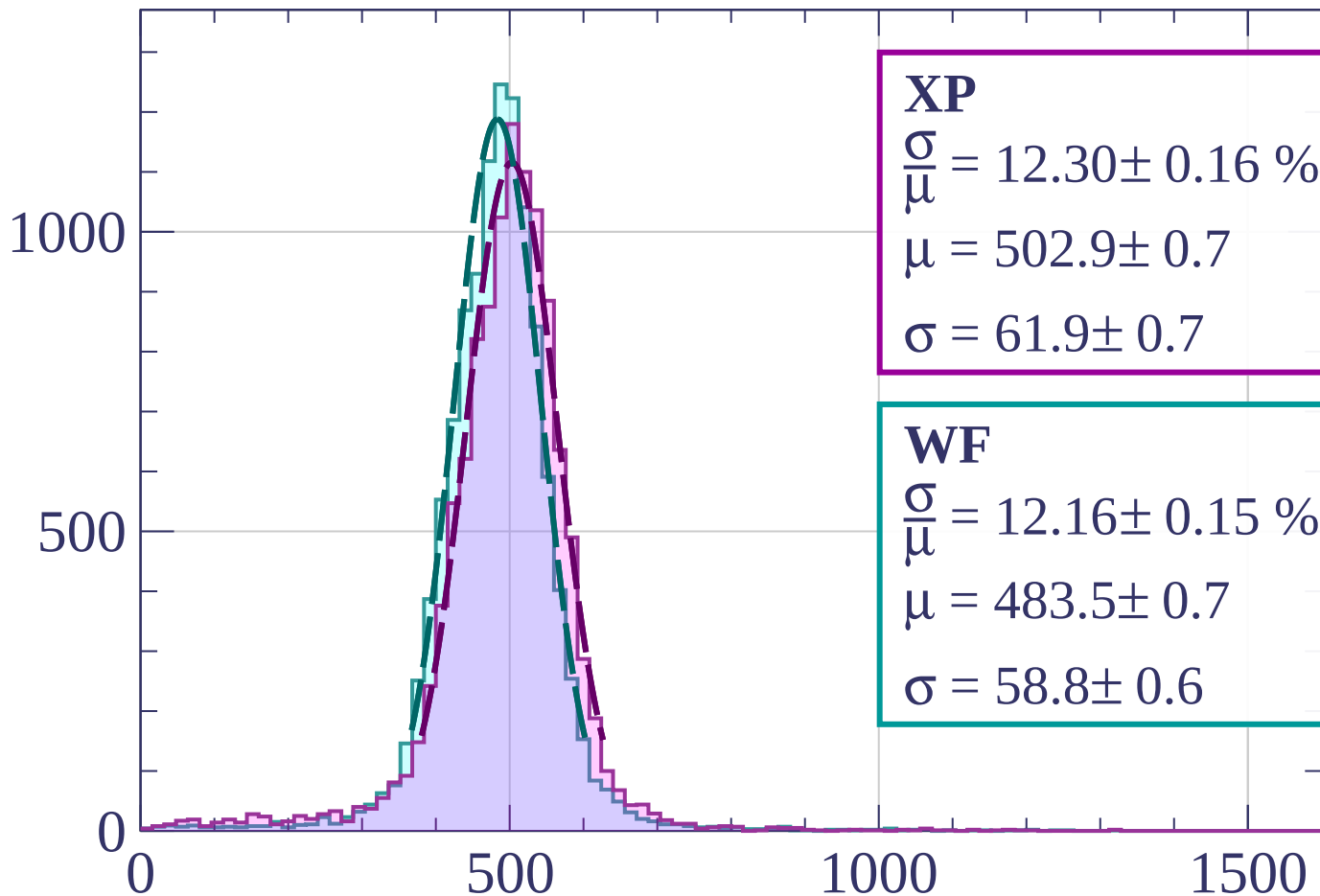


Energy loss | $45 < \phi < 48$



Energy loss | $48 < \phi < 51$

Count



XP

$$\frac{\sigma}{\mu} = 12.30 \pm 0.16 \%$$

$$\mu = 502.9 \pm 0.7$$

$$\sigma = 61.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 12.16 \pm 0.15 \%$$

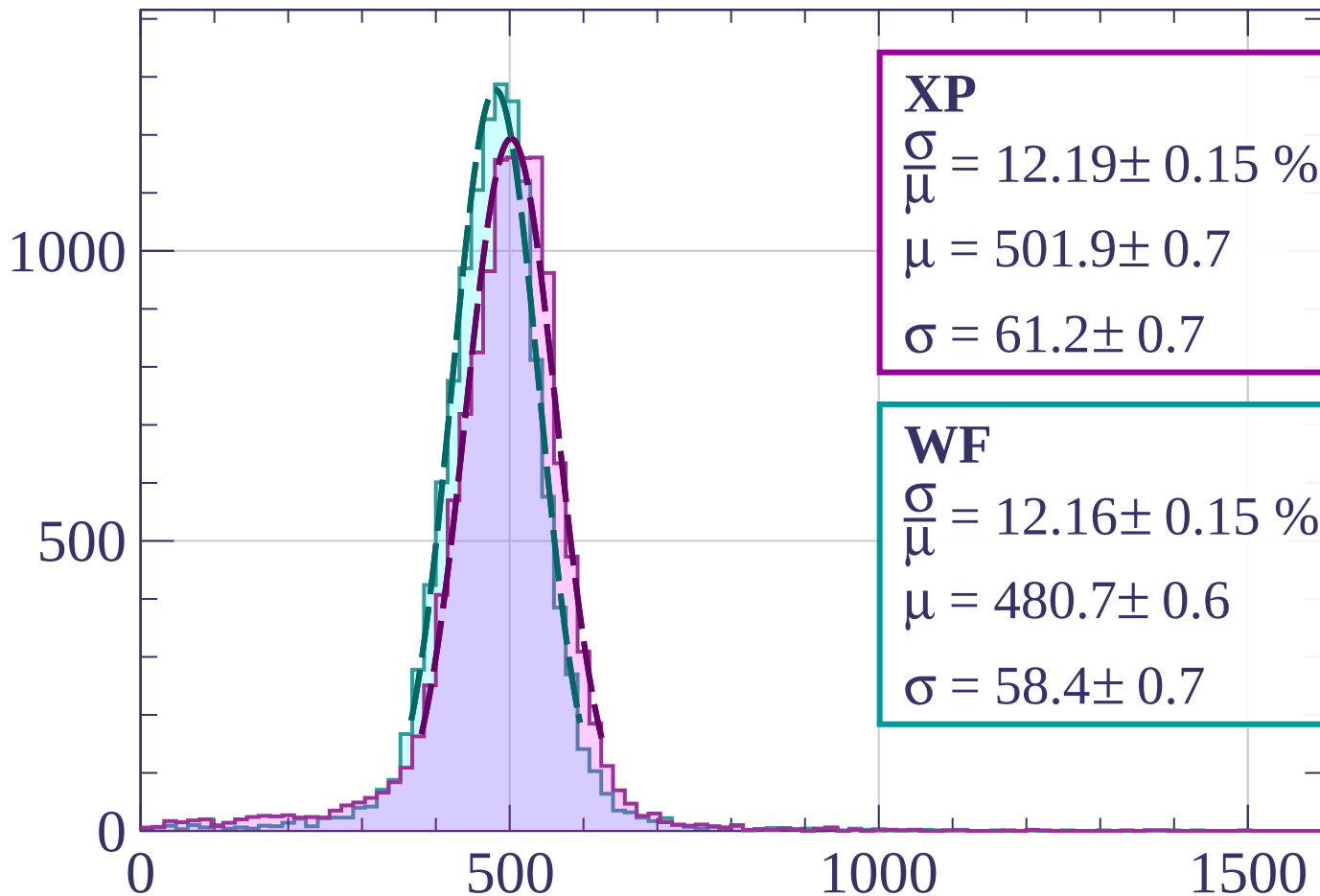
$$\mu = 483.5 \pm 0.7$$

$$\sigma = 58.8 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \phi < 54$

Count



XP

$$\frac{\sigma}{\mu} = 12.19 \pm 0.15 \%$$

$$\mu = 501.9 \pm 0.7$$

$$\sigma = 61.2 \pm 0.7$$

WF

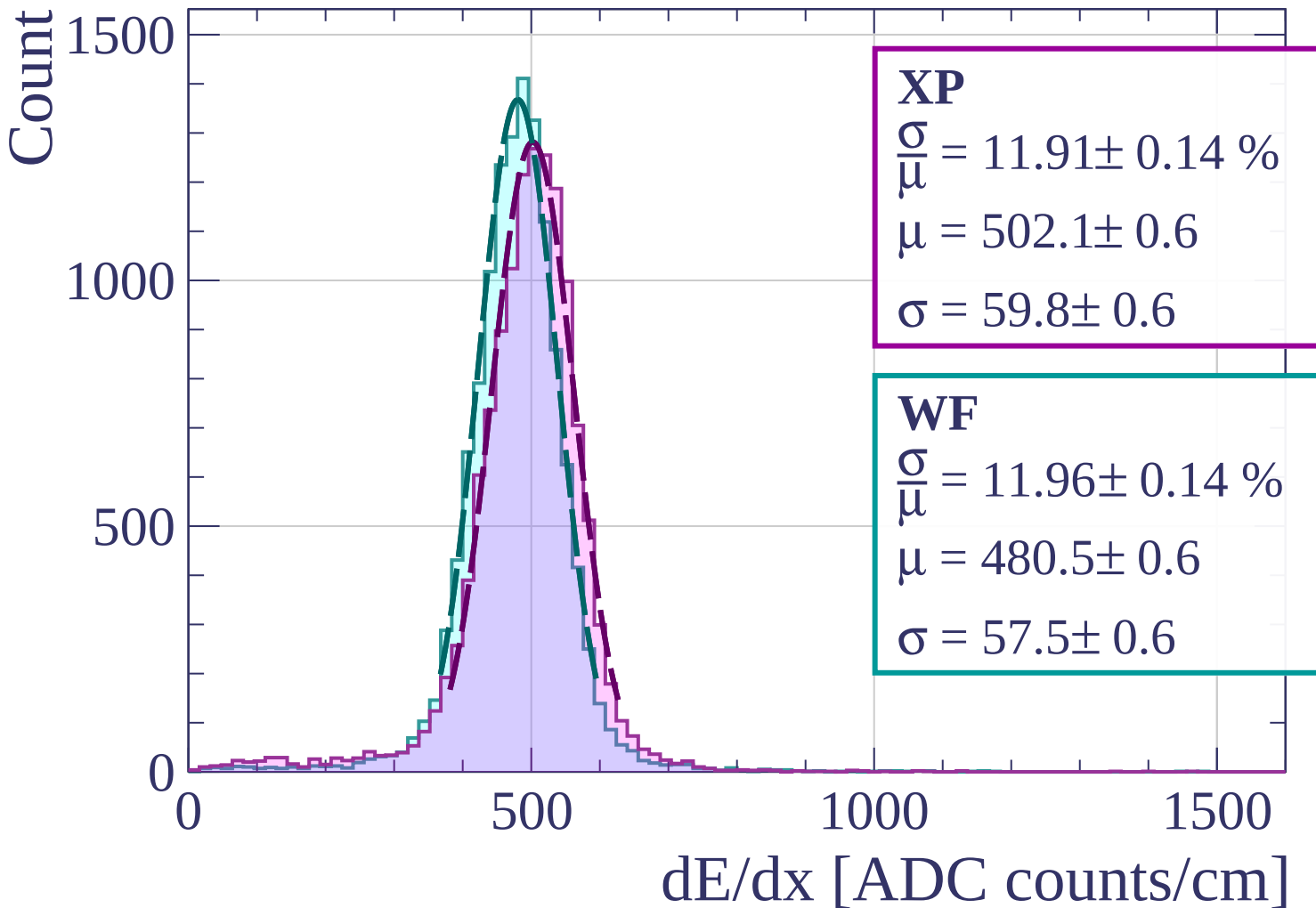
$$\frac{\sigma}{\mu} = 12.16 \pm 0.15 \%$$

$$\mu = 480.7 \pm 0.6$$

$$\sigma = 58.4 \pm 0.7$$

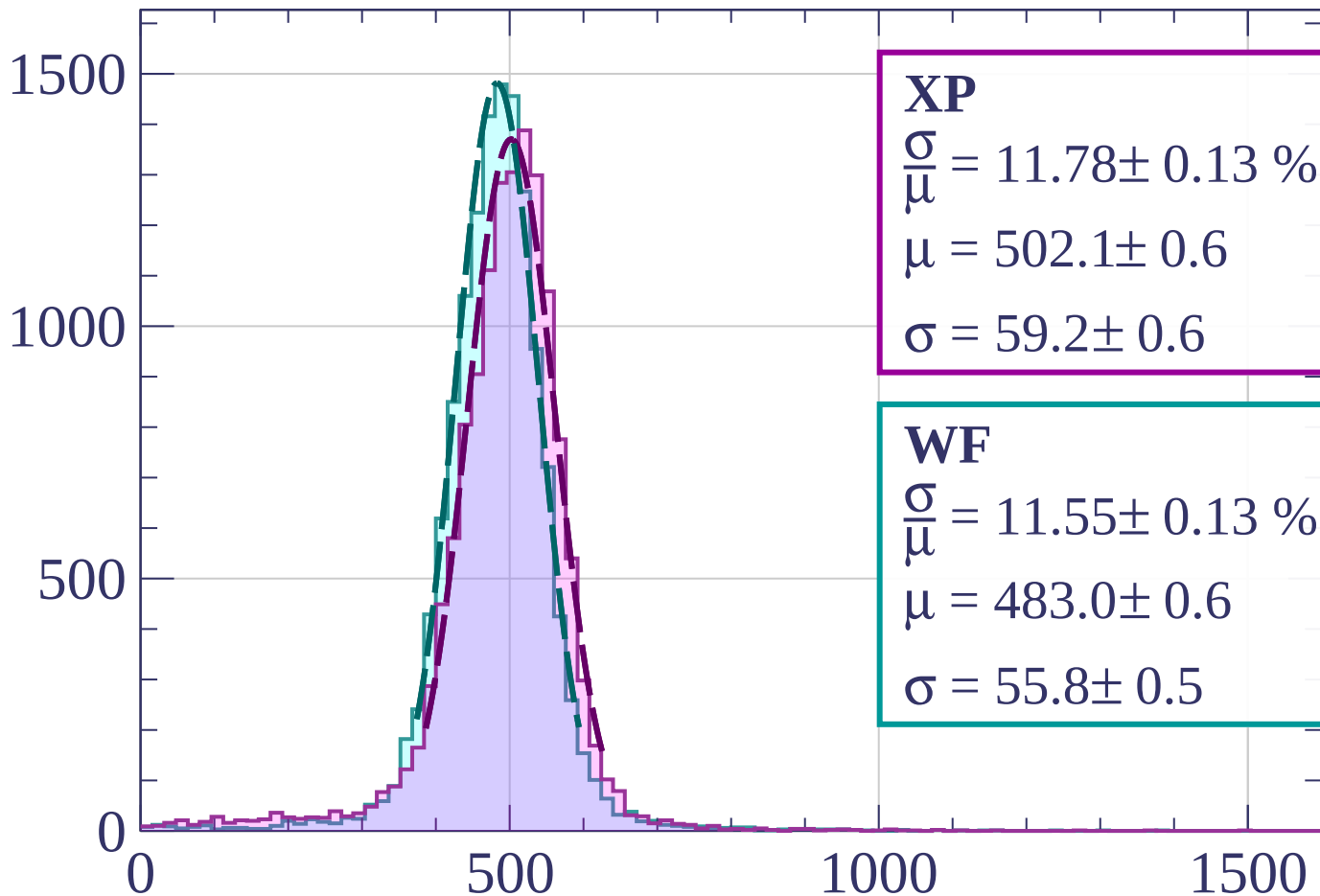
dE/dx [ADC counts/cm]

Energy loss | $54 < \phi < 57$



Energy loss | $57 < \phi < 60$

Count



XP

$$\frac{\sigma}{\mu} = 11.78 \pm 0.13 \%$$

$$\mu = 502.1 \pm 0.6$$

$$\sigma = 59.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 11.55 \pm 0.13 \%$$

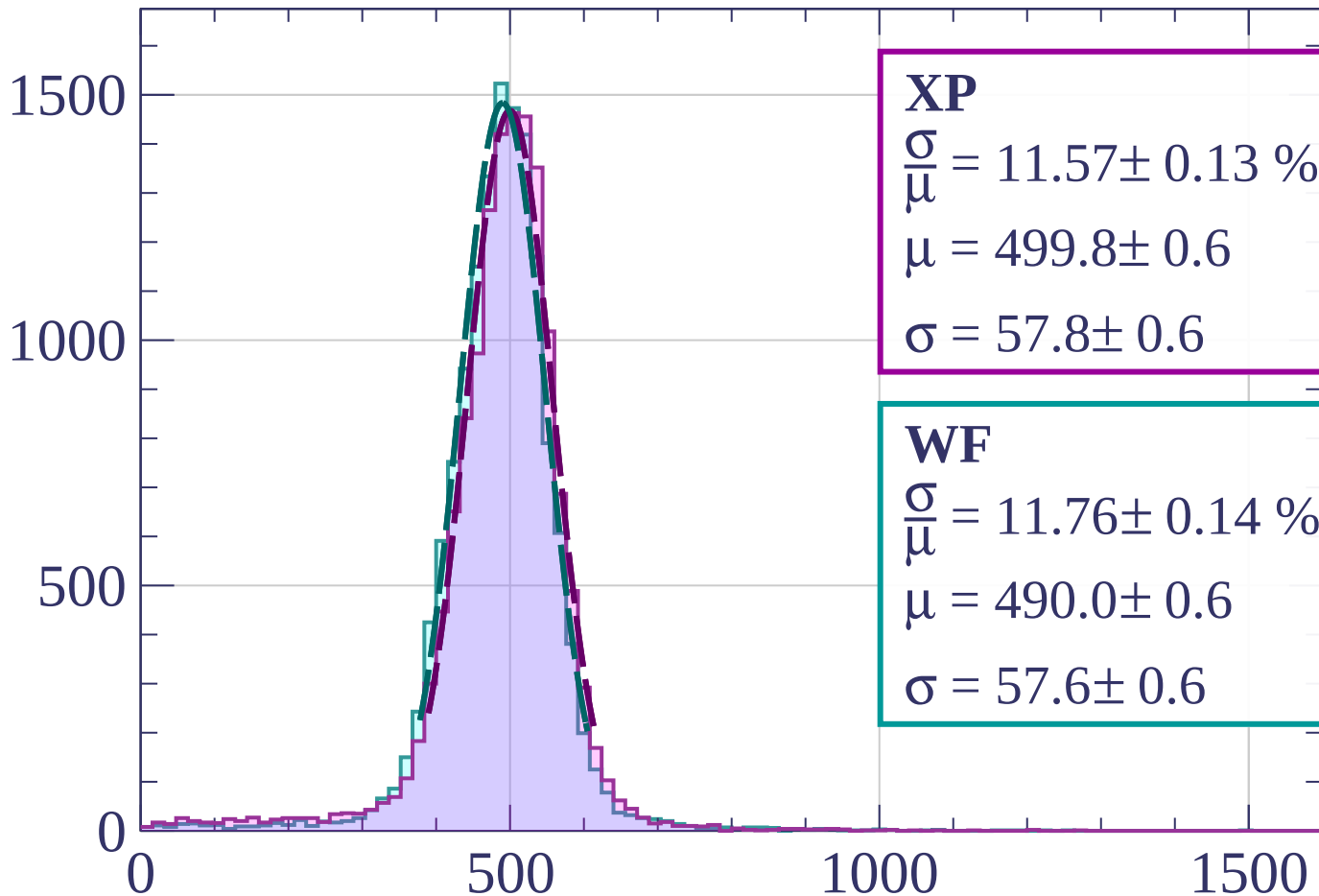
$$\mu = 483.0 \pm 0.6$$

$$\sigma = 55.8 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \phi < 63$

Count



XP

$$\frac{\sigma}{\mu} = 11.57 \pm 0.13 \%$$

$$\mu = 499.8 \pm 0.6$$

$$\sigma = 57.8 \pm 0.6$$

WF

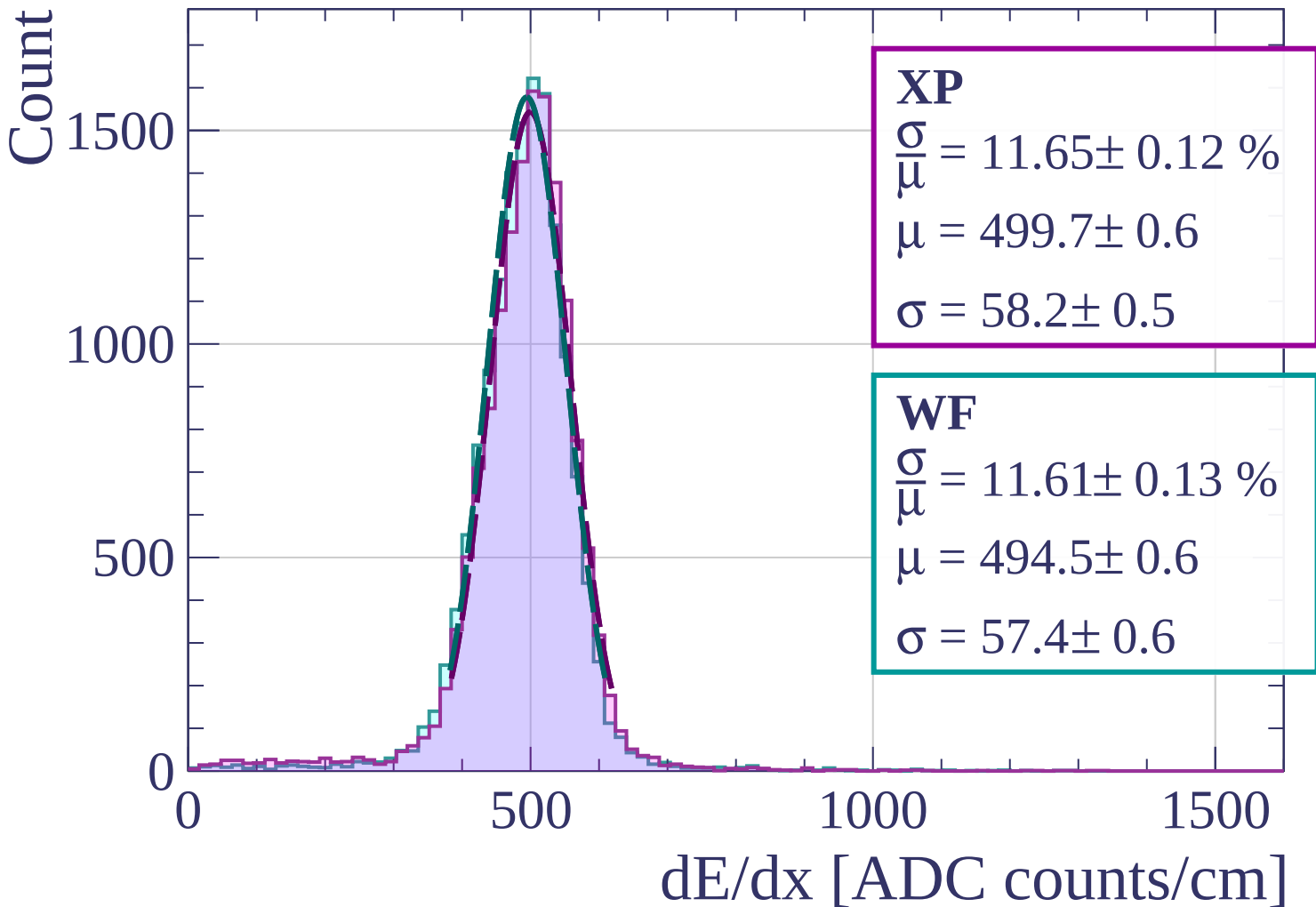
$$\frac{\sigma}{\mu} = 11.76 \pm 0.14 \%$$

$$\mu = 490.0 \pm 0.6$$

$$\sigma = 57.6 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $63 < \phi < 66$



Energy loss | $66 < \phi < 69$

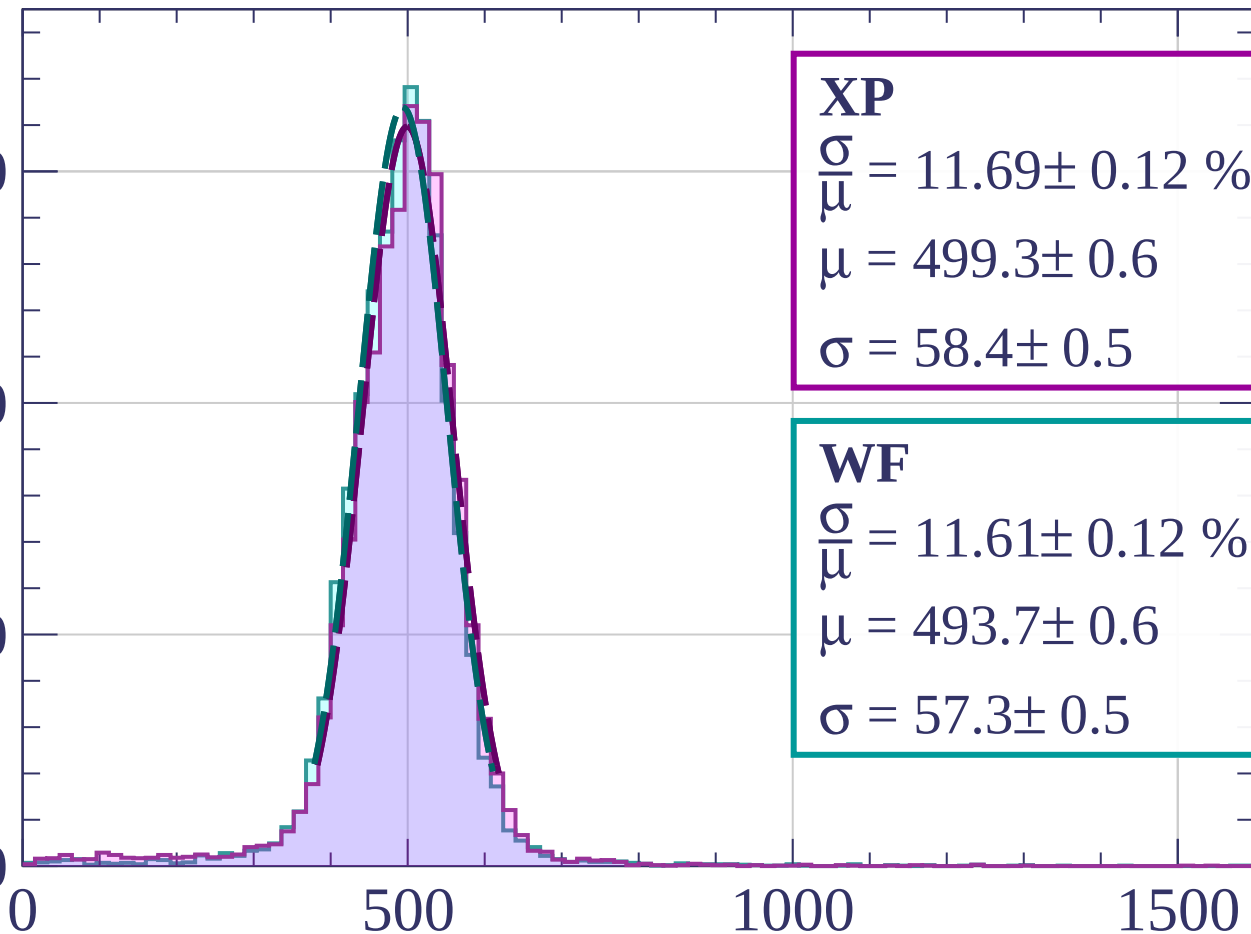
Count

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 11.69 \pm 0.12 \%$$

$$\mu = 499.3 \pm 0.6$$

$$\sigma = 58.4 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 11.61 \pm 0.12 \%$$

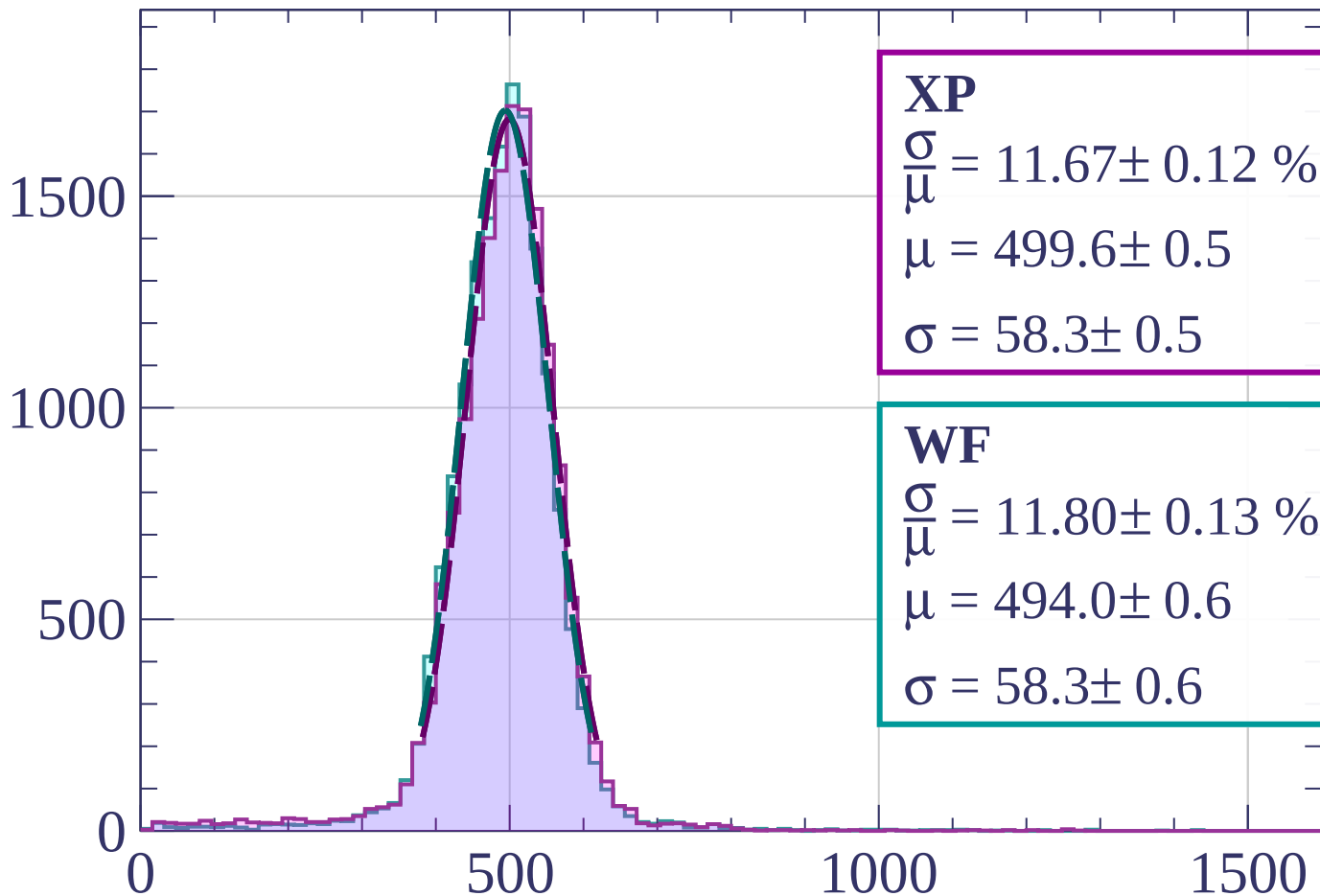
$$\mu = 493.7 \pm 0.6$$

$$\sigma = 57.3 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $69 < \phi < 72$

Count



XP

$$\frac{\sigma}{\mu} = 11.67 \pm 0.12 \%$$

$$\mu = 499.6 \pm 0.5$$

$$\sigma = 58.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 11.80 \pm 0.13 \%$$

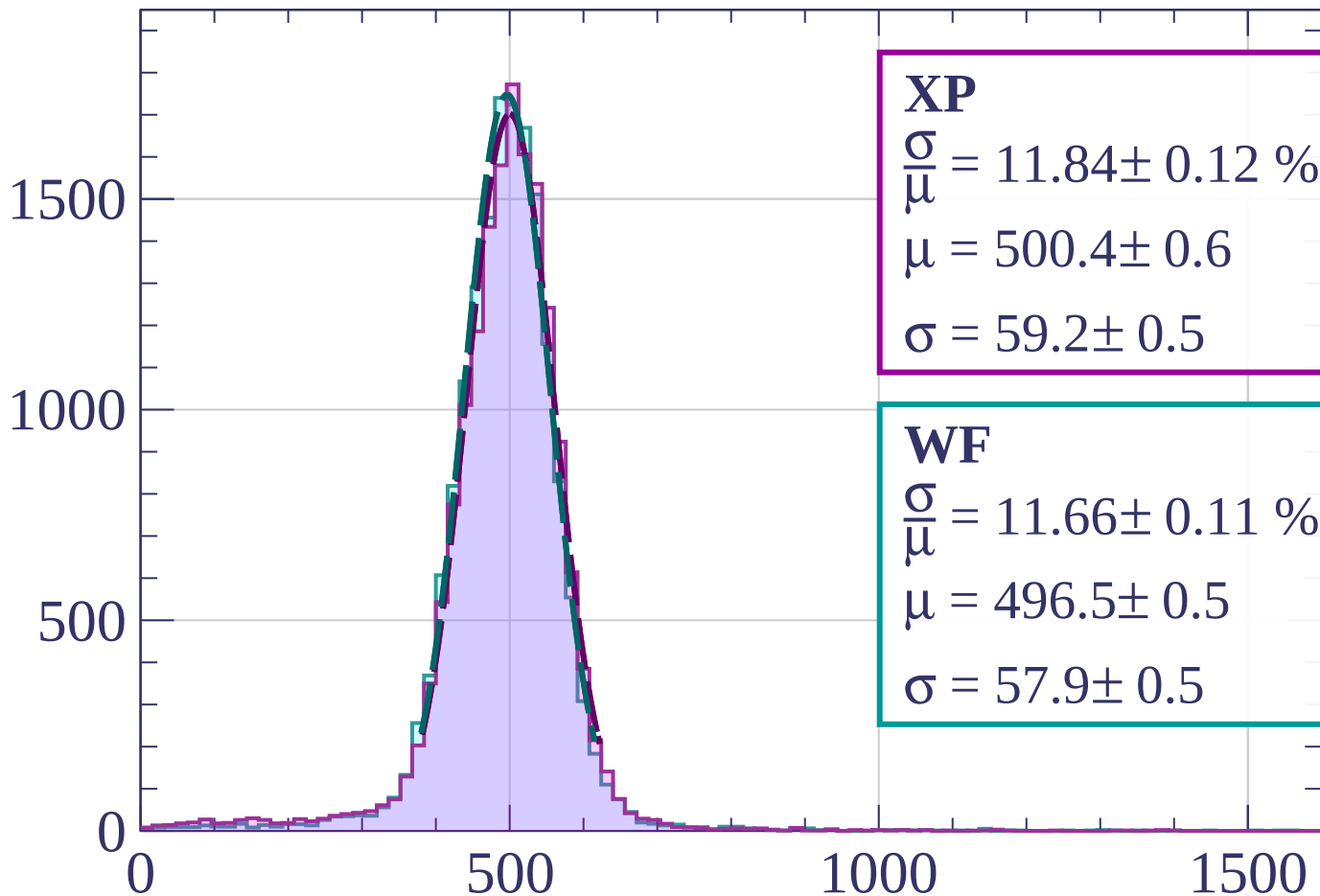
$$\mu = 494.0 \pm 0.6$$

$$\sigma = 58.3 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $72 < \phi < 75$

Count



XP

$$\frac{\sigma}{\mu} = 11.84 \pm 0.12 \%$$

$$\mu = 500.4 \pm 0.6$$

$$\sigma = 59.2 \pm 0.5$$

WF

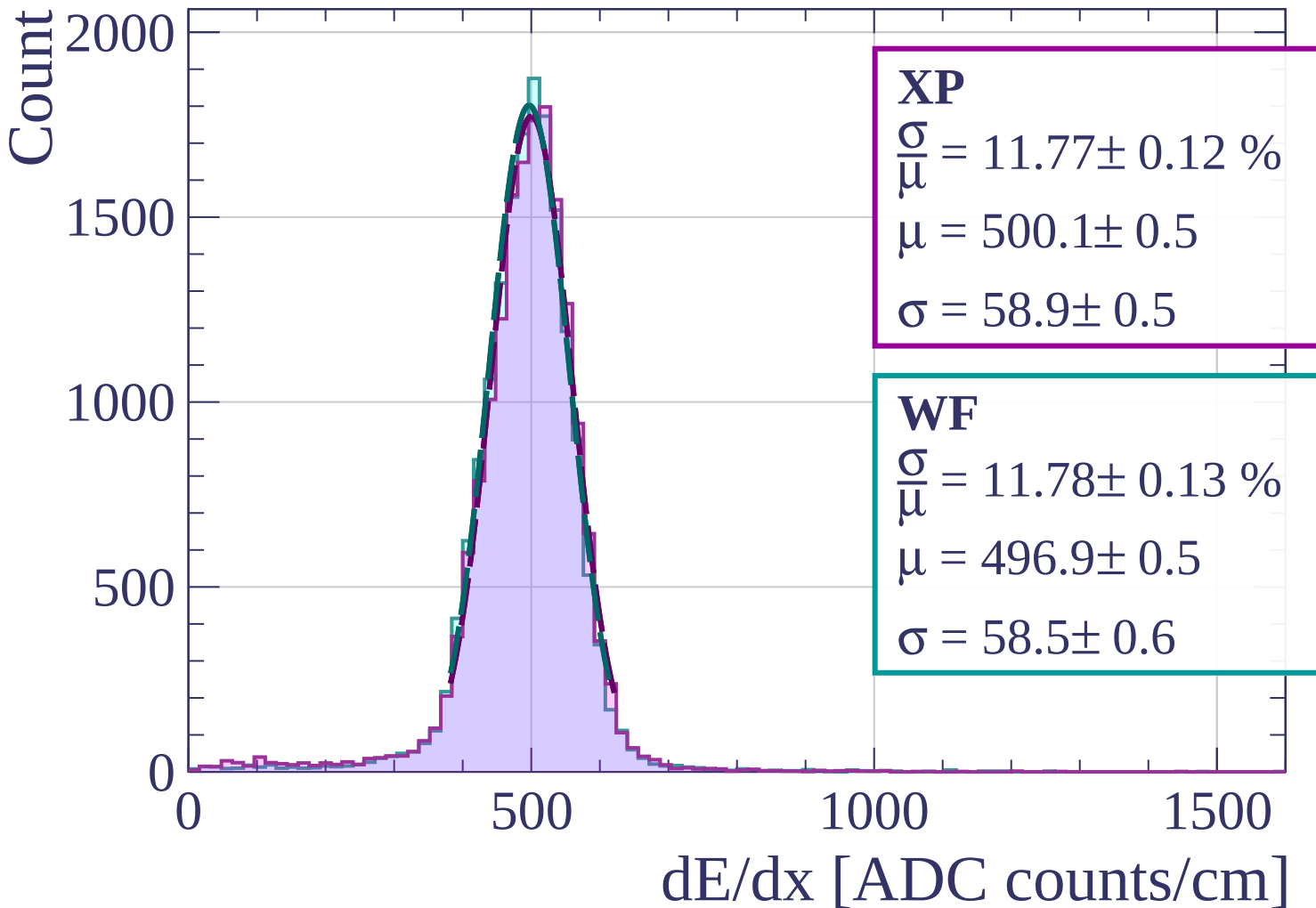
$$\frac{\sigma}{\mu} = 11.66 \pm 0.11 \%$$

$$\mu = 496.5 \pm 0.5$$

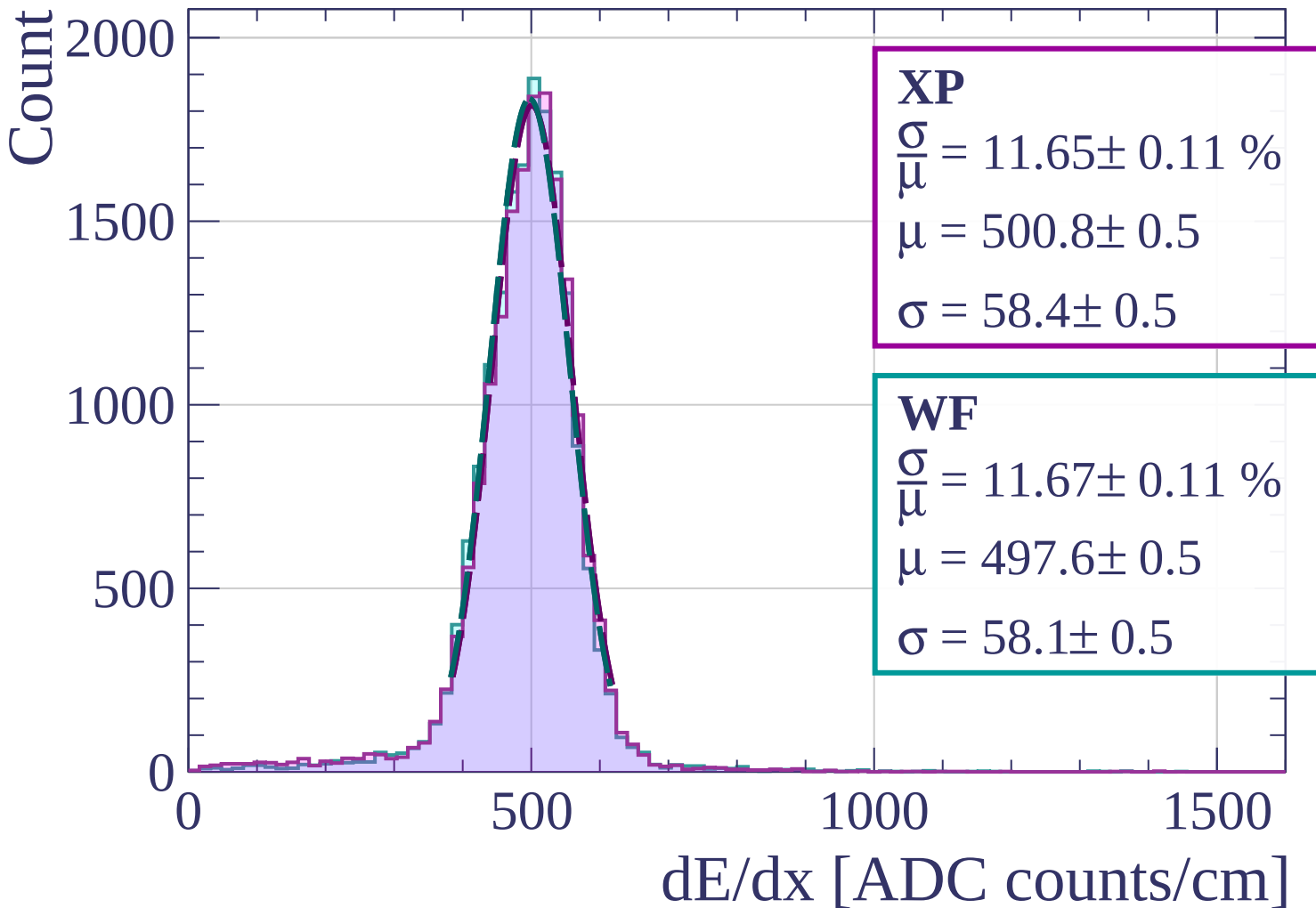
$$\sigma = 57.9 \pm 0.5$$

dE/dx [ADC counts/cm]

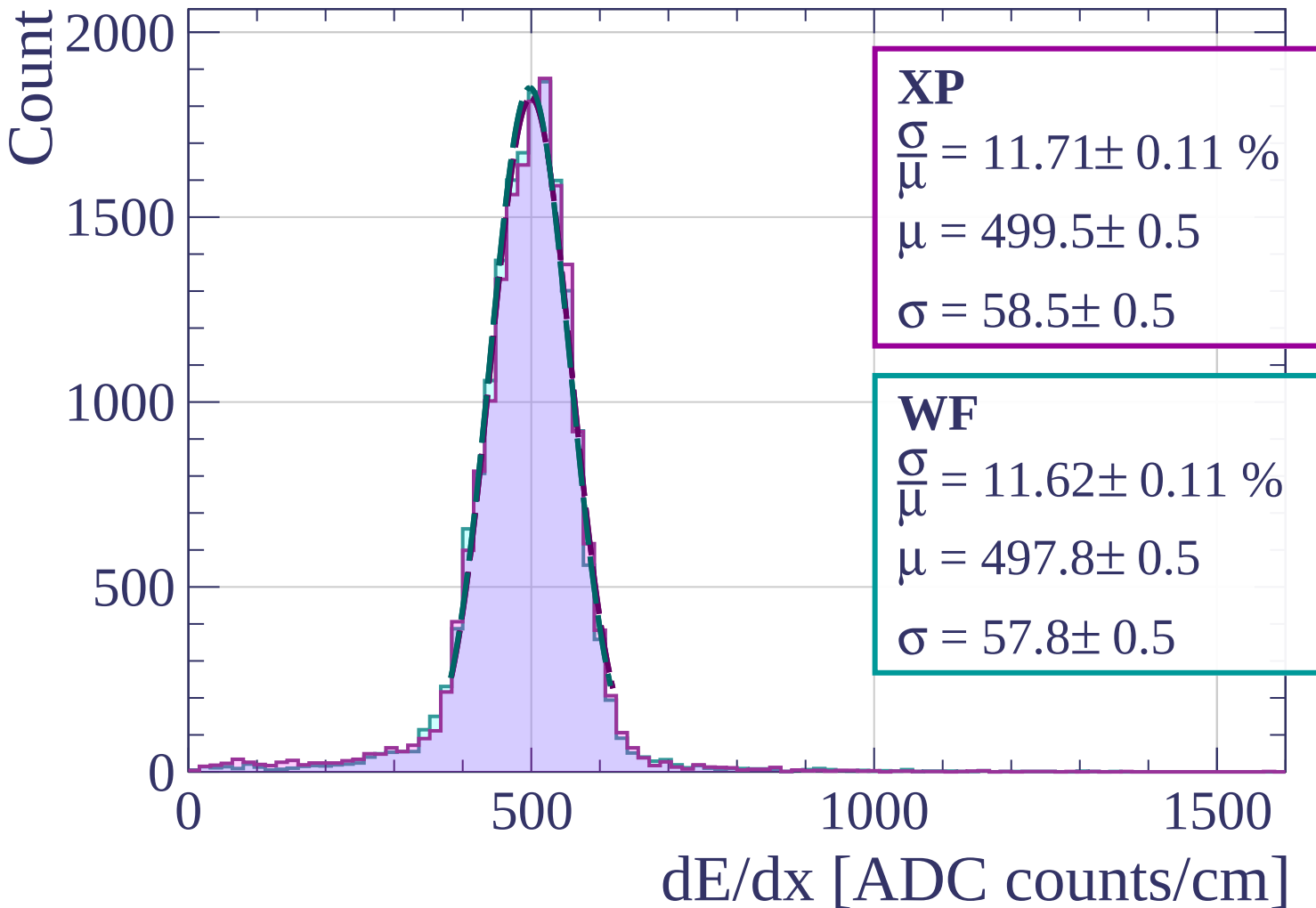
Energy loss | $75 < \phi < 78$



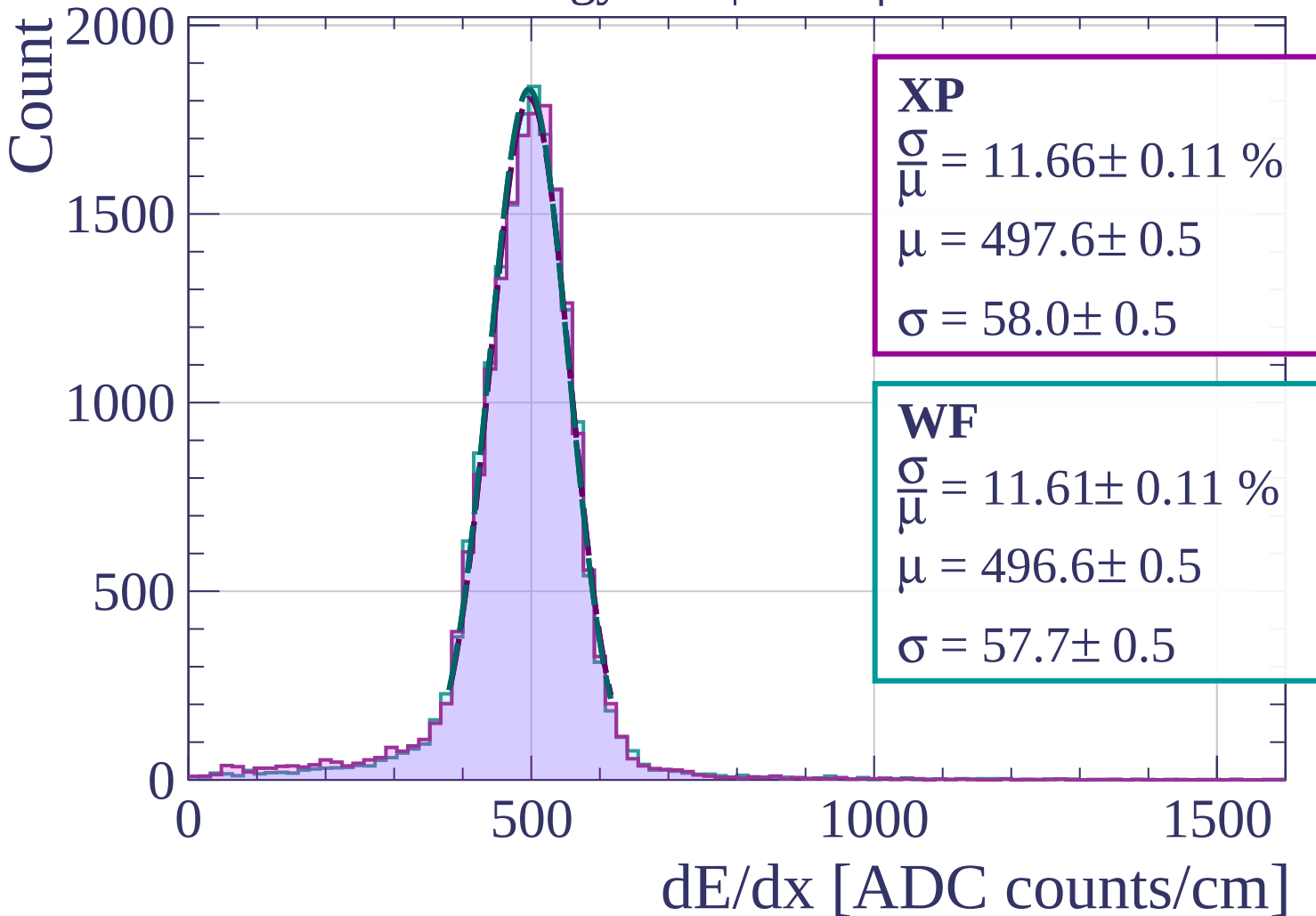
Energy loss | $78 < \phi < 81$



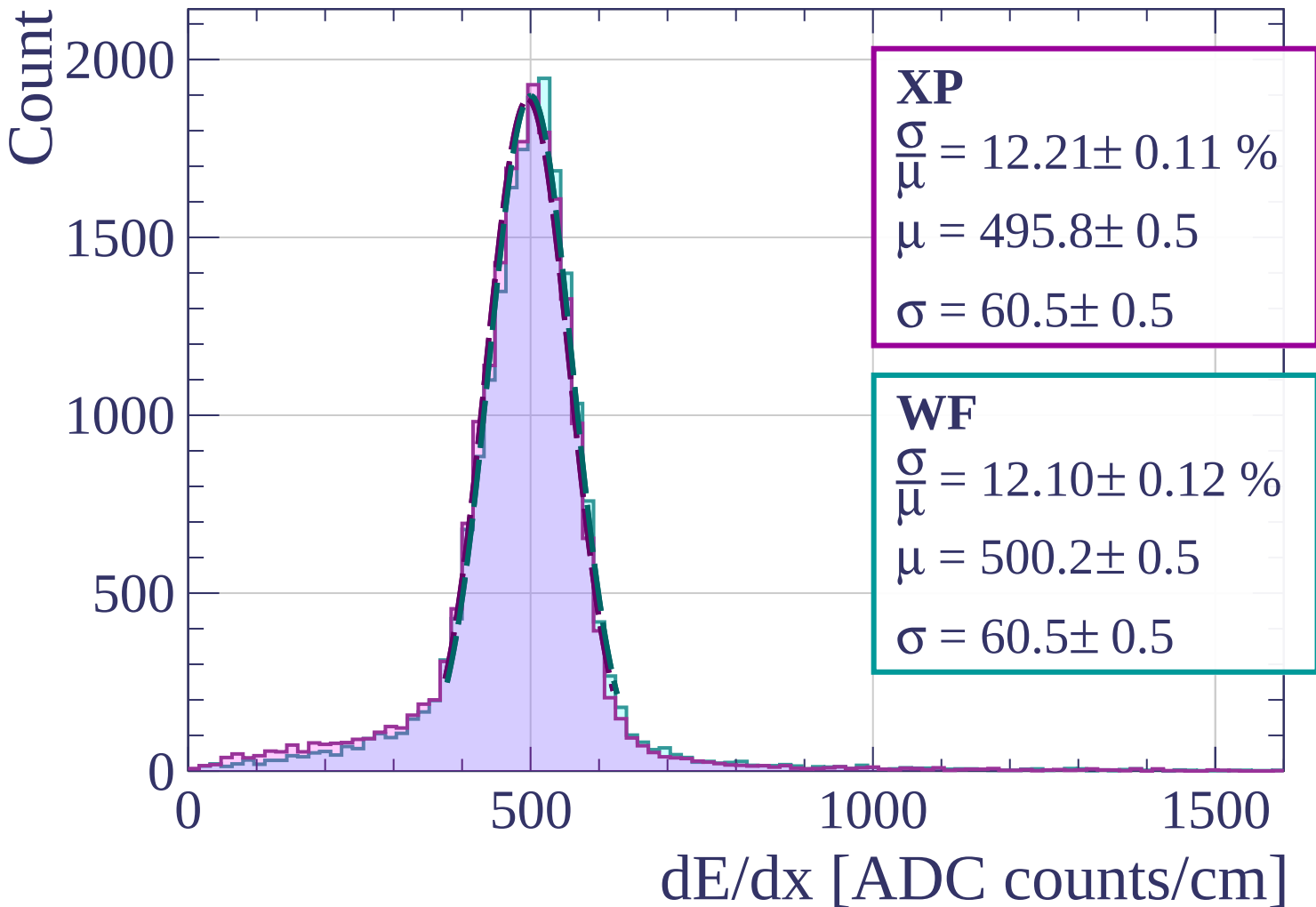
Energy loss | $81 < \phi < 84$



Energy loss | $84 < \phi < 87$

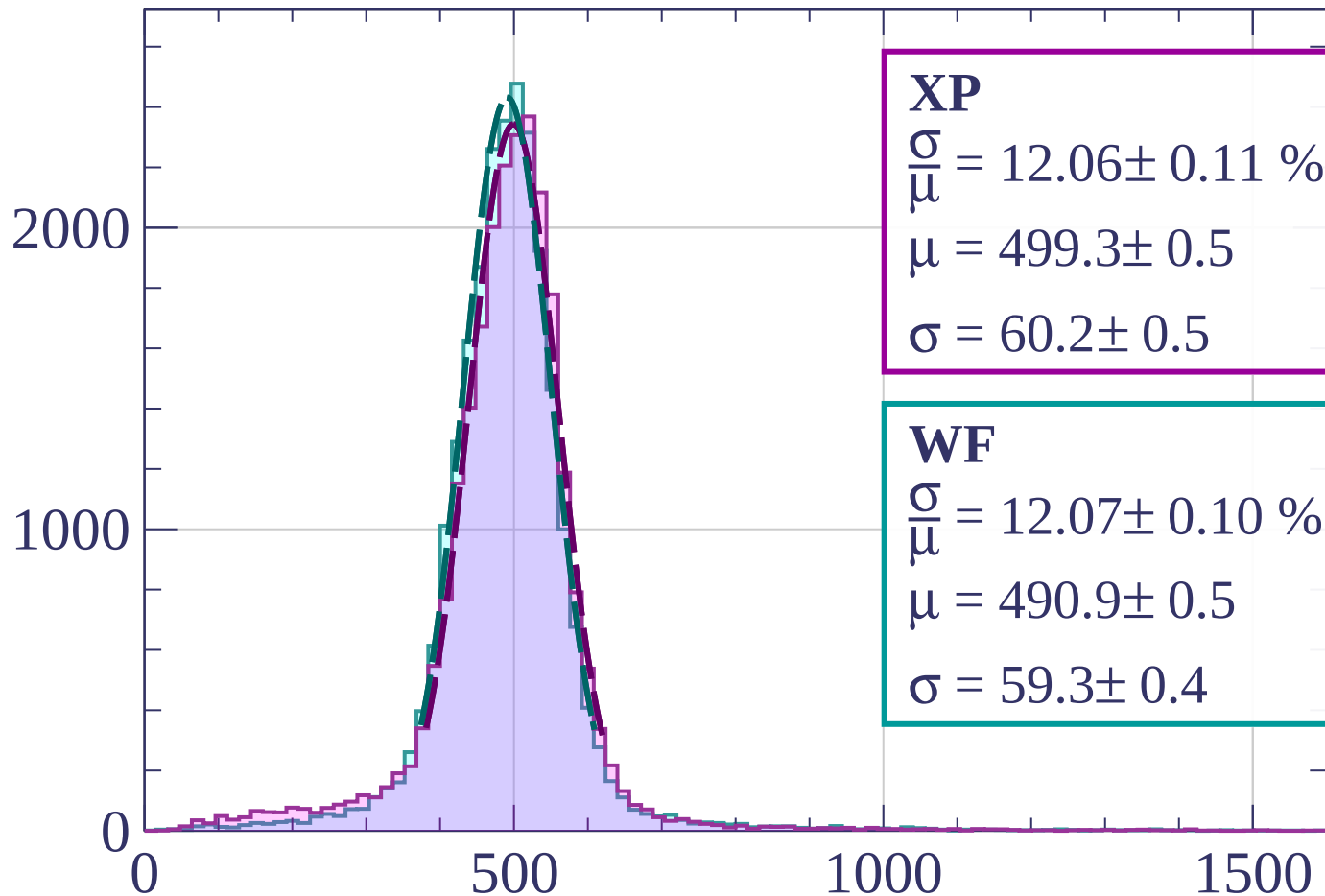


Energy loss | $87 < \phi < 90$



Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 12.06 \pm 0.11 \%$$

$$\mu = 499.3 \pm 0.5$$

$$\sigma = 60.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 12.07 \pm 0.10 \%$$

$$\mu = 490.9 \pm 0.5$$

$$\sigma = 59.3 \pm 0.4$$

dE/dx [ADC counts/cm]

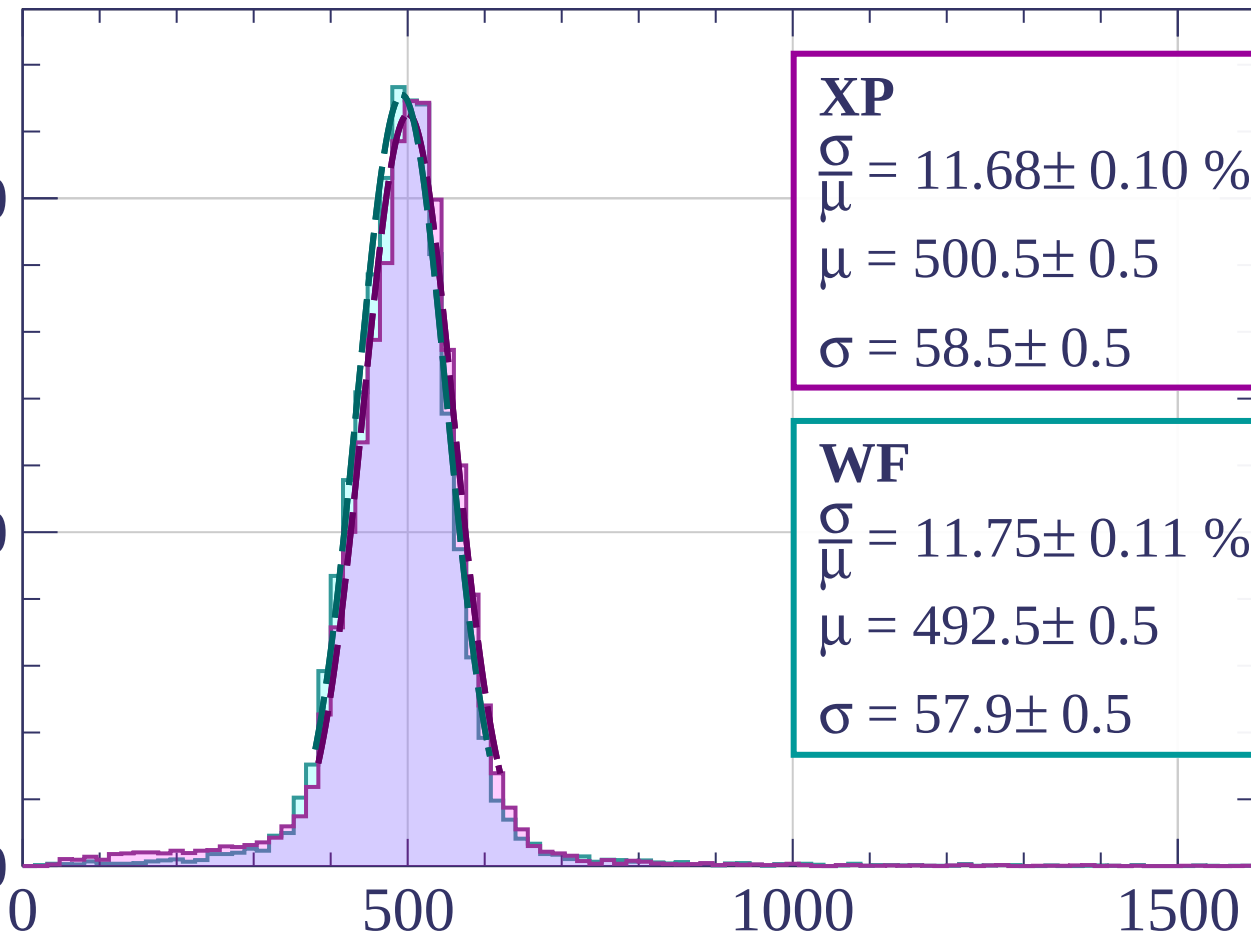
Energy loss | $3 < \theta < 6$

Count

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 11.68 \pm 0.10 \%$$

$$\mu = 500.5 \pm 0.5$$

$$\sigma = 58.5 \pm 0.5$$

WF

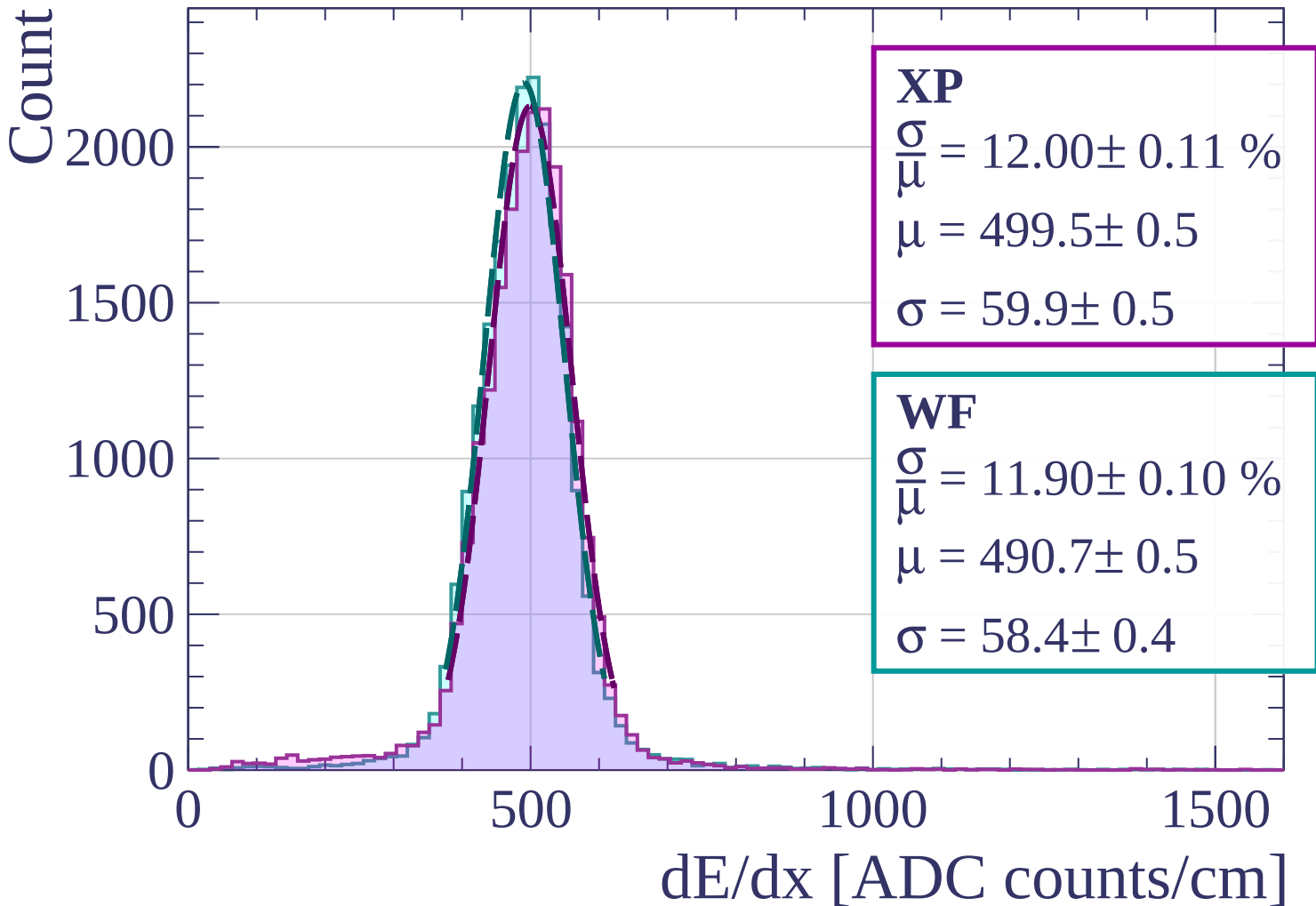
$$\frac{\sigma}{\mu} = 11.75 \pm 0.11 \%$$

$$\mu = 492.5 \pm 0.5$$

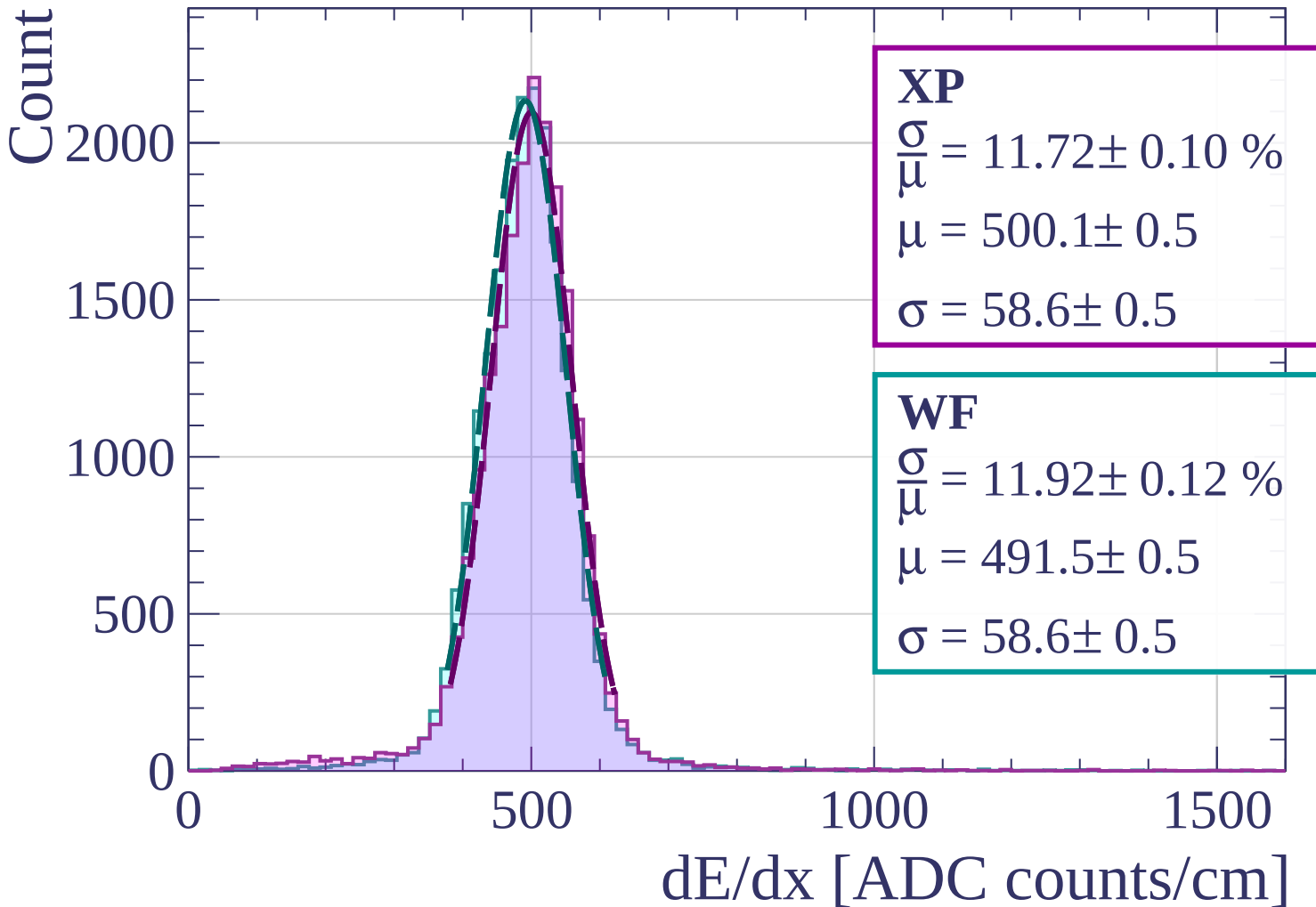
$$\sigma = 57.9 \pm 0.5$$

dE/dx [ADC counts/cm]

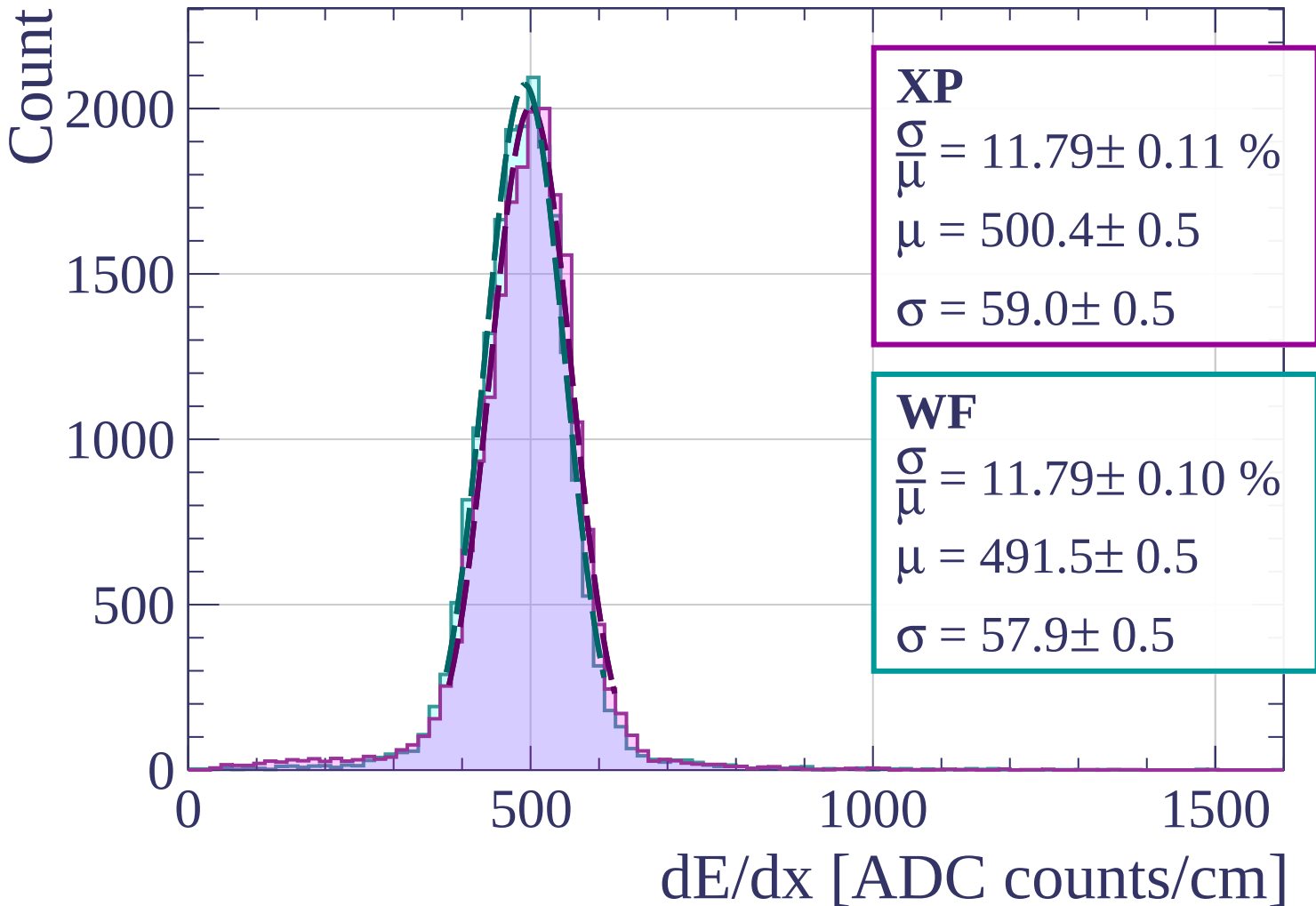
Energy loss | $6 < \theta < 9$



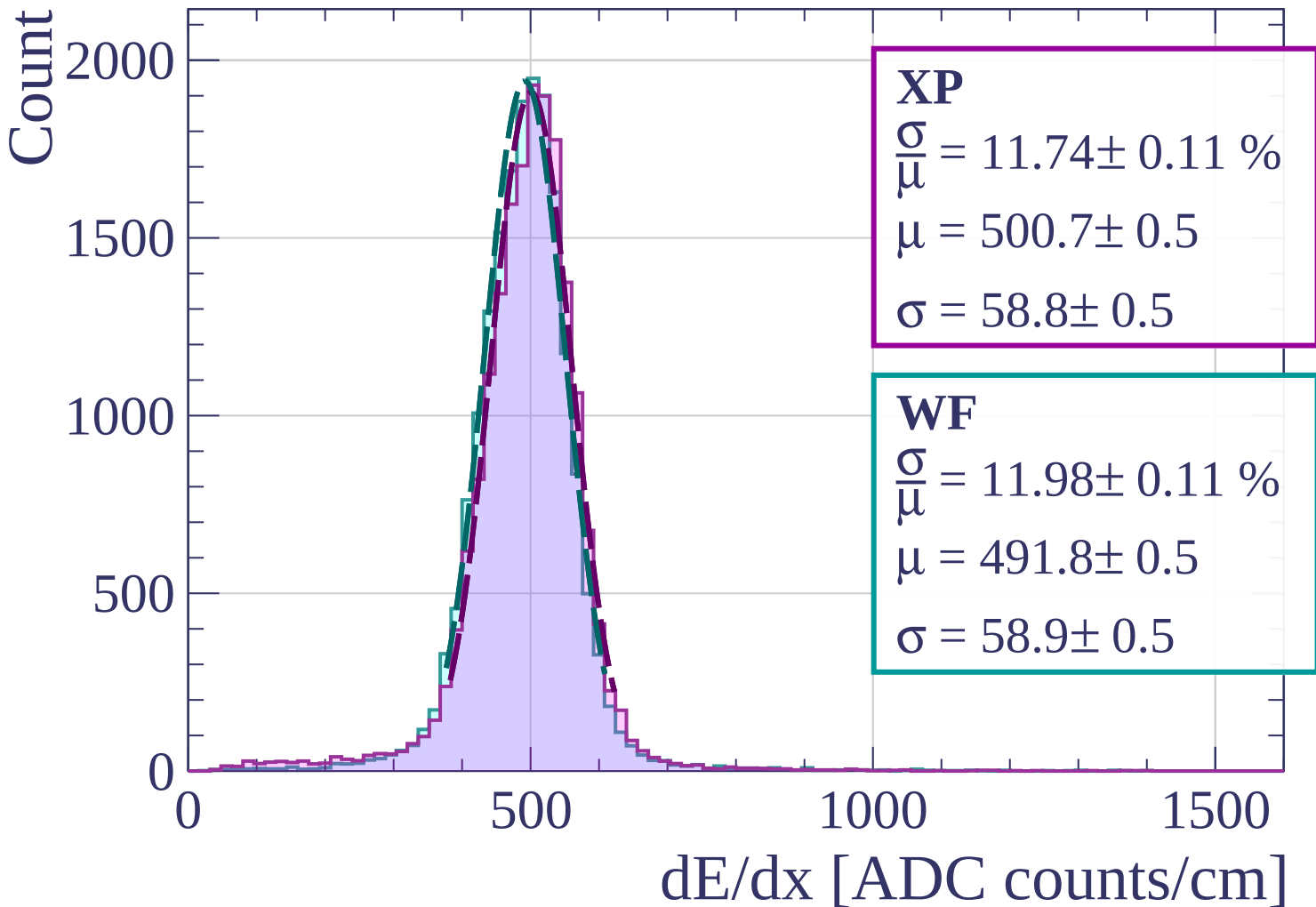
Energy loss | $9 < \theta < 12$



Energy loss | $12 < \theta < 15$



Energy loss | $15 < \theta < 18$



Energy loss | $18 < \theta < 21$

Count

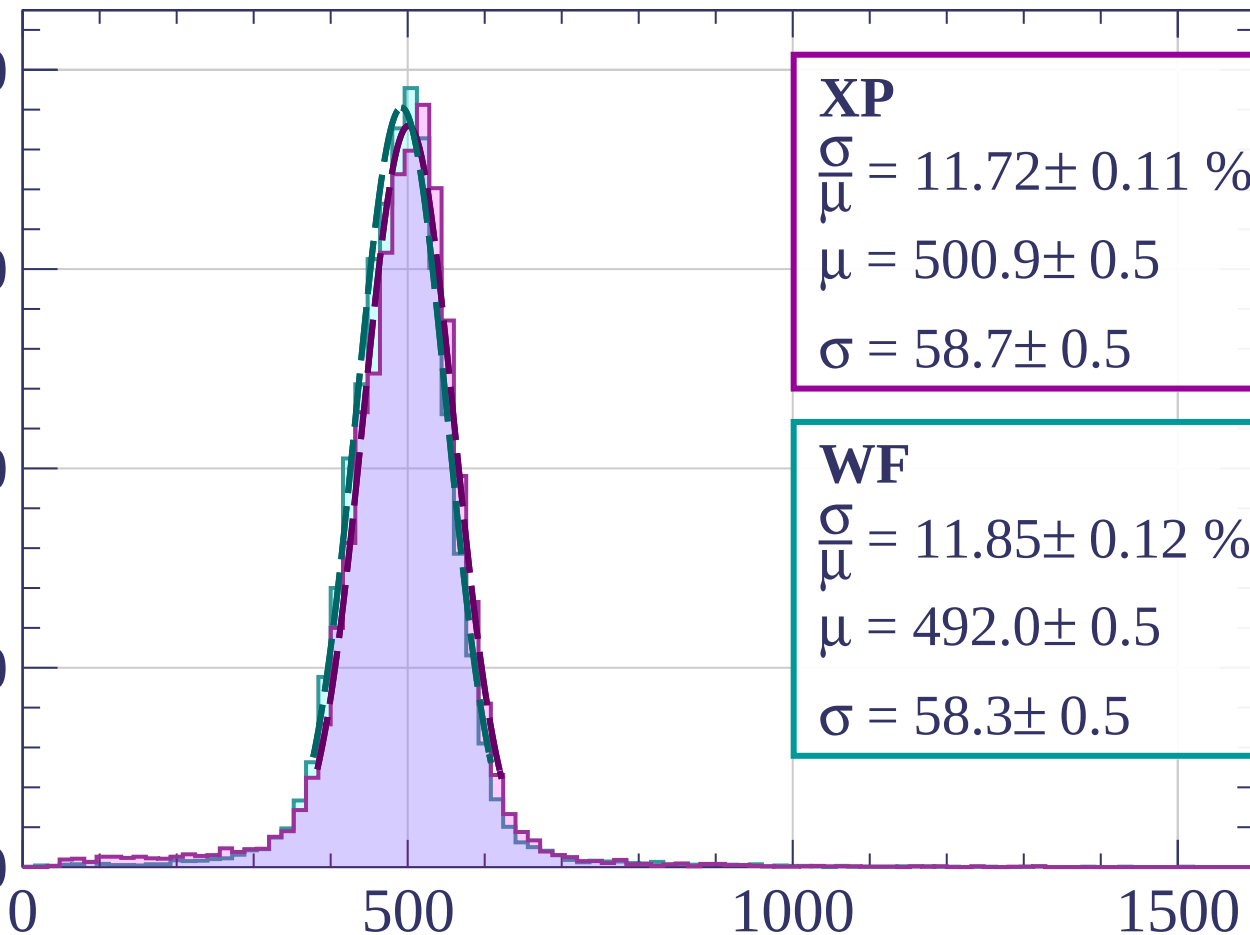
2000

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 11.72 \pm 0.11 \%$$

$$\mu = 500.9 \pm 0.5$$

$$\sigma = 58.7 \pm 0.5$$

WF

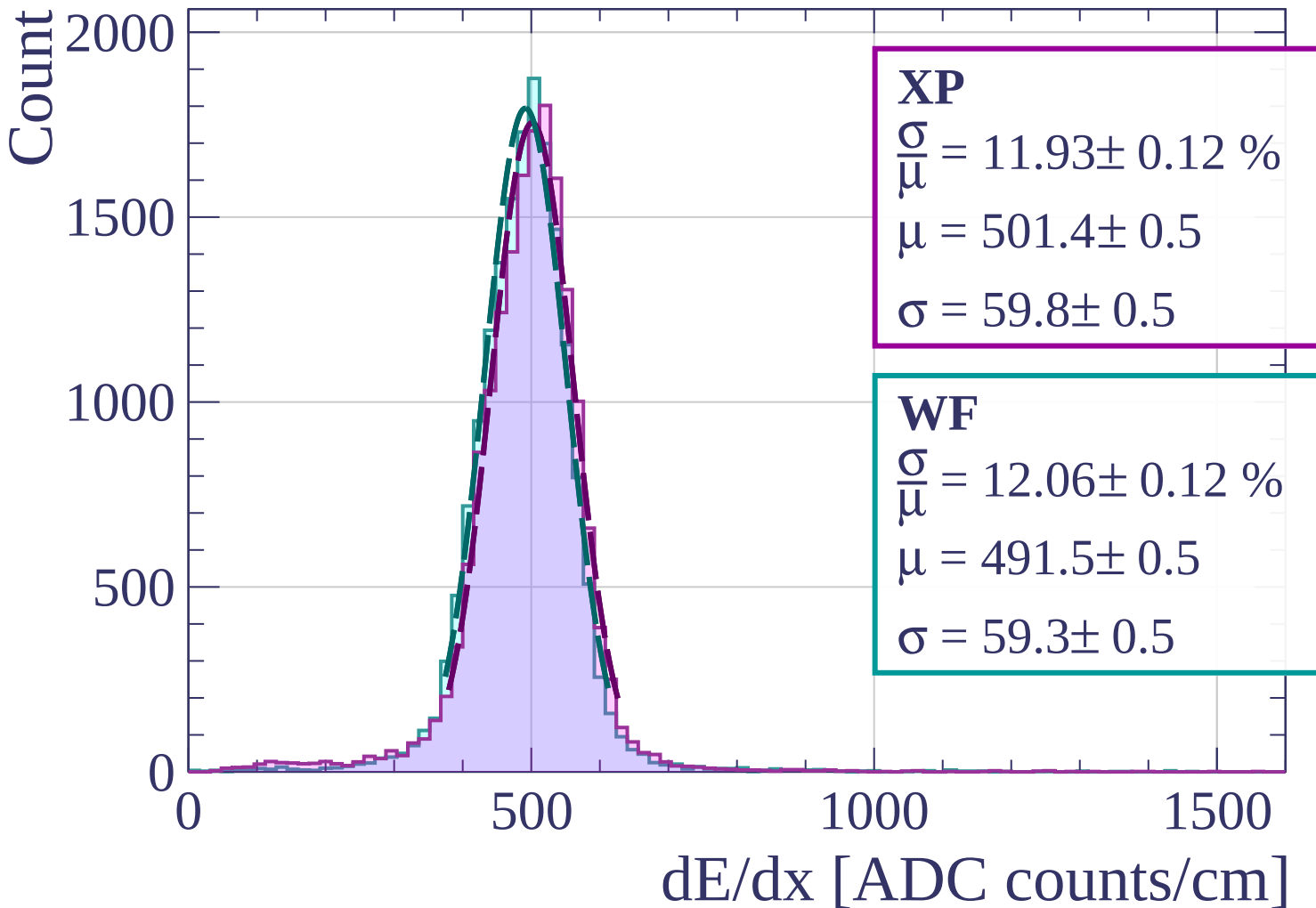
$$\frac{\sigma}{\mu} = 11.85 \pm 0.12 \%$$

$$\mu = 492.0 \pm 0.5$$

$$\sigma = 58.3 \pm 0.5$$

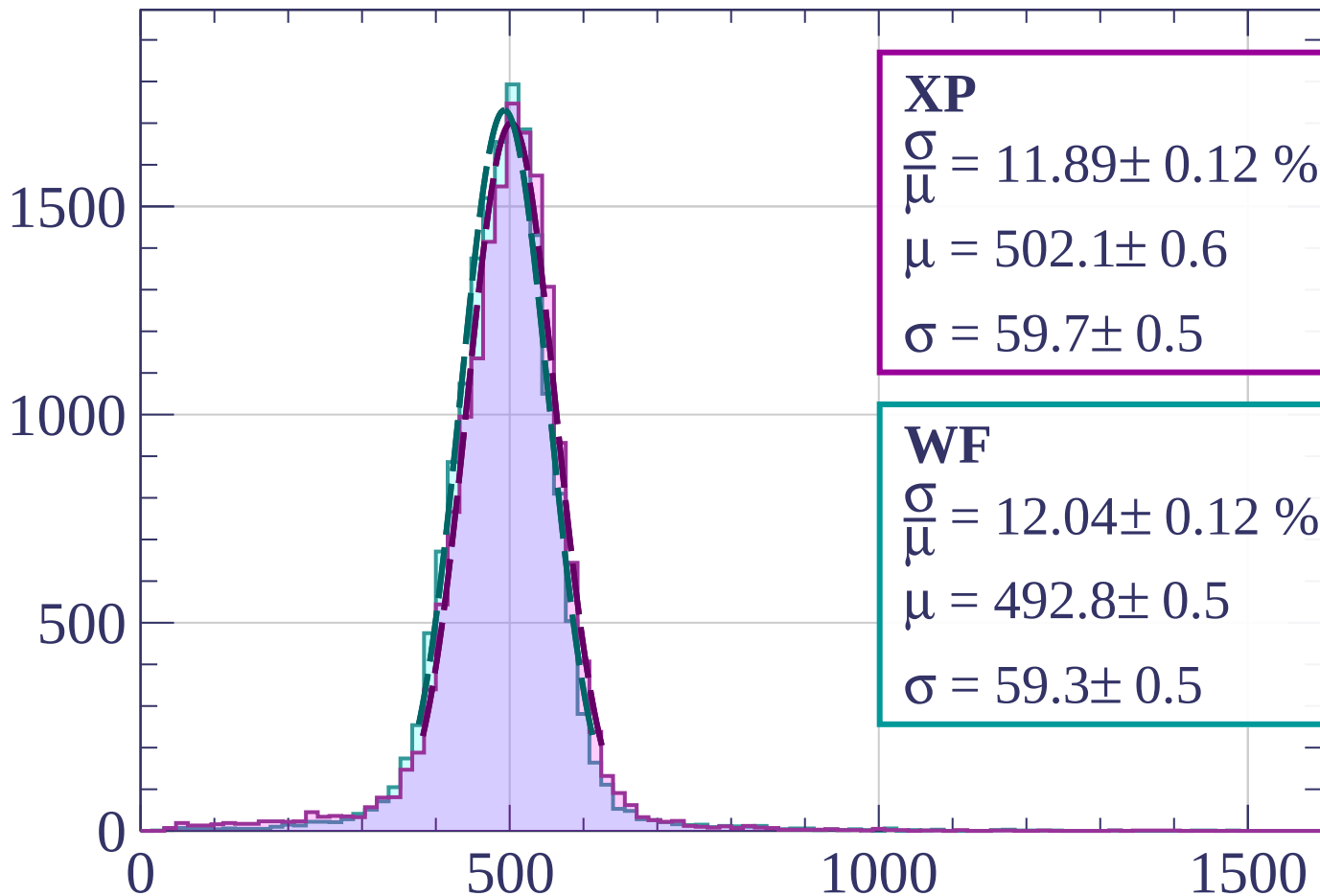
dE/dx [ADC counts/cm]

Energy loss | $21 < \theta < 24$



Energy loss | $24 < \theta < 27$

Count



XP

$$\frac{\sigma}{\mu} = 11.89 \pm 0.12 \%$$

$$\mu = 502.1 \pm 0.6$$

$$\sigma = 59.7 \pm 0.5$$

WF

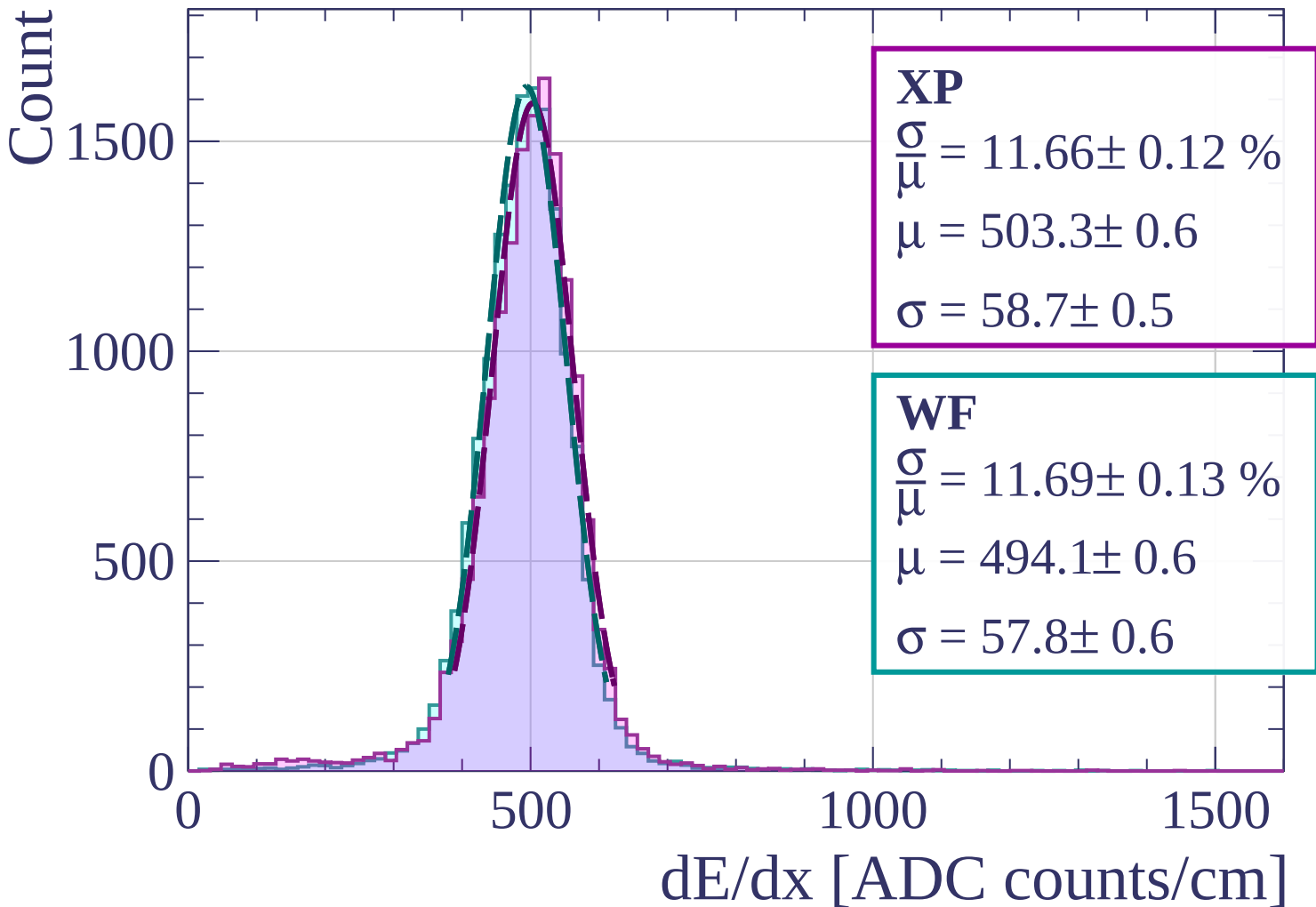
$$\frac{\sigma}{\mu} = 12.04 \pm 0.12 \%$$

$$\mu = 492.8 \pm 0.5$$

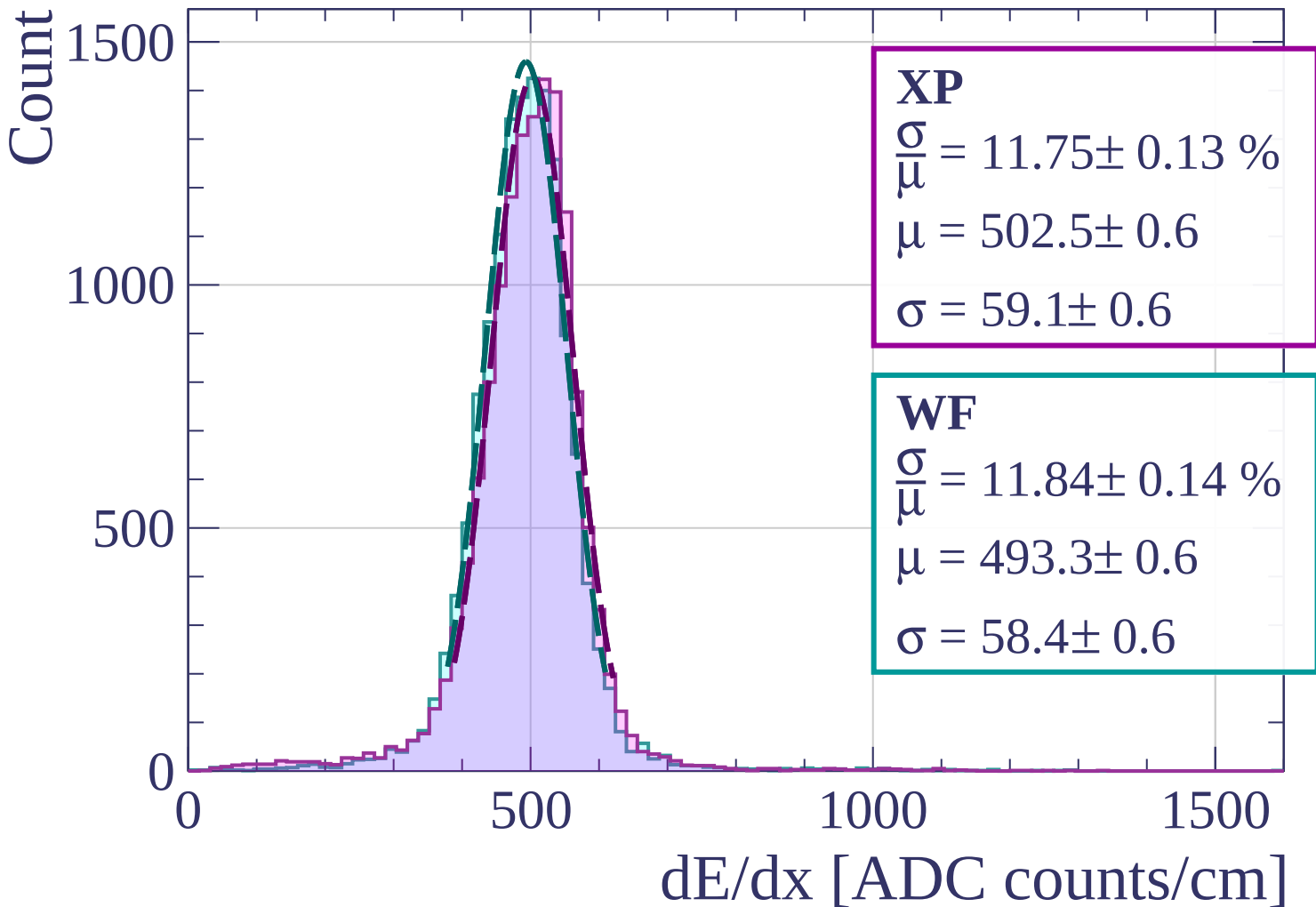
$$\sigma = 59.3 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \theta < 30$

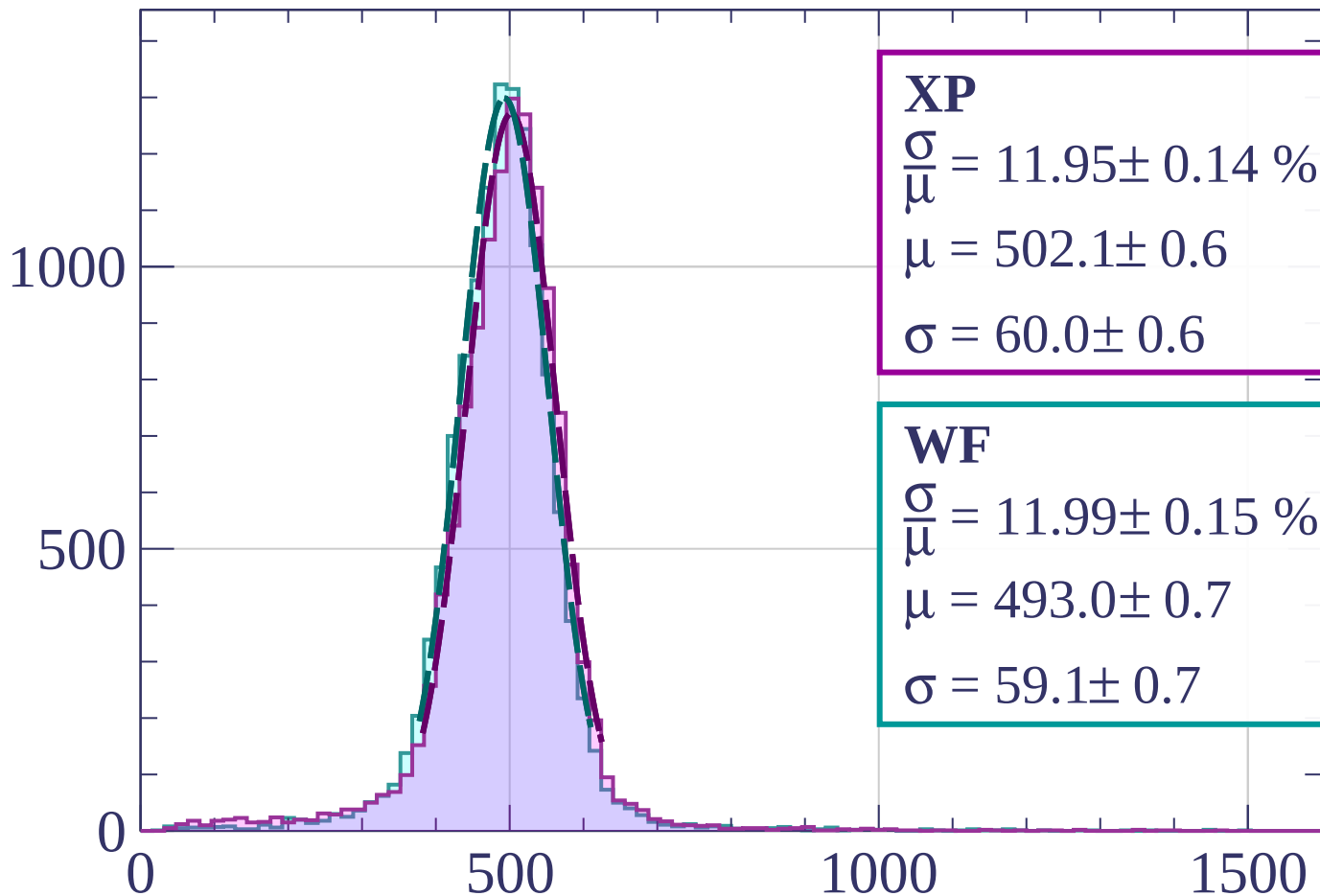


Energy loss | $30 < \theta < 33$



Energy loss | $33 < \theta < 36$

Count



XP

$$\frac{\sigma}{\mu} = 11.95 \pm 0.14 \%$$

$$\mu = 502.1 \pm 0.6$$

$$\sigma = 60.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 11.99 \pm 0.15 \%$$

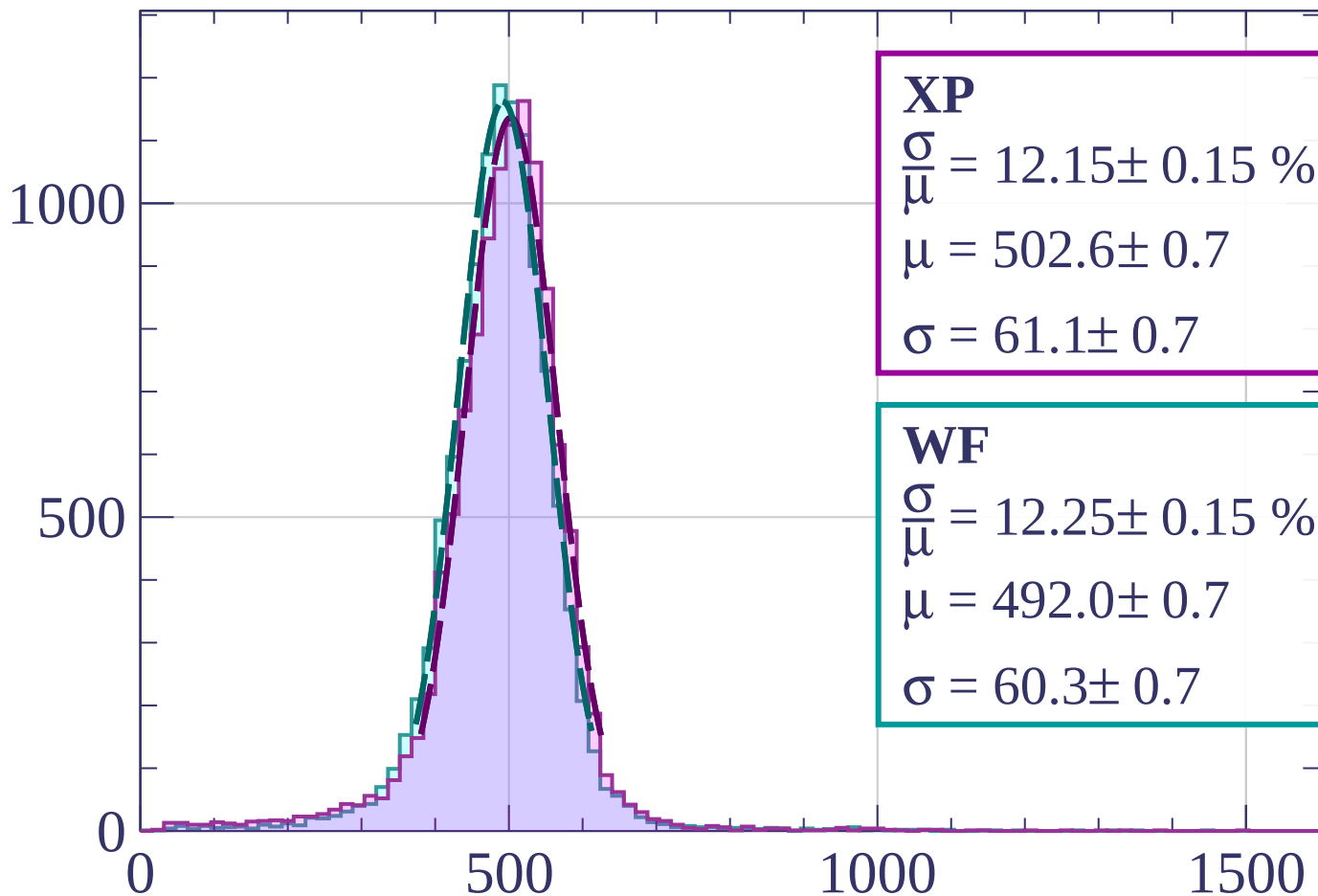
$$\mu = 493.0 \pm 0.7$$

$$\sigma = 59.1 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \theta < 39$

Count



XP

$$\frac{\sigma}{\mu} = 12.15 \pm 0.15 \%$$

$$\mu = 502.6 \pm 0.7$$

$$\sigma = 61.1 \pm 0.7$$

WF

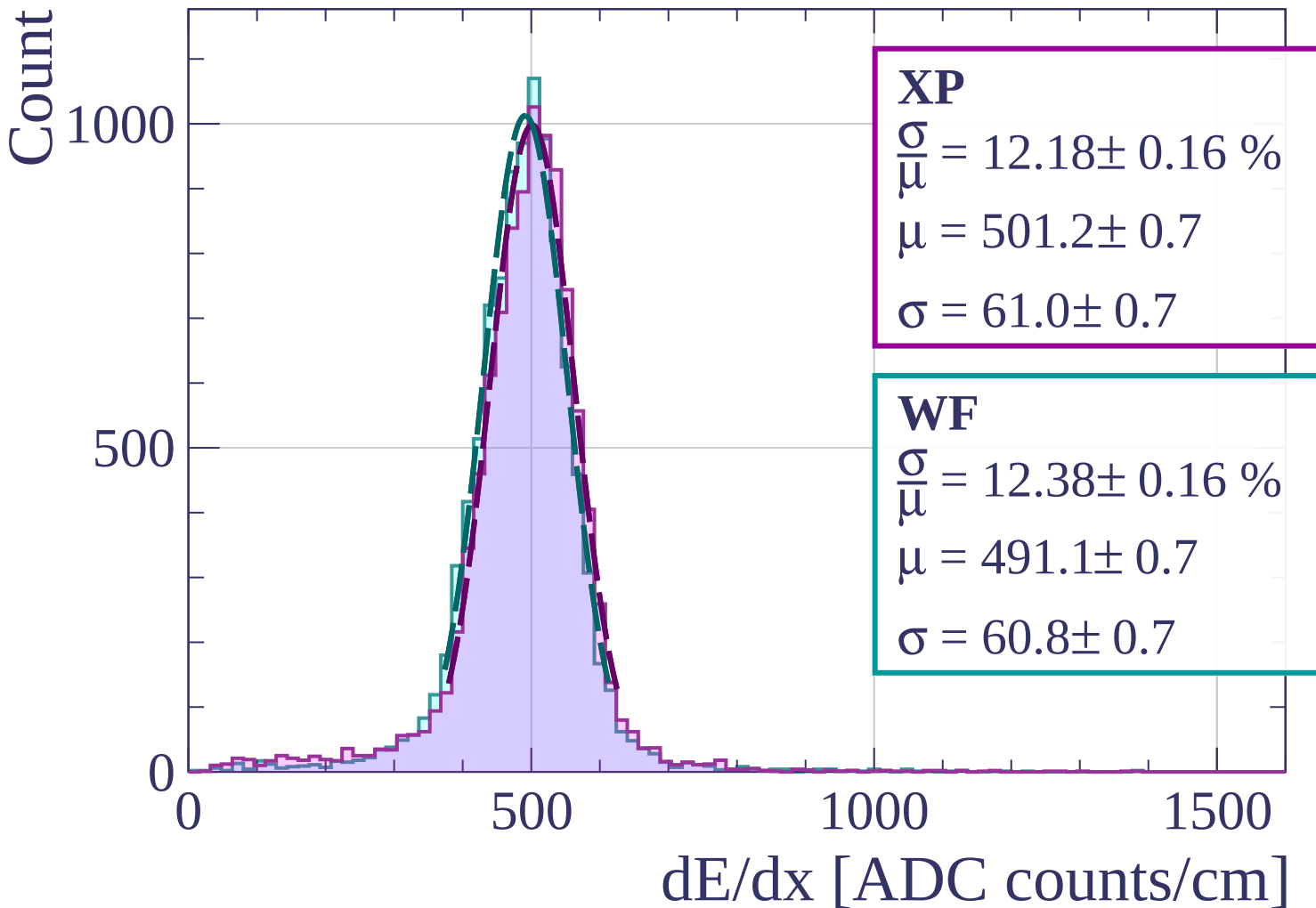
$$\frac{\sigma}{\mu} = 12.25 \pm 0.15 \%$$

$$\mu = 492.0 \pm 0.7$$

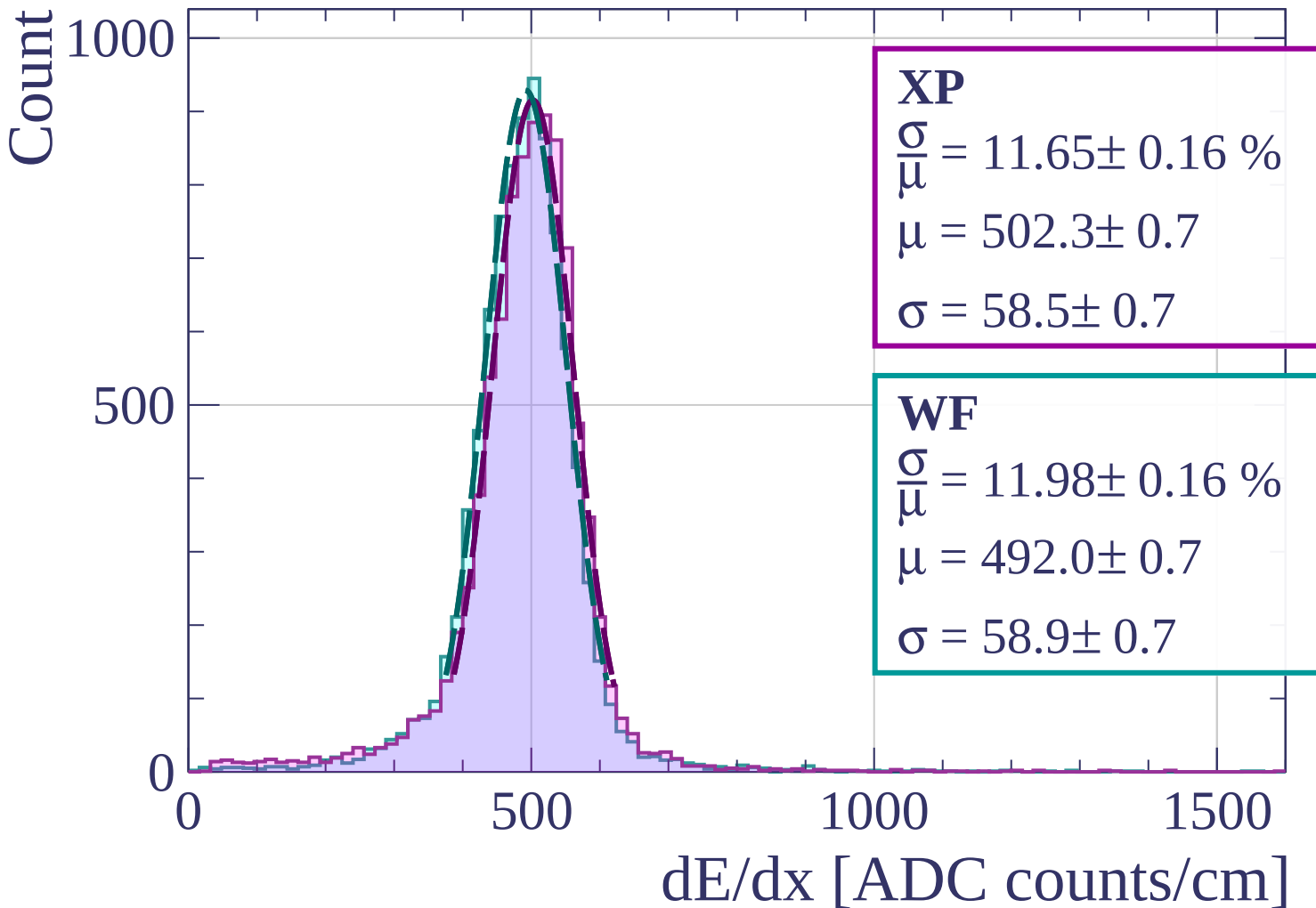
$$\sigma = 60.3 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \theta < 42$

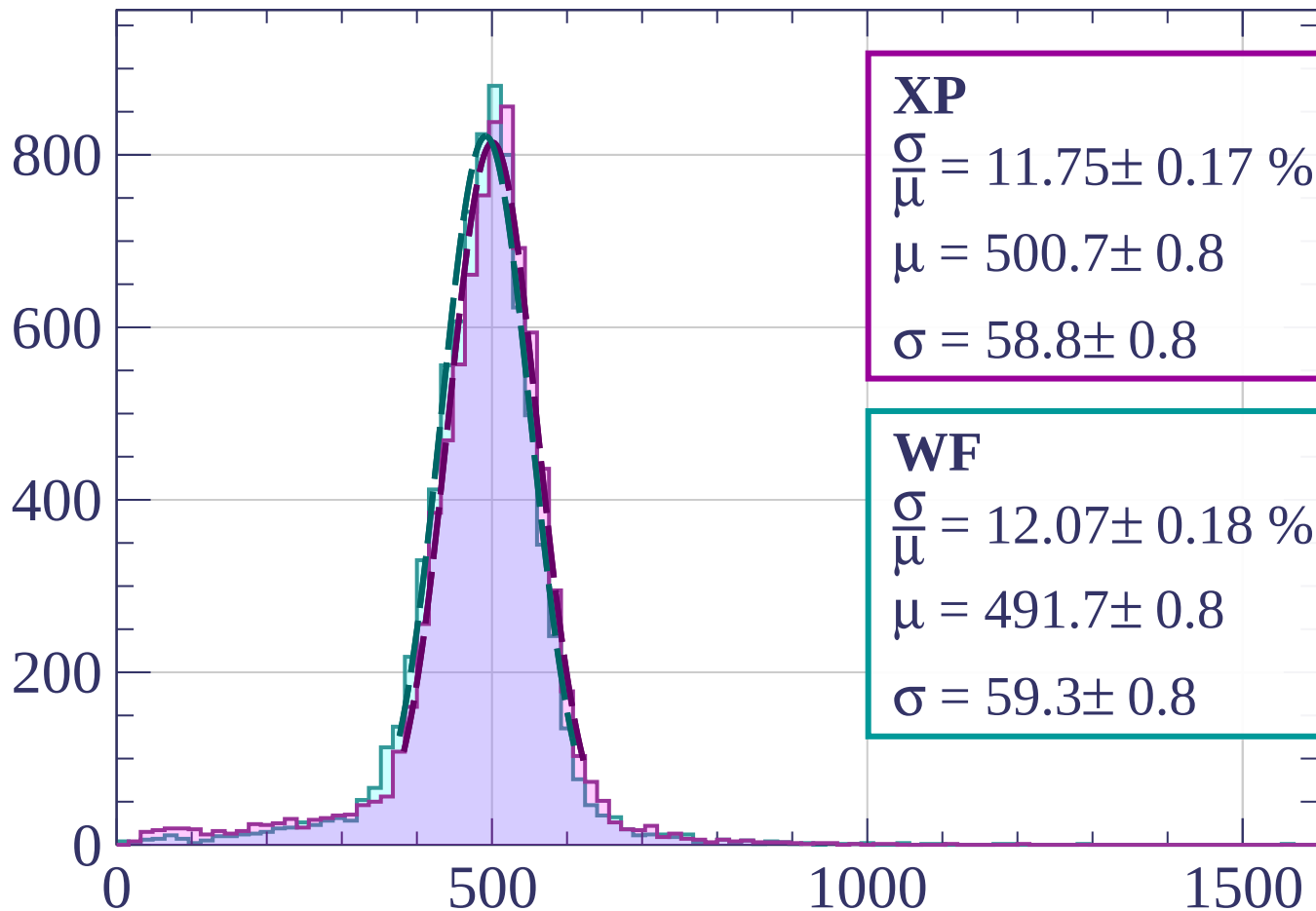


Energy loss | $42 < \theta < 45$



Energy loss | $45 < \theta < 48$

Count



XP

$$\frac{\sigma}{\mu} = 11.75 \pm 0.17 \%$$

$$\mu = 500.7 \pm 0.8$$

$$\sigma = 58.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 12.07 \pm 0.18 \%$$

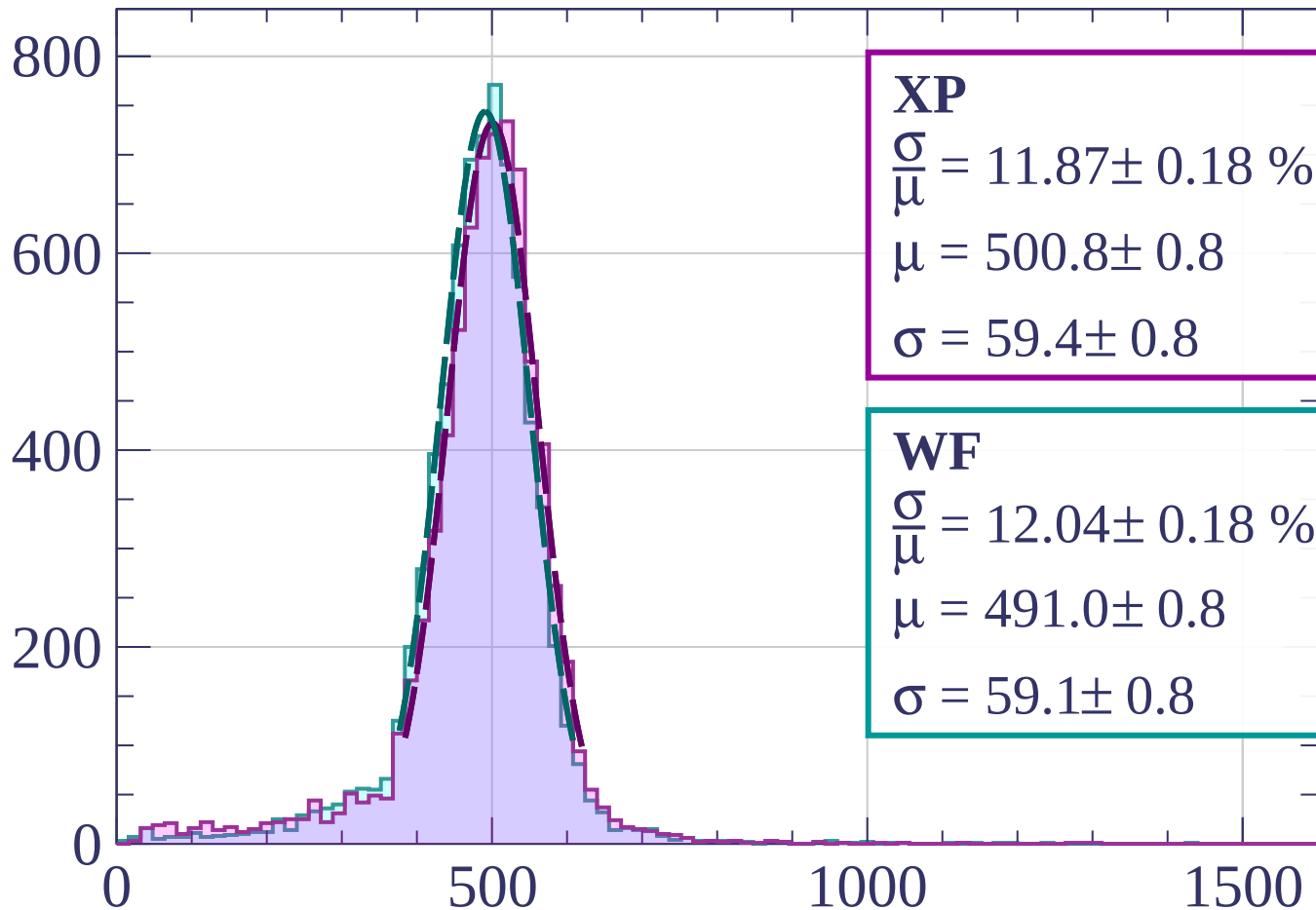
$$\mu = 491.7 \pm 0.8$$

$$\sigma = 59.3 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $48 < \theta < 51$

Count



XP

$$\frac{\sigma}{\mu} = 11.87 \pm 0.18 \%$$

$$\mu = 500.8 \pm 0.8$$

$$\sigma = 59.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 12.04 \pm 0.18 \%$$

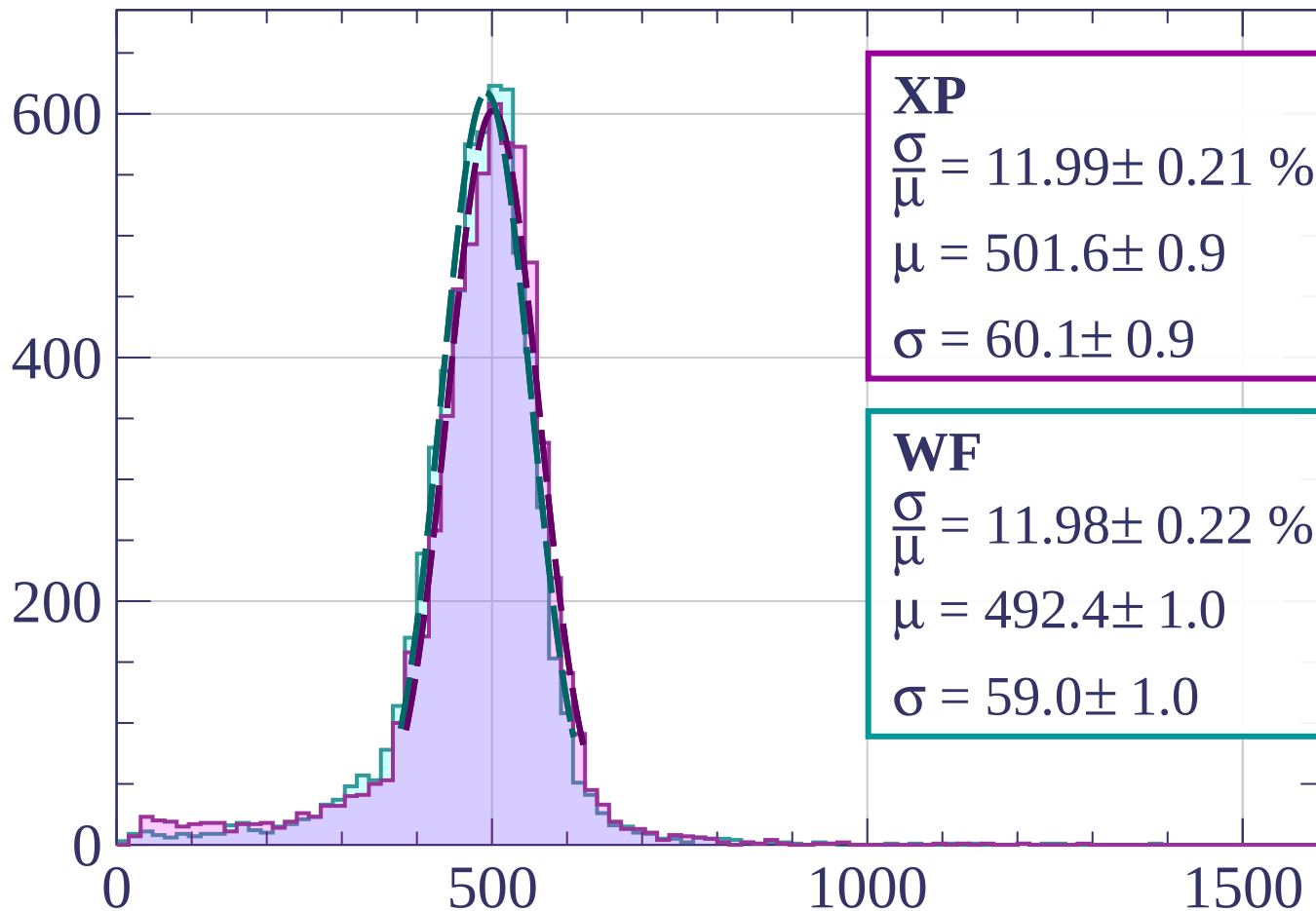
$$\mu = 491.0 \pm 0.8$$

$$\sigma = 59.1 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \theta < 54$

Count



XP

$$\frac{\sigma}{\mu} = 11.99 \pm 0.21 \%$$

$$\mu = 501.6 \pm 0.9$$

$$\sigma = 60.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.98 \pm 0.22 \%$$

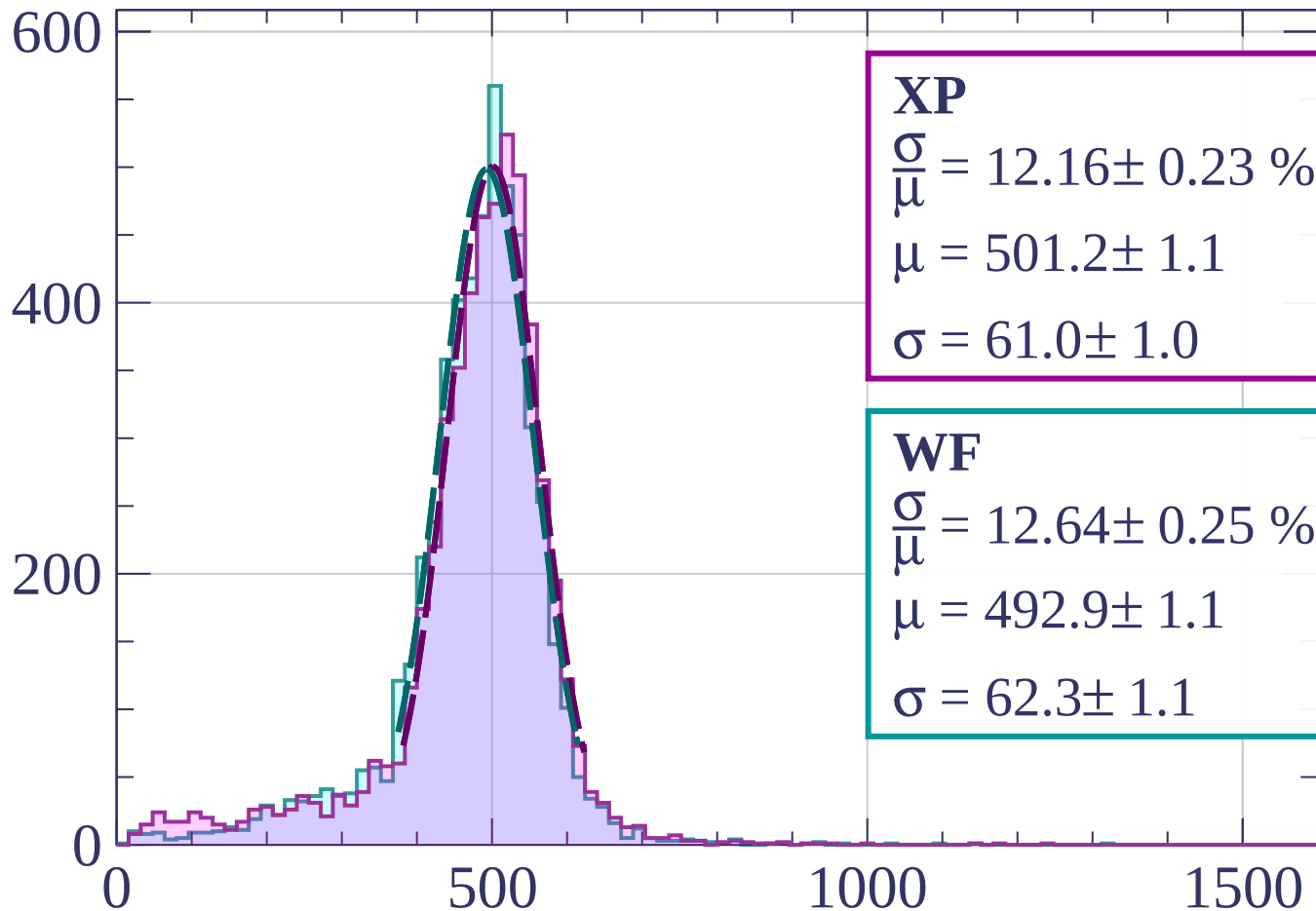
$$\mu = 492.4 \pm 1.0$$

$$\sigma = 59.0 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \theta < 57$

Count



XP

$$\frac{\sigma}{\mu} = 12.16 \pm 0.23 \%$$

$$\mu = 501.2 \pm 1.1$$

$$\sigma = 61.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 12.64 \pm 0.25 \%$$

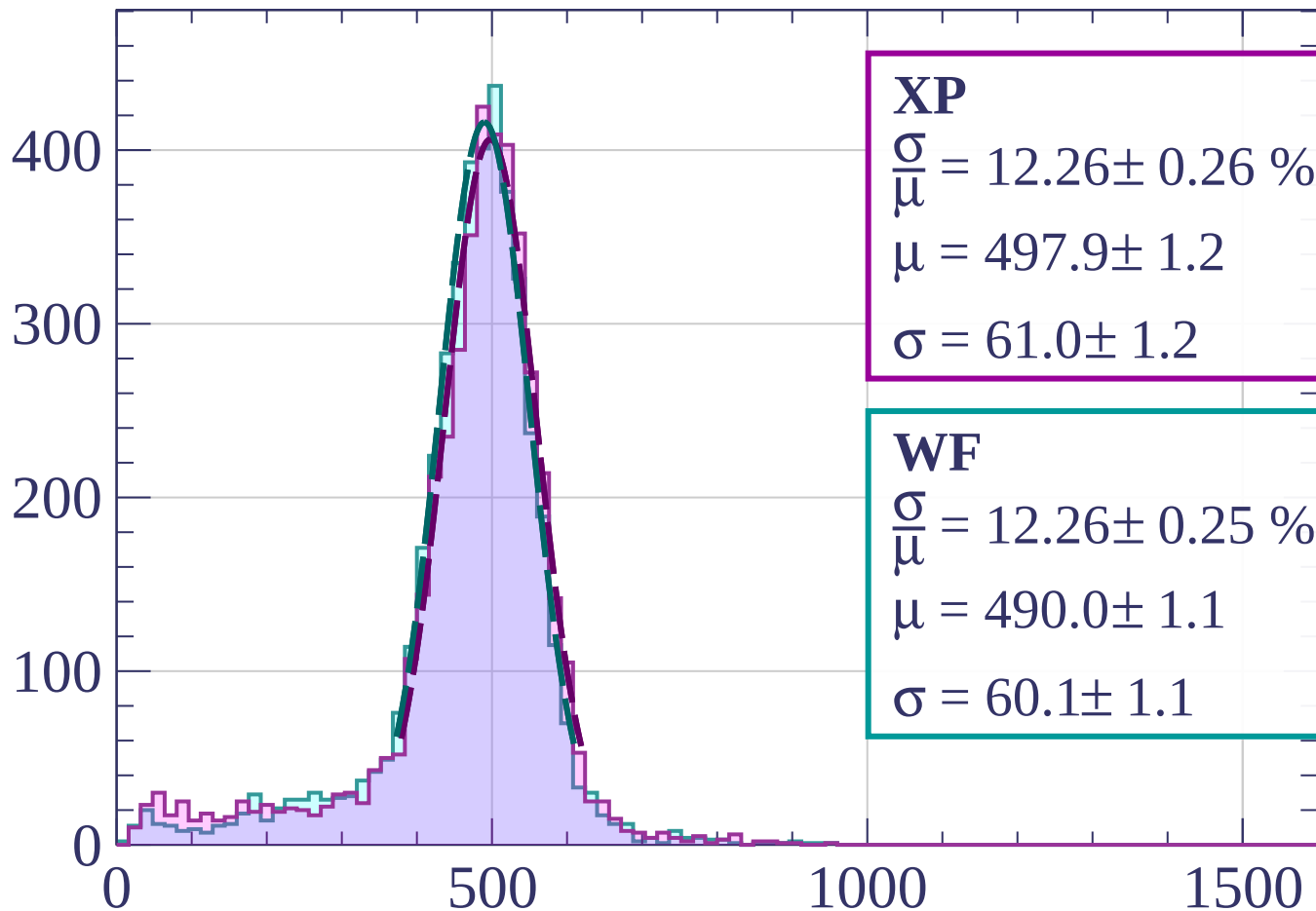
$$\mu = 492.9 \pm 1.1$$

$$\sigma = 62.3 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \theta < 60$

Count



XP

$$\frac{\sigma}{\mu} = 12.26 \pm 0.26 \%$$

$$\mu = 497.9 \pm 1.2$$

$$\sigma = 61.0 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 12.26 \pm 0.25 \%$$

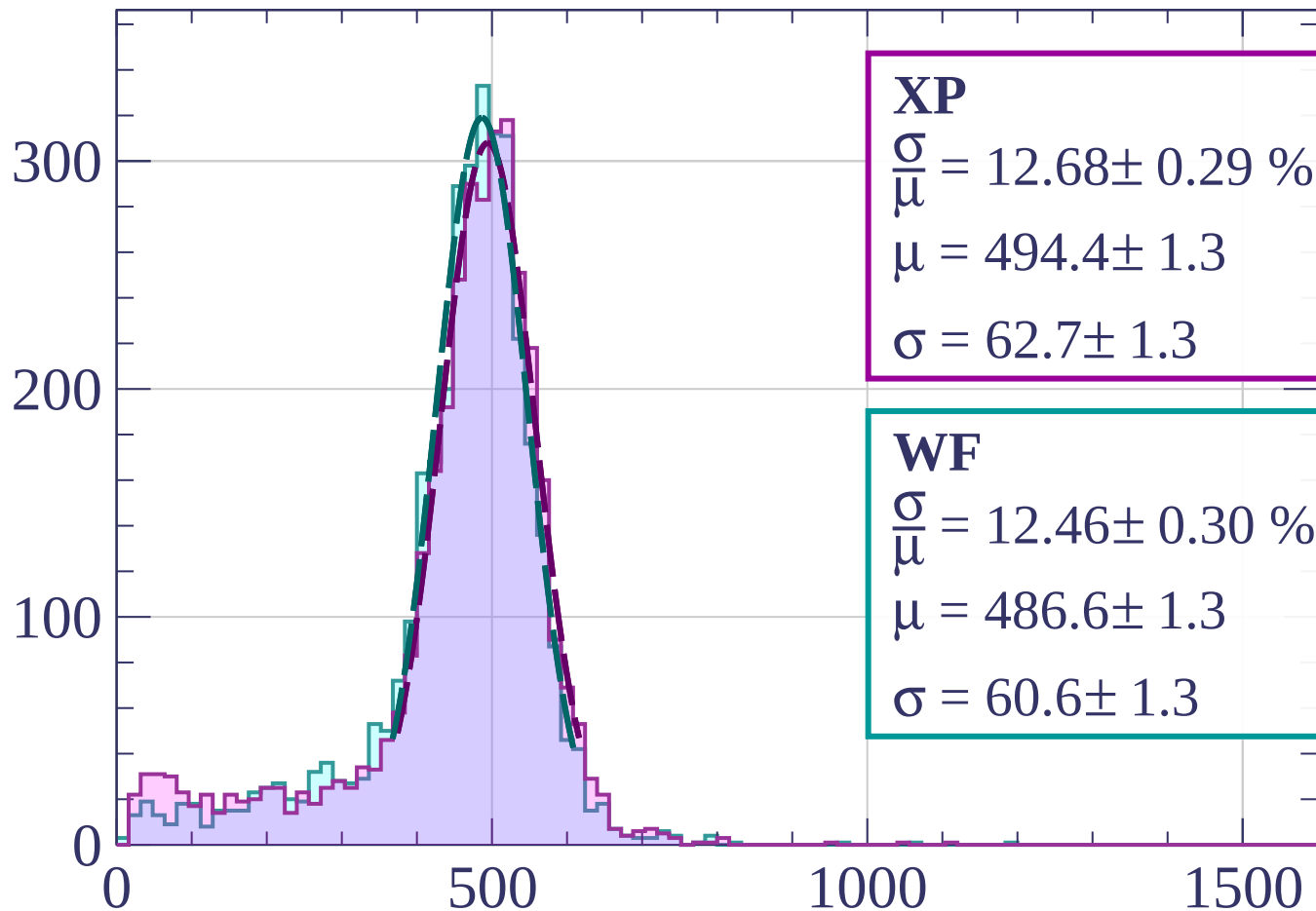
$$\mu = 490.0 \pm 1.1$$

$$\sigma = 60.1 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \theta < 63$

Count



XP

$$\frac{\sigma}{\mu} = 12.68 \pm 0.29 \%$$

$$\mu = 494.4 \pm 1.3$$

$$\sigma = 62.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 12.46 \pm 0.30 \%$$

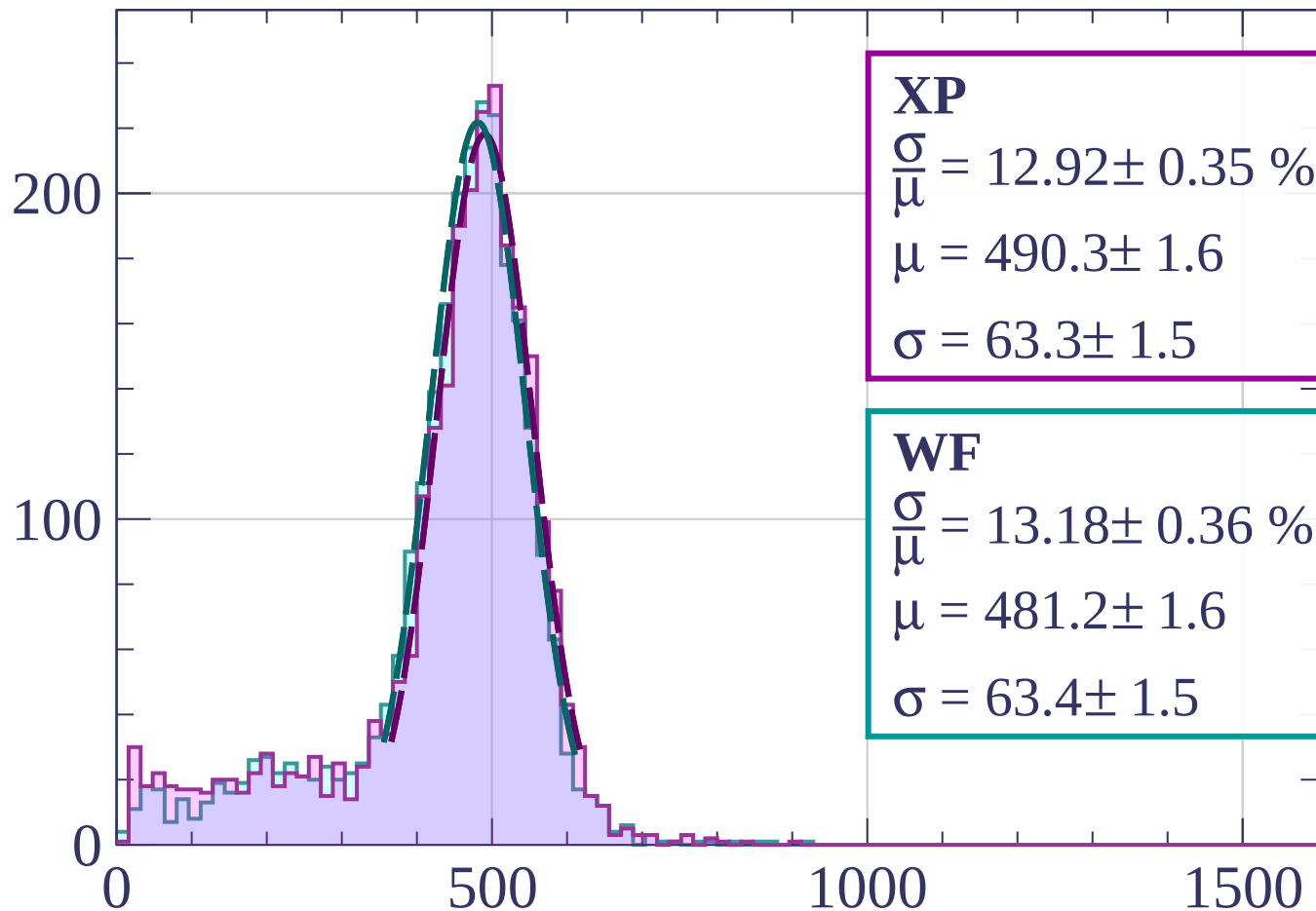
$$\mu = 486.6 \pm 1.3$$

$$\sigma = 60.6 \pm 1.3$$

dE/dx [ADC counts/cm]

Energy loss | $63 < \theta < 66$

Count



XP

$$\frac{\sigma}{\mu} = 12.92 \pm 0.35 \%$$

$$\mu = 490.3 \pm 1.6$$

$$\sigma = 63.3 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 13.18 \pm 0.36 \%$$

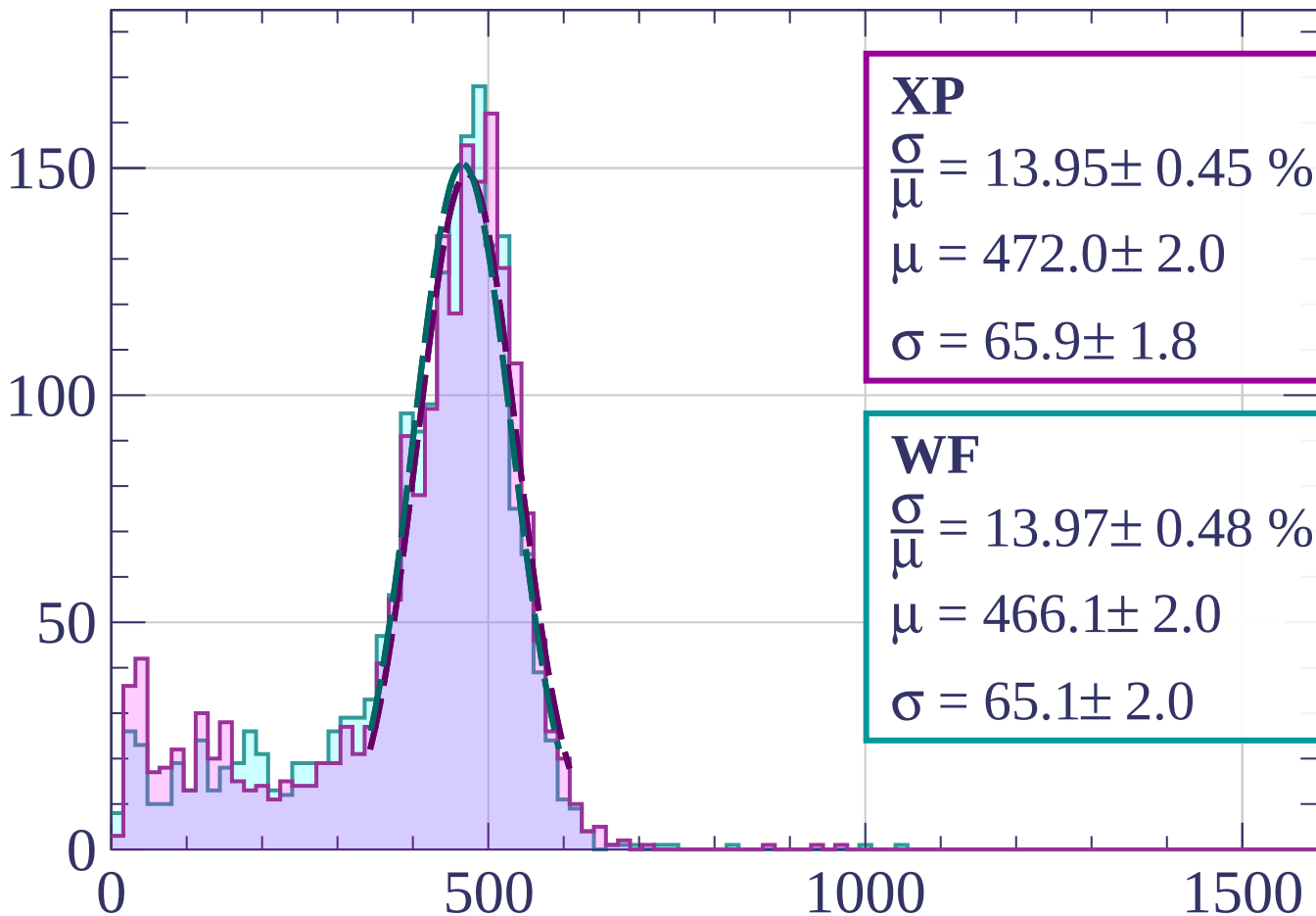
$$\mu = 481.2 \pm 1.6$$

$$\sigma = 63.4 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $66 < \theta < 69$

Count



XP

$$\frac{\sigma}{\mu} = 13.95 \pm 0.45 \%$$

$$\mu = 472.0 \pm 2.0$$

$$\sigma = 65.9 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 13.97 \pm 0.48 \%$$

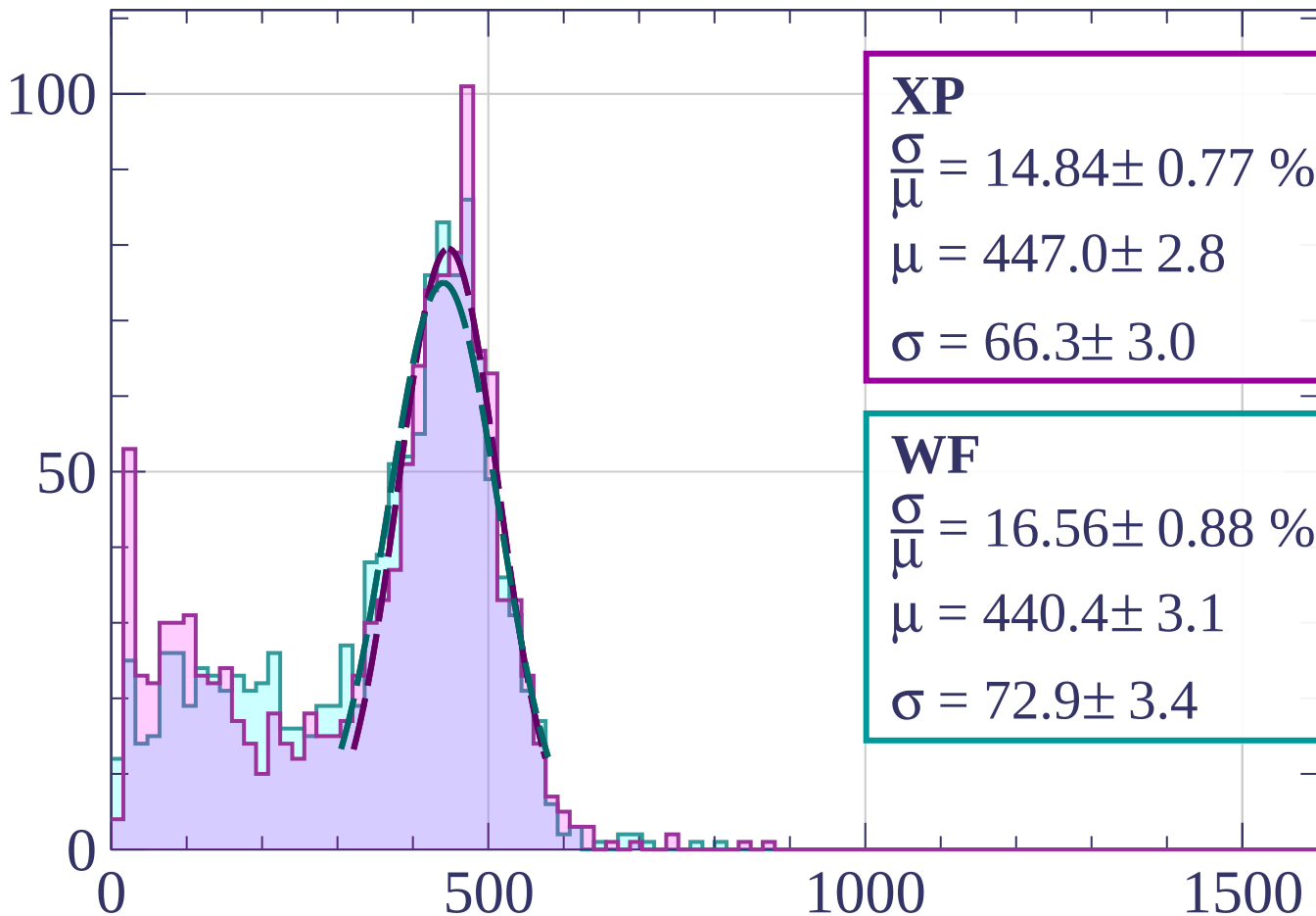
$$\mu = 466.1 \pm 2.0$$

$$\sigma = 65.1 \pm 2.0$$

dE/dx [ADC counts/cm]

Energy loss | $69 < \theta < 72$

Count



XP

$$\frac{\sigma}{\mu} = 14.84 \pm 0.77 \%$$

$$\mu = 447.0 \pm 2.8$$

$$\sigma = 66.3 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 16.56 \pm 0.88 \%$$

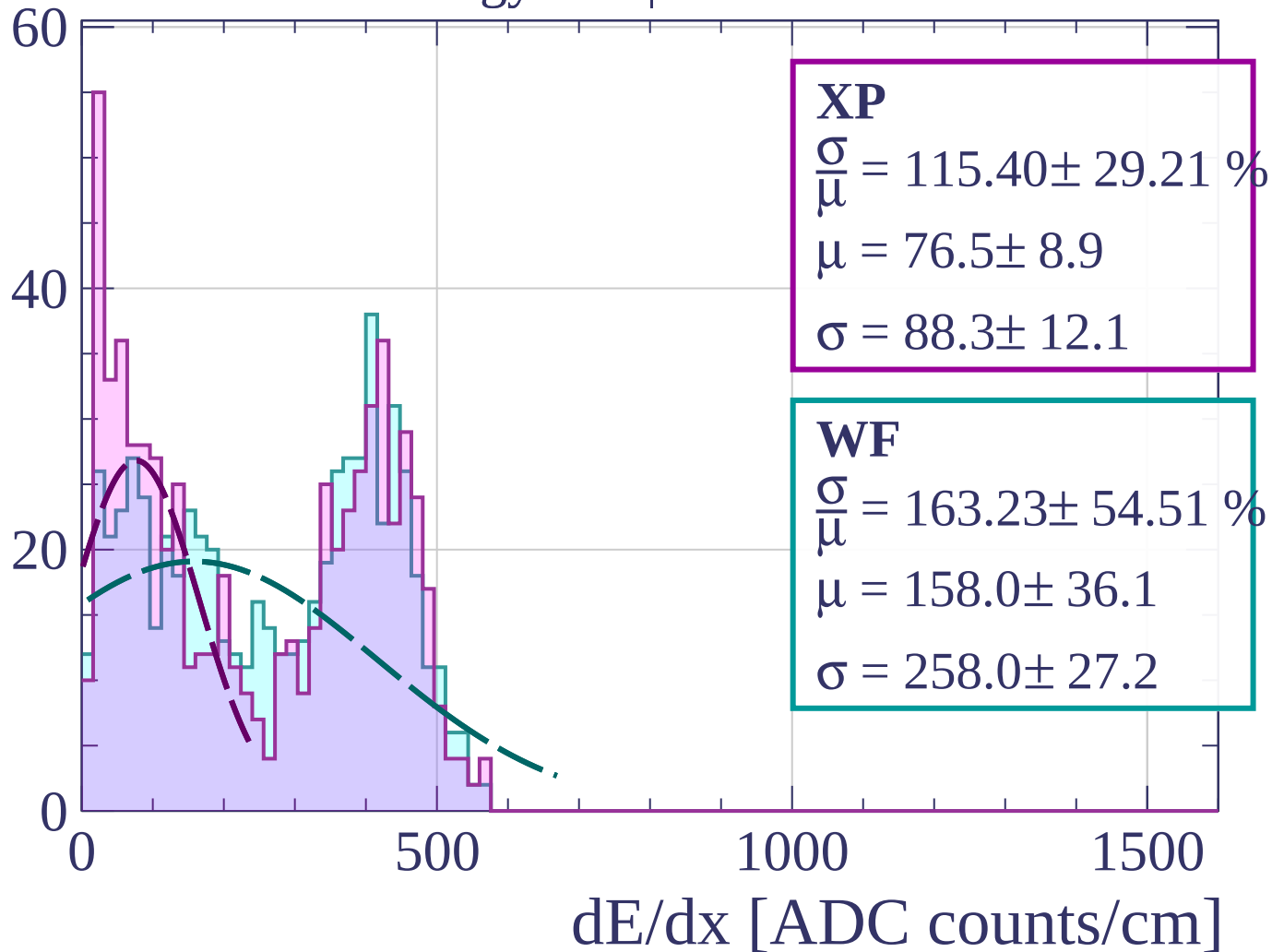
$$\mu = 440.4 \pm 3.1$$

$$\sigma = 72.9 \pm 3.4$$

dE/dx [ADC counts/cm]

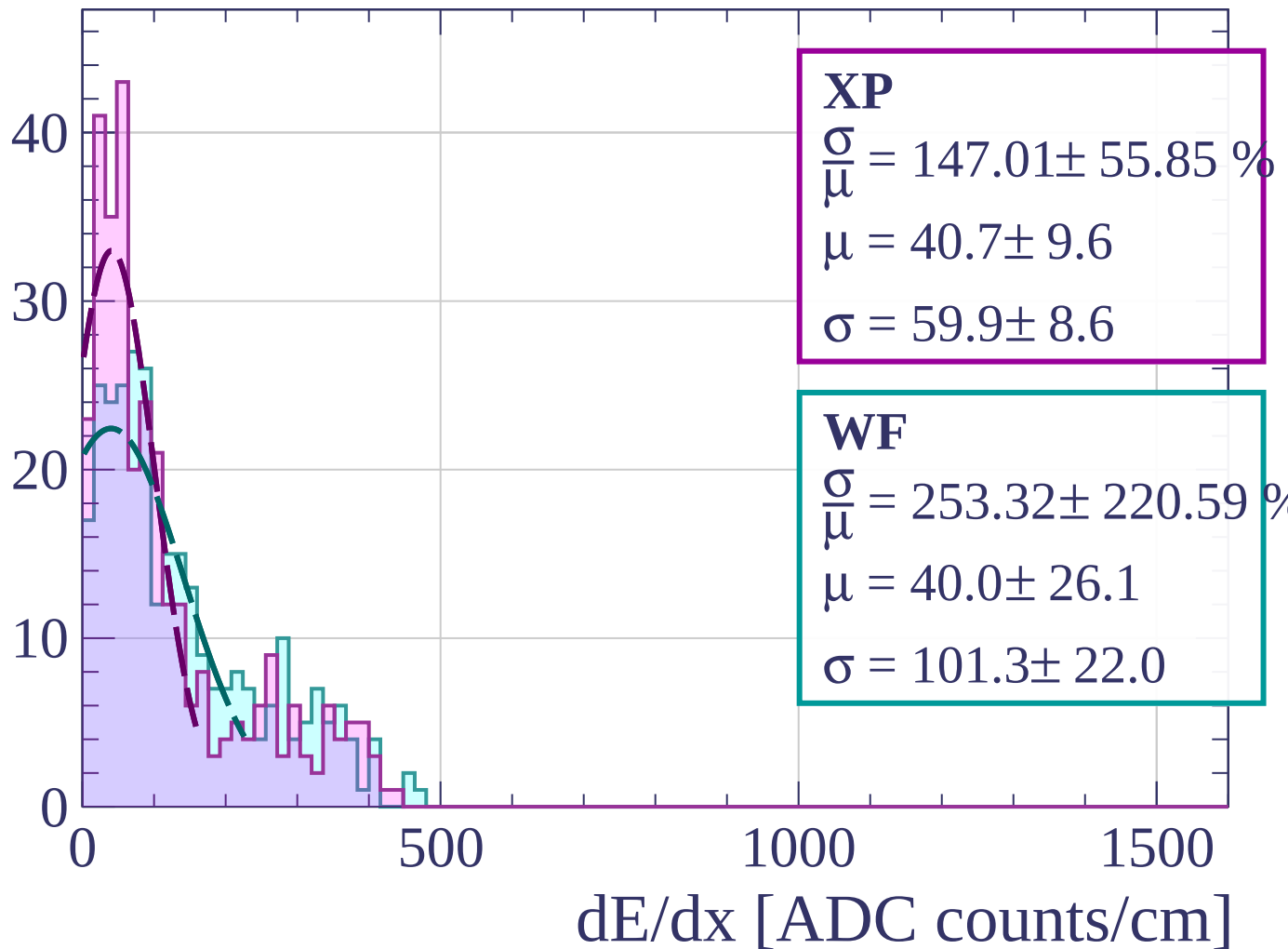
Energy loss | $72 < \theta < 75$

Count



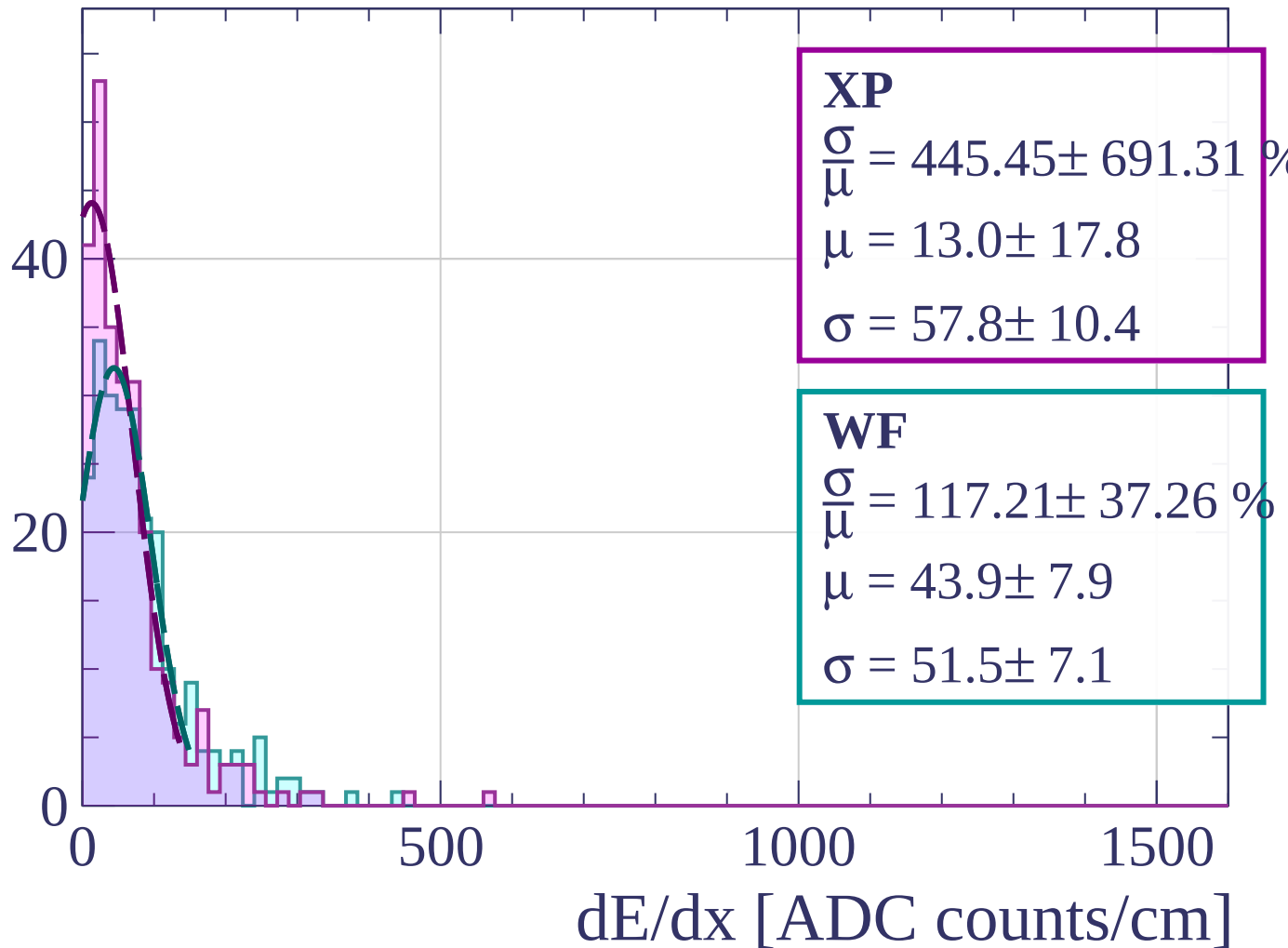
Energy loss | $75 < \theta < 78$

Count



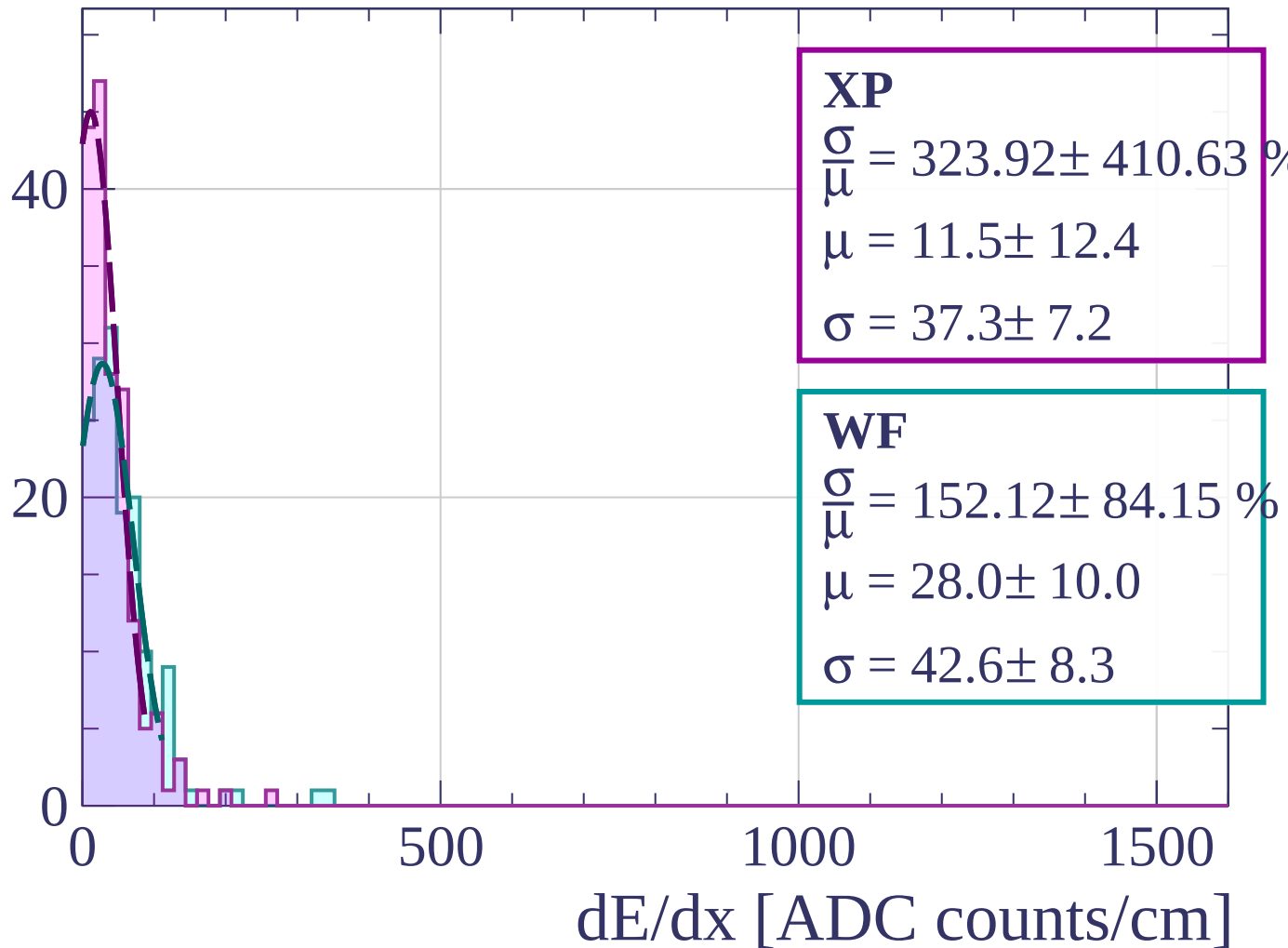
Energy loss | $78 < \theta < 81$

Count



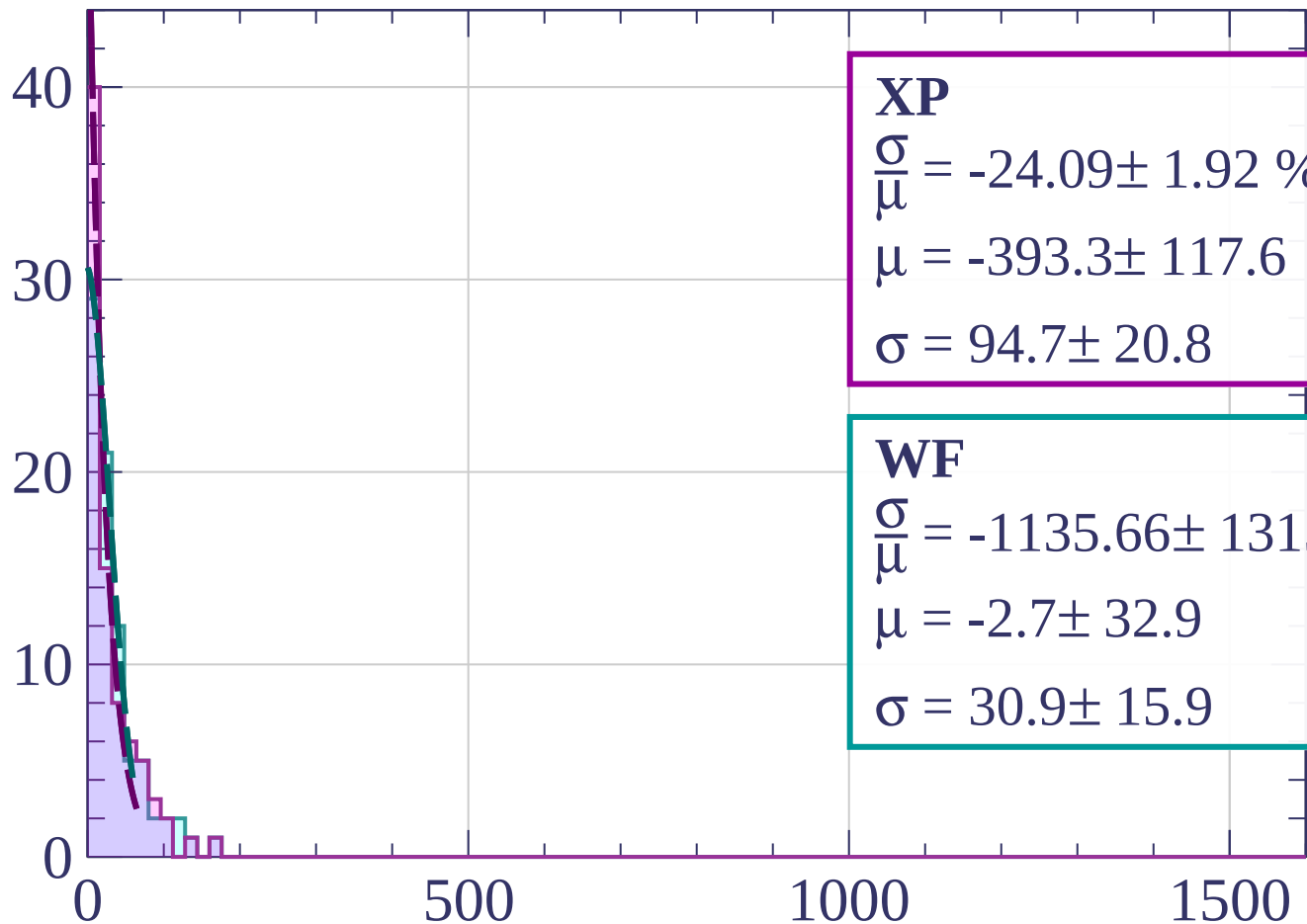
Energy loss | $81 < \theta < 84$

Count



Energy loss | $84 < \theta < 87$

Count



XP

$$\frac{\sigma}{\mu} = -24.09 \pm 1.92 \%$$

$$\mu = -393.3 \pm 117.6$$

$$\sigma = 94.7 \pm 20.8$$

WF

$$\frac{\sigma}{\mu} = -1135.66 \pm 13159.$$

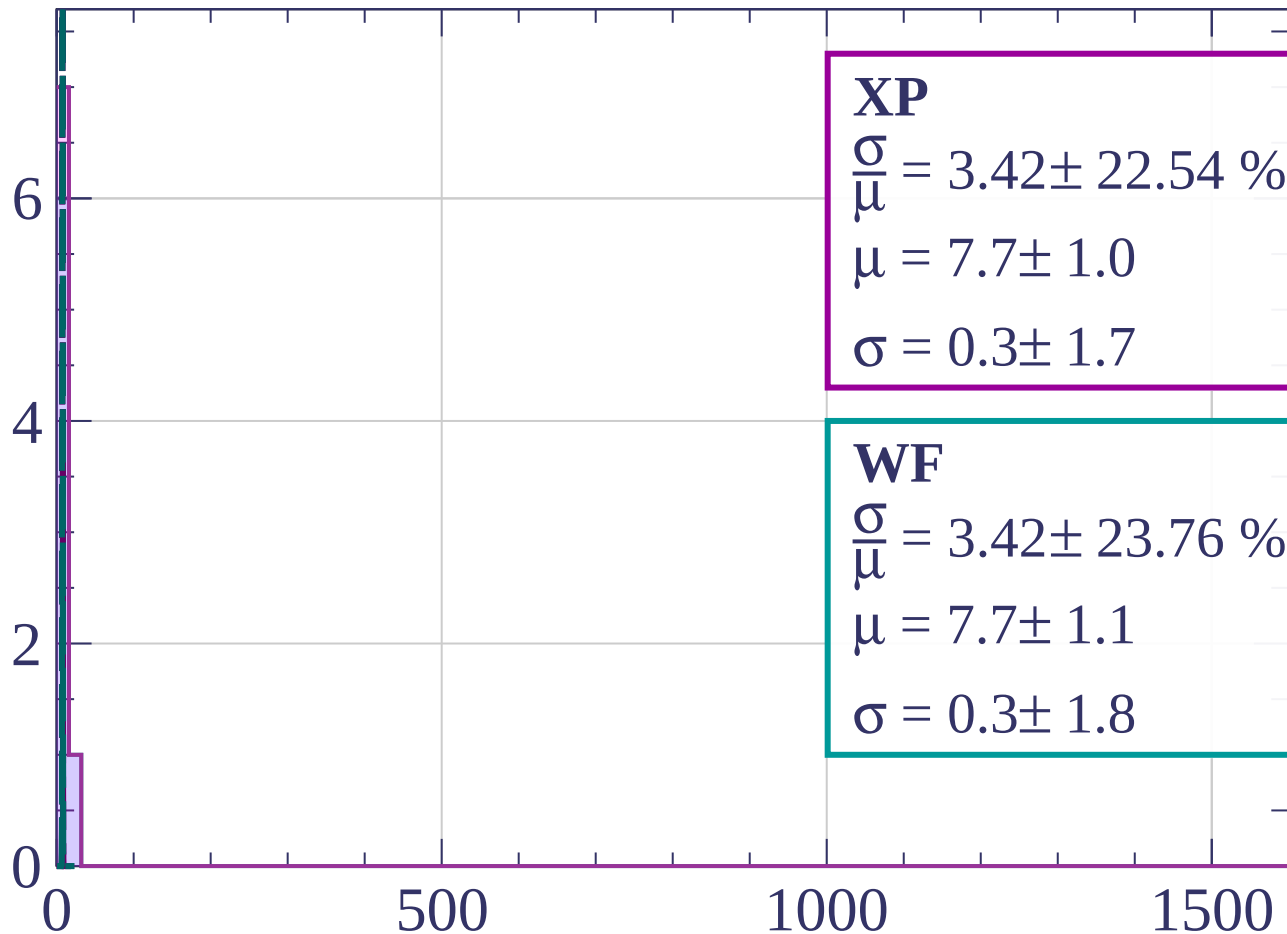
$$\mu = -2.7 \pm 32.9$$

$$\sigma = 30.9 \pm 15.9$$

dE/dx [ADC counts/cm]

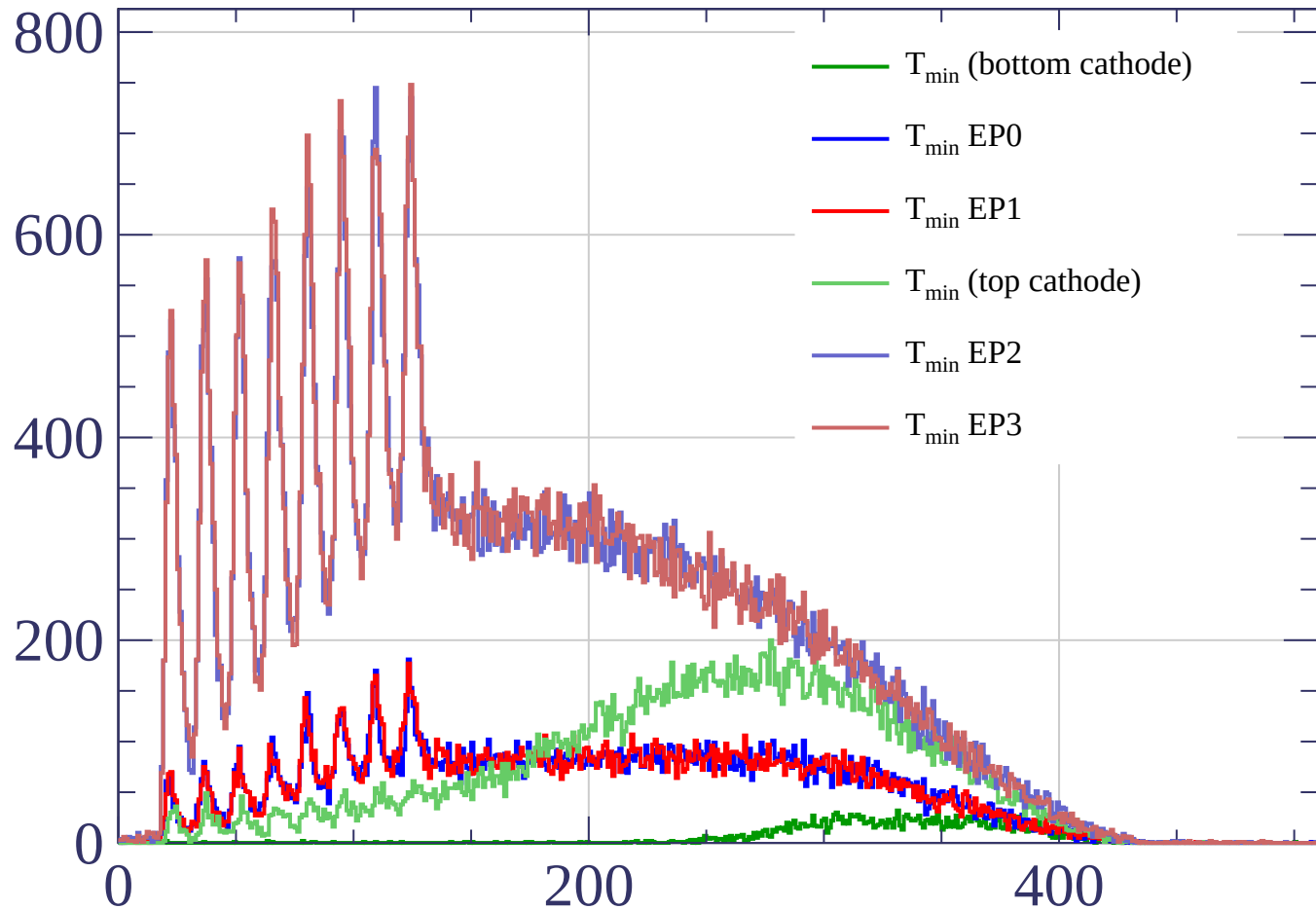
Energy loss | $87 < \theta < 90$

Count

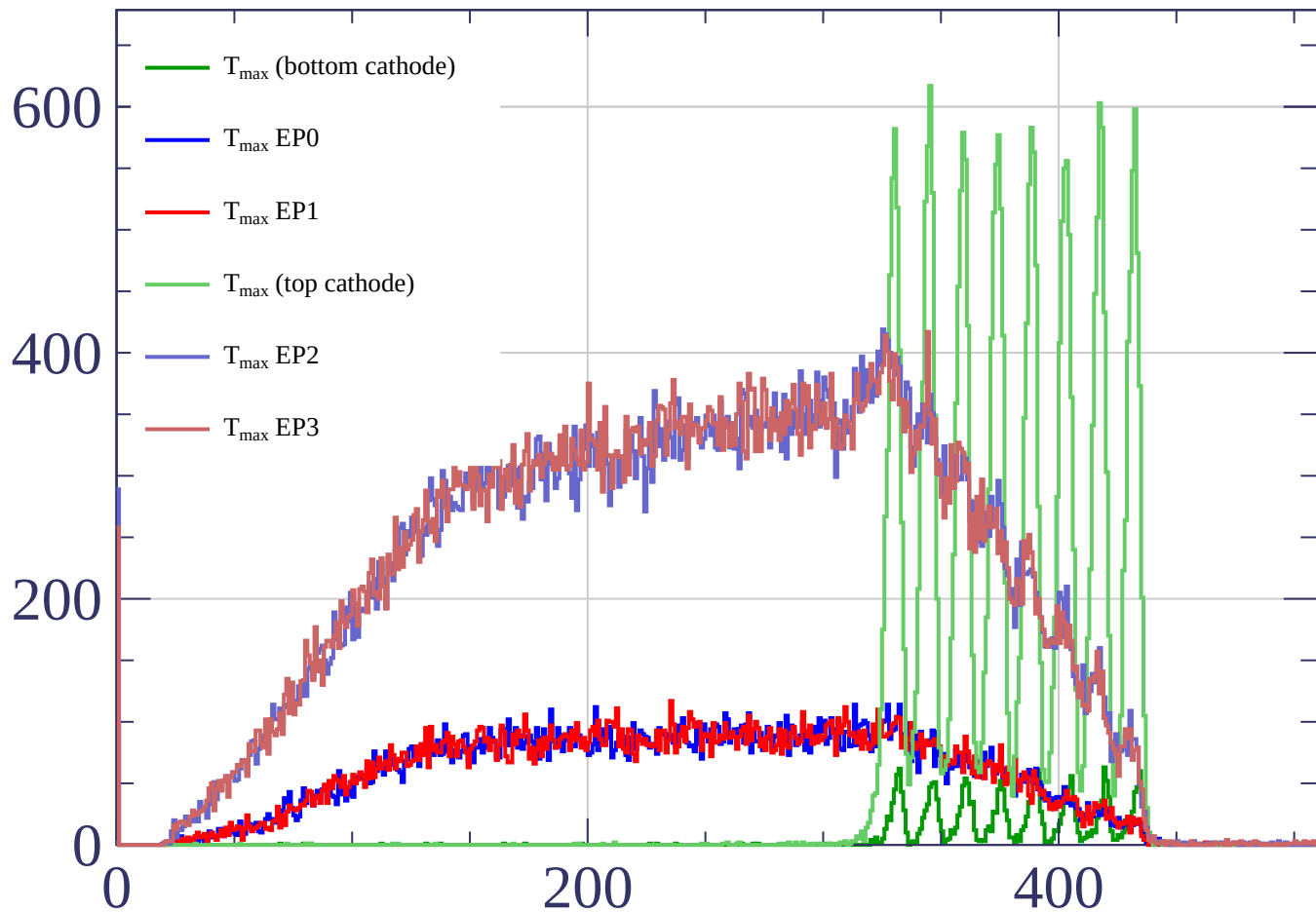


dE/dx [ADC counts/cm]

Count



Count



time bin